

THEORIA

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UVODNA REČ

Osnovan 1976. godine, časopis *Theoria* pripada grupi domaćih filozofskih časopisa sa najdužom tradicijom. Ove godine, *Theoria* obeležava pedeset godina svog postojanja i, shodno tome, posvećuje svoje brojeve različitim filozofskim oblastima. U periodu kada, čini se, postoji sve uža specijalizacija između i unutar različitih naučnih oblasti, *Theoria* nastavlja da bude jedan od časopisa koji ističu prednosti interdisciplinarnog naučnog i filozofskog istraživanja. S tim u vezi, velika mi je čast da budem gostujući urednik ovog broja koji objedinjuje čak tri filozofske oblasti: epistemologiju, filozofiju jezika i filozofiju nauke.

Iako je teško utvrditi kada su ove filozofske oblasti nastale, čini se da možemo ispratiti njihov istorijski razvoj. Termin „epistemologija“ je zvanično formulisao 1854. godine od strane škotskog metafizičara Džejmisa Frederika Fierera (1808-1864), ali koreni ove filozofske oblasti dosežu čak do antičkog perioda u kojem se susrećemo sa dva dobro poznata termina: *episteme* (znanje) i *logos* (misao, nauka, a u nekim širim kontekstima i moć razmišljanja). Shodno tome, epistemologija predstavlja nauku o znanju i, ukoliko postoji ličnost koja se u antičkom periodu najviše povezuje sa ovom naukom, to bi verovatno bio Platon – Sokratov učenik i Aristotelov učitelj. Filozofija jezika je mlađa disciplina koja se razvila tokom 19. i 20. veka, a čije osnivanje je inspirisano revolucionarnim delom Fridriha Ludviga Gotloba Fregea (1848-1925) pod nazivom „O smislu i referenciji“ (*Über Sinn und Bedeutung*, 1892). No, zanimljivo je da su još Platon i Aristotel diskutovali o pitanju odnosa reči prema stvarima u prirodi ili društvenim navikama. Konačno, čini se da je razvoj filozofije jezika pratio i razvoj filozofije nauke. Teško je utvrditi koja od ove dve discipline je „mlađa“, ali možemo reći da je filozofija nauke postala zasebna akademska disciplina u 20. veku pod uticajem Bečkog kruga i logičkih pozitivista. Ipak, kao i u slučaju filozofije jezika, tragovi filozofije nauke se mogu pronaći u antičkom periodu, posebno Aristotelovim radovima o logici, prirodi i načinima na koji stičemo naučno znanje.

Nesumnjivo je da nam je antički period podario mnogo značajnih filozofa čiji radovi su uticali na potonji razvoj filozofije i nauke. Epistemologija, filozofija jezika i filozofija nauke se danas prepoznaju kao odvojene filozofske oblasti, ali ne treba prevideti činjenicu da se njihovo poreklo pronalazi u istom „izvoru“ – antičkoj filozofiji, te da su sve tri discipline usmerene na isti cilj: unapređenje (izvesnosti) našeg znanja. Ostvarenje ovog cilja, čini se, počiva na međusobnoj „saradnji“ ove tri oblasti. Iako se filozofija nauke fokusira na razmatranje različitih naučnih teorija, takva istraživanja obuhvataju i nezaobilazna pitanja epistemologije i filozofije jezika: koje su osnovne pretpostavke određene naučne teorije? Kakav je njen metod? Koliko je određeni naučni metod pouzdan? Šta znači određeni naučni termin i da li se njegovo značenje iscrpljuje u njegovoj referenciji? Pitanja je mnogo, a odgovori se ne mogu naći unutar jedne zasebne naučno-filozofske oblasti. Možemo reći da se upravo u ovome ogleda

značaj *Theorie* kao interdisciplinarnog časopisa i ovog broja: izgleda da se odgovori na fundamentalna naučna i filozofska pitanja pronalaze u povezivanju i ujedinjenju različitih naučno-filozofskih oblasti. Ovaj broj će, nadam se, pokazati kako se pitanja epistemologije, filozofije jezika i filozofije nauke prepliću, a odgovori na njih zajedno unapređuju našu kritičku misao i, naravno, znanje.

Gostujući urednik

Bogdana Stamenković Jajčević

**HOLISTIČKA SINTEZA: EPISTEMOLOGIJA, FILOZOFIJA JEZIKA
I FILOZOFIJA NAUKE**



Živan Lazović

MODELI U NAUČNOJ PRAKSI

Apstrakt: Glavna tema ovog rada jesu uloge koje modeli igraju u naučnoj praksi. Iako su mehanistički modeli bili upotrebljavani još od nastanka njutnovske klasične mehanike u 17. veku, filozofi nauke se nisu slagali oko njihovog značaja za naučne teorije. Logički empiristi između dva rata težili su racionalnoj rekonstrukciji nauke i izgradnji normativne metodologije nauka u kojoj modeli nisu imali svoje mesto. Tokom 50-ih godina prošlog veka odigrao se kopernikanski obrt u filozofiji nauke, pomeranje fokusa sa racionalne rekonstrukcije na aktualnu naučnu praksu u kojoj su modeli široko prisutni. U radu su prikazani način njihove izgradnje, osnovne karakteristike (medijacija između apstraktnih naučnih teorija i konkretnih fizičkih fenomena, interpretativna i reprezentujuća funkcija) i kognitivni značaj (deskriptivna adekvatnost, eksplanatorna i prediktivna moć, doprinos razumevanju i istraživanju). Na nizu primera pokazano je da su naučni modeli ipak toliko raznovrsni da je gotovo nemoguće formirati objedinjujuću teoriju o njima - oni ne dele neka bitna zajednička svojstva, već su povezani mrežom vitgenštajnovski shvaćenih "porodičnih" sličnosti.

Ključne reči: naučna praksa, teorije, modeli, posredovanje, interpretacija, reprezentovanje, porodične sličnosti

U jednom razgovoru koji je 1948. vodio sa svojim učenikom M. Drurijem (M. Drury), L. Vitgenštajn (L. Wittgenstein) je izneo sledeći utisak o Hegelovoj filozofiji: „Čini mi se da Hegel uvek želi da kaže kako su stvari koje izgledaju različito zapravo iste. Mene, pak, zanima da pokažem kako su stvari koje izgledaju isto zapravo različite.” dodavši da je pomišljao da kao moto za *Filozofska istraživanja* uzme citat iz Šekspirovog *Kralja Lira*: „Naučiću vas razlikama”.¹ Pokušavajući da utvrde univerzalno važeće norme naučnog istraživanja i izgrade jednu opštu metodologiju nauka, tradicionalni filozofi nauke su ispoljavali hegelovsku ambiciju. U savremenoj filozofiji nauke je, međutim, sve uticajniji vitgenštajnovski stav da su naučne teorije i naučna praksa mnogo složenije nego što se to na prvi pogled čini. Taj stav svoju potvrdu nalazi u raznovrsnim modelima koje naučnici koriste kako bi produbili, primenili, proverili i razvili svoje teorijske uvide u prirodu sveta koji nas okružuje.²

¹Drury 1996: 157.

²Osim donekle u završnom 7. odeljku, ovaj članak je pisan bez velikih teorijskih ambicija. Glavni cilj mi je bio da ukratko rekonstruišem istorijat rasprave o ulozi modela u nauci i, navodeći primere, prikažem reprezentativne tipove modela, njihove funkcije i karakteristike. U tome sam se uglavnom oslanjao na izvore dostupne u opsežnoj savremenoj literaturi koja se bavi ovom tematikom.

1.

Podstaknuta Njutnovim kapitalnim delom *Philosophiae Naturalis Principia Mathematica* (1687), u Engleskoj je među fizičarima sve do kraja 19. veka dominirala upotreba mehanističkih modela. Značaj koji im je pripisivan izrazio je lord Kelvin u svom čuvenom geslu: "Provera da li u fizici razumemo neku pojavu svodi se na to da li možemo da izgradimo mehanistički model za nju!"³ Prve rasprave o ulozi modela u nauci začete su potkraj tog veka kada se pokazalo da mehanistički modeli etra nisu u stanju da objasne prirodu svetlosti i elektromagnetizma. Ubrzo je došlo do razilaženja između, uslovno rečeno, kontinentalnog i ostrvskog pristupa. U svom poznatom delu *La Théorie physique* (1906) Pjer Dijem (Pierre Duhem) je tvrdio da se engleski fizičari oslanjaju na mehanističke modele zbog toga što su skloni imaginaciji i vizuelnom prikazivanju uzročnih mehanizama koji su u osnovi prirodnih zbivanja. Dijem nije osporavao upotrebu modela, ali je smatrao da oni nisu logički neophodni za nastanak ili strukturu naučnih teorija već su samo pomoćno, psihološki korisno sredstvo za njihovu ilustraciju. Poznato je, recimo, da je model vodenih talasa poslužio za objašnjenje prirode zvuka ali su, prema tom tumačenju, do matematički izražene talasne teorije zvuka naučnici mogli da dođu čak i da u prirodi uopšte nema vodenih talasa. Na drugoj strani, pristup kontinentalnih fizičara Dijem je okarakterisao kao formalistički, sistematičan i koherentan jer oni naučne teorije tumače po uzoru na Euklidovu geometriju, kao formalne deduktivne sisteme apstraktnih formula ili rečenica unutar kojih fundamentalni principi imaju status aksioma, iz njih se dedukuju hipoteze opadajućeg stepena opštosti i, kao poslednja karika u tom logičkom lancu, empirijski proverljivi opservacioni iskazi.

Dijemovom sudu suprotstavio se engleski fizičar i filozof N. R. Kembel (N. R. Campbell) u knjizi *Physics, the Elements* (1920). On je smatrao da modeli počivaju na analogijama koje "nisu puka 'pomagala' za uspostavljanje teorije", nego su "u potpunosti suštinski deo teorija..."⁴, naglasivši posebno dve stvari: prvo, pošto nam je u naučnom istraživanju cilj da objasnimo prirodne fenomene, za tu svrhu nam je pored matematičke forme neophodna interpretacija teorijskih zakona, što se postiže upravo modelima koji reprezentuju pojave; i drugo, teorije nisu dovršen, statički inventar ljudskog duha, one imaju dinamički karakter, razvijaju se i menjaju tokom novih istraživanja, u čemu modeli takođe igraju bitnu ulogu.

Dijemovsko gledište bilo je usvojeno od strane većine tadašnjih filozofa nauke, posebno zastupnika logičkog empirizma čija glavna ambicija je bila da racionalno rekonstruišu strukturu naučnog istraživanja i izgrade opštu metodologiju nauka. Jedan

³Citat preuzet iz: Bailer-Jones 2009: 29.

⁴Citat preuzet iz: Hesse 1966: 4. Hese navodi doprinos modela bilijarskih kugli razvoju kinetičke teorije gasova: sagledavanje sistema kao što je idealan gas u svetlu tog modela podstaklo je kreativnu imaginaciju naučnika, rodilo ideje za teorijski napredak i otvorilo mogućnost da se model po analogiji proširi na druge slične sisteme.

od vodećih predstavnika tog pravca, R. Carnap (R. Carnap), složio sa Dijemom da konstruisanje modela ima "... samo estetsku ili didaktičku ili u najboljem slučaju heurističku vrednost, ali uopšte nije bitno za uspešnu primenu fizičke teorije".⁵ U toj tradiciji, teorije su shvaćene kao formalni sistemi koji, uz osnovne zakone, matematičke formule i logička pravila neophodna za dedukciju, uključuju iskaze koji imaju empirijsko poreklo (sadržaj) i podložni su proveru. Veza između teorije i fizičkog sveta obezbeđena je semantički, pomoću pravila korespondencije koja dodeljuju empirijski sadržaj teorijskim terminima tako što specifikuju iskustvene ili eksperimentalne okolnosti u kojima su oni primenljivi na fizičke objekte. Kada se upuštamo u naučno istraživanje fenomena koje želimo da objasnimo, prvo prikupljamo i sistematizujemo podatke o njima, izvodimo univerzalnije iskaze i formulišemo hipoteze relativno visokog stepena opštosti. Opravdanje teorije isključivo je stvar logike jer se, zahvaljujući pomenutim pravilima korespondencije, iz opštih zakona i hipoteza deduktivno izvode opservacioni iskazi koji se iskustveno proveravaju.

To je, ukratko, hipotetičko-deduktivni obrazac koji bi, po pretpostavci tradicionalnih filozofa nauke, svako naučno istraživanje trebalo da sledi. Metodološki gledano, nisu neophodna nikakva dodatna sredstva koja bi posredovala u izgradnji ili proveru naučnih teorija. Primer iz istorije nauke kojim se taj obrazac može ilustrovati jeste istraživanje putanja planeta u solarnom sistemu. U prvom koraku na osnovu sistematskog posmatranja formulisan je opis kretanja planeta oko Sunca. Da bi se objasnilo zašto se planete tako kreću, uvedena je hipoteza da interakcija između njih i Sunca podleže zakonima klasične mehanike (Njutnovi zakona kretanja i zakon univerzalne gravitacije). U odnosu na prikupljenu empirijsku evidenciju hipoteza je tačna ako se, uzimajući u obzir (a) masu Sunca i bilo koje konkretne planete i (b) uzajamno dejstvo njihovih gravitacionih sila, predviđena putanja (dedukovane opservacione posledice) podudara sa opaženom (stvarnom) putanjom planete.

V. Kvajn (W. Quine) je među prvim "iznutra" načeo logičko-empiristički projekat, ukazao na njegov krah⁶ i pozvao filozofe da, namesto zaludnih pokušaja racionalne rekonstrukcije, proučavaju stvarne konstrukcije naučnog znanja. Tom pozivu odazvali su se mnogi savremeni filozofi. Analizirajući aktualnu naučnu praksu, oni su došli do argumenata za spoljašnju kritiku tradicionalnog pristupa: naučne teorije imaju dinamičan karakter, uglavnom ne mogu da se aksiomatizuju, retko zadovoljavaju hipotetičko-deduktivni obrazac, tipovi naučnih objašnjenja variraju zavisno od oblasti istraživanja i metodoloških pretpostavki itd.⁷

Po svedočenju D. Bajler-Džons (D. Bailer-Jones) odziv je "rezultirao proučava-

⁵ Citat je preuzet iz: Gelfert 2016: 11.

⁶ U člancima "Two Dogmas of Empiricism" (1951) i "Epistemology Naturalized" (1969).

⁷ Primer su brojne biološke, ekonomske ili socijalne teorije - zakonitosti koje su prisutne u živom svetu ili društvenim zajednicama nisu univerzalne ni nužne, fenomeni kojima se bave veoma su kompleksni i promenljivi, njihov odnos prema iskustvu složeniji je nego što su logički empiristi tvrdili.

njem naučnih modela kao delom 'humanističke' (ili naturalističke) epistemologije koja se usredsredila na naučne istraživače i njihovu potragu za znanjem".⁸ Već početkom 50-ih godina prošlog veka, R. Brejtvajt (R. Braithwaite), M. Hese (M. Hesse) i E. Hatén (E. Hutten) su posvetili pažnju naučnoj aktivnosti izgradnje, primene i promene teorija. Oni su se i dalje držali stava da su teorije piramidalne hipotetičko-deduktivne građevine - u podnožju su opservacione datosti, pri vrhu opšte hipoteze - koje se grade "odozdo naviše" i proveravaju "odozgo naniže", sučeljavanjem sa empirijskim podacima. Nisu osporavali ni tezu da je za proveru teorija neophodno da se apstraktnim terminima i zakonima osigura interpretacija, ali su umesto pravilima korespondencije tu interpretativnu ulogu dodelili modelima. Dok ih je Brejtvajt (1953) shvatao formalno, uglavnom u duhu logičkog empirizma, Hese (1953) i Hatén (1954) su detaljnije analizirali aktivnost i potrebe naučnika, ilustrujući interpretativnu funkciju modela primerima iz naučne prakse. Njihovi radovi pokrenuli su niz zanimljivih pitanja - korišćenje analogija i metafora, mogućnost čisto matematičkih modela, veza sa eksperimentima i kreativnim formulisanjem novih hipoteza i dr. - koja su početkom 60-ih godina pobudila interesovanje ne samo za logičke nego i za druge vrste modela.

2.

U okviru dijemovske tradicije iskristalisala su se dva shvatanja teorija, sintaksičko i semantičko. Sve do kraja sedme decenije prošlog veka uticajnije je bilo sintaksičko gledište koje teorije tumači kao aksiomatizovane lingvističke sisteme, odnosno skupove rečenica povezanih logičkim i semantičkim relacijama. Prema tom shvatanju, modeli mogu da posluže za interpretaciju teorijskih termina i zakona samo u logičkom smislu, s tim što je ta uloga sporedna, nebitna za status same teorije. Tako su, recimo, fizičari matematički formulisane rečenice kinetičke teorije gasova interpretirali pomoću modela bilijarskih kugli u kojem se one pokazuju kao istinite. Model nije suštinski deo teorije - štaviše, nju je moguće interpretirati i pomoću nekih drugih modela - ali je didaktički koristan jer nam apstraktne formule približava na primeru konkretnih objekata sa kojima smo upoznati, ili je heurističko sredstvo koje nam daje naznake kakav uzročni mehanizam reguliše ponašanje čestica ne samo u gasu već i u tečnim ili čvrstim telima.⁹

Same teorije su, međutim, skupovi aksioma i njihovih deduktivnih posledica. Jedan od glavnih zastupnika sintaksičkog gledišta, Hempel, smatrao je da aksiome čine unutrašnji teorijski principi koji utvrđuju relacije između bazičnih pojmova i premošćujući principi koji apstraktne teorijske pojmove povezuju sa terminima sa kojima smo već iskustveno upoznati. Premošćujući principi su neophodni za primenu i proveru teorije diktirane, uprošćeno rečeno, teoremama oblika "Ako nastupi situacija u kojoj su ispunjeni početni uslovi za javljanje A, desiće se B". Teorijski zakoni

⁸Bailer-Jones 1999: 38.

⁹Karnap i Hempel (Hempel) su isticali didaktički, a Brejtvajt i Nejgel (Nagel) heuristički momenat.

oblika "Za svako x , ako Fx onda Gx " važe univerzalno, nezavisno od toga da li imamo premošćujuće principe na osnovu kojih bismo mogli da predvidimo šta će se desiti na pojavnom planu, dok za primenu teorije moramo da raspoložemo opisima koji odgovaraju opaženim početnim uslovima. Klasična mehanika, recimo, sadrži zakone koji između ostalog opisuju kretanje nekog manjeg objekta na koji deluje gravitaciona sila većeg. Ako taj opis pokušamo da primenimo na kretanje bilo koje planete oko Sunca, nećemo biti načisto koja teorema važi zato što se osobine koje njutnovska teorija specifikuje ne podudaraju sa time kako u običnom jeziku opisujemo opaženo kretanje planeta. Odgovarajući Njutnov zakon glasi "Ako na telo u pokretu mase m deluje sila GmM/r^2 , onda će se to telo kretati eliptičnom putanjom $1/r = 1 + e \cos \theta$ (e je odstupanje od centra)" - da bismo uočili njegovu relevantnost za sistem koji posmatramo potreban nam je premošćujući princip koji nam kaže da je GmM/r^2 veličina gravitacione sile između većeg tela mase M i manjeg tela mase m . Bez tog principa, oslanjajući se samo na teoriju, ne bismo mogli da predvidimo eliptičnu putanju planeta oko Sunca.¹⁰

Pristalice semantičkog gledišta svojim protivnicima zameraju to što teorije poistovećuju sa njihovim jezičkim formulacijama, gubeći pritom iz vida da je u nauci glavni cilj objašnjenje pojava u realnom svetu. Čisto formalni sistemi imaju ograničenu ekspresivnu moć i nisu u stanju da ispune taj cilj, pa je neophodno da budu dopunjeni odgovarajućim nejezičkim strukturama, odnosno modelima koji ih interpretiraju. Koncept interpretacije preuzima se iz logike. Teorija kao formalni sistem sastoji se od formula apstraktnog kalkulusa, valjano formiranih rečenica koje, da bi se učinile smislenim i procenjivale kao istinite ili lažne, moraju da se interpretiraju u nekom domenu objekata. Protumačeni kao relacije strukture konkretnih objekata, modeli nemaju lingvistički karakter; oni obezbeđuju značenje za promenljive, veznike i konstante kalkulusa, zahvaljujući čemu mogu teorijske iskaze da čine istinitim. Iskaz kinetičke teorije gasova " P (pritisak) $\cong V$ (zapremina) $= 1/3n$ (broj molekula) $\cong m$ (masa molekula) $\cong c^2$ (srednji kvadratni koren brzine molekula)" je, na primer, tačan za roj čestičnih molekula. Interpretacija formule " $pv = 1/3nmc^2$ " kao rečenice koja omogućuje navedeni iskaz iziskuje da se promenljivama i konstantama dodele značenja (vrednosti) iz "sveta" zamišljenih materijalnih entiteta, odnosno modela. U raznim slučajevima interpretacija daje različite iskaze iste forme; ako je neki od njih lažan, onda to znači da predloženi model nije dobar.

Modeli u tom smislu predstavljaju interpretacije skupa neinterpretiranih teorijskih termina i zakona. Odnos između apstraktnog kalkulusa, interpretiranih rečenica i semantičkog modela isti je u fizici (i ostalim empirijskim naukama) kao i u logici ili matematici.¹¹ P. Supis (P. Suppis) podvlači da je "teorija lingvistički entitet koji

¹⁰Primer iz: Cartwright 1999: 245–6.

¹¹Hare (Harré) primećuje da je razlika u tome što u fizici po pravilu model prethodi izgradnji teorije i njenoj matematičkoj formulaciji. Tako je model za kinetičku teoriju po svojoj prilici prvo formulisao F. Bejkon (F. Bacon), a matematički ga je kasnije izrazio Maksvel. Takode, fizičari uzimaju u obzir neka

se sastoji od skupa rečenica, dok su modeli nelingvistički entiteti koji zadovoljavaju tu teoriju”¹² i dodaje da je model formalno gledano ”skupovno-teorijski entitet koji je neka vrsta uređenog t -pla sastavljenog od skupa objekata i relacija i operacija koje se vrše nad tim objektima”.¹³ Tako shvaćeni, modeli nisu izdvojene samostalne konstrukcije; metaforično govoreći, oni su gradivni blokovi teorija.

Kada je reč o primeni teorija, semantičko gledište se ne razlikuje puno od sintaksičkog. Da bismo neku naučnu teoriju primenili na stvarni svet, oslanjamo se na modele i posmatramo gde se početni uslovi precizirani u njima podudaraju sa početnim uslovima pojave koju istražujemo. U igri su ponovo premošćujući principi. Prikazujući Girovo (Gieryn) stanovište, Morgan (Morgan) konstatuje da opšti fizički zakoni - npr. Njutnovi u klasičnoj mehanici ili Šredingerova jednačina u kvantnoj mehanici - ”nisu opisi realnih sistema već su pre sastavni deo karakterizacije modela koji onda mogu da reprezentuju različite vrste realnih sistema. Ali zakon $F = ma$ sam po sebi ne definiše model ničega; potrebno je da dodamo specifične funkcije sile, granične uslove, aproksimacije itd. Tek kad su mu ti uslovi pridodati model može da se upoređuje sa nekim realnim sistemom.”¹⁴

Prednost semantičkog tumačenja u odnosu na sintaksičko je u tome što logički koncipirani modeli kao nejezički entiteti bolje ekspliciraju strukturu i primenu naučnih teorija. Ipak, naučnici u praksi koriste mnoštvo modela koji nemaju logički karakter. S tim u vezi, N. Kartrajt (N. Cartwright) duhovito primećuje da oba gledišta, fokusirana na relaciju teorija-pojave, primenu teorije na fizičke objekte zamišljaju po uzoru na samouslužne aparate: ako ubacite odgovarajuću novčanicu ili prislonite bankovnu karticu pa pritisnete dugme za željeni artikal (input), posle par sekundi čangrljanja artikal će izaći iz mašine kao gotov proizvod (output) - ”kao Atina iz Zevsove glave”. Njena zamerka je da takva zamisao nije u skladu sa aktualnom naučnom praksom jer ne ostavlja prostor za kreativnost - građenje novih teorija je dopušteno, ali je udeo ljudske aktivnosti svoden na dva ne baš stvaralačka koraka. Prvi je pažljivo posmatranje pojava, beleženje podataka i kombinovanje onih koji se mogu ubaciti u automat kao input; pošto ipak nismo u stanju da konstruišemo mašinu koja će nam prosto isporučiti teorijski očekivane autpute, u drugom koraku krećemo u zamorne dedukcije ili dovtljive aproksimacije kako bismo dobili bar faksimil željenog outputa. Zaključak da su oba koraka lišena istinskog stvaralaštva Kartrajtova ilustruje na otkriću fenomena superprovodljivosti: ”Proizvesti model neke novotkrivene pojave kao što je superprovodljivost neverovatno je teška i kreativna aktivnost. Tako su dobijene Nobelove nagrade.”¹⁵

spoljašnja, empirijska ograničenja zato što bi model trebalo da odgovara opaženim fizičkim sistemima. (Harré, p. 22)

¹² Suppes 1961: 166.

¹³ *Ibid.*, p. 167.

¹⁴ Morgan *et al.* (eds.) 1999: 5.

¹⁵ Cartwright 1999: 247. Za otkriće superprovodljivosti 1911. godine, holandski fizičar H. K. Onnes (H.

Naučna praksa nam otkriva da su formiranje, razvoj i primena teorija daleko složeniji i raznolikiji procesi nego što se u dijemovskoj tradiciji pretpostavljalo. Slikovita potvrda te činjenice jeste široka zastupljenost raznih tipova modela, kako u prirodnim tako i u društvenim naukama.¹⁶ Ne dovodeći direktno u pitanje ambiciju tradicionalnih filozofa nauke, D. Bajler-Džons konstatuje da je jedna važna lekcija iz istorije filozofije nauke "... da analiza nauke iziskuje više od racionalne rekonstrukcije i retrospektivnog opravdanja naučnog znanja" i da "proučavanje naučne prakse razotkriva značaj koji modeli imaju u naučnom procesu."¹⁷ Slično zapažanje iznosi A. Gelfert (A. Gelfert): "Moderna nauka i tehnologija prožeti su modelima - u toj meri da teško možemo da zamislimo na šta bi moderni naučni svet ličio bez upotrebe modela... Modeli su, dakle, svugde oko nas, kako u prirodnim tako i u društvenim naukama, pa bi svaki pokušaj razumevanja naučnog procesa trebalo da uzme u obzir i razjasni tu bazičnu činjenicu o naučnoj praksi."¹⁸ Nešto direktniji i isključiviji je fon Nojman (von Neumann): "Nauke ne pokušavaju da objasne, one teško da teže čak i tome da rastumače, one uglavnom konstruišu modele."¹⁹ Konačno, sa stavom da je konstrukcija teorija pre svega gradnja modela slaže se i Harei koji je okretanje od formalno shvaćenih teorija aktualnoj naučnoj praksi okvalifikovao kao svojevrsni kopernikanski obrt u filozofiji nauke.²⁰

Zaista, modeli u nauci su brojni i toliko raznovrsni da se ne mogu svesti na interpretaciju deduktivne strukture teorija. Pored apstraktnih modela koji zadovoljavaju formalne zahteve (matematički ili kompjutacioni modeli), ima konkretnih, najčešće materijalnih modela koji su izrađeni uz dosta truda, dovijanja, ponekad čak i improvizacije kao što je slučaj sa modelom brane u delti San Franciska koji ćemo nešto kasnije prikazati. Većina njih ima reprezentujući karakter koji se ogleda u tome što prikazuju neke izdvojene, za teorijsko objašnjenje relevantne segmente ili aspekte fenomena koji su predmet istraživanja. Reprezentujuću ulogu naučnih modela svakako valja razlikovati od pomenute interpretativne funkcije prisutne kod logičkih modela. Ipak, ponekad se dešava da jedan isti model igra obe uloge, kao što je slučaj sa njutnovskim imaginarnim modelom kretanja planeta u kojem dva savršeno sferična homogena

K. Onnes) je 1913. dobio Nobelovu nagradu. Kreativnost se ne manifestuje samo u stvaranju novih teorija. Postojeće, prihvaćene teorije čije deduktivne posledice su bitne za izgradnju modela koji reprezentuju pojave, koliko god bile uspešne, podložne su poboljšanjima i promenama (koje ponekad mogu biti radikalne u Kunovom smislu) u svetlu novih opaženih slučajeva, što iziskuje prilagođavanje modela posebnim ograničenjima koje ti slučajevi sa sobom nose.

¹⁶Pomenimo ovde samo neke: njutnovski model kretanja planeta oko Sunca, model molekula gasa kao bilijarskih kugli, Borov model atoma, model DNK kao dvostruke isprepletene trake, skalarni modeli u inženjerskoj tehnologiji, ekonomski modeli racionalnih individua, model opšte cirkulacije u klimatskim naukama i mnogi drugi.

¹⁷Bailer-Jones 2009: 15.

¹⁸Gelfert 2016: v.

¹⁹von Neumann 1961: 492.

²⁰Rothbart, ed. 2004: 10.

tela - položaj krupnijeg je fiksiran dok se manje kreće - u potpuno praznom prostoru i bez ikakvog spoljnog uticaja deluju jedno na drugo svojim gravitacionim silama. Model je istovremeno i interpretativan u logičkom smislu i reprezentujući u odnosu na fizički sistem. Kada se osnovni principi klasične mehanike tumače tako da se odnose na model, on ih potvrđuje (čini ih istinitim), pri čemu takođe reprezentuje dinamički odnos dva stvarna nebeska tela, Sunca i bilo koje planete: "Njutnovski model je logički model jednačine kretanja zato što su termini jednačine - F , m i a - interpretirani tako da referiraju na obeležja modela i tako interpretirana jednačina je tačna. Njutnovski model je, prema tome, za ' $F = ma$ ' isto što i londonski toranj za ' $Fx \& Gx$ '. Ali on sadrži i nešto više. Interpretacija neke jednačine pomoću dve homogene savršene lopte ne stvara model ni za šta drugo, pa ni za planetarno kretanje. Da bi se njutnovski model *odnosio na* solarni sistem, on mora biti i reprezentujući model: dve lopte moraju biti uzete kao da reprezentuju Sunce i jednu od planeta. Dve lopte se, stoga, prema Suncu i toj planeti odnose kao minijatura replika prema RMS Lakonija. Primer njutnovskog modela nam, dakle, pokazuje kako jedan isti objekat može u isti mah da bude i logički i reprezentujući model."²¹

3.

Teorije su, dakle, lingvistički entiteti, formalno uređeni skupovi iskaza različitog stepena opštosti koji se ne odnose direktno na fizičke objekte. Na drugoj strani, modeli ne moraju da imaju lingvistički karakter - mogu biti materijalni ali i mentalni, u vidu vizuelnih prikaza ili apstraktnih predstava, jednačina, dijagrama i sl. Za većinu njih bitno je da su primenljivi na fizičke pojave zahvaljujući tome što ih reprezentuju na neki od načina koje ćemo uskoro prikazati.²² Takođe, iako su obično napravljeni kao sastavni deo teorija, to ne mora biti slučaj; važnije je to da posreduju između teorija i realnog sveta tako što konkretizuju apstraktne teorijske pojmove i zakone. U toj medijaciji oni ipak poseduju izvestan stepen autonomije, ne umeću se kao poseban nivo u piramidalnu hijerarhiju datosti-teorija, već predstavljaju svojevrsna epistemička oruđa pomoću kojih dovodimo u vezu teorije i fizičke pojave. Njihova reprezentujuća funkcija je besumnje pragmatički uslovljena: članovi naučne zajednice ih grade u želji da pomoću njih primene postojeće ili dođu do novih teorijskih uvida u uzročne mehanizme koji vladaju prirodom. Kako ističe Gelfert, "reprezentovanje pojava pomoću modela jeste u osnovi tromesna relacija između samog modela, pojave koja je predmet istraživanja i naučnika koji taj model koriste. Vrednost i funkcija modela proističu iz njihove uloge u istraživanju, tj. iz načina na koji oni korisnicima

²¹Frigg 2023: 49. RMS Lakonija je britanski prekomorski brod koji je tokom II svetskog rata korišćen za vojni transport; njegovo torpedovanje 1942. dovelo je do poznatog incidenta i promene pravila ratovanja na moru.

²²Priroda relacije reprezentovanja je posebna i prilično zahtevna tema, više o njoj u: Nguyen and Frigg 2022.

omogućuju da izvede zaključke o sistemima koje istražuju, formiraju predviđanja ili olakšavaju sebi neke druge poteze.”²³

N. Kartrajt je mišljenja da su nam modeli neophodni zbog toga što između teorije i prirodnih fenomena postoji jaz koji se ispoljava u činjenici da fundamentalni naučni zakoni ne oslikavaju u potpunosti fizičke fenomene. Prema njenom mišljenju, objasniti neku pojavu znači naći model koji je uklapa u bazični okvir teorije, što nam omogućuje da izvodimo konkretne instance zamršenih i složenih fenomenoloških zakona koji za nju važe.²⁴ Za ostvarenje te uloge, dovoljno je da se u model unesu bar neke osobine modelovanog objekta, dok preostala svojstva mogu biti stvar invencije, konvencije i fikcije u cilju uspešnog uključenja objekta u domen teorije. Ilustrovaćemo tu premošćujuću ulogu na dva primera o kojima je već bilo reči.

(1) Prvi korak ka gradnji *njutnovskog modela kretanja planeta* bio je grubi opis planeta koji počiva na posmatranju i sadrži opservacione termine: planete opažamo kao loptaste materijalne objekte koji menjaju svoj položaj na nebu, okreću se oko sopstvene ose, pri čemu su fenomeni plime i oseke vidljivi efekti gravitacionih sila bar onih nebeskih tela koja su dovoljno blizu Zemlje. Kinematički opis nam ne otkriva uzročni mehanizam koji je u osnovi tih pojava. Na temelju posmatranja, Galilej (Galileo) i Kepler (Kepler) su bili u mogućnosti da utvrde položaj planeta u određenom trenutku i pravilan geometrijski oblik njihovih putanja; pravilnost je izražena pomoću matematičkih jednačina (Keplerovi zakoni), bez odgovora na pitanje zašto je putanja planeta eliptična. Praveći imaginarni model, Njutn je koristio kalkulus koji je smislio, teorijske pojmove i zakone klasične mehanike. U model je uneo samo relevantne osobine planeta i formulisao modifikovan opis u kojem su - navešću samo najbitnije momente - one prikazane kao savršene lopte sa središtima homogene mase, položaj Sunca se fiksira, a gravitaciona interakcija između pojedinačnih planeta i Sunca posmatra izolovano od bilo kog drugog tela u univerzumu. Na taj način polazni kinematički opis biva dopunjen dinamičkim razjašnjenjem: kretanje, položaj i oblik putanje planeta instanciraju Njutnove zakone i objašnjavaju se delovanjem gravitacionih sila između Sunca i svake od planeta pojedinačno.

Tako konstruisan model igra obe funkcije koje su bitne za posredovanje između teorije i objekata: *reprezentuje* pojavu koju želimo da objasnimo (kretanje planeta) i *interpretira* pojmove i zakone teorije koja je objašnjava (klasična mehanika). Apstraktni pojmovi i zakoni klasične mehanike nisu primenjeni direktno na fizičke fenomene, nego na idealizovan sistem prikazan u modelu. Iako u stvarnosti nijedna planeta solarnog sistema nije savršena lopta, nema homogenu masu i nije izolovana od dejstva gravitacionih sila drugih nebeskih tela, kao što ni Sunce nije fiksirano nego se okreće oko sopstvene ose, model nam omogućuje da prikažemo i objasnimo njihovu stvarnu

²³ Gelfert 2016: 25.

²⁴ Cartwright 1983: 152. Ona zastupa antirealističko tumačenje modela kao simulakruma i smatra da ne treba ni težiti tome da potpuno odgovaraju pojavama, da previše realizma čak može biti prepreka.

putanju.²⁵ Interpretativnu funkciju, pak, on ispunjava tako što konkretizuje Njutnov zakon koji apstraktan pojam sile dovodi u vezu sa masom i ubrzanjem tela. Pojmu sile u realnosti ne odgovara nikakav opažljivi korelat - ona se u fizičkom svetu ispoljava kod stvari koje poseduju neke druge osobine (masu i ubrzanje), postoji samo u nekim specifičnijim oblicima (gravitacija ili elektrostatika), pripisuje se fizičkim objektima tek posredstvom konkretnijih pojmova sadržanih u modelu (gravitaciono privlačenje dva tela ili elektrostatička interakcija).²⁶

Njutnov model bio je plodotvornim u proširenju na druge fizičke sisteme čije ponašanje je u skladu sa zakonima klasične mehanike (klatno, opruge, struktura atoma i sl). Ipak, fizičari su ubrzo naišli na prepreke. Nedugo pošto je N. Bor (N. Bohr) po uzoru na planetarni model konstruisao model strukture atoma (1913), niz krupnih otkrića u periodu 1914-1927. - fine strukture spektralnih linija, njihovo cepanje u jakom magnetnom ili električnom polju, dualna priroda fotona, Šredingerova talasna jednačina i Hajzenbergov princip neodređenosti - podrio je ideju da na subatomsom planu važe Njutnovi zakoni. Smišljena je nova teorija u kojoj je Borov model zamenjen kvantno-mehaničkim.

(2) U nekim drugim oblastima mehanistički modeli su i dalje bili veoma plodni. To je bio slučaj u jednoj od važnijih epizoda savremene fizike, nastanku kinetičke teorije gasova. Njeni tvorci pošli su od pretpostavke o korpuskularnoj konstituciji gasova pa su makroskopske pojave kao što su temperatura, gustina ili pritisak objašnjavali procesima koji se odvijaju na mikroskopskom nivou (kretanjem molekula), a odgovarajuće jednačine - npr. termodinamičke - formulisali su na osnovu zakona klasične mehanike. Ulogu medijatora odigrao je *model bilijarskih kugli*: molekuli gasa u nekom zatvorenom prostoru (kontejneru) ponašaju se kao bilijarske kugle koje se haotično kreću i pri sudaru razmenjuju linearni momentum (mv).²⁷

Poput prethodnog, i taj model je podstakao kreativnu imaginaciju fizičara i otvorio prostor za širu primenu na druge slične sisteme. L. Bolcman (L. Boltzmann) ga je 1872. primenio u statističkoj termodinamici, tumačeći pritom atomističku hipotezu samo kao teorijsku, ne i metafizičku. U statističkoj interpretaciji Drugog zakona termodinamike, korpuskularna hipoteze i model bilijarskih kugli ilustruju uzročni mehanizam koji je u osnovu fenomena entropije. Kao i njutnovski model planetarnog sistema, ni model bilijarskih kugli nije zamišljen kao potpuno verna slika stvarnosti. Gas je prikazan kao skup molekula koji se nalaze u potpuno izolovanom kontejneru, slobodno se kreću u raznim pravcima i sudaraju uz očuvanje energije i momentuma. Mehanička interakcija

²⁵ Frigg 2023: 364-5.

²⁶ *Ibid.*, p. 373.

²⁷ M. Hese je kao glavni primer uzela upravo taj model, razlikujući tri vrste analogija na kojima on počiva: pozitivnu (osobine koje model deli sa objektom ili pojavom na koju se odnosi), negativnu (osobine po kojima se razlikuju) i neutralnu (osobine za koje ne znamo da li ih dele, ali mogu biti predmet novih otkrića). Konkretno, model bilijarskih kugli sa česticama gasa deli kretanje, momentum i sudaranje, dok na primer boja kugli spada u negativnu analogiju (Hesse 1966: 10).

odvija se po principima klasične mehanike i dovodi gas u ravnotežno, sasvim neuređeno stanje, pri čemu je najverovatniji ishod uvećanje entropije kao ireverzibilan proces - entropija može samo da se uvećava i, kad se ostvari stanje termodinamičke ravnoteže, ostane konstantna. Fenomen entropije i pitanje zašto univerzum još nije postigao termodinamičke ravnoteže okupirali su Bolcmanovu pažnju i, budući da nije našao odgovor, mučili ga sve do njegove tragične smrti.²⁸

Zanimljiv je širi naučni kontekst u okviru kojeg se Bolcman odlučio da oprezno izloži svoje ideje. Naime, na prelazu iz 19. u 20. vek sukob između atomista i energetista - prema kojima je materija u osnovi jedno kontinuirano polje energije koja se transformiše i, kao u termodinamici, ispoljava u raznim oblicima - bio je na vrhuncu. Većina vodećih nemačkih fizičara toga doba, na čelu sa Mahom (Mach) i Ostvaldom (Ostwald), odbacivala je atomizam. Svestan njihovog uticaja u naučnoj zajednici, Bolcman je predložio pomirljivo rešenje prema kojem bi obe strane u sporu mogle da se bave fizikom bez polemisanja oko ontoloških pitanja. Sledeći Herca (Hertz), on je atome tumačio kao *slike* (Bilder)²⁹ u našem duhu; za atomiste su te slike verodostojne, dok suprotna strana može da ih koristi kao fiktivni model (o takvim modelima biće nešto više reči kasnije). Energetisti nisu pristali na kompromis, odbacujući i atomizam i Bolcmanovu statističku interpretaciju Drugog zakona termodinamike.

Rasprava je predstavljala uvod u razvoj teorijske fizike podstaknut otkrićima na subatomsom nivou. Nekoliko godina posle Bolcmanove smrti (1908-1909) Ž.B. Peren (J.B. Perrin) je, proučavajući fenomen Braunovog kretanja u koloidnim suspenzijama,³⁰ iskoristio Ajnštajnove teorijske zaključke iz 1905. i potvrdio vrednosti Avogadrovog broja i Bolcmanove konstante, što je išlo u prilog korpuskularnoj pretpostavci. Peren je 1926. dobio Nobelovu nagradu, ali se ubrzo pokazalo da je struktura materije daleko kompleksnija. Iako je korpuskularna hipoteza uspešno proširena na elektricitet i magnetizam, ona nije mogla da objasni kvantne skokove elektrona i dualnu prirodu svetlosti³¹. Dinamički karakter nauke ponovo je došao do izražaja - fizičari su krenuli u potragu za novom teorijom koju su pronašli u kvantnoj mehanici.

²⁸ Bolcman je 1906. godine izvršio samoubistvo, a neki biografi njegov duševni slom delom pripisuju tome što je bio mentalno skrhan bezuspešnim pokušajima da reši probleme entropije i vremenskog toka.

²⁹ "Mi za sebe formiramo slike (*Bilder*) ili simbole spoljašnjih objekata; a forma koju im dajemo takva je da su nužne posledice slika u mislima uvek i nužne posledice u prirodi oslikanih stvari" (Hertz 1894: 1)

³⁰ Reč je o smešama u kojima je jedna supstanca u vidu raspršenih nerastvorljivih mikroskopskih čestica uronjena u drugu.

³¹ K. Hajgens (C. Huygens) je u 17. veku predložio teoriju svetlosti po uzoru na teoriju zvučnih talasa; danas bismo mogli reći da su mu zvučni talasi poslužili kao dobar teorijski model za razumevanje prirode svetlosti. (Cf. Gelfert 2016: 6)

4.

Da bi uspešno posredovali između teorije i stvarnosti, modeli bi trebalo da usklade odabrane empirijske podatke i teorijske pretpostavke. Građeci prve mehanističke modele, fizičari su uočili koliko su u tom poslu važne analogije. Polazeći od distinkcije između pozitivnih, negativnih i neutralnih analogija, njihov značaj podvukla je M. Hese (1966). Za što verniji prikaz fenomena ključne su pozitivne analogije, što je očigledno u primeru bilijarskih kugli koje sa molekulima gasa dele osobine kompaktnosti, mase i brzine kretanja. Neutralne analogije - svojstva zastupljena u modelu za koja još ne znamo da li spadaju u pozitivnu ili negativnu analogiju - otvaraju prostor za nova predviđanja o očekivanom ponašanju pojava.³² Kombinovanje te dve vrste analogija Hese je ilustrovala na primeru talasnog modela:

VODENI TALASI	ZVUK	SVETLOST
Proizvodi ih kretanje čestica vode	Kretanje nekog izvora zvuka (gong)	Pokretanje izvora svetlosti (plamen)
Svojstva odbijanja	Eho	Odbijanje od ogledala
Svojstva prelamanja	Čuje se iza ugla	Prelama se kroz proreze
Amplitude	Glasnost	Sjaj
Frekvencija	Visina	Boja
Medijum je voda	Vazduh	(Pretpostavljeni) medijum "Etar"

Prve tri sličnosti uočljive su na pojavnom planu i one bi, kako Hese primećuje, bez teškoća mogle da se uvrste u Bekonove tablice ili Milov (Mill) metod slaganja: poput vodenih talasa, zvuk i svetlost se odbijaju od prepreka i, prelamajući se, zaobilaze njihove obline. Opažene pozitivne analogije nagoveštavaju da su fenomeni slični na fundamentalnijem nivou. Da bi to otkrili, naučnici su proučavali opservaciono najpristupačniju pojavu, ponašanje vodenih talasa. Pošto su uočili da se remećenje jedne čestice prenosi na susedne u vidu talasa koji se u koncentričnim krugovima šire od centra poremećaja, amplitude i frekvencije talasa su prikazali sredstvima elementarne matematike jednostavnog harmoničnog kretanja i potom izveli zakone odbijanja i prelamanja. Tako je formirana teorija o vodenim talasima koja se sastoji od jednačina tipa $y = a \sin 2\pi f x$, gde je y visina vode na tački x merena horizontalno, a je maksimalna visina ili amplituda talasa, a f njihova frekvencija. Iz ove matematički izražene teorije mogu da se dedukuju zakoni koji vladaju opisanim procesom (npr. jednakost upadnog i izlaznog ugla).³³

Polazeći od opažene (pozitivne) analogije i talasnog modela, fizičari su došli do teorije o zvuku kao fenomenu koji nastaje kretanjem čestica vazduha u koncentričnim

³²Hesse 1966: 8.

³³*Ibid.*, pp. 11-12.

kružnim talasima od središnjeg izvora. Što je centralni poremećaj snažniji, veće su amplitude sa kojima je povezana jačina zvuka, dok je visina zvuka u korelaciji sa frekvencijom zvučnih talasa. Na tragu analogije, između opažljivih svojstava zvuka i modela vodenih talasa uspostavljena je korespondencija 1:1 koja je omogućila da se matematički formulisana teorija talasa primeni na zvuk. Na sličan način model je, uz pretpostavku o etru kao medijumu, proširen i na objašnjenje kretanja svetlosti. U svakom od tih primera, analogije i model su samo *nagoveštavali* teoriju čija tačnost je bila stvar provere - priroda zvuka je uspešno objašnjena, ali ne (bar delimično) i svetlosti.³⁴

Istorija nauke vrvi od epizoda u kojima su modeli utemeljeni na analogijama pobudili kreativnu imaginaciju naučnika i doprineli nastanku novih teorija. M. Morgan (M. Morgan)³⁵ navodi primer iz oblasti ekonomije. I. Fišer (I. Fischer) je početkom 20. veka izgradio model koji počiva na analogiji između ujednačavanja robne razmene i balansiranja mehaničke vage, pri čemu mu je osciliranje vage kao deo neutralne analogije pomoglo da izvede zaključke o cikličnim procesima u ekonomiji. On je tako došao do teorije o monetarno izazvanim ciklusima u komercijalnoj aktivnosti, koji se ogledaju u odnosu između kupovne moći novca i prosečne visine cena - veća količina novca u opticaju uravnotežuje se povećanjem cena i obrnuto. Zakoni sadržani u njegovom modelu prikazuju samo tendencije u ekonomskim zbivanjima i u primeni su ograničeni uslovom *ceteris paribus*, što samo po sebi ne predstavlja problem zato što slična ograničenja postoje u fizici. Iako ima statistički karakter, Fišerov model je dovoljno veran prikaz realnih dešavanja koji nam omogućuje sticanje znanja o ekonomskim procesima. Zakon o proporcionalnoj zavisnosti količine novca i visine tržišnih cena "jednako je fundamentalan u ekonomiji kao što je Bojlov zakon o proporcionalnoj zavisnosti između pritiska i gustine gasa jedan od fundamentalnih zakona fizike, pod pretpostavkom da se temperatura gasa ne menja. Poznato je, međutim, da su u praksi stope i razmena retko konstantni, baš kao što je to slučaj i sa temperaturom gasa. Ipak, *tendencija* koju prikazuje kvantitativna teorija i dalje je prisutna, šta god da se dešava sa ostalim pridruženim elementima, kao što je i *tendencija* reprezentovana u teoriji o gustini tačna šta god da se dešava sa temperaturom."³⁶

Lotka-Voltera model u populacionoj biologiji demonstrira kako se selektuju svojstva koja su bitna za objašnjenje pojava. Neočekivano saznanje do kojeg je došao na osnovu svedočenja ribara i statističkih podataka o vrstama riba prodavanim na pijacama u Trstu, Veneciji i Fjumiju, da je smanjenje aktivnosti ribarenja tokom I svetskog rata favorizovalo populaciju grabljivaca a ne plena, začudilo je biologa U. Dankonu

³⁴ Zanimljivo je da je još Aristotel formulisao hipotezu da se zvuk talasno prostire kroz vazduh, poredeći to sa opaženim kretanjem vodenih talasa. Pretpostavka da i svetlost mora imati fizički medijum kroz koji se talasi prostiru bila je jedna od najvećih zabluda fizičara u 19. veku. Hipoteza o etru definitivno je odbačena zahvaljujući čuvenom Majkelson-Morlijevom eksperimentu (1887).

³⁵ Morgan *et al.* (eds.) 1999, ch. 12.

³⁶ *Ibid.*, p. 367.

(U. d'Ancona). On se za pomoć obratio svom tastu, matematičkom fizičaru V. Volteri (V. Volterra) koji je 1926. formulisao par linearnih diferencijalnih jednačina prvog reda koje prikazuju populacionu dinamiku sistema u kojem su samo dve životinjske vrste, grabljivci i plen.³⁷ Model je pokazao da se dinamika sistema sastoji od ciklusa koji se odvijaju po istom obrascu: kada se smanji broj grabljivaca, populacija plena brzo raste (čak i dok se populacija grabljivaca tek oporavlja), da bi potom populacija grabljivaca počela da raste sve dok ne počne da se smanjuje brojnost plena (čak i ispod nivoa potrebnog za održavanje uvećane populacije grabljivaca).

Izgrađen za dinamički sistem grabljivci-plen, model uzima u obzir samo veličinu te dve populacije, prosečne podatke o njihovom priraštaju, te prosečan broj istrebljenog plena koji je potreban da bi došlo do množenja grabljivaca, a zanemaruje prostorne relacije, zgusnutost, klimatske uslove ili interakcije sa drugim životinjskim vrstama. Model je ujedno i primer koji potvrđuje da izgradnja može da započne takoreći *ab ovo*. Opažanje neobičnog fenomena za koji nije bilo objašnjenja podstaklo je konstruisanje matematičkog modela, ali je nije diktiralo. Za razliku od naučnika u epizodi sa talasnim modelom, Voltera nije raspolagao nikakvim ranijim biološkim modelom na koji bi se ugledao, već je krenuo takoreći ni od čega, vođen samo intuicijom i pretpostavkama o uočenoj zavisnosti između brojnosti dve biološke populacije.³⁸

Ne treba izgubiti iz vida da, uz oslonac na naučne teorije, modeli mogu imati i praktičnu svrhu. Primer u inženjerskim naukama je materijalni model brane u delti zaliva San Franciska.³⁹ Da bi rešio problem gradskog vodosnabdevanja, Dž. Reber (J. Reber) - inače amaterski muzičko-pozorišni producent - osmislio je grandiozni projekat (1907) koji je između ostalog predviđao velike brane na dva kraka zaliva u koja su se ulivale reke Sacramento i San Hoakin. Predlog su zdušno podržali gradski čelnici i građani, dok su kritičari upozoravali na moguće devastirajuće posledice - nametnulo se pitanje kako ga proveriti? Rešenje je nađeno u izgradnji skalarnog hidrauličnog modela zaliva - napravljen je u prostranom skladištu na površini od preko 40 ari, sa cevima, pregradama koje su sprečavale mešanje slane i slatke vode i pumpama koje su simulirale dejstvo plime i rečnih tokova. Za uspešnu primenu bilo je neophodno prikupiti obilja podataka o nivou okeana u periodima plime na raznim pozicijama u zalivu, salinitetu, snazi talasa i rečnih tokova i dr, kalibrirati sve te parametre i proveriti ih novim merenjima. Model je, srećom, poslužio svrsi jer je pomoću njega dokazano

³⁷Nezavisno od Voltere, A. Lotka (A. Lotka) je 1920. na primeru relacije između biljojeda i biljki došao do istog skupa jednačina - otuda naziv modela.

³⁸Weisberg 2013: 75. Potvrdu za to da izgradnja modela ne mora biti diktirana teorijom imamo i u modelu superprovodljivosti koji su 1934. konstruisala braća London (Fritz i Heinz London). Oni su došli do jednačine za superprovodni strujni tok koja je objašnjavala pojavu a nije se uklapala ni u jednu od tada prihvaćenih teorija (elektromagnetsku, teoriju o superprovodnicima i dr); štaviše, njihov model je pružio polazište za novo teorijsko razumevanje pojave (cf. Morgan et al. 1999: 13–4). Model je bio prihvaćen sve dok 1951. nije dokazano da je superprovodljivost u osnovi kvantni fenomen.

³⁹Cf. Weisberg 2013: 7–10.

da bi realizacija projekta imalo razorne posledice po zaliv i ekosistem.

Primer potvrđuje da modeli imaju određen stepen autonomnosti kako u odnosu na realne pojave (imaginarni modeli za predmet objašnjenja imaju zamišljene sisteme) tako i u odnosu na teorijsko zaleđe (mogu biti izgrađeni bez ikakvog teorijskog oslonca). Njihov autonoman status Morgan opisuje sledećim rečima: "Upravo zahvaljujući tome što su delimično nezavisni i od teorija i od sveta, modeli sadrže autonomnu komponentu i mogu biti upotrebljeni kao instrumenti istraživanja u oba ta domena (...) Mada je odvojen i od eksera i od zida, čekić je napravljen da bi se ekser pomoću njega ukucao u zid. Slično je sa modelima. Oni funkcionišu kao alatke ili instrumenti koji, premda su nezavisni, posreduju između stvari; kao i alatke, i oni često mogu biti iskorišćeni za brojne različite zadatke ..."⁴⁰

Nameće se zaključak da, za razliku od naučnih postupaka kao što su eksperimentisanje ili merenje, za izgradnju modela ne postoje ustanovljene opšte procedure. U nastojanju da objasne pojave koje istražuju naučnici po pravilu polaze od uočenih analogija i teorijskih hipoteza, ali su im za konstruisanje efikasnih modela često potrebne zavidne veštine, dovitljivost i kreativnost. Čini se da je Baumans (Boumans) u pravu kada kaže: "Izgradnja modela je poput pravljenja kolača bez recepta. Sastojci su teorijske ideje, strateški pogledi, matematizacija ciklusa, metafore i empirijske činjenice."⁴¹

5.

Naučni modeli se upadljivo razlikuju. Jedni su preuzeti iz prirode (organistički modeli u biologiji koriste neke organizme za reprezentovanje drugih, npr. u medicinskim ispitivanjima) ili artificijelnog sveta (model bilijarskih kugli); drugi su konstruisani u stvarnosti (modeli aviona čija aerodinamičnost se proverava u specijalnim tunelima sa vazдушnim strujanjem); treći su izmišljeni ili izgrađeni takoreći "u glavi", sa namerom da reprezentuju stvarne fizičke sisteme (njutnovski model kretanja planeta) ili kao čisto formalni (matematički i kompjutacioni modeli), itd.⁴² Tako, recimo, na jednoj strani imamo fiktivni model zaliva San Franciska kao rezultat krajnje zahtevnog inženjerskog poduhvata lokalizovanog u prostoru i vremenu, napravljen od cevi, pumpi i pregrada, na osnovu opsežnih merenja terenskih karakteristika sprovedenih radi sticanja relevantnog znanja o ispitivanom fizičkom sistemu (projektovanoj brani). Na drugoj strani, pak, Lotka-Voltera model je iziskivao samo papir, olovku i odgovarajuće matematičko znanje - on se neposredno odnosio na matematički strukturisan sistem o kojem se znanje stiče manipulisanjem jednačinama koje ga opisuju, ali je posredno

⁴⁰Morgan 1999: 10.

⁴¹Boumans 1999: 67.

⁴²Ostavićemo po strani ontološko pitanje o prirodi modela. Neki autori, na primer Frig, smatraju da su modeli fikcije poput literarnih ili filmskih, dok ih drugi kao Vajsberg tumače kao realne (konkretne ili apstraktne) objekte koji samo u nekim slučajevima sadrže fiktivne scenarije.

bio primenljiv na biološku relaciju između populacija grabljivaca i plena.

Uprkos mnoštvu razlika, moguće je izdvojiti neke glavne karakteristike naučnih modela.

(a) Na prvom mestu je svakako njihovo *posredovanje* između naučnih teorija i fizičkog sveta. De Regt upravo tako opisuje upotrebu modela bilijarskih kugli: "... koristeći kinetičku teoriju, mi razumemo neke bazične pojave kod gasova pomoću objašnjenja koje počiva na modelu. Taj primer lepo ilustruje relaciju između teorije, modela i pojava: teorija pruža principe (na pr. zakone njutnovske mehanike i identifikaciju toplote sa kinetičkom energijom molekula) i prikazuje generički model gasova koji je neophodno artikulirati da bi se konstruisala objašnjenja pojava."⁴³ Ta uloga proistekla je iz potrebe da se premosti jaz između apstraktnih teorijskih zakona i prirodnih fenomena na koji je, videli smo, ukazala N. Kartrajt. Naučnici obično ne konstruišu teorije da bi ih direktno primenili na konkretne objekte, dok se modeli uglavnom grade sa namerom da adekvatno reprezentuju fizičke sisteme i interpretiraju teorijske pojmove i zakone.

Mišljenja su podeljena oko pitanja kako modeli *reprezentuju* pojave. Paralelno sa raznolikošću naučnih modela, i relacija reprezentovanja varira od sličnosti do pragmatički uslovljenog zastupanja pojave u istraživačkom kontekstu.⁴⁴ Pošto je asimetrično, nerefleksivno i netranzitivno, reprezentovanje je nesvodljivo na sličnost ili izomorfizam. Da te dve relacije nisu ni nužan ni dovoljan uslov za reprezentujuću funkciju modela M. Suarez (M. Suárez) ilustruje na primerima slikarskih dela. Bez ikakve sličnosti, Pikasova *Gernika* reprezentuje dvostruko: patnju streljanih stanovnika tog baskijskog grada tokom španskog građanskog rata i pretnju od nadirućeg fašizma. Takođe, dok se za poznati Velaskeзов portret pape Inoćentija X može reći da reprezentuje papu, Bejkonove opsesivne replike tog Velaskeзовog dela uprkos zadržanoj sličnosti to ne čine; u najboljem slučaju one reprezentuju Velaskeзовu sliku.⁴⁵

S obzirom na to da mnoge tipovi modela (posebno matematički) nemaju slikovni karakter, relacija reprezentovanja može da se odredi pomoću tri aspekta: denotiranja, dokazivanja i instanciranja. To gledište se, po početnim slovima reči, označava skraćenicom DDI.⁴⁶ pomoću elemenata modela naučnici *denotiraju* elemente sistema koji istražuju (npr. feromagnetizma, superprovodljivosti ili sistema grabljivac-plen), interna dinamika modela im omogućuje da izvedu teorijske zaključke o unutrašnjem mehanizmu sistema koje *dokazuju* u modelu i potom ih *instanciraju* na samom sistemu. Dodatna prednost DDI gledišta je što je ono spojivo sa pragmatičkim shvatanjem reprezentujuće funkcije modela koje brani Gelfert: "Reprezentovanje pomoću modela, tako shvaćeno, u osnovi je tromesna relacija između samog modela, pojave koja je

⁴³de Regt 2017: 34.

⁴⁴O prirodi relacije reprezentovanja više u: Nguyen, J. & Frigg, R. 2022.

⁴⁵Suárez 1999: 77–8.

⁴⁶Zastupa ga Hjuz (Hughes), cf. Gelfert 2016, sect. 2.3.

predmet istraživanja i njegovog korisnika. Vrednost funkcije modela izvodi se iz njihove uloge u istraživanju, tj. iz toga kako korisnicima omogućuju da na osnovu njih izvedu zaključke o ciljanom sistemu, formiraju predviđanja, ili olakšaju neke druge istraživačke korake.⁴⁷ Prema tom tumačenju, ne postoji objektivno reprezentovanje nezavisno od naučno-istraživačkog konteksta i aktivnosti naučnika, ili šire: *"Nema reprezentovanja osim u tom smislu što su neke stvari upotrebljene, napravljene ili uzete tako da na ovaj ili onaj način reprezentuju neke druge stvari."*⁴⁸

Kontekstualna zavisnost i pragmatički momenat su očito presudni za reprezentujući funkciju modela. Na kraju krajeva, naučnici su akteri koji promišljeno prave modele i po potrebi im dodeljuju ulogu reprezentovanja predmeta čiju prirodu istražuju. Kao što primećuje Gelfert, "... kada teorijski biolog zapisuje Lotka-Voltera jednačine i stipuliše da bi one trebalo da reprezentuju populacionu dinamiku sistema grabljivac-plen, on namerava da čitaoci prepoznaju njegovu nameru da kod njih aktivira verovanje da bi jednačine trebalo da se shvate kao zamena za stvarni sistem."⁴⁹ To svakako ne znači da u naučnom kontekstu bilo šta može biti upotrebljeno kao model koji reprezentuje bilo šta. Naučno istraživanje ipak podleže ograničenjima koja proističu kako iz teorijskog zaleđa, tako i iz prirode predmeta istraživanja, zbog čega izbor modela nije prepušten pukoj volji naučnika. Jedan od najvažnijih ograničavajućih faktora jeste mogućnost *interpretacije* teorije za koju se model konstruiše. Taj faktor prisutan je čak i kod najapstraktnijih matematičkih modela koji predstavljaju normu u egzaktnim naukama. Bilo da se koriste u fizici, biologiji ili ekonomiji, osim toga što su resurs za izvođenje deduktivnih zaključaka i predviđanja, promenljivama u njihovim jednačinama dodeljuju se vrednosti iz realnog sveta, kvaliteti i relacije između fizičkih objekata.

Videli smo ranije da su za interpretativnu funkciju modela bitni premošćujući principi koji apstraktne teorijske termine povezuju sa konkretnim fizičkim pojavama.⁵⁰ Prema mišljenju N. Kartrajt, tradicionalni filozofi nauke (posebno fizike) se nisu detaljnije upuštali u analizu tih principa zato što su se, prvo, fokusirali na same teorije u nastojanju da rekonstruišu njihovu formalnu, pre svega matematičku strukturu, i drugo, pošto su bili platonistički orijentisani ka jednom uređenijem svetu apstraktnih objekata i relacija, prikazanom u fundamentalnim jednačinama i instanciranom u interpretativnim logičkim modelima, zanemarivali su svet opažljivih fizičkih fenomena. S jedne strane, tako shvaćene teorije nemaju gotovo nikakve veze sa aktualnom naučnom praksom, odnosno sa tehnikama, hipotezama ili nagađanjima kojima se naučnici (i inženjeri) služe kako bi konstruisali ne-logičke modele pomoću kojih su u mogućnosti da kontrolišu prirodne pojave. Sa druge pak strane, platonistički "svet"

⁴⁷Boumans 1999: 125.

⁴⁸van Fraassen 2008: 23 (kurziv u originalu).

⁴⁹Gelfert 2016: 32.

⁵⁰Cf. Cartwright 1999: 243.

kakav je prikazan u tradicionalno shvaćenoj formalno strukturisanoj teoriji veoma je udaljen od realnog sveta u čije istraživanje se naučnici upuštaju. Vezu između ta dva sveta uspostavljaju premošćujući principi - većina najvažnijih deskriptivnih termina u naučnim teorijama primenljiva je na fizičke objekte samo onda kada im se pridruže specifični opisi koji sadrže konkretnije pojmove.

Apstraktni termin "sila" u klasičnoj mehanici, recimo, iziskivao je odgovarajuće modele da bi uopšte bio instanciran u telima. Njegova primena posredovana je opisima koji uključuju tradicionalne mehanističke pojmove kao što su "pozicija", "ekstenzija", "masa" ili "kretanje". Neka kompaktna masa m može biti opisana kao *izložena sili* GMm/r^2 kada je *locirana na distanci r od neke druge mase, M* . Prvi opis je apstraktan, dok je drugi konkretniji i mora biti primenljiv kad god je prvi tačan. To je, kako ističe Kartrajtova, razlog zbog kojeg su "interpretativni modeli i premošćujući principi toliko važni u fizici. Apstraktni termini se prilagođavaju pomoću konkretnih opisa koje pružaju modeli. A premošćujući principi pridodaju konkretne interpretativne modele apstraktnim pojmovima fizičkih teorija."⁵¹

(b) Na primerima izgradnje modela zapazili smo da se naučnici služe *apstrakcijom* - izdvajaju osobine koje smatraju relevantnim a zanemaruju irelevantne (npr. izostavljanje međumolekularnih sila iz zakona o idealnom gasu), i *idealizacijom* - neka relevantna svojstva uzimaju u savršenijem obliku koji ne odgovara realnosti (elektron se, recimo, prikazuje kao tačkasta čestica).⁵² Ta dva postupka rezultiraju karakteristikama koje modele manje ili više udaljuju od stvarnosti olakšavajući pritom njihovu posredničku ulogu.

Efekte apstrakcije i idealizacije detaljnije ću prikazati na modelu klatna.⁵³ U konstrukciji tog modela fizičari su pošli od drugog Njutnovog zakona $F = ma$ te su, primenjujući ga na harmonične oscilatore, vrednost sile odredili kao proporcionalnu izmeštanju tela $F = -kx$ i izveli jednačinu kretanja $md^2x/dt^2 = -kx$ koja opisuje ponašanje klatna. Ako, pak, želimo da dobijemo Galilejev zakon prema kojem je period oscilovanja klatna proporcionalan kvadratnom korenu njegove dužine i nezavisan od njegove mase, u model moramo da unesemo dodatna ograničenja, pa ćemo silu definisati jednačinom $F = -(mg/l)x$ u kojoj je l dužina klatna a $-mg$ uniformna gravitaciona sila koja na njega deluje. I taj dorađeni model još uvek je idealizacija stvarnog klatna zato što dvodimenzionalan prikaz oscilacija na x - y ravni zamenjuje jednodimenzionalnim

⁵¹ *Ibid.*, p. 255. Kartrajtova smatra da uvođenje premošćujućih principa nije u sukobu sa naučnim realizmom zato što se teorije potvrđuju svojim empirijskim uspesima i protežu dokle i njihovi interpretativni modeli. (*Ibid.*, p. 266)

⁵² Vajsberg razlikuje tri tipa idealizacije: galilejevsku (uprošćavanje radi lakše primene matematičkih oruđa), minimalističku (ograničava se samo na glavne uzroke pojave, izostavljaju se manje bitni) i idealizaciju u mnoštvenim modelima (nesaglasni su, nude različite tvrdnje o prirodi i uzročnom mehanizmu iste ciljane pojave) (Weisberg 2013: 98; *passim*)

⁵³ *Cf.* Giere, 1988. Prikaz Girovog opisa idealizovanja i deidealizovanja modela klatna rekonstruisaćemo po: Cartwright 1999: 249–250.

prikazom na x -osi, a samo klatno prikazuje kao tačkastu česticu mase čije je kretanje, izloženo samo dejstvu gravitacije, svedeno na mali ugao. Tako zamišljeno idealno klatno sada zadovoljava jednačinu kretanja $F=ma$ u njenom modifikovanom obliku $m d^2x/dt^2 = -kx$.

U modelu je, dakle, pretpostavljeno da je gravitaciono polje uniformno i da se klatno mase m kreće bez otpora, dok je u stvarnosti sila Zemljine teže delom balansirana napetošću duž žice S koja se može razložiti u vertikalnu i horizontalnu komponentu. Idealizovan model zanemaruje otpor vazduha, izostavlja vertikalnu komponentu i uzima u obzir jedino horizontalno kretanje klatna; s obzirom na to, zakon koji opisuje harmonično oscilovanje klatna ne reprezentuje u punoj meri kretanje fizičkog klatna. To je potvrda da zakoni klasične mehanike nisu direktno primenljivi na obično klatno kakvo je, recimo, u mehanizmima zidnih satova, već samo na idealizovan model. Ipak, kako Gir ističe, uz nužne korekcije - unošenje dodatnih parametara i, paralelno tome dopunu jednačine kretanja klatna dodatnim faktorima sile - model možemo da "deidealizujemo" i prilagodimo ga što vernijem reprezentovanju realnog klatna, toliko vernom da smo u mogućnosti da izmerimo lokalno gravitaciono ubrzanje do četvrt decimalne tačnosti.⁵⁴ Koliko će korigovani modeli biti verni reprezenti pojave zavisi od specifičnih ograničenja koja se unose, pa samim tim i od toga kako ćemo interpretirati apstraktne termine u zakonima koji ih određuju. Time modele i zakone činimo krajnje složenim - što su realističniji, to su složeniji. Međutim, gledano iz perspektive naučnih ciljeva, bitno je da se oni i dalje uklapaju u polaznu teoriju (klasičnu mehaniku) i odgovaraju njutnovskim zakonima - oni opstaju u svom osnovnom obliku, samo se modifikuju dodatnim ograničenjima kako bi se dobila što realističnija reprezentacija konkretnog fizičkog fenomena.

Koja svojstva će biti apstrahovana ili idealizovana zavisi od vladajućih naučnih paradigmi, usvojenih teorijskih pretpostavki i interesa naučnika. S obzirom na to često se dešava da se kod iste pojave, pogotovo ako je ona krajnje kompleksna i ako za nju nije formirana objedinjujuća teorija, različita svojstva uzimaju kao relevantna pa se konstruišu različiti, međusobno saglasni ili čak isključivi modeli. Morison (Morrison) nas podseća na to da se u literaturi navodi više od 30 uzajamno nesaglasnih modela atomskog jezgra koji objašnjavaju različite eksperimentalne rezultate (npr. dok neki opisuju rasipanje, drugi opisuju cepanje jezgra).⁵⁵ Delom je to posledica nedostatka jedne koherentne teorije o nukleusu, ali i složenosti fenomena - nukleus ispoljava različite tipove ponašanja, od klasičnog u kojem se ponaša kao kap tečnosti, do kvantno-mehaničkog u kojem manifestuje ljuskastu strukturu sličnu strukturi atoma. Na slične primere mnoštva modela koji se odnose na istu pojavu nailazimo i u klimatskim naukama, ekonomiji ili biologiji.⁵⁶

⁵⁴Giere 1988: 47–8.

⁵⁵Cf. Morrison 1999: 45.

⁵⁶U biologiji raka, gde takođe nema objedinjujuće teorije, napravljeni su razni modeli njegovog

Apstrahovanje i idealizacija učestvuju i u konstruisanju fiktivnih modela. Pribe-
gavajući tim postupcima i služeći se imaginacijom, naučnici neretko prave modele
za teorijske sisteme kojima u realnom svetu ništa ne odgovara - opisuju zamišljene
biološke populacije, neuronske mreže ili društvene zajednice savršeno racionalnih
individua. Koliko god to bilo neočekivano, pravljene su i modeli koji se odnose na
nemoguće sisteme, npr. Fajmanov (Feynman) model čegrtaljke (zamišljen kao misaoni
eksperiment) koji bi trebalo da reprezentuje *perpetuum mobile*.⁵⁷

6.

Epistemološkog gledano, modeli su obezbedili mesto u kontekstu naučnog istraži-
vanja zahvaljujući svom kognitivnom značaju.

(a) Od modela koji reprezentuju fizičke sistema pre svega očekujemo da su *deskrip-
tivno adekvatni*. U igri su, pak, i drugi parametri kao što su generalnost, detaljnost ili
preciznost opisa, koji najčešće moraju da se balansiraju zavisno od teorijskih pretpo-
stavki i naučnih ciljeva. Potreba za apstrahovanjem i idealizacijom delom je povezana
sa tim ograničenjima jer su nam ponekad uvidi i razumevanje pojava važniji od pot-
pune tačnosti opisa i prediktivnog uspeha modela, i obrnuto. "U kontekstu aktualnog
naučnog istraživanja visoka cena kompromisa - inače uglavnom apstraktnog - između
preciznosti i opštosti je, dakle, uslovljena stepenom heterogenosti u skupu sistema
koji su predmet istraživanja: što je raznorodnija klasa tih sistema, to će biti teže da
se u isti mah uvećaju preciznost i opštost, na primer tako što bi se svi oni objasnili
pomoću istog modela."⁵⁸

Sa takvim kompromisima susrećemo se, recimo, u pojedinim oblastima biologije
(ekologiji, populacionoj biologiji ili teoriji evolucije). Dok je u fizici moguće konstrui-
sati veoma precizan opšti model ponašanja elektrona zato što svi elektroni imaju ista
inherentna svojstva i svaki od njih istovetno reaguje na magnetno polje, organizmi
su složeni adaptivni sistemi čije reakcije na spoljašnje stimuluse variraju. Usled kom-
pleksnosti i heterogenosti jedinki, njihove evolucije i izrazitih varijacija relevantnih
osobina ekosistema, biolozi teško mogu da izgrade modele u kojima će istovremeno
maksimizirati sve osnovne parametre (verodostojnost, opštost i preciznost); zavisno
od cilja koji sebi postavljaju, naučnici su prinuđeni da neke od njih žrtvuju. Zbog
toga su zakonolike generalizacije retke ne samo u biologiji, nego i u mnogim drugim
(posebno društvenim) naukama.⁵⁹

(b) Od uspešnog balansiranja navedenih parametara u velikoj meri zavisi *ekspla-
natorni uspeh* modela. Njihov kognitivni značaj najvećim delom proizlazi upravo iz

nastanka i razvoja - atavistički, socijalni, model selekcije na više nivoa i dr. Cf. Kamerer 2024.

⁵⁷Korišćenje takvih modela spada u etablirane naučne strategije. Argumente u prilog tvrdnji da oni
mogu biti eksplanatorno korisni možemo naći u: Bokulich 2011.

⁵⁸Gelfert 2016: 63. Cf. Weisberg 2013: 156–7.

⁵⁹Cf. Gelfert 2016: 63.

njihove eksplanatorne uloge.⁶⁰ Većina modela, posebno onih koji reprezentuju uzročne mehanizme u prirodi, ne ograničava se na opis već ima za cilj da nam razjasni zašto i kako se nešto dešava. Imajući to u vidu, N. Kartrajt im je s pravom dodelila središnje mesto u naučnom objašnjenju: "objasniti neku pojavu znači konstruisati model koji je uklapa u teoriju".⁶¹ Potvrdu za to imamo u njutnovskom modelu kretanja planeta i modelu bilijarskih kugli - oba modela preuzela su relevantne opservacione termine iz opisa posmatranih fizičkih fenomena, što im je osiguralo deskriptivnu adekvatnost, ali su povrh toga konkretizovala zakone klasične mehanike i eksplanatorno značajne teorijske pojmove sile i mase.⁶²

(c) Na deskriptivnoj adekvatnosti i eksplanatornom uspehu počiva i *prediktivna moć* modela. Taj kognitivni značaj posebno je isticala M. Hese, dovodeći ga u vezu sa doprinosom pozitivnih i neutralnih analogija u izgradnji modela. Zahvaljujući tome što su deskriptivno adekvatni i eksplanatorno uspešni, modeli kretanja planeta i bilijarskih kugli se odlikuju i prediktivnim potencijalom: na osnovu poznavanja uzročnih mehanizama koji su im u osnovi, možemo predvideti gde će se planete naći u određenom vremenskom trenutku i kako će se gasovi ponašati u raznim novim okolnostima.

(d) Filozofi nauke su naglasak stavljali na eksplanatorne i prediktivne resurse teorija, zanemarijući potrebu za *razumevanjem* pojava kao jednim od naučnih ciljeva. Razumevanje kao specifično kognitivno postignuće nije rezervisano samo za pojedine vannaučne sfere (npr. tumačenje umetničkih dela) ili neke oblasti društvenih nauka u kojima se bavimo pojedinačnim događajima i u kojima ne vladaju strogi zakoni (istorija), ono je zastupljeno u svim naukama. Konkretno objašnjenje neke pojave u vidu odgovora na pitanje zašto se ona dešava predstavlja najkrupniji korak ka razumevanju prirode te pojave zbog toga što omogućuje njeno podvođenje pod principe teorije, uključanje u širi teorijski kontekst i sistematizaciju naučnog znanja: ako smo objasnili putanju planeta pozivajući se na Njutnove zakone, mi smo u isti mah razumeli kako funkcioniše solarni sistem na pozadini šireg teorijskog okvira klasične mehanike, zato što se fundamentalni principi te teorije mogu primeniti i na druge slične fizičke sisteme. De Regt u tom duhu piše: "Objašnjenje neke pojave koje se zasniva na modelu postiže se konstruisanjem specifičnog modela (...) koji reprezentuje istraživani sistem (pojavu) tako da su na njega primenljivi teorijski principi. Ako je takvim objašnjenjem pojava uspešno obuhvaćena teorijom, postigli smo njeno naučno razumevanje."⁶³

⁶⁰Cf. Bokulich 2011: 33.

⁶¹Cartwright 1983: 17.

⁶²U tom pogledu ni fiktivni modeli nisu izuzetak. Iako se kod njih pitanje deskriptivne adekvatnosti uopšte ne postavlja, oni mogu biti od eksplanatorne koristi tako što pružaju osnov za protivčinjeničke iskaze tipa "kad bi se (da se) desilo A, desilo bi se i B". Upravo tako je na modelu zaliva San Franciska pokazano do kakvih posledica bi dovela realizacija ambicioznog projekta izgradnje brane od kojeg se na kraju odustalo.

⁶³de Regt 2017: 32.

(e) Još jedan važan kognitivni značaj modela ogleda se u njihovom *istraživačkom potencijalu*. Slično eksperimentima, modeli mogu da vode otkriću novih činjenica i posluže kao polazište: za dalje istraživanje, uključujući i izgradnju novih modela (Maksvelov model vrtloga); za dokazivanje polaznih načela i ispoljavanja očekivanog ponašanja fenomena (Lotka Voltera model); za generisanje mogućih objašnjenja opaženih pojava (talasni model se pokazao efikasnim u istraživanju prirode zvuka i svetlosti) i dr.

Prvi modeli harmoničnih oscilatora dizajnirani su za predviđanje periodičnog kretanja fizičkih sistema ali su kao matematički ostali apstraktni, bez očiglednih sličnosti sa svojstvima žica, molekula, pa čak i klatna. Zahvaljujući toj opštosti i analogijama, hemičari su ih proširili na modelovanje vibracija u hemijskim vezama tako da reprezentuju atomičke pozicije kao tačke u koordinatnom sistemu i opisuju periodični pomak tih tačaka koji korespondira molekularnim vibracijama. Sličan napredak u istraživanju postigao je R. Gudvin (R. Goodwin) primenom matematičke strukture Lotka-Voltera modela u oblasti ekonomije.⁶⁴ Umesto dve suprotstavljene populacije, Gudvin je uključio dve makroskopske ekonomske varijable: udeo radnika u nacionalnom dohotku (nadnice) i stopu zaposlenosti pa je, uz nekoliko dodatnih pretpostavki, po obrascu Lotka-Voltera modela formulisao jednačine koje prikazuju dinamičke protivrečnosti kapitalizma, doprinoseći razumevanju fenomena na koje je ukazao K. Marks (K. Marx). Gudvinov model predviđa nestabilni ekvilibrijum sličan onom u sistemu grabljivci-plen: pad stope zaposlenosti dovodi do uvećanja, a porast do smanjenja nadnica; kada se nadnice spuste ispod određene granice, to uzrokuje smanjenje stope zaposlenosti i, posledično, porast nadnica.

Gelfert ovako opisuje istraživački doprinos modela u okolnostima u kojima naučnici ne raspolazu polaznom teorijom: "ako objasniti neku pojavu znači konstruisati model pomoću kojeg se ona može uklopiti u teoriju, postavlja se pitanje šta činiti kada nemamo nijednu fundamentalnu teoriju na raspolaganju. U tim okolnostima istraživački modeli igraju važnu ulogu: oni nam pomažu da smislimo moguća objašnjenja, na primer tako što ćemo zamisliti scenarije koji će, ako su istiniti, dovesti do onih vrsta pojava koje konstituišu eksplanandum."⁶⁵ On nas podseća na to kako je Maksvel (Maxwell) posle Faradejevog (Faraday) eksperimentalnog otkrića veze između elektriciteta i magnetizma pomoću modelovanja razvio elektromagnetnu teoriju - analogija sa nesputanim razlivanjem tečnosti poslužila mu je za istraživanje fizičke geometrije linija sila koje se šire u praznom prostoru. Maksvel nije svodio elektromagnetske pojave na mehaničke, već je samo izdvojio matematičke formule koje je smatrao neophodnim za proučavanje elektriciteta. Na osnovu tih pretpostavki napravio je model etra pomoću kojeg je vizuelno predstavio kako se šire linije magnetnih sila oko magnetna. Sam model pokazao se kao korisno oruđe za istraživanje tzv. molekularnih vrtloga,

⁶⁴Weisberg 2013: 77–8.

⁶⁵Gelfert 2016: 87.

izgradnju modela vrtloga i nastanak nove teorije o tom fenomenu.

Zaključićemo ovaj odeljak sa dva citata koja sažimaju zapažanja o raznolikosti naučnih modela i njihovoj višestrukoj ulozi u naučnom istraživanju:

”Modeli mogu biti fizički objekti, matematičke strukture, dijagrami, kompjuterski programi ili bilo šta drugo, ali svi oni služe kao instrument za istraživanje sveta, naših teorija, pa čak i drugih modela.”⁶⁶

”Modeli, dakle, nisu prosto neutralne alatke koje upotrebljavamo s namerom da reprezentujemo aspekte sveta; oni u isti mah sputavaju i omogućuju naše znanje i iskustvo o svetu koji nas okružuje: modeli istovremeno posreduju, doprinose i omogućuju naučno znanje.”⁶⁷

7.

Opsežna literatura o naučnim modelima obiluje pokušajima filozofa da se oni sistematizuju. Razni autori nude različite klasifikacije. Prema jednoj od njih, modele možemo grupisati u *skalarne* (napravljeni po uzoru na objekat ali u veličini pogodnoj za primenu ili proveru teorije); *analoške* (oslanjaju se na istovetnost ili sličnost relevantnih svojstava) i *idealizujuće* (odstupaju od stvarnosti tako što se neka relevantna svojstva zamišljaju kao savršena). Nešto precizniju podelu, uz priznanje da ni ona nije iscrpna, napravio je Vajsberg - na *konkretne*, *matematičke* i *komputacione* modele. Utisak je, međutim, da i ta tema deli sudbinu filozofskih problema na koje je Vitgenštajn, distancirajući se od hegelovskog pristupa, skrenuo pažnju: u nastojanju da se pronađe jedinstvena priroda naučnih modela, zamagljuje se ili potpuno gubi iz vida koliko su oni po mnogo čemu različiti.

Pomalo paradoksalno, ali s obzirom na brojnost i raznolikost modela filozofi koji se njima bave našli su se u situaciji koja nalikuje onoj u kojoj se, usled kompleksnosti predmeta i nedostatka objedinjujuće teorije, za istu pojavu gradi mnoštvo često uzajamno nesaglasnih modela. Stvarnost koju naučno istražujemo enormno je složena, radi proučavanja njenih slojeva tokom proteklog veka došlo je do diversifikacije naučnih oblasti i rađanja novih naučnih disciplina, što je praćeno pravom ekspanzijom modela. Danas je u upotrebi toliko različitih tipova naučnih modela da je formiranje objedinjujuće filozofske teorije o njima postalo gotovo nemoguća misija.⁶⁸ Ta okolnost ipak ne mora da nas obeshrabri, u njoj čak možemo naći pozitivan momenat jer ”snaga modela kao istraživačkih oruđa leži upravo u njihovoj raznovrsnosti i fleksibilnosti”.⁶⁹ Možda baš zato pojedini autori odustaju od traganja za jednoznačnim određenjem prirode modela i poručuju da bi trebalo da se zapitamo čime se opravdava to što

⁶⁶Morgan *et al.* (eds) 1999: 32.

⁶⁷Gelfert 2016: 127.

⁶⁸Frig navodi podatak da je u literaturi zabeleženo više od 120 različitih tipova modela i da se taj broj samo uvećava (Frigg 2023: 11).

⁶⁹Gelfert 2016: vii.

nešto prihvatamo kao model. Gelfert navodi odgovor B. Mara (B. Mahr): "ako svi naučni modeli imaju išta zajedničko, to nije njihova priroda nego njihova funkcija".⁷⁰ Na žalost, ta karakterizacija je tautološka - pošto razni modeli imaju razne uloge, a ponekad jedan isti model može da igra više različitih uloga, njihova *de facto* jedina objedinjujuća funkcija jeste to što se koriste kao modeli.

Činjenica je da naučni modeli nemaju nijednu bitniju zajedničku osobinu. Ako pogledamo najbolje kandidate za takva svojstva, uverićemo se da postoje modeli koji ne služe kao medijatori između teorije i realnog sveta već samo za proveru novih teorijskih alata ili matematičkih tehnika - takav je u kvantnoj teoriji tzv. ϕ^4 model koji za cilj nema interpretaciju teorijskih pojmova i ne reprezentuje nijedan stvarni fizički sistem već se, pošto je matematički jednostavan, među fizičarima koristi za razvijanje intuicija o kompleksnijim teorijama. Neki modeli ne mogu ni da igraju interpretativnu funkciju zato što su izgrađeni *ab ovo* ili su potpuno autonomni u odnosu na postojeće teorije; pored Lotka-Voltera modela o kojem je bilo dosta reči, takav je i model superprovodljivosti braće London - pošto tadašnja kvantna mehanika nije bila u stanju da objasni fenomen, jednačine modela nisu ni imale za cilj da interpretiraju njene principe nego su formulisane na temelju opaženih makroskopskih elektromagnetnih pojava.⁷¹ Takođe, nisu svi modeli ni artifičijelni (organistički modeli u medicini ili talasni model u fizici preuzeti su iz prirode) niti materijalni (apstraktni modeli u fizici su npr. skup parcijalnih diferencijalnih jednačina koje prikazuju protok toplote u čvrstim telima ili Isingov model feromagnetizma u izvesnim metalima koji ne interpretira kvantnu teoriju magnetizma već se koristi kao jednostavnija matematička zamena za proučavanje faznih prelaza feromagnetizma).

Budući da nemaju opšta zajednička obeležja i ne podležu jednoznačnoj iscrpnoj klasifikaciji, najviše što bi se iz filozofskog ugla moglo reći o prirodi naučnih modela jeste da su povezani vitgenštajnovski shvaćenom relacijom porodičnih sličnosti. Vitgenštajnova pouka u vezi sa mnogim pojmovima izraz je njegovog generalnog stava prema filozofskim problemima i u duhu je citata iz *Kralja Lira* ("Naučiću vas razlikama") koji smo naveli na samom početku teksta: to što neke stvari podvodimo pod isti opšti pojam ne znači da one dele zajednička svojstva. Omiljena Vitgenštajnova ilustracija su igre koje su povezane jedino razuđenom mrežom sličnosti - neke su prosto zabavne, neke takmičarske; jedne zahtevaju više veštine, a druge sreću; neke su grupne, neke individualne itd. Modeli u nauci su poput igara u svakodnevnom životu. Sličnost između njih nije tranzitivna, oni ne poseduju nijednu zajedničku (netrivijalnu) karakteristiku, povezani su samo mrežom preklapajućih i ukrštenih sličnosti - modeli M_1 i M_2 mogu nalikovati u pogledu materijalne konstitucije, M_2 i M_3 u pogledu načina

⁷⁰ *Ibid.* 2016: 3–4.

⁷¹ Decenijama kasnije (1957) kvantno-mehaničko objašnjenje objavili su američki fizičari Dž. Bardin (J. Bardeen), L. Kuper (L. Cooper) i D. R. Šrifer (J. R. Schrieffer), po čijim inicijalima se ono naziva BCS teorijom.

reprezentovanja, M_3 i M_4 u pogledu apstraktne strukture, itd.

Čuveno je Poperovo (Popper) poređenje naučnih teorija i hipoteza sa mrežama koje ribari zabacuju u more nadajući se bogatom ulovu. Popperova metafora primenljiva je i na naučne modele - pored teorija i hipoteza, i oni su sastavna okca prefinjenih mreža koje članovi naučnih zajednica, angažujući sve svoje intelektualne, imaginativne i kreativne moći, tkaju i bacaju u okean stvarnosti sa željom da dođu do što bogatijeg ulova u vidu objašnjenja, razumevanja i uspešnog predviđanja prirodnih pojava.

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Živan Lazović

Models in Scientific Practice*(Summary)*

The main topic of this paper is the roles that models play in scientific practice. Although mechanistic models have been used since the emergence of Newtonian classical mechanics in the 17th century, philosophers of science have disagreed about their importance for scientific theories. Logical empiricists between the two wars strove for a rational reconstruction of science and the construction of a normative methodology of science in which models had no place. During the 50s of the last century, the Copernican turn in the philosophy of science took place, shifting the focus from rational reconstruction to actual scientific practice in which models are widely employed. The paper shows the ways of their construction, basic characteristics (mediation between abstract scientific theories and concrete physical phenomena, interpretative and representative function) and cognitive significance (descriptive adequacy, explanatory and predictive power, contribution to understanding and research). A number of examples have shown that scientific models are so diverse that it is almost impossible to form a unifying theory about them - they do not share any essential common properties, but are connected by a network of Wittgensteinian "family resemblances".

KEYWORDS: scientific practice, theories, models, mediation, interpretation, representation, family resemblances.

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NOMINALISM ABOUT REFERENCE CLASSES AND THE ERROR OF REIFICATION

Abstract. *Anthropic reasoning is impossible without the concept of the reference class. However, this basic concept is often unclear and even misunderstood. Here I bring attention to one particular error often committed in the area, namely reification of the reference class. While the reference class is a theoretically crucial concept, it corresponds to little in reality. On the contrary, a nominalist approach to reference classes promises the best – or at least most palatable – results. The analogy with the treatment of potentials in classical field theory could be instructive in both similarities and dissimilarities.*

Keywords: *anthropic reasoning; reification error; nominalism; Bayesianism; Doomsday Argument; human extinction; existential risk; personal identity*

1. Introduction: anthropic reasoning and reference classes

Anthropic reasoning is a philosophical and scientific framework stating that the universe's observed physical constants and laws must be compatible with the existence of life, because if they were not, we would not exist to observe them. It acts as a selection effect, suggesting our existence explains why the universe and its various subsystems appear “fine-tuned” for life and observership (Hogan 2000; Bostrom 2002; Jevtić 2011; Barnes 2012; Ćirković 2016; Lacki 2023).

The locution “anthropic principle” comes from the study of Carter (1974), which defined anthropic principles and went some way to impose order on the nature and quality of anthropic arguments used in cosmology thus far. For most people and sources, Carter's study is the ultimate origin of anthropic reasoning, although we know that it only followed from previous considerations (detailed, although hardly unbiased, historical survey could be found in Barrow and Tipler 1986).

From the start the debate revolved about the infamous problem of fine-tuning of fundamental constants of nature and cosmological parameters. While Dirac argued that some numerical relationships between those (understood as pure numbers, in order to avoid confusion stemming from man-made units of measurement) hold in the absolute sense throughout the history of the universe, Dicke has turned the explanatory issue “upside down” by asking when and how such approximate relationships could be *observed*. For example, we observe stars of age similar – within an order of a magnitude – to the age of the universe itself, since we are observers dependent on stars and stellar evolution for our existence. Either much earlier or much later, it

would be impossible that observers like us evolve in the universe, thus there is no surprise that we observe approximate relationship between these timescales. This conclusion of Dicke could be generalized to other fine-tuning “coincidences”, as has been subsequently clarified. In any case, Dicke pioneered an approach which is *evolutionary* in nature, relying on the concept of observership, rather than (at least partly) metaphysical concepts of absolute laws. The price to be paid was introduction of “observers like us” – a reference class – into the discourse of physics.

So, historically, anthropic reasoning *has* indeed emerged from cosmological considerations. But it went much farther than this circumscribed domain in the 40+ years since Carter’s seminal work. One of the most stunning applications of the anthropic reasoning has been the so-called Doomsday Argument (henceforth DA).

Consider first the classical exposition of DA (cf. Leslie 1989, 1996; Bostrom 2002). The first phase is the urn-and-balls thought experiment. Suppose two large urns are put in front of you, and you know that one of them contains ten balls and the other a million, but you are ignorant as to which is which. The balls in each urn are numbered 1, 2, 3, 4 ... etc. Now you take a ball at random from the left urn, and it shows the number 7. Clearly, this is a strong indication that that urn contains only ten balls. If originally the odds were fifty-fifty (identically-looking urns), an application of Bayes’ theorem gives you the posterior probability that the left urn is the one with only ten balls as $P_{\text{post}}(n=10) = 0.99999$.

In the second phase, consider the situation where instead of the urns you have two possible models of humanity, and instead of balls you have human individuals, ranked according to birth order. One model suggests that human race will soon go extinct (or at least that the number of individuals will be greatly reduced), and as a consequence, the total number of humans that will have ever existed is about 100 billion. The other model indicates that humanity will colonize other planets, spread through the Galaxy, and continue its existence for many future millennia; consequently, we can take that number of humans in this model is of the order of, say, 10^{20} . As a matter of fact, you happen to find that your rank is about 80 billion. According to Carter and Leslie, we should reason in the same way as we did with the urn balls. That you should have a rank of sixty billion or so is much more likely if only 100 billion persons will ever have lived than if there will be 10^{20} persons. Therefore, by Bayes’ theorem, you should update your beliefs about mankind’s prospects and realize that *an impending doomsday is much more probable than you have previously thought*.

The conclusion bothers many thinkers, and for good reason. It is at best extremely counterintuitive to draw such an all-important conclusion from a meager and almost trivial evidence such as one’s birth rank. Thus many regard DA as a paradox to be resolved. Two broad families of approaches are in play in the search for resolution:¹

¹ Here I neglect the third option, namely that one rejects Bayesianism *tout court*. While it is a lively tradition in contemporary philosophy (e.g., Haack 2012), such a contrarianism seems irrational when

- A. Uncertainties about the relevant *reference class*.
- B. Additional *anthropic assumptions* one needs when dealing with intelligent observers/human beings/conscious minds.

In other words, we may counter the DA reasoning by either asking “Who do we count as observer and how do we count them?” (option A) or “How do we reason about ourselves once we accept our position as observers?” (option B) In the latter category, we find the well-known anthropic assumptions such as the *Self-Sampling Assumption* (SSA: One should reason as if one were a random sample from the set of all observers in one’s reference class) and the *Self-Indication Assumption* (SIA: Given the fact that you exist, you should favor hypotheses according to which many observers exist over hypotheses on which few observers exist, *ceteris paribus*).²

The option B has been far more popular in philosophical literature on the paradox thus far, irrespectively of which additional assumption various authors prefer (e.g., Dieks 1992; Kopf, Krtous and Page 1994; Bostrom 2001; Olum 2002; Adelstein 2024). Analyses of the reference class have, however, been considerably rarer. Are there any extra-philosophical reasons for such a situation is an interesting culturological question, I shall briefly return to in the concluding section below. First, we shall see how redefining reference class can help resolve the apparently paradoxical aspects of DA, while leaving its substantial reasoning intact.

The plan of exposition in the rest of this paper is as follows. After I briefly outline how nominalist thinking about reference classes resolves the apparently paradoxical aspects of DA (Section 2), I discuss two possible objections to this view (Section 3). Natural next step is to ascribe instability and vagueness to the concept of reference class in general (Section 4). A partial analogy with the concept of potentials in the classical field theory is then outlined (Section 5), before some summarizing remarks and prospects for further research are given in the concluding Section 6. In contrast to what a superficial reading might suggest, the present paper has no ambition to explain “how to do anthropic reasoning” or something such, which is a far more complex and expansive topic.

2. Nominalism about reference classes resolves the Doomsday Argument

In contrast to simplistic dichotomies often offered in the media and popular accounts, it is entirely possible to *accept the reasoning* behind DA and yet to *reject the (“apocalyptic”) interpretation* of its results. The simplest, and arguably the most realistic way to do this is just to cut the Gordian knot of the reference class and reject the idea that future (post)humans belong to the same reference class as we do. There

spectacular successes of the Bayesian approaches in modern science are taken into account.

²Other locutions for SIA used in the literature (e.g. “one is more likely to find oneself in the long-lived race” – Olum 2002; “given that one exists, one should think that the universe has many people” – Adelstein 2024) are equivalent to this.

are many operational models of the idea. Let me outline two of them, since that will demonstrate how modifying our intuitions about the reference class can help solve DA and similar problems:

1. *The Day-Million people*: From a future point τ onward, there are only statistically dominant posthumans, constituting a different reference class from the present-day humans. A person living in the “Day Million” CE³ may look like me at a glance, but they decidedly do not belong to “our” reference class if agential properties, physical and moral capacities, or biological morphology represent the defining characteristics of the reference class.
2. *The Hive-mind*: From a future point τ onward, all or most hitherto human observers are joined into a kind of single planetary superorganism. Irrespective of the technical, ontological or even theological properties of such a superorganism (envisioned by the likes of Vladimir Vernadsky and Pierre Teilhard de Chardin⁴), it is only rational to conclude that it will be sufficiently different in any and all respects from the present-day human observers to not be in the same reference class with respect to our anthropic reasoning.

While the details of such scenarios are a matter for futures studies, neuroscience and other disciplines, it is clear that they “resolve” DA – or whatever exactly there is to resolve there. The Bayesian shift is indeed real, but it pertains to specific reference class, namely current humans. The total human observer tally is indeed about 100 billion by this day (call it t_0), plus all those which will have lived at any time t between now and the transition epoch, $t_0 < t < \tau$. If the transition epoch occurs in the near future, say in 2100 CE, the Bayesian shift will be very small: it is indeed likely that we find ourselves exactly where we are in the birth ranks of humanity.

Although it is beyond the scope of the present paper, notice in passing that the number of human observers need not drop exactly to zero at for the reasoning to work. Suppose that in the *Hive Mind* scenario “only” 99.99% of humans decide to join the superorganism; the remaining human population (say about a million individuals) may continue to exist for many hundreds of generations into the future without significantly shifting the probability of our finding ourselves at the birth rank of about 80-100 billion. Whether we should find such an outcome ethically valuable or not is a separate question; however, it can hardly be described as “doomsday” or even “extinction”. We surely ought not be so arrogant and self-centered as to presume that *the end of people like us* necessarily means the end of the world.⁵

³Hence about a thousand years in our future, to pay homage to a famous short story of Frederick Pohl (1970).

⁴E.g., Vidal (2024).

⁵Parenthetically, this is also the point in which the seriousness of DA diverges from far-fetched –

Consider, for instance, Adelstein's (2024) compact formulation of DA: "if a world with more people is no more likely to contain me, and a world with fewer people entails that I'd be early, while a world with more people does not, then my existence at some early time is very strong evidence the future won't have very many people." (p. 6) So, it suffices to accept that in both scenarios 1 and 2 for any time $t > \tau$ there are few like Adelstein – so the chance for those worlds "to contain [him]" is either zero or very small, depending on the size of the residual human population. His acknowledgement that this nominalist view of the reference class resolves the paradoxical element of DA is followed with the comment that "[t]he view that avoids the DA, therefore, isn't merely one that claims future people will be very different, but instead one that claims that future people are categorically *barred* from being in one's reference class."⁶ The locution used indicates the truly contentious issue here: the *reification* of the concept of reference class.

People who argue that the DA-inspired thought experiments should be accepted as legitimate no matter how weird, like Bostrom (2001) or Adelstein (2024), tend to treat the reference class as a given or even as a part of the physical reality, past, present and future. Hence talk such as that "the reference class contains ABC" or that "PQR are barred from the reference class" – as if the reference class is something like the Reform Club or an NBA game or the National Socialist Party or the Gambino crime family or the Harvard Law School. I shall call such a view *reference class realism*. Reference class realists hold that the reference class invoked in a particular anthropic problem (e.g., DA) is an independently existing thing into which one could be inducted or barred from, and that it could be investigated for content and properties existing independently of us as observers and utterers of statements about it.

However, this is unwarranted, since the reference class is not an entity, not a "thing" in the world; it is simply an abbreviation for the set of actors we apply our reasoning to. In other words, it is a shorthand, a compact expression for a (potentially long) list of entities with certain properties, the same as e.g., the Taylor Swift fandom, the readership of the *Atlantic* magazine, the set of humans who have their first names ending in the letter "n," or the set of living creatures on Proxima b. There is nothing else there. Saying that "X and Y belong to the same reference class" or that "Z is in my reference class" without any additional explication does not convey useful information. Without exactly specifying which properties constitute the set we cannot really conclude much about its properties. The set itself is *flatus vocis*, as the nominalists of old would say (Klima 2022). We cannot empirically discover a reference class, or examine it

and somewhat frivolous – thought-experiments like Bostrom's (2001) "Adam & Eve" or Adelstein's (2024) "Graham's number machine". It is an unsupported assertion of these authors that such, completely fanciful situations do highlight something of importance for the very real issues of the future of humanity and its possible extinction (see the analysis in Ćirković 2026; the point is also presciently emphasized by Leslie 1996).

⁶Adelstein (2024), p. 7; emphasis added.

independently of our anthropic thinking and assumptions as we can do with minerals or political parties or Hilbert spaces. It is not a thing in the world and should not be treated as such. The reference class realists are therefore committing the *error of reification*.

The error is surprisingly common in both philosophy and everyday life. We have already seen that the reifying tendency is common for thinkers of radically different views on the anthropic assumptions such as SSA and SIA or about DA, who draw very different conclusion from the same thought experiments (those of Bostrom 2001 are particularly illuminating in this regard). Hence, clearing the confusion is necessary and even urgent.

4. Two realist objections

Suppose that a reference class realist argues: *It is well-known that one who adopts SSA should not think there are many other people in their reference class—for if one's reference class is larger, the probability that they would be themselves, rather than another person in the reference class, is extremely low.* (The converse is true for adepts of SIA, of course.) This is a kind of argumentation often encountered in, for instance, discussions of the Sleeping Beauty problem (Elga 2000).

However, this complaint is either wrong or *not even wrong*, to use Pauli's immortal phrase. Let us first consider this second horn, which comes from denying that "the probability that they would be themselves" is intelligible in the first place. How could we ever assign probability to such a state-of-affairs? Obviously, frequentists cannot help us here. Bayesians could at best state that the prior probability of "evidence", that is the speaker's existence, is 1. Outside of the New Age/Oprah Winfrey/self-help gurus/extreme postmodernist circles, the identitarian notion of "being oneself" makes no (non-tautological) sense. What could it mean? Of course any speaker is herself/himself/itself in the sense of Aristotelian essential properties. Anthropic reasoning revolves around the notions of exactly those properties of an observer; it cannot and should not deal with the situations in which, for example, an observer is drunk or high on LSD or brainwashed by the Ministry of Truth, so that she is "not herself" in the vernacular sense.⁷ A relevant probability and a legitimate usage of the anthropic reasoning is, for instance, the probability that an observer is land-based vs.

⁷Not to say that these are necessarily uninteresting states – but they belong to other fields, like psychiatry or criminology or neurochemistry or political science, not to anthropic reasoning. While I have been surprised to hear different intuitions on this subject from time to time, upon more careful investigation it has always turned out to be some of these vernacular states. For instance, stating that "Ana is not herself whenever she drinks too much" simply means that an extraordinary input (consuming a large quantity of liquor) correlates with extraordinary/unusual behavioural output. That statement being true does not in any way impinge upon Ana's personal identity. Alternatively, one could state that her personal identity necessarily includes the capacity for untypical or extraordinary behaviour under such-and-such conditions.

marine organism, or that is an evolved mind vs. AI vs. Boltzmann brain mind.⁸ These properties do not and cannot constitute an observer's identity ("being oneself"). It makes no sense – at least outside of what could be dubbed *police metaphysics* – to ask for a probability that a particular observer named John Smith is indeed *the true* John Smith.

Could we charitably construe the reference class realist's claim so that is just plain wrong? Perhaps, if we were to regard it as an instance of what is sometimes called the problem of old evidence (e.g., Glymour 1980; Juhl 2007). In brief, it appears when a piece of evidence E is already known – hence fully certain, $p(E) = 1$ – before a hypothesis H is formulated. Bayesian evaluation of H cannot logically raise its probability on the basis of such ("old") evidence. Nobody gave this problem a more beautiful exposition than Stanisław Lem in one of his mock essays, a very real review of two fictional books, *De Impossibilitate Vitae* and *De Impossibilitate Prognoscendi*, by a fictional author Cezar (or perhaps Benedykt) Kouska who uses his own ancestry to "prove" that the theory of probability is unsound.⁹

A certain army doctor, during the First World War, ejected a nurse from the operating room, for he was in the midst of surgery when she entered by mistake. Had the nurse been better acquainted with the hospital, she would not have mistaken the door to the operating room for the door to the first-aid station, and had she not entered the operating room, the surgeon would not have ejected her; had he not ejected her, his superior, the regiment doctor, would not have brought to his attention his unseemly behavior regarding the lady (for she was a volunteer nurse, a society miss), and had the superior not brought this to his attention, the young surgeon would not have considered it his duty to go and apologize to the nurse, would not have taken her to the café, fallen in love with her, and married her, whereby Professor Benedykt Kouska would not have come into the world as the child of this same married couple.

After elaborating in some humorous (an understatement!) detail on other such apparent coincidences leading to the author's birth, and their generalization from human genealogy to the very evolution of our species, Lem poses the key ironic question:¹⁰

Each man is, as it were, the first prize in a lottery, in the kind of lottery, moreover, where the winning ticket is a teragigamegamulticentillion-to-one shot. Why, then, do we not daily feel the astronomically monstrous minuteness of the chance of our own

⁸Boltzmann brains are hypothetical observers arising purely through random fluctuations under fairly general cosmological conditions in contrast to evolved observers (like humans) or intentionally constructed ones (AIs). Although the idea itself has many predecessors – including eponymous Ludwig Boltzmann – it was only after Dyson, Kleban and Susskind (2002) showed that random fluctuations are a generic feature of cosmological models with dark energy that Boltzmann brains entered mainstream physical cosmology.

⁹Lem [1971] (1999), pp. 144-5.

¹⁰*Ibid.*, p. 158.

or another's coming into the world?

Not only is it impossible in retrospect to prove that a particular evolutionary outcome was particularly probable or improbable *in isolation*, but we would need to know probabilistic causes to non-trivially answer the identity question.¹¹ The “old evidence” such as the speaker's existence always has the probability 1.

Our reference class realist friend may try to pose another challenge to nominalism about reference classes. She may argue that “excluding” future people from the reference class is indeed parochial in the temporal sense. “[W]hat's so special about current people?” (Adelstein 2024: 7) Again, there are two ways to counter this complaint. One may notice that “temporal egalitarianism” or “temporal typicality” or “temporal Copernicanism” is *not* a reliable guide, at least in some cosmological and astrobiological applications where it has been studied (Ćirković and Balbi 2020, and references therein). If a reference class realist wishes to use it for bolstering her position, she better provide some *independent* evidence for it. Prior to obtaining such evidence, we are entitled to treat *any* temporal cross-section of a population as special *enough*.

Alternatively, one could simply argue that it is the *capacities* of current humans – and not really *all* humans at that – which are “special” in the sense that they define the abstract reference class used in DA and similar applications of anthropic reasoning. Obviously, some humans are indeed excluded from this reference class, although the reference class realists would like to hide this awkward fact: humans in a coma, very small children or those who as a consequence of grave illnesses or brain injuries lack the capacity of conceiving concepts such as their birth rank. To use the idiom of Bostrom and Adelstein, while one obviously can fall into a coma as a consequence of sickness or injury, one cannot assign (non-zero) probability of *currently being* a comatose human speaker, just as one cannot assign (non-zero) probability of *currently being* a mosquito or a lobster. So, the answer of a reference class nominalist to his fellow realist is: there is nothing really problematic about conceiving the reference class in this exclusionary manner. And besides, there are independent reasons to believe that future posthumans will be sufficiently different in their capacities (e.g., Bostrom 2008) to justify such exclusion.¹²

¹¹ It is an entirely separate and very difficult problem to establish how many properties, both necessary and contingent, determine an individual's identity; the answer may as well be infinitely many, but the present manuscript certainly cannot answer that.

¹² Recall that the reference class nominalism do not see reference classes as “real” – so that exclusion from them is purely an abstract operation, which should alleviate all kinds of postmodernist politically-correct worries about the alleged evils of exclusion!

3. There is no stable reference class, not even one's own

In addition, I submit that there is no stable “X’s reference class” even when X is the speaker. It is a purely linguistic matter to use this locution as a shorthand (and *only* as a shorthand!), while all kinds of vicious nonsense arise when it is taken to mean something real, non-nominalist. Each one of us belongs to many – perhaps infinitely many – reference classes, just as there are many sets to which one could belong: the set of humans, the set of people with brown eyes, the set of people who collect books by Jorge Luis Borges, the set of people who never read anything by Jorge Luis Borges, the set of people who were spanked as children, the set of university administrators, the set of beings, terrestrial or else, who at least once observed a quasar through a telescope, the set of inhabitants of a neutron star (cf. Forward 1980), etc. etc. etc.

I do not mention Borges merely accidentally or whimsically; at least two of his short stories are very relevant for the present topic (both are contained in Borges 1999). In *Tlön, Uqbar, Orbis Tertius*, originally published in 1940, the Argentinian magician makes gentle fun of platonists, realists about the universals and other people who reify all kinds of arbitrary/linguistic sets. Academia of the fictional world of Tlön finds the doctrines of materialism and nominalism incomprehensible. This is reflected in their language: “One does not say ‘moon’; one says ‘aerial-bright above dark-round’ or ‘soft-amberish-celestial’ or any other string.”¹³ In *The Congress* (1971) Borges goes a couple of steps further, showing how wasteful and even potentially dangerous it is to reduce excessively complex individual identity to a couple of historically contingent properties. In trying to create a universally representative body, the story’s protagonists face a tough problem:¹⁴

Designing a body of men and women which would represent all humanity was akin to fixing the exact number of Platonic archetypes, an enigma that has engaged the perplexity of philosophers for centuries. He suggested, therefore, that (to take but one example) don Alejandro Glencoe might represent ranchers, but also Uruguayans, as well as founding fathers and red-bearded men and men sitting in armchairs. Nora Erfjord was Norwegian. Would she represent secretaries, Norwegians, or simply all beautiful women? Was one engineer sufficient to represent all engineers, even engineers from New Zealand?

Clearly, there is a vast number – perhaps infinity – of possible reference classes, dependent on the problem situation (or the way we ask questions). Once that is accepted, the error of reification becomes truly egregious: the suggestion that even a minuscule part of physical reality like, e.g., the city of Montevideo, contains infinity of reference classes each as real as roses and palms in the Parque Prado amounts to obvious nonsense. There cannot be any “one’s reference class” any more than there

¹³Borges (1999), p. 73.

¹⁴*Ibid.*, pp. 426-7.

can be “one’s pronouns”; both are empty rhetorical devices running amok.

Beyond this conceptual instability, reference classes are also unavoidably *vague*. Their boundaries are certainly very hard to sharply delineate. I have already mentioned borderline cases: comatose humans, babies and toddlers, adults with extreme cognitive impairments or disabilities.¹⁵ These are obviously human beings with all human rights and moral worthiness, but cannot be fairly considered members of the reference class at present. Artificial intelligence in current and especially in future advanced versions poses even more extreme problem, since it could be made to simulate some features of human cognition with high fidelity – but does it qualify for the reference class?

Generally speaking, there are three possible sources of vagueness in practically any philosophical problem situation:

- A. vagueness as a property of human language (semantic vagueness);
- B. vagueness as a consequence of human ignorance (epistemic vagueness); and
- C. vagueness as a property of the world (ontological vagueness).

(Of course, any combination of the two or all three can be true as well in a particular case.) I’d argue that contributions of all three are in fact legitimate sources of vagueness of reference classes *in general*, while the reference class nominalist will retain A and B, discarding C. Ontological vagueness is something indicated by our best physical theories (including quantum field theory, general relativity and the modern synthetic theory of biological evolution); the anthology of Syropoulos and Papadopoulos (2021) highlights many cases of vagueness in the so-called exact sciences. Of course, if one accepts reference class nominalism, we need not worry about it in the first place.

Epistemic vagueness (e.g., Williamson 2002) is unavoidable in any case, thus applying to reference classes as well – and, paradoxically, if somehow removed (by acquiring “divine knowledge”) would have led us *out* of the current human reference class. Finally, one may think that semantic vagueness could be dealt with by improving our language – except that one quickly realizes that even within formalized mathematical language one encounters the same phenomenon.

As John Leslie cogently argued in his book on human extinction and DA, there must be a cognitive threshold for participation in the reference class, since one could not persuasively have been born as a being below that threshold and then use one’s birth rank as an argument for the Bayesian shift (Leslie 1996: 205–236, 256–263). Without such a restriction, we could in principle include our closer or more distant evolutionary ancestors. Are the Neanderthals in our reference class, as used by the reference class realists? Are *Homo floresiensis*? If distant common ancestors of both us and chimps are not counted in the reference class for birth-ranking in DA, why should comatose humans (or university administrators) with even less self-awareness count?

¹⁵And then there are university administrators.

4. Potentials in classical field theory as an analogous concept

In classical electrodynamics of Maxwell, an extremely important role is played by electromagnetic potentials: the scalar potential ϕ and the vector potential $\vec{A}$. By applying vector operators to these potentials, one easily obtains the observable quantities, namely electric field $\vec{E}$ (= gradient of the scalar potential ϕ) and magnetic field $\vec{B}$ (= curl of the vector potential $\vec{A}$). In Maxwell's theory, the potentials themselves are not observable; it is at best the *difference* between potentials in two points in space or two moments in time which could be measured by our empirical devices.¹⁶

Early logical empiricism/instrumentalism tended to treat potentials as mere *calculational aids*, whose symmetries enable much easier computation of various practical problems than using sources and their fields only. However, this view has gradually become untenable for two main reasons: (1) potentials often not just simplify formulations, but predict new phenomena and offer a unification platform for the entire theory (e.g., via Lagrangian/Hamiltonian frameworks). (2) In quantum theory, potentials appear to have direct physical significance even where fields vanish (notably the Aharonov–Bohm effect).

The second point is less important for the present analogy. The historical fact, however, is that as a result of reflection upon these reasons, many philosophers of physics now adopt some kind of structural realism: potentials are not straightforwardly “real objects,” but neither are they eliminable. They indicate that the true underlying ontology has to be *relational*, reflecting the shift from local properties to global structural features of the gauge field. What matters is the network of relations preserved under gauge transformations. The work of Richard Healey is an excellent illustration of this approach (Healey 2001, 2007). To simplify, the debate about potentials has thus shifted from asking: “Are potentials real?” to asking: “What level of structure incorporating potentials in a field theory carries physical content?”

There appears to be an apt analogy with the role of reference classes in anthropic reasoning. Reference classes may not be real, and yet there are indispensable in being part of the overall explanatory and even heuristic structure of anthropic reasoning. Of course, philosophy of physics is much older and more developed than the anthropic reasoning, so it is not surprising that it has reached deeper and richer insights into its foundational concepts. On the other hand, one could regard this circumstance as a fruitful challenge for more foundational work in the field of anthropic reasoning.

¹⁶One can combine scalar and vector potential into a single 4-dimensional vector potential which then enables particularly compact and elegant formulation of the field equations, as Feynman brilliantly showed in the Volume 2 of his *Lectures* (Feynman, Leighton and Sands 1964).

5. Discussion

Reference class is a much more demanding concept than it looks at the first glance and more than most philosophers are willing to admit, but not because it is like superconductivity or reverse transcriptase or the government of Eritrea. I have argued for a reference class nominalism, which denies reality, as well as irreducibility, stability and sharpness of reference classes in anthropic reasoning. Besides resolving the paradoxical aspect of DA (admittedly the issue of great practical import), this approach has the advantage of bringing anthropic reasoning into closer harmony with both materialist ontology of physical science and our transhumanist expectations.

Why is this controversial in the first place? One may suspect that some extra-scientific reasons are at work here, notably the current culturally fashionable low ebb of materialist and nominalist philosophies generally, but also the still powerful influence of the postmodernist “critical thinking”. Especially bizarre are attempts to make reference classes “more inclusive” or to shun “exclusion” in this context; this can hardly be described as anything other than virtue signaling (Miller 2019). Nature itself is massively exclusionary when agencies and capacities are concerned (otherwise, evolution through natural selection would have never worked), so I do not think anthropic theorists are in any special danger.

Further research is clearly necessary in order to elaborate the causal structure of anthropic claims, especially in view of the inflationary multiverse (cf. Lewis, Barnes and Goff 2026). This work at the confluence of cosmology, astrobiology and philosophy of science offers a unique *objective* window on the place of mind in physical universe. At the same time, the research in philosophy of artificial intelligence and SETI studies will enable us to study problem situations involving different kinds of minds, and therefore radically different reference classes in comparison to the mundane “all humans” reference class. Only in the fight against undue anthropocentrism, which is still so prevalent in our societies, our philosophies and our institutions, can we hope to better understand this evolved and disteleological universe.

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Milan M. Ćirković

Nominalizam u pogledu referentnih klasa i greška reifikacije

(Rezime)

Antropičko rasuđivanje je nemoguće bez pojma referentne klase. Ovaj temeljni pojam je, međutim, često nejasan, pa i pogrešno shvaćen. U ovom radu se ukazuje na jednu konkretnu grešku na koju se u ovoj oblasti često nailazi: reifikaciju referentne klase. Iako je referentna klasa teorijski ključan pojam, ona ne odgovara ničemu u realnosti. Naprotiv, nominalistički pristup referentnoj klasi obećava najbolje – ili barem najprihvatljivije – rezultate. Analogija sa tretmanom potencijala u klasičnoj

teoriji polja može biti poučna u pogledu kako sličnosti, tako i razlika.

KLJUČNE REČI: antropičko rasuđivanje; greška reifikacije; nominalizam; Bajesianizam; Argument sudnjeg dana; izumiranje čovečanstva; egzistencijalni rizik; lični identitet

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HOW DO WE KNOW WHAT WE KNOW WHEN WE KNOW THE AXIOMS?*

Abstract: *Tyler Burge reconstructs Frege's epistemology as Euclidean rationalism, treating primitive logical laws as self-evident and proofs as displaying an objective order of justificatory dependence. This paper argues that this proposition-first reconstruction misses a central feature of Frege's logic. Frege's demand for gapless proof and the structural machinery reconstructed by Schroeder-Heister in Grundgesetze §48 require deductions to be distinguishable, licensed and composable. General proof theory and categorical proof theory make explicit the mathematical structure: deductions are arrows, their composition corresponds to cut, and their equality calls for independent criteria. Philosophy of logic must answer to the mathematics of deduction.*

Keywords: *Frege, axioms, structural rules, general proof theory*

1. Burge's Euclidean Frege

Frege does not use “axiom” for any sentence selected to begin a theory. An axiom is a basic truth. It is certain without proof, whereas a theorem receives its full justification through proof from truths that are prior to it. Burge calls this conception Euclidean, and the comparison is apt. Frege's foundational project asks for the ultimate grounds of arithmetical truths and for a deductive arrangement that displays their dependence. In the preface to the *Begriffsschrift*, he distinguishes how one happens to arrive at a judgment from how that judgment is finally grounded. In *The Foundations of Arithmetic*, he insists on proofs even where nobody seriously doubts the result, because certainty does *not* by itself reveal what the result depends upon (Frege 1879: Preface; Frege 1884: §§1–3; Burge 1998: 305–11).

Burge's account, however, is *not* a simple foundationalism according to which primitive truths announce their self-evidence and everything else follows mechanically. Frege explains his notation, compares it with ordinary and mathematical language, constructs long derivations, revises his formulations, and repeatedly draws attention to the range and fruitfulness of the system. Burge takes this practice to be epistemically central. A law may be objectively self-evident without being obvious to a particular thinker. The obstacle may lie *not* in the law but inadequate understanding. Working within a theory may expose that inadequacy. It may clarify the content of a principle,

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reveal its relations to other principles, and produce the understanding under which its self-evidence becomes available. The later essays collected in *Truth, Thought, Reason* make the same point. Understanding is not a quasi-perceptual inspection of an isolated thought. It is achieved through systematic reflection within a theory; insight is the culmination of articulated rational capacities. Deduction therefore does more than transmit warrant from premises to conclusion. Deductive practice partly constitutes the understanding through which primitive warrant can be recognised (Burge 2005: 62–64).

Proof itself is more than a convenient presentation. In “Frege on Knowing the Foundation”, Burge interprets Fregean proof as displaying a natural or canonical ordering of justification. This order is objective. It does not depend on which theory we happen to formulate or on our actual ability to discover the relevant proofs. Axioms require no proof because no more basic truths stand before them in the canonical order. Derived truths receive their grounding justification from their place within this order. Burge’s later “Postscript” makes the termination at basic laws a matter of principle: analysis reaches basic logical laws; it does *not* analyse them. They are warranted, but *not* by proof from prior truths (Burge 1998: 305–47; Burge 2005: 360–62, 389). He interprets Frege as giving deduction an objective role in the ordering of reasons and a constitutive epistemic role in the acquisition of adequate understanding. The remaining question is *what Fregean practice forms as deductive mastery*. Burge’s answer points mainly to an order of truths and reasons. Deductive practice enables us to grasp propositions adequately and recognise their justificational relations. That account does not capture the whole of Frege’s logic.

2. Logic as object and as tool

Extensional adequacy as a presentation of logical consequence is not the same thing as adequacy as an analysis of deduction. Two calculi may prove exactly the same formulas and validate exactly the same consequence relation while differing radically as theories of deduction. One may expose discharged assumptions, structural transformations and normal forms; another may conceal all of them. One may permit a substantive account of equality of proofs and their composition; another may register only that a conclusion is obtainable. Coextensiveness of theoremhood does not imply sameness of deductive structure. From Aristotle onwards logic has been called the science of deduction. If logic is reduced to determining which propositions follow from which while the transitions themselves are treated as dispensable, it becomes a tool. Another mathematical discipline may require that restriction. Logic must still ask what deductions are, what makes them deductions, and how they are structured.

Došen formulates this contrast as one between categorical and hypothetical notions. A proposition is categorical; a deduction from assumptions is hypothetical. In “Inferential Semantics”, he makes the structural consequence of this contrast explicit.

If an inference from A to B is reduced to the bare fact that B is a consequence of A, the inferential term that distinguishes one deduction from another disappears. In categorial language, the category of deductions collapses into a preorder: there is at most one arrow between a given source and target. Došen calls this an *extensionalization of inference* (Došen 2015: 153–57). Modern semantics, including much model theory, characteristically gives primacy to the categorial side. The issue is not whether asserting is the most important linguistic activity in general, but whether logic, given its own subject matter, can afford to treat deducing as secondary (Došen 2015: 149–57; Došen 2016: 65–77). The point is methodologically prior to semantic questions. One may fully accept Boolean semantics for the classical connectives and still insist that it gives an impoverished account of their role in mathematical practice. A truth table tells us the extension of a truth-function. It does not by itself tell us how an assumption containing that connective may be used in a proof, what introduction and elimination patterns are associated with it, how such patterns interact with structural principles, or what counts as the same deduction. For those purposes, knowing the truth-function is not yet knowing how to reason with the connective. Došen's analysis gives a precise form to the distinction stated above: a consequence relation preserves source and target information, but the passage from a category of deductions to its preorder forgets the deductions that witness consequence (Došen 2015: 155–57). The paths of deductions are lost.

3. Gapless proof and the irreducibility of rules

This is not yet proof-theoretic semantics - it does not establish that the meanings of logical constants are constituted by rules. The weaker claim is enough: rules cannot be eliminated from an adequate account of logical practice by redescribing them as further truths. A theory of deduction must distinguish what is asserted from what licenses or constitutes a transition. Došen's "Inferential Semantics" raises the same point from the side of general proof theory. An inference is not exhausted by its type, its premise and conclusion. In the categorial notation $f : A \vdash B$, the arrow term f records the operations by which the deduction is built. On this view a theory of inference is, in part, an algebra of inference rules: the rules are operations for constructing deductions, and equations state when the resulting deductions are equal. To reduce inference to consequence does not merely abbreviate the theory. It forgets the operations whose behaviour the theory of deduction studies (Došen 2015: 157–58).

Frege's own practice moves in this direction, even though his explicit philosophy does not formulate the point in Gentzen's terms. In *Begriffsschrift* §13 he says that complex judgments are derived from simpler ones not in order to make them more certain, but in order to bring their connections into view. Knowing laws is not the same thing as knowing them together with their connections. Burge rightly stresses this. Primitive warrant therefore does not by itself determine the best deductive

articulation of a body of truths. The structural component moves more to the centre in Schroeder-Heister's reconstruction of *Grundgesetze* §48. Frege's logical system there can be regarded as a sequent calculus of a specific form. His two-dimensional "implications" distinguish an asserted conclusion from antecedent information on which it depends. On Schroeder-Heister's reconstruction, the relevant permissions can be represented, in modern terms, as operations corresponding to exchange, contraction, and cut. Antecedents may be permuted. Repeated antecedent formulae may be contracted. Deductions may be glued together through an intermediate judgment (Schroeder-Heister 2014: 233–45).

These principles are not additional logical truths about implication. They govern deductions. Exchange concerns the order of assumptions. Contraction concerns their number and reuse. Cut concerns the composition of derivations. They belong to the structural framework within which formulas occur. This is exactly the sort of distinction that an ordering of propositions cannot represent by itself. Frege did not possess Gentzen's general proof theory, including its general account of structural rules. He did not define the meanings of the logical constants through introduction or elimination rules. He did not make normalization or equality of proofs foundational for semantics. His explicit philosophy remained tied to thought, truth, judgment and assertion. In that respect Frege was not a Gentzenian before Gentzen, and it would be bad history to make him one.

But, Frege's explicit philosophy must not determine in advance what counts as philosophically significant in his formal practice. Otherwise, the material isolated by Schroeder-Heister disappears from view. Frege's demand for gaplessness forced him to regulate deductions at a level not exhausted by the truth of their endpoints. His practice anticipates a distinction that Hertz and Gentzen later made central, even though Frege did not draw its full philosophical moral (Schroeder-Heister 2002). Frege anticipated structural features that later figured in general proof theory. He did not endorse the proof-theoretic conception of logic to which those features later contributed. His explicit philosophy of logic was, in this respect, inadequate to his own formal achievement.

4. Gentzen and the mathematics of deduction

Došen's work provides a means for saying what is missing from a proposition-first reconstruction. In "Logical Consequence: A Turn in Style", he emphasises that Gentzen's notation makes structural features of consequence available for direct investigation. Sequents do not merely record that a proposition is a consequence of others. They build premise and conclusion structure into the formal judgment itself. This shift of notation made questions such as cut elimination visible in a way that a purely extensional representation of consequence (such as Tarski's) did not (Došen 1997: 289–311). In "Inferential Semantics" the point is the same: if the

inference itself is suppressed in favour of the ordered pair of its endpoints, the result is a preorder of consequence, not a theory of the possibly many deductions connecting those endpoints (Došen 2015: 155–58). Gentzen’s general proof theory made deductions themselves objects of sustained mathematical investigation. Cut elimination concerns transformations of derivations. The normalization theory that grew from natural deduction concerns reductions of derivations. Admissibility, permutation of rules, the subformula property and the use or non-use of structural principles are likewise questions about the internal organisation of proofs. Gentzen’s consistency proof for arithmetic showed that the analysis of proofs could yield results whose mathematical significance extends well beyond the presentation of a logical calculus (Gentzen 1934–35; 1936).

The analysis of proofs produced results that show that proofs carry enough structure to sustain an autonomous mathematical theory. A view according to which proof structure is merely a dispensable bureaucracy of an antecedently fixed consequence relation must confront mathematics whose theorems concern that structure. Substructural logics provide the best example for this. Once exchange, contraction, weakening and related principles are isolated as structural, they can be varied rather than silently built into the background. Changing the structural environment while retaining logical vocabulary can yield different logics. The structural environment of deduction helps determine a logic; it is not a typographical shroud around a fixed stock of truths. Gentzen’s separation of logical from structural rules supplied the conceptual machinery for posing such questions clearly.

Developments after Gentzen make it harder to regard deductive structure as a technical detail. The argument concerns structure, not a single line of historical influence: Gentzen’s analysis made the internal organisation of deductions mathematically explicit, and categorical proof theory provided the right tools for studying such a structure. Roughly speaking, it begins from deductive syntax and organises it as a freely generated mathematical structure. Logic and theoretical computer science often pass from generators to free structures: the finite words over an alphabet form the free monoid on that alphabet. The categorical construction provides a richer instance of the same mathematical practice.

Došen’s simplest example already blocks the extensional collapse. From $p \wedge p$ to p there are two projections: one takes the first conjunct and the other the second. They have exactly the same premise and conclusion, yet in the free category with binary products the corresponding arrows π_1 and π_2 are distinct. The difference is not an accidental feature of notation; coherence gives us mathematical models in which the two arrows remain distinct. Hence the ordered pair $(p \wedge p, p)$ does *not* determine the inference (Došen 2015: 155–56).

There is a methodological moral here that should precede, rather than follow from, philosophical allegiance. We do *not* first adopt inferentialism and *then* seek a

mathematics that vindicates it. We observe that the analysis of deduction produced cut elimination, normalization, consistency proofs, substructural logics, freely generated categories of proofs, adjunctions and coherence theorems. We *then* ask what conception of logic is adequate to this mathematics. This is why Došen's distinction between categorical and hypothetical approaches is more than a terminological preference. A proposition is categorical; a deduction from assumptions is hypothetical. If logic attends only to propositions, truth and the extension of consequence, it can miss mathematical structure residing in the deductions themselves. In a free syntactic category formulas supply objects, but proofs supply arrows, and there can be many non-equal arrows between the same objects. Collapsing the category to the preorder is a mathematically intelligible operation *precisely because* there was richer structure there to forget.

5. What this does to Burge's reconstruction

Frege's mathematical and logical practice implicitly contains structural rules. Schroeder-Heister's reconstruction of *Grundgesetze* §48 makes this plain: Frege's calculus permits moves corresponding to exchange, contraction and cut (Schroeder-Heister 2014: 233–45). Once a proof is required to be gapless, such moves cannot remain unanalysed. Gaplessness is not merely a demand that a conclusion stand at the right place in an order of truths. It requires that inferential steps be distinguishable, licensed and composable. An analysis of deductions is therefore a necessary condition for the very kind of proof that Frege demands.

Frege's proposition-first philosophy of logic does not fully account for this practice. It can identify basic and derivative truths, and it can order their justificational dependence. It cannot, merely as an ordering of truths, represent the structural operations that make a deduction a gapless deduction: the use and repetition of assumptions, the chaining of derivations, and the equality or inequality of deductions with the same endpoints. Burge does not notice this discrepancy. His reconstruction treats deduction chiefly as the means by which finite thinkers come to grasp truths and their order; it leaves out the structure of deductions which Frege's own calculus already has to regulate.

Došen's categorical analysis states exactly what is missing. When the arrow term in $f: A \vdash B$ is replaced by the pair (A, B) , the result preserves consequence but forgets the deduction: its composition, its transformations and the question of its identity (Došen 2015: 155–58). The comparison does not identify Burge's canonical order with a preorder, nor does it make Frege a Gentzenian. It shows why the later mathematics of deduction can expose a limitation in a philosophy which takes propositions and their ordering as its primary logical objects.

The methodological order is not negotiable. First comes the mathematics of deduction: the rules, the proof transformations, the composition of derivations and the

criteria for equality of proofs. Then comes philosophy of logic. First comes logic and general proof theory; then comes a philosophy of logic adequate to them. Frege's formal practice supplies the mathematics, and it shows that his explicit philosophy of logic does not completely comprehend its own achievement. Burge's reconstruction inherits that blind spot. The philosophy of mathematics must answer to the mathematics, not determine in advance what the mathematics is allowed to contain.

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Kako znamo ono što znamo kada znamo aksiome?

(Rezime)

Tajler Berdž rekonstruiše Fregeovu epistemologiju kao složen oblik euklidskog racionalizma: osnovni logički zakoni su očigledni i ne zahtevaju dokaze, a dokazi pokazuju objektivan poredak opravdavajuće zavisnosti među istinama. U radu se tvrdi da takva filozofija logike, koja polazi od iskaza, ne zahvata važan deo Fregeove formalne prakse. Zahtev za dokazima bez praznina, zajedno sa strukturalnim mehanizmom koji Šroeder-Hajster izdvaja u Grundgesetze §48, zahteva analizu dedukcija. Opšta teorija dokaza i kategorijalna teorija dokaza rasvetljavaju matematiku o kojoj je reč: dedukcije su strelice, njihova kompozicija odgovara pravilu sečenja, a njihova jednakost zahteva posebne kriterijume. Relacija logičke posledice čuva izvodivost, ali zanemaruje dedukcije koje svedoče toj izvodivosti. Filozofija logike mora da odgovara matematici dedukcija, a ne da unapred određuje njen predmet.

KLJUČNE REČI: Frege, aksiome, strukturalna pravila, opšta teorija dokaza

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MODE 3 KNOWLEDGE PRODUCTION, HELIX MODELS AND HOSPESISM AS A STRUCTURAL ELEMENT OF INNOVATION

ABSTRACT: *This paper explores the value of considering together two bodies of literature that are rarely brought into relation with one another, namely Mode 3 knowledge production and Helix frameworks, on the one hand, and the literature on epistemic trespassing (particularly in the context of the idea of hospesism), on the other. It suggests that, while Mode 3 and Helix frameworks help explain why cross-domain engagement is often necessary and valuable, the literature on epistemic trespassing helps specify the conditions under which such engagement is epistemically legitimate, reliable, and beneficial. Together, these perspectives offer a stronger way of understanding responsible innovation in complex knowledge settings that depend on both deep specialization and interdisciplinary collaboration.*

KEYWORDS: *epistemic boundary crossing, expertise, universities, entrepreneurship education*

Introduction

Understanding the role of universities in innovation and entrepreneurship requires more than the traditional view of the university as a site of teaching and research. Although that view remains historically important, it is no longer sufficient to capture the range of functions universities perform today. Contemporary universities are embedded in social and economic processes that demand more from them than the production and transmission of disciplinary knowledge alone. If we are to understand their present role adequately, we must examine the changing context in which they act. This requires attention to at least three interrelated developments: the emergence of the university's "third mission", the transition from linear models of innovation toward nonlinear and networked forms of knowledge production, and the growing importance of innovation ecosystems conceptualized through the Triple, Quadruple, and Quintuple Helix models (each of which will be further discussed in the following sections). Taken together, these developments reveal that the contemporary university must be understood not only as a producer of knowledge, but also as an institutional mediator among heterogeneous forms of expertise.

Let us begin with what is often described as the university's third mission. This refers to the growing expectation that universities should contribute not only through teaching and research, but also through broader forms of economic and social engagement. Beyond their established missions, universities are increasingly expected

to support equitable economic and social development through innovation, knowledge transfer, social engagement, civic education, and wider processes of inclusion (Etzkowitz & Leydesdorff 2000; Carayannis & Campbell 2012; OECD 2022; European Commission and OECD 2023). Within this expanded role, one important set of competences universities are increasingly expected to foster is entrepreneurial competence, since entrepreneurial skills and literacy are often understood as significant drivers of innovation and economic development, as Baggen and Kaffka (2022: 23) argue. Entrepreneurial competence matters not only for venture creation, but also as a broader capacity for adaptability, value creation, and resilience in changing social and economic conditions (Bacigalupo et al., 2016; Baggen & Kaffka, 2022; OECD, 2022; European Commission & OECD 2023). Yet universities often struggle to foster these capacities effectively, since innovation increasingly takes place in complex, multi-actor, and interdisciplinary settings that demand new forms of effective collaboration (Tolstykh, Gamidullaeva & Shmeleva 2021; Jurgelevičius et al. 2025).

Different modes of knowledge production

Earlier conceptions of knowledge production emphasized disciplinary inquiry conducted within relatively autonomous academic settings, with limited concern for the practical applicability of research findings. Traditional approach to academic research reflects what is often described as the linear model of innovation, or so-called “Mode 1” knowledge production, according to which ideas originate within academic settings and then move outward to firms and other market actors, who transform them into commercially or socially useful outcomes. From this view, innovation follows a one-way sequence: discovery comes first, application follows, and the main economic value is realized outside the university itself. Thus “Mode 1” (and the linear model of innovation) has a sequential “cause-effect” relationship between knowledge production and application. The sequential relationship between university research (conducted without any consideration for applicability) and the eventual commercialization of new knowledge by the economy cannot keep up with the demands of the market. Under intensifying global competition, the time horizons for research and development cycles to reach the market are shrinking. Additionally, this model of innovation is unfit to properly communicate user preferences and knowledge from the market back to the production of basic research. Carayannis and Campbell (2012) argue that systems relying solely on such a linear model of innovation would be at an obvious disadvantage compared to systems engaged in what is described as nonlinear innovation. These insights have accelerated the shift of universities (and the systems of which they are a part) from “Mode 1” knowledge creation toward the aforementioned nonlinear models of innovation. Contrary to the assumption that the relationship between research, its broader social role, and its practical application proceeds through a single linear transfer, Carayannis and Campbell (2012) state that

nonlinear models are better suited for fast-moving markets with high competition, pointing to Mode 2 knowledge production as a form of research carried out in light of its practical relevance. Universities engaged in “Mode 2” research do not see research and innovation as successive steps, but as equally important, simultaneous steps, and therefore are more actively involved in innovation. Such research can be adequately pursued only if it is accompanied by transdisciplinarity, heterogeneity, diversity, reflexivity and social responsibility.

While Mode 2 emphasizes the context of application, regardless of whether universities have established mechanisms for capturing value from their discoveries, Mode 3 extends this perspective by incorporating society and the natural environment into the innovation process. Within this framework, social and ecological consequences are no longer treated as external constraints imposed after the fact, but as epistemically relevant to judgments of innovation themselves. The active environment is expected to support the exploitation of the economic potential of research results better than “Mode 1” and “Mode 2” of knowledge production. Mode 3 thus presents knowledge production as a pluralistic and evolving process, shaped by the interaction of different knowledge and innovation modes through co-specialization, co-opetition, cross-learning and hybrid networks. In this sense, it points to a more dynamic model of innovation, in which universities collaborate with a wider range of stakeholders and share resources across interconnected research and innovation initiatives. Given the range of actors with whom universities are expected to interact as they move toward Mode 3 innovation processes, they are likely to have a positive impact regardless of local context by initiating and sustaining the involvement of other stakeholder groups.

Helix models of innovation

In line with broader shifts in knowledge production and innovation, the following section will introduce Helix models of innovation. Helix models are conceptual frameworks that explain innovation as a result of interactions between different societal actors rather than the activity of a single institution. The most well-known is the Triple Helix model, which focuses on the dynamic relationships between universities, industry, and government in the production and application of knowledge (Etzkowitz & Leydesdorff 2000; Carayannis & Campbell 2012). Rather than treating these as separate domains with fixed roles, the model presents them as closely intertwined and often overlapping in practice. Their interaction generates new forms of coordination and hybrid arrangements, including spin-offs and other collaborative structures. On this view, universities do not merely teach and conduct research, firms do not simply commercialize results, and the state does not act only as an external regulator. Each sphere can assume new functions, and innovation arises from their repeated exchange, coordination, and mutual reshaping rather than from a linear sequence of clearly separated stages. This picture becomes broader with the move to the Quadruple and Quintuple

Helix models. The Quadruple Helix expands the earlier focus on universities, business and government by giving a more explicit place to the wider public. Innovation, on this view, is not just a matter of institutional coordination among formal actors, but also of public communication, cultural context and social uptake. The Quintuple Helix extends the model further by bringing the natural environment into the picture, treating ecological conditions and sustainability not as external limits on innovation, but as concerns internal to it. Together, these expanded models suggest that innovation unfolds within a wider social and environmental field than the Triple Helix alone captures. Within such ecosystems, universities are not simply one stakeholder among many, but often occupy a particularly important position. As Tolstykh, Gamidullaeva, and Shmeleva (2021) note, actors in an innovation ecosystem may take on five roles: pacemaker, integrator, developer, implementer, and promoter; and among these the role of integrator is especially significant. Universities are frequently as playing an integrative role, since they bring different actors together, help identify and cultivate relevant competences, and sustain the balance of interests needed for continued collaboration. Although the precise configuration of these relations varies across local contexts, the literature consistently suggests that universities can strengthen innovation ecosystems by building partnerships, circulating knowledge, and fostering the skills needed for participation in increasingly complex forms of innovation. These expanded models suggest that innovation ecosystems are not only multi-actor, but also socially and environmentally embedded. As a result, they require more heterogeneous forms of expertise, greater interdisciplinarity, and a stronger ability to connect perspectives that would otherwise remain isolated.

From an epistemological point of view, however, this immediately raises a further question. If contemporary innovation depends on crossing disciplinary and institutional boundaries, how can educational settings aimed at fostering entrepreneurial competence preserve the benefits of such crossings without also encouraging problematic forms of epistemic overreach? Mode 3 and Helix approaches make clear that boundary crossing should not be treated as a deviation from proper knowledge production, but as one of its normal conditions that capture the innovative and open spirit of educational practice. Yet that insight alone does not tell us how to distinguish epistemically fruitful forms of cross-domain engagement from cases in which authority is extended too far and misused, evidence is misread and distorted, or competence is presumed or ascribed where it has not been earned. It is precisely here that the literature on epistemic trespassing becomes useful. It provides conceptual tools for asking and examining when boundary crossing generates epistemic value, and when it instead gives rise to distortion, misplaced deference, or harmful overreach.

Toward a Conceptual Convergence

Epistemic trespassing occurs when experts move beyond the boundaries of their own domain and make judgments in areas where they lack either the relevant evidential base or the interpretive competence required for reliable assessment. In such cases, expertise in one field does not automatically justify deference in another (Ballantyne 2019; Gerken 2018; DiPaolo 2022; Pavličić et al. 2024). At first glance, epistemic trespassing may seem to be simply a matter of error, in the sense that people who are not sufficiently competent in a given domain fail to grasp the complexity of the issue and therefore misinterpret the evidence before them. But the problem is not exhausted by misunderstanding or faulty interpretation alone. It is also a problem of misplaced authority, since outsiders may be inclined to trust the speaker on the strength of expertise that no longer warrants such trust. Subsequent work has both sharpened and complicated this diagnosis. DiPaolo (2022), for example, argues that what is wrong in central cases of epistemic trespassing is not merely that experts overstep, but that they do so in ways that take advantage of the vulnerability of non-experts, who depend on expert guidance precisely because they cannot independently assess highly specialized claims. For that reason, the problem is not merely that individuals sometimes speak beyond their expertise, but that such interventions can become socially and epistemically dangerous when they are received as authoritative by audiences unable to assess properly the limits of the speaker's knowledge (Gerken 2018; DiPaolo 2022). On this view, epistemic trespassing can be socially harmful even when it is well intentioned. More recent contributions, however, resist treating all boundary crossing as inherently suspect. Watson (2022) argues that disciplinary crossings are justified more often than Ballantyne allows, because many such cases are better understood as exercises of neighboring expertise rather than as genuine instances of epistemic trespassing, and are therefore not *prima facie* epistemically problematic. Similarly, Pavličić and her co-authors (2024) adopt a more charitable approach, emphasizing that some cross-disciplinary movements have historically advanced scientific inquiry and that an overly rigid condemnation of trespassing risks undermining scientific progress. What follows from this is that the central question is not whether boundary crossing as such should be permitted, but how it should be governed. One lesson of this debate is that epistemic trespassing should not be understood solely in terms of excluding those who lack formal authorization. Such a view risks conflating institutional credentialing with epistemic competence and, in doing so, obscuring cases in which valuable judgment emerges from neighboring or otherwise non-standard positions. Yet the issue is not simply one of openness either, since credentials continue to structure trust, deference, and authority in ways that can mislead non-experts and reinforce social hierarchy (see also Fuller 2023). What is needed, therefore, is a more adequate account of gatekeeping, one capable of blocking unwarranted authority claims without foreclosing the kinds of boundary-

crossing criticism and conceptual friction on which inquiry and innovation often depend. Framed in this way, the key question becomes who is in a position to judge boundary crossings well, by what standards such judgments should be made, and with what awareness that both overly permissive and overly restrictive forms of gatekeeping can distort knowledge production, especially in interdisciplinary and transdisciplinary settings.

This debate becomes especially relevant once we turn to Mode 3 and Helix environments, where epistemic collaboration across domains is not exceptional but structurally built into the very structure of knowledge production and organization of innovation. In their account of Mode 3, Carayannis and Campbell (2012) emphasize the coexistence and co-evolution of different knowledge and innovation paradigms, linked through forms of adaptive integration rather than kept apart in fixed institutional compartments. Their later elaboration of the Triple, Quadruple, and Quintuple Helix models further expands this picture: innovation emerges through the interaction of universities, industry and government, but also through the involvement of civil society, media, culture, and, ultimately, the natural environment as a constitutive horizon of sustainable development (Carayannis & Campbell 2012). More recent work continues to refine and extend these models into new domains, including digitalization and sustainability, while preserving the basic idea that innovation depends on heterogeneous actors working across boundaries rather than within them. This is precisely why the notion of hospesism developed by Pavličić et al. (2024) is so helpful. Hospesism captures a mode of entry into another epistemic domain that neither dismisses expertise nor seeks to replace it. A simple example would be a chemist contributing to geography by identifying patterns of soil contamination that reshape understandings of land use and environmental risk, or by proposing new ways of mapping pollution diffusion that go beyond established geographical models. In such cases, the contribution is not grounded in applying chemical methods directly, but in using an external epistemic standpoint to reframe geographical problems in a different way. Hospesism names a form of contribution made possible by partial exteriority – one brings something valuable precisely because one is not wholly enclosed within the assumptions of the field one enters. In this respect, hospesism provides a more refined normative vocabulary than a mere contrast between “experts” and “non-experts”. It allows us to see how external perspectives may be epistemically useful without implying that all boundaries of expertise should be dissolved. This point creates a natural bridge to the literature on entrepreneurship education.

Such a move is of special importance once we acknowledge the literature on Mode 3 and Helix settings, tends to treat boundary crossing primarily as something to be fostered, coordinated, and institutionally designed. Cross-sectoral exchange appears here as a condition of innovation, not first and foremost as a source of epistemic risk. For that reason, Mode 3 and Helix approaches have much to gain from engagement with

the literature on epistemic trespassing. If the former explain why boundary crossing is often necessary, the latter helps clarify when such crossings remain epistemically responsible and when they become forms of overreach. That is precisely why bringing these literatures together seems not only productive, but conceptually natural and well motivated.

A further brief clarification helps explain why. Epistemic trespassing is a normative and role-sensitive notion. It concerns who is entitled to speak or act with authority, under what evidential and skill-related conditions, and how non-experts should revise their judgments when they suspect such trespassing. Mode 3 and Helix frameworks, by contrast, are descriptive and programmatic. Often used to guide policy and institutional design, they treat boundary crossing as a central mechanism of innovation, with co-evolution, co-specialization, pluralism and cross-learning cast as positive features. Entrepreneurship education, understood here not simply as training for starting businesses, but more broadly as the capacity to turn ideas and opportunities into action by drawing on and organizing relevant resources (Bacigalupo et al. 2016), often unfolds precisely in these hybrid spaces, through cross-disciplinary cohorts, external entrepreneurs, incubators, third-space professionals, and competence-based forms of teaching that value initiative and experimentation. Insights from the literature on epistemic trespassing can normatively deepen Mode 3 and Helix theory by showing that successful entrepreneurship education and innovation ecosystems require not only more boundary crossing, but better epistemic norms for crossing boundaries. Helix and Mode 3 scholarship offers a rich vocabulary for collaboration, coordination, and institutional design, but it has less to say about epistemic legitimacy in cross-domain teaching and mentoring: who is entitled to claim what, how uncertainty should be signaled, how learners calibrate deference, and how they identify genuine expertise (e.g. Grundmann 2025) rather than relying too heavily on social markers or formal credentials. What is needed, therefore, is a framework capable of explaining how the same form of boundary crossing can be epistemically productive under some institutional conditions and harmful under others. Such a cross-disciplinary move would connect competence-oriented research in entrepreneurship education with normative discussions of expertise in social epistemology and philosophy of science, while also opening the way to a more sustained engagement between these literatures. In doing so, it may sharpen and enrich the broader family of innovation-systems frameworks.

Concluding remarks

In this paper, we have argued that Mode 3 and Helix frameworks can be productively read in conjunction with the literature on epistemic trespassing. The former shows that boundary crossing is not an anomaly in contemporary knowledge production, but one of its defining structural features, in the sense that universities, industry, government, civil society, and environmental actors increasingly interact through

hybrid and overlapping forms of collaboration. For that reason, epistemic trespassing should not be judged in simple binary terms, as either inherently desirable or inherently suspect. Precisely because such crossings are now so central to innovation, the literature on epistemic trespassing becomes normatively important. It helps us distinguish productive cross-domain engagement from harmful overreach by clarifying the conditions under which authority claims remain legitimate, reliable, and beneficial. Therefore, we can conclude that Mode 3 and Helix approaches explain why boundary crossing is often necessary, while the literature on epistemic trespassing helps specify how it can be carried out responsibly. Taken together, they offer a stronger basis for thinking about responsible innovation in highly specialised yet increasingly interdisciplinary knowledge settings.

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Strahinja Đorđević, Ákos Horváth i Jelena Pavličić Cerović

„Mode 3” proizvodnja znanja, heliks modeli i hospesizam kao strukturni element inovacije

(Rezime)

Ovaj rad razmatra značaj povezivanja dva korpusa literature koji se retko dovode u međusobnu vezu, pri čemu se s jedne strane razmatraju koncept „Mode 3” proizvodnje znanja i Helix okviri, a s druge literatura o epistemičkom prekoračenju granica stručnost (posebno u kontekstu ideje hospesizma). U radu se sugerše da, dok Mode 3 i Helix okviri pomažu da se objasni zašto je epistemički angažman koji podrazumeva prelaženje granica stručnosti često neophodan i vredan, od literature o epistemičkom prelaženju granica stručnosti očekujemo da razjasni uslove pod kojima je takav angažman epistemički legitiman, pouzdan i koristan. Zajedno, ove perspektive doprinose dubljem razumevanju odgovorne inovacije u složenim okruženjima proizvodnje znanja koja zavise kako od duboke specijalizacije tako i od interdisciplinarne saradnje.

KLJUČNE REČI: epistemičko prelaženje granica stručnosti, ekspertiza, univerziteti, preduzetničko obrazovanje



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STRUKTURA, STANJE I NOSIVOST: ONTOLOŠKI OKVIR STABILNOSTI U STRATIFIKOVANIM SISTEMIMA

Apstrakt: Rad razvija ontološki okvir za analizu stabilnosti i promene u složenim, stratifikovanim sistemima. Polazi se od razlike između strukture i stanja, pri čemu stanje označava konkretnu konfiguraciju ispoljavanja, dok struktura označava poredak odnosa i mehanizama koji određuju prostor mogućih stanja. U radu se uvodi pojam nosivosti kao kriterijum kvaliteta strukturnih odnosa, odnosno sposobnosti sistema da održava stabilne režime bez stalnih kompenzacionih intervencija. Time se stabilnost razlikuje od puke ponovljivosti ishoda, a promena od varijacije stanja. Predloženi okvir omogućava preciznije razlikovanje promena stanja, adaptacija i strukturnih promena prema njihovom ontološkom dometu.

Ključne reči: ontološka vertikala; struktura; stanje; nosivost; stabilnost; stratifikovani sistemi

1. Uvod: Stabilnost, struktura i pitanje nosivosti

Savremeni pristupi stabilnosti i promeni u složenim sistemima često se oslanjaju na performativne kriterijume. Sistem se smatra stabilnim ukoliko proizvodi očekivane ishode, održava funkcije i uspešno se prilagođava poremećajima. Takav okvir ima jasnu pragmatičnu vrednost u inženjerskim, organizacionim, biološkim i društvenim kontekstima, ali se u teorijama kompleksnosti i mehanističkim pristupima objašnjenju naglašava značaj unutrašnje organizacije sistema i mehanizama koji proizvode posmatrano ponašanje, kao i ograničenja pristupa koji se oslanjaju isključivo na ishode (Simon 1962; Craver 2007; Wimsatt 2007). Time se otvara pitanje u kojoj meri stabilnost posmatranog ponašanja zaista odražava kvalitet uzročnog ili organizacionog poretka koji ga proizvodi. Drugim rečima, performanse mogu ostati prihvatljive čak i onda kada je strukturna osnova sistema postala krhka ili zavisna od trajnih kompenzacija, što pokazuje da evaluacija zasnovana isključivo na ishodima može prikriti strukturno relevantne razlike u kvalitetu organizacije sistema.

Ova teškoća postaje naročito izražena u stratifikovanim sistemima, u kojima različiti nivoi organizacije stoje u asimetričnim odnosima određivanja. Rasprave o nivoima objašnjenja i stratifikaciji realnosti naglašavaju da fenomeni na jednoj ravni ne moraju biti iscrpno objašnjivi terminima druge ravni, kao i da su veze među nivoima često nelinearne i metodološki zahtevne (Bhaskar 1975; Craver 2007; Wimsatt 2007). U tom smislu, stabilizacija stanja na nivou ispoljavanja može prikriti promene (ili deficite) na nivou strukture. Ishodi mogu ostati stabilni dok se uslovi njihove mogućnosti postepeno menjaju. Upravo zato je za analizu stabilnosti i promene neophodno eksplicitno

razdvajanje nivoa na kojima se fenomen manifestuje od nivoa na kojima je uzročno ukorenjen (Kim 1992; Craver & Bechtel 2007).

Ovaj rad polazi od navedene metodološke teškoće i predlaže teorijski okvir zasnovan na tri povezana pojma: ontološkoj vertikali, distinkciji između strukture i stanja, i nosivosti kao kvalitetu strukturnih odnosa. Cilj nije uvođenje nove domenske teorije, već preciziranje pojmovnog aparata pomoću kojeg se stabilnost i promena mogu razlikovati na strukturno relevantan način, u skladu sa uvidima o stratifikaciji i mehanističkom objašnjenju (Craver & Darden 2013; Ladyman & Ross 2007). Ontološka vertikala omogućava da se nivoi realnosti posmatraju u asimetričnom odnosu određivanja; distinkcija struktura/stanje razdvaja poredak odnosa od njegovih pojava realizacija; pojam nosivosti uvodi kriterijum za procenu da li struktura može održavati stabilnost bez stalne proizvodnje kompenzacija.

Doprinos ovog rada može se sažeti u tri povezana tačke. Prvo, sistematski se razrađuje distinkcija između strukture i stanja i pokazuje zašto je ona neophodna za razumevanje stabilnosti i promene u složenim sistemima (Craver 2007; Glennan 2002). Drugo, uvodi se pojam nosivosti kao kriterijum kvaliteta poretka odnosa i ograničenja koja određuju prostor mogućih stabilnih režima. Ovaj pojam se razvija u osloncu na uvide o značaju organizacije i ograničenja u složenim sistemima (Simon 1962; Wimsatt 2007). Treće, ontološka vertikala se koristi da se precizira asimetrija između nivoa i objasni kako su mogući režimi stabilnosti bez nosive strukture. U nastavku rada najpre se razrađuje pojam ontološke vertikale, zatim distinkcija između strukture i stanja, potom pojam nosivosti i, konačno, razlikovanje tipova promene prema njihovom strukturnom dometu.

Iako u savremenoj literaturi postoje brojni uvidi o značaju strukture, mehanizama i ograničenja u razumevanju ponašanja složenih sistema, oni često ostaju raspršeni ili implicitni. U tom smislu, doprinos ovog rada nije u uvođenju potpuno novih elemenata, već u njihovom sistematskom povezivanju i uvođenju pojma nosivosti kao kriterijuma koji omogućava da se razlika između stabilnosti zasnovane na strukturi i stabilnosti održavane kompenzacijama učini pojmovno eksplicitnom i metodološki operativnom. U najužem smislu, doprinos rada sastoji se u pojmovnom razlikovanju stabilnosti koja se prepoznaje na nivou stanja i stabilnosti kao posledice nosive strukture.

2. Ontološka vertikala i asimetrija nivoa

Polazna pretpostavka ovog rada jeste da se složeni sistemi ne mogu adekvatno razumeti ukoliko se posmatraju kao homogena celina, već da njihova struktura podrazumeva stratifikaciju na različite ontološke nivoe. Ova ideja ima dugu tradiciju u filozofiji nauke i ontologiji, gde se govori o nivoima objašnjenja, stratifikaciji realnosti i višeslojnoj organizaciji sistema, pri čemu se naglašava da različite ravni mogu imati različita svojstva i uzročno relevantnost (Bhaskar 1975; Wimsatt 2007; Craver 2007). Ključni uvid tih pristupa jeste da fenomeni na jednoj ravni ne mogu uvek biti iscrpno

objašnjeni terminima druge ravni, kao i da odnosi među nivoima nisu nužno simetrični.

U tom kontekstu, pojam ontološke vertikale uvodi se kako bi se naglasila upravo ta asimetrija. Viši nivoi ispoljavanja zavise od nižih nivoa strukture i mehanizama, dok obrnuta zavisnost ne važi u istom smislu. Drugim rečima, promene u dubljim slojevima sistema određuju prostor mogućih stanja i ponašanja na višim slojevima, ali promene na nivou ispoljavanja ne moraju nužno implicirati promene u uzročnom poretku koji ta ispoljavanja proizvodi. Ovakvo razumevanje odnosa između nivoa oslanja se na savremene rasprave o mehanističkom objašnjenju i višeslojnoj uzročnosti, u kojima se naglašava da objašnjenje fenomena zahteva referencu na organizaciju delova i njihove interakcije, a ne samo na opis vidljivih ishoda (Craver 2007; Craver & Bechtel 2007; Craver & Darden 2013).

Pojam ontološke vertikale ovde se koristi za označavanje asimetrične stratifikacije nivoa organizacije, u kojoj dublji nivoi određuju uslove mogućnosti i prostor mogućih stanja viših nivoa. Vertikalu pritom ne treba razumeti kao normativnu hijerarhiju vrednosti, već kao strukturu zavisnosti između različitih ravni organizacije i ispoljavanja.

Rasprave o nivoima objašnjenja dodatno pokazuju da se različiti nivoi ne razlikuju samo po "stepenu apstrakcije", već i po tipu svojstava i objašnjenja koja su na njima relevantna (Kim 1992; Wimsatt 2007). U tom smislu, ontološka vertikala ne označava hijerarhiju vrednosti ili važnosti, već mrežu odnosa u kojima jedan nivo postavlja uslove mogućnosti za drugi. Ovakvo shvatanje oslanja se na uvide kritičkog realizma, koji insistira na stratifikaciji realnosti i na razlici između pojavnih događaja i dubljih generativnih mehanizama (Bhaskar 1975).

U teorijama kompleksnosti i sistemskom mišljenju sličan uvid pojavljuje se kroz razlikovanje između arhitekture sistema i njegovog ponašanja. Sajmonova (Simon) klasična analiza arhitekture kompleksnosti pokazuje da stabilni obrasci ponašanja zavise od organizacije sistema i od načina na koji su podsistemi povezani, a ne samo od trenutnih stanja varijabli (Simon 1962). Wimsat (Wimsatt) dodatno naglašava da su sistemi "razloživi" na načine koji zavise od nivoa opisa i da ograničenja na jednom nivou mogu fundamentalno oblikovati mogućnosti na drugom (Wimsatt 2007). Ovi uvidi ukazuju na značaj razmatranja odnosa između različitih nivoa organizacije u analizi stabilnosti i promene.

Ontološka vertikala, kako se ovde uvodi, zato ima dvostruku ulogu. Prvo, ona omogućava da se jasno razdvoje ravan ispoljavanja i ravan uzročnog ukorenjenja fenomena. Drugo, ona uvodi kriterijum za razlikovanje tipova promene – promene koje se dešavaju na nivou stanja ne moraju zahvatiti nivo strukture, dok promene na nivou strukture nužno redefinišu prostor mogućih stanja. Ova distinkcija je u skladu sa mehanističkim shvatanjem uzročnosti, prema kojem intervencije mogu delovati na različitim nivoima organizacije sistema, pri čemu razumevanje njihovih efekata zahteva uvid u to da li zahvataju samo manifestacije ili i mehanizme koji ih proizvode

(Woodward 2003; Craver & Darden 2013).

Važno je naglasiti da ontološka vertikalna ne podrazumeva da su nivoi potpuno izolovani ili da među njima ne postoje povratne sprege. Naprotiv, savremene rasprave o “top-down” i “bottom-up” odnosima pokazuju da su interakcije između nivoa složene i često posredovane (Craver & Bechtel 2007). Ipak, čak i u prisustvu takvih povratnih veza, može se tvrditi da je zavisnost između nivoa asimetrična u ontološkom smislu. Bez odgovarajućeg poretka mehanizama i odnosa na dubljem nivou, određeni obrasci ispoljavanja na višem nivou ne bi bili mogući.

Upravo iz tog razloga, ontološka vertikalna predstavlja važan konceptualni okvir za analizu stabilnosti. Ako se stabilnost posmatra isključivo na nivou ispoljavanja, zanemaruje se pitanje na čemu ta stabilnost počiva. Tek kada se stabilnost sagleda u odnosu na vertikalnu strukturu nivoa, postaje moguće postaviti pitanje da li je ona utemeljena u nosivoj strukturi ili se održava putem kompenzacija i adaptivnih mehanizama koji ne menjaju uzročni poredak. Ova perspektiva priprema teren za naredni korak analize – preciziranje ontološke distinkcije između strukture i stanja, koja omogućava da se razlika između stabilnosti ispoljavanja i kvaliteta strukture učini pojmovno operativnom.

3. Struktura i stanje: ontološka distinkcija

Razlikovanje između strukture i stanja predstavlja ključnu ontološku distinkciju za analizu stabilnosti i promene u složenim sistemima. Pod stanjem se ovde podrazumeva konkretna konfiguracija varijabli ili elemenata sistema u datom trenutku, dok struktura označava poredak odnosa i mehanizama koji određuju koje su konfiguracije uopšte moguće, verovatne ili stabilne. Ova razlika nalazi snažan teorijski oslonac u mehanističkim pristupima objašnjenju, u kojima se ponašanje sistema objašnjava kroz organizaciju delova i njihovih interakcija, a ne samo kroz opis trenutnih ishoda (Craver 2007; Craver & Darden 2013; Glennan 2002). Slično tome, u teorijama kompleksnosti naglašava se da se stabilni obrasci ponašanja ne mogu razumeti isključivo na nivou pojava stanja, već zahtevaju analizu arhitekture sistema i ograničenja koja ta arhitektura nameće (Simon 1962; Wimsatt 2007).

U ontološkom smislu, ova distinkcija se može povezati sa idejom stratifikacije realnosti, prema kojoj različiti nivoi mogu imati različite tipove svojstava i uzročnu relevantnost u različitim teorijskim pristupima (Bhaskar 1975; Kim 1992). Stanja pripadaju ravni ispoljavanja, dok struktura pripada ravni uslova mogućnosti tog ispoljavanja. Posledica toga jeste da promena stanja ne mora implicirati promenu strukture. Sistem može prelaziti iz jednog stanja u drugo, a da poredak odnosa koji određuje prostor mogućih stanja ostane neizmenjen. Upravo iz tog razloga, stabilnost posmatrana samo na nivou stanja može prikriti strukturne deficite, dok analiza strukture omogućava da se postavi pitanje o kvalitetu samog uzročnog poretka koji proizvodi stabilna ili nestabilna ispoljavanja.

Mehanistički okvir dodatno precizira ovu razliku. Prema tom pristupu, objašnjenje fenomena zahteva identifikaciju mehanizama, organizovanih sklopova delova i aktivnosti, koji generišu posmatrane ishode (Craver 2007; Craver & Darden 2013). Stanja su, u tom smislu, manifestacije rada mehanizama u datim okolnostima, dok struktura obuhvata samu organizaciju mehanizama i njihovih međusobnih odnosa. Glenanov (Glennan) naglasak na organizaciji i stabilnosti mehanizama pokazuje da isti mehanizam može proizvoditi različita stanja, kao i da različite organizacije mogu proizvoditi slična stanja, što dodatno potvrđuje potrebu za njihovim razlikovanjem kao analitički distinktnih nivoa (Glennan 2002).

Sistemska i kompleksna perspektiva dopunjuju ovu sliku razlikovanjem između strukture sistema i njegovog ponašanja. Sajmonova analiza arhitekture kompleksnosti pokazuje da stabilnost ponašanja zavisi od načina na koji su podsistemi povezani i od ograničenja koja proizlaze iz te organizacije, a ne samo od trenutnih vrednosti varijabli (Simon 1962). Vimsat dodatno naglašava da su sistemi organizovani na načine koji uvode robusna ograničenja i selektivne puteve razlaganja, što znači da promene na nivou stanja često ostaju unutar istih strukturnih granica (Wimsatt 2007). Ovi uvidi ukazuju na značaj razumevanja poretka odnosa u analizi stabilnosti.

Iz perspektive ontološke vertikale, razlika između strukture i stanja dobija dodatnu težinu. Ako su nivoi organizovani asimetrično, onda promene na dubljem nivou (strukтури) redefinišu prostor mogućih stanja na višem nivou, dok promene na nivou stanja ne moraju zahvatiti strukturu. To znači da je moguće imati bogatu dinamiku stanja, uključujući prilagođavanja, oscilacije i stabilizacije, bez promene u poretku odnosa koji tu dinamiku proizvodi. U tom slučaju, promene su realne i empirijski vidljive, ali njihov ontološki domet ostaje ograničen na ravan ispoljavanja.

Ova distinkcija ima neposredne metodološke posledice za analizu stabilnosti i promene. Ako se stabilnost meri isključivo kroz ponovljivost ili održavanje stanja, tada se gubi iz vida pitanje da li struktura koja ta stanja proizvodi ostaje nosiva ili se, naprotiv, održava putem sve složenijih kompenzacija. Razlikovanje strukture i stanja omogućava da se stabilnost sagleda u odnosu na uslove svoje mogućnosti, a ne samo u odnosu na ishode. Time postaje moguće preciznije razlikovanje između promena koje zahvataju samo konfiguracije ispoljavanja i onih koje menjaju sam poredak odnosa.

U tom smislu, distinkcija struktura/stanje predstavlja konceptualni most između ontološke vertikale i kriterijuma stabilnosti koji se uvode u nastavku rada. Tek kada se jasno razdvoje ova dva nivoa, postaje smisleno postaviti pitanje kvaliteta strukture i njene sposobnosti da dugoročno održava stabilne režime bez oslanjanja na trajne kompenzacije. Upravo to pitanje vodi ka pojmu nosivosti, koji se razrađuje u narednom poglavlju kao kriterijum analize stabilnosti.

4. Nosivost kao kriterijum stabilnosti

Ako se stabilnost posmatra isključivo na nivou ispoljavanja, tada ona ostaje vezana za ponovljivost stanja i očuvanje performansi. Međutim, iz perspektive ontološke vertikale i distinkcije između strukture i stanja, postaje jasno da se pitanje stabilnosti mora preformulisati. Ključno nije samo da li sistem održava određena stanja, već na čemu ta stabilnost počiva. Ovaj rad uvodi pojam nosivosti kako bi označio kvalitet strukturnih odnosa koji omogućavaju generisanje i održavanje stabilnih režima bez oslanjanja na trajne kompenzacije. Time se analiza stabilnosti pomera sa ravni ishoda na ravan uslova mogućnosti.

U literaturi o složenim sistemima i inženjerskoj biologiji, srodni problemi razmatraju se kroz pojmove robusnosti i rezilijentnosti. Robusnost se često definiše kao sposobnost sistema da zadrži funkciju uprkos poremećajima, dok rezilijentnost naglašava sposobnost sistema da se oporavi ili reorganizuje nakon šoka (Holling 1973; Kitano 2004). Iako su ovi pojmovi izuzetno korisni, oni ostaju pretežno performativni. Fokusirani su na očuvanje ili povratak funkcije, a ne nužno na kvalitet poretka odnosa koji tu funkciju omogućava. Karlson i Dojl (Carlson and Doyle) dodatno pokazuju da sistemi mogu biti projektovani ili evoluirani tako da budu robusni u odnosu na jedne klase poremećaja, a istovremeno krhki u odnosu na druge, upravo zbog načina na koji su njihova ograničenja i arhitektura organizovani (Carlson & Doyle 2002).

Pojam nosivosti, kako se ovde uvodi, pomera akcenat sa otpornosti ishoda na kvalitet strukture. Struktura je nosiva ukoliko omogućava da stabilni režimi nastaju i opstaju bez potrebe za stalnim korekcijama i kompenzacijama ad hoc karaktera. U tom smislu, nosivost je bliska ideji ograničenja (engl. constraints) u teorijama sistema. Stabilni obrasci ponašanja nisu samo rezultat trenutnih stanja, već su oblikovani trajnim ograničenjima koja proizlaze iz organizacije sistema (Ashby 1956; Meadows 2008). Kada su ta ograničenja koherentna i međusobno usklađena, sistem može održavati stabilnost bez potrebe za intenzivnim dodatnim intervencijama; kada nisu, stabilnost se mora „proizvoditi“ kroz sve složenije intervencije.

Ilustrativan primer pružaju organizacioni sistemi koji uspevaju da kratkoročno održavaju performanse uprkos rastućim strukturnim deficitima. Institucija može zadržavati funkcionalnost kroz stalno uvođenje novih procedura, korektivnih mehanizama i regulatornih slojeva, pri čemu se stabilnost ishoda održava po cenu rastuće kompleksnosti upravljanja. U takvim slučajevima performanse mogu ostati stabilne, dok sama struktura postaje sve manje nosiva, jer stabilnost više ne proizlazi iz koherentnog poretka odnosa, već iz kontinuirane potrebe za kompenzacionim intervencijama.

Ova razlika može se dodatno precizirati kroz prizmu arhitekture kompleksnosti. Sajmon je pokazao da hijerarhijska i modularna organizacija omogućava stabilnost i adaptivnost time što uvodi strukturna ograničenja koja lokalizuju poremećaje i sprečavaju njihovo nekontrolisano širenje (Simon 1962). U takvim slučajevima, stabilnost ponašanja nije samo rezultat stalnih korekcija, već izraz arhitekture koja poseduje

nosivost. Nasuprot tome, sistemi sa loše usklađenim ili kontradiktornim ograničenjima mogu održavati funkcionalnost samo uz cenu kontinuiranih korektivnih intervencija, odnosno onoga što se u organizacionoj praksi često opisuje kao „gašenje požara”.

Važno je naglasiti da nosivost nije isto što i nepromenljivost. Sistem može biti nosiv i istovremeno sposoban za promene stanja i adaptacije. Razlika je u tome što se u nosivim sistemima adaptacije odvijaju unutar okvira koji ostaje koherentan, dok u nenosivim sistemima adaptacije sve više funkcionišu kao zamena za strukturnu koherenciju. U literaturi o rezilijentnosti često se ističe da adaptacija može uključivati i transformaciju, ali bez jasnog kriterijuma otežano je razlikovanje između promene koja redefiniše strukturu i promene koja samo produžava funkcionisanje unutar istih ograničenja (Holling 1973; Kitano 2004). Pojam nosivosti uvodi upravo takav kriterijum: da li struktura ostaje sposobna da dugoročno održava stabilnost bez eskalacije kompenzacija.

Iz perspektive distinkcije struktura/stanje, nosivost se odnosi na kvalitet poretka odnosa, a ne na trenutne konfiguracije. Sistem može prolaziti kroz širok spektar stanja, a i dalje biti nosiv, ukoliko se ta stanja odvijaju unutar koherentnog strukturnog okvira. Obrnuto, sistem može pokazivati stabilna ili čak poboljšana stanja, a da pri tom njegova struktura gubi nosivost, što znači da se stabilnost održava uz rastuću zavisnost od korektivnih mehanizama.

Ova razlika ima neposredne metodološke posledice. Ako se stabilnost procenjuje isključivo kroz performanse, tada se gubi iz vida pitanje da li se stabilnost održava ili proizvodi. Pojam nosivosti omogućava jasno razlikovanje ove dve situacije. Nosiva struktura održava stabilnost kao posledicu svoje organizacije, dok nenosiva struktura zahteva stalnu proizvodnju stabilnosti kroz intervencije, prilagođavanja i kompenzacije. Time se stabilnost razdvaja na stabilnost kao svojstvo strukture i stabilnost kao rezultat stalnog upravljanja posledicama poremećaja.

U tom smislu, nosivost postaje ključni kriterijum za razumevanje ontološkog dometa promene. Promene koje poboljšavaju performanse, ali ne menjaju kvalitet strukturnih odnosa, ostaju ograničene na ravan stanja. Nasuprot tome, promene koje zahvataju poredak odnosa i unapređuju ili narušavaju nosivost strukture menjaju same uslove mogućnosti stabilnosti. Ova distinkcija uvodi naredni korak analize – razlikovanje tipova promene prema tome da li deluju na nivou stanja, adaptacije ili strukture.

5. Tipovi promene i ontološki domet

Ako se stabilnost sagleda kroz prizmu ontološke vertikale, distinkcije između strukture i stanja i pojma nosivosti, postaje metodološki opravdano razlikovati tipove promene prema njihovom ontološkom dometu. U praksi intervencija često se govori o „promeni“ bez njenog preciznijeg određivanja, bez razjašnjenja da li se radi o promeni konfiguracije ispoljavanja, o adaptivnom prilagođavanju unutar postojećeg okvira ili o

promeni samog poretka odnosa. Ovakva neodređenost vodi ka metodološkoj konfuziji, jer različiti tipovi promene uključuju različite uzroke, mehanizme i posledice po dugoročnu stabilnost sistema.

Polazeći od distinkcije struktura/stanje, najpre se izdvaja promena stanja. Pod tim se podrazumevaju promene konkretnih konfiguracija varijabli ili elemenata sistema, dok poredak odnosa koji određuje prostor mogućih stanja ostaje nepromenjen. U mehanističkom okviru, ovakve promene odgovaraju promenama u radu istih mehanizama pod različitim početnim ili graničnim uslovima (Craver 2007; Craver & Darden 2013). Iz perspektive ontološke vertikale, njihov domet je ograničen na ravan ispoljavanja. Sistem može pokazivati bogatu dinamiku, oscilacije ili čak poboljšane performanse, a da pri tom uzročni poredak ostaje isti. Takve promene su realne i empirijski relevantne, ali ne zahvataju uslove mogućnosti stabilnosti.

Drugi tip promene predstavlja adaptacija. Adaptivne promene ne svode se na prostu promenu stanja, već podrazumevaju prilagođavanje načina funkcionisanja sistema u okviru postojećeg strukturnog okvira. U istraživanjima kompleksnih sistema i rezilijentnosti adaptacija se često razume kao sposobnost sistema da modifikuje svoje odgovore na poremećaje kako bi očuvao funkciju (Holling 1973; Kitano 2004). U mehanističkim terminima, to može značiti rekonfiguraciju tokova aktivnosti, promenu u raspodeli uloga među delovima ili uvođenje novih regulatornih petlji, pri čemu osnovna organizacija ostaje prepoznatljiva (Craver & Darden 2013). Ontološki domet adaptacije je širi od puke promene stanja, ali i dalje ne zahvata nužno sam poredak odnosa koji određuje nosivost strukture.

Treći tip promene predstavlja strukturna promena. Ona zahvata nivo na kojem se definišu uslovi mogućnosti stabilnosti: poredak odnosa, organizaciju mehanizama, i ograničenja koja oblikuju prostor mogućih stanja. U okviru uzročne analize intervencija, to odgovara promenama koje menjaju same uzročne mehanizme ili organizaciju odnosa među njima, a ne samo njihove izlaze (Woodward 2003; Craver 2007). Sa stanovišta ontološke vertikale, strukturna promena redefiniše odnos između nivoa, tj. menja šta sistem može stabilno da bude, a ne samo kako se ponaša unutar datog režima.

Ova trodelna podela dobija punu analitičku jasnoću kada se poveže sa pojmom nosivosti. Promene stanja mogu poboljšati ili pogoršati performanse bez direktnog uticaja na nosivost strukture. Adaptacije mogu privremeno ili dugoročnije stabilizovati sistem, ali ostaju ontološki dvoznačne. One mogu ili rasteretiti strukturu i otvoriti prostor za obnovu nosivosti, ili, naprotiv, povećati zavisnost sistema od kompenzacija i time prikriti strukturni deficit. Strukturne promene, s druge strane, jedine neposredno menjaju nosivost, jer zahvataju sam poredak odnosa i ograničenja koji određuju da li se stabilnost održava kao posledica organizacije ili se proizvodi kroz stalne korekcije.

Ovi tipovi promene ne razlikuju se primarno po „veličini“ ili „dramatičnosti“ ishoda, već po nivou na kojem deluju. Mala intervencija na strukturnom nivou može imati

dalekosežne posledice, dok velike promene stanja mogu ostaviti strukturu netaknutom. Ovaj uvid odgovara mehanističkom i intervencionističkom shvatanju uzročnosti, prema kojem je ključno pitanje na kom mestu u uzročnom lancu intervencija deluje, a ne samo kakav efekat proizvodi (Woodward 2003; Craver & Darden 2013).

Sa metodološkog stanovišta, razlika između ova tri tipa promene omogućava da se izbegne česta greška poistovećivanja poboljšanja ishoda sa korekcijom uzroka. Sistem može pokazivati bolje performanse zahvaljujući promenama stanja ili adaptacijama, a da pri tom ostaje unutar istog strukturnog okvira. U takvim slučajevima, ontološki domet promene ostaje ograničen, čak i ako su empirijski rezultati impresivni. Nasuprot tome, strukturne promene mogu kratkoročno destabilizovati sistem, ali dugoročno redefinisati uslove njegove stabilnosti.

Ova razlika omogućava razumevanje odnosa između adaptivne stabilizacije i korekcije. Adaptivna stabilizacija odnosi se na procese koji održavaju ili poboljšavaju funkcionisanje unutar postojećeg okvira, dok korekcija, u strogom smislu, zahvata sam poredak odnosa i menja nosivost strukture. Bez jasnog kriterijuma ontološkog dometa, ove dve vrste promene lako se mešaju, što vodi ka precenjivanju efekata intervencija koje deluju samo na nivou ispoljavanja ili adaptacije.

U tom smislu, tipologija promene zasnovana na ontološkom dometu omogućava preciznije razlikovanje promena koje deluju na ravni ispoljavanja, adaptivnih reorganizacija unutar postojećeg okvira i promena koje zahvataju sam poredak odnosa. Time se uspostavlja osnova za razvoj kriterijuma identifikacije ontološkog dometa intervencija i razlikovanje adaptivne stabilizacije od strukturne korekcije. Detaljnija razrada ovog metodološkog aparata može se razvijati kao zasebna metodološka analiza.

Zaključak: Nosivost kao kriterijum stabilnosti

U ovom radu razvijen je teorijski okvir za analizu stabilnosti i promene u složenim sistemima, zasnovan na tri međusobno povezana pojma: ontološkoj vertikali, distinkciji između strukture i stanja, i nosivosti kao kriterijumu kvaliteta strukturnih odnosa. Polazeći od uvida o stratifikaciji realnosti i mehanističkom objašnjenju, pokazano je da stabilnost posmatrana isključivo na nivou ispoljavanja ne predstavlja pouzdan indikator kvaliteta uzročnog poretka koji ta ispoljavanja proizvodi (Bhaskar 1975; Craver 2007; Wimsatt 2007). Time je uspostavljena razlika između stabilnosti kao performativnog svojstva i stabilnosti kao posledice nosive strukture.

Uvođenjem pojma ontološke vertikale precizirana je asimetrija između nivoa organizacije – promene na dubljim nivoima redefinišu prostor mogućih stanja na višim nivoima, dok promene na nivou stanja ne moraju zahvatiti uzročni poredak. Distinkcija između strukture i stanja omogućila je da se ova asimetrija učini pojmovno preciznom i analitički operativnom, razdvajajući promene koje zahvataju samo konfiguracije ispoljavanja od onih koje menjaju same uslove njihove mogućnosti. U tom kontekstu, mehanistički pristupi objašnjenju i intervencionističko shvatanje uzročnosti pružaju

teorijski oslonac za razlikovanje nivoa na kojima intervencije i promene deluju (Craver & Darden 2013; Woodward 2003).

Centralni doprinos rada jeste uvođenje pojma nosivosti kao kriterijuma stabilnosti zasnovanog na kvalitetu strukturnih odnosa. Pokazano je da pojmovi poput robusnosti i rezilijentnosti, iako izuzetno korisni u analizi performansi i adaptacije, ostaju ograničeni ukoliko se ne povežu sa analizom kvaliteta strukture i ograničenja koja određuju prostor mogućih stabilnih režima (Holling 1973; Kitano 2004; Carlson & Doyle 2002). Nosivost omogućava da se stabilnost sagleda kao svojstvo strukture, a ne samo kao rezultat stalnih kompenzacionih intervencija, čime se uvodi jasna razlika između održavanja stabilnosti i njenog privremenog „proizvođenja“.

Na osnovu uvedenih distinkcija razvijena je tipologija promene prema njenom strukturnom dometu: promene stanja, adaptacije i promene strukture. Ova podela pokazuje da se promene ne razlikuju primarno po intenzitetu ili vidljivosti ishoda, već po nivou organizacije na kojem deluju. Time se uspostavlja metodološka osnova za procenu intervencija, ne samo prema njihovim efektima, već prema tome da li zahvataju ravan ispoljavanja, način funkcionisanja unutar postojećeg okvira ili sam poredak odnosa koji određuje nosivost strukture.

U tom okviru, ovaj rad ne nudi novu domensku teoriju, već teorijski i metodološki okvir za preciznije lociranje problema stabilnosti i promene. Njegov doprinos sastoji se u razjašnjavanju i sistematizaciji pojmovnih razlika koje su često implicitne ili zamašnjene u interdisciplinarnim raspravama o kompleksnim sistemima. Time se postavlja teorijski temelj za naredne korake: detaljniju analizu režima patološke stabilnosti i razvoj kriterijuma za njihovu dijagnostiku i korekciju.

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Dušan Stojanović and Katarina Stekić Stojanović

Structure, State and Bearing Capacity: An Ontological Framework of Stability in Stratified Systems

(Summary)

This article develops an ontological framework for analysing stability and change in complex, stratified systems. It starts from the distinction between structure and state, whereas a state denotes a concrete configuration of manifestation, whereas structure denotes the order of relations and mechanisms that determines the space of possible states. The article introduces the concept of bearing capacity as a criterion for assessing the quality of structural relations, understood as the capacity of a system to sustain stable regimes without continuous compensatory interventions. Stability is thereby distinguished from the mere repetition of outcomes, and change from the variation of states. The proposed framework makes it possible to differentiate changes of state, adaptations, and structural changes according to their ontological scope.

KEYWORDS: ontological vertical; structure; state; bearing capacity; stability; stratified systems

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The Razor's Edge: On the Uniqueness of Truths

Abstract. *In this paper, we argue against the view according to which there are multiple senses in which something may be true. Our argument against this view can be summarized as follows. One can chose arbitrary sense in which something may be true and express the same in all the other senses. In accordance with Occam's razor, there is no need for multiple senses in which something may be true.*

Keywords: *truth pluralism; alethic pluralism; truth; Occam's razor; monism about truth.*

It is wise to hearken, not to me, but to my Word, and to confess that all things are one. Heraclitus

1. Introduction

According to truth pluralism, there is more than one way for a proposition to be true. By a “sense of truth” I mean, provisionally, a way in which propositions are said to be true. To name some simple examples, “ $2 + 2 = 4$ ” is true in a different sense than the proposition “This stone weighs two kilograms”, if they are true at all – one is mathematical truth and the other is truth about the physical world, and these ways of being true are completely different by assumption.¹

Our aim in this paper is to present an argument against this view, which can be summarized as follows. One can chose arbitrary sense in which something may be true and express the same in all the other senses. In accordance with Occam's razor, there is no need for multiple senses in which something may be true.

Firstly, we will briefly present pluralism about truth as our target. Then, the main argument will be presented. Finally, we will provide the reader with the proof of the vital premise of the argument.

2. The target

Following the terminology used in the contemporary debate, we may call this view

¹Note that the immediate reason for writing this paper stems from the problems regarding the philosophy of mathematics. Namely, Gödel once suggested that the truths of mathematics may be regarded as true in the same sense as some well-established physical theory (Gödel 1990). On the other hand, for example, there are conceptions according to which the truths of mathematics are truths about fictional entities. Such is the conception of Hartry Field (Field 1989). According to that conception, the truth of the sentence “ $2 + 2 = 4$ ” is more similar to the truth of the sentence: “Sherlock Holmes lives in Baker Street” than to the truth of the sentence: “Every two bodies with mass attract each other”.

alethic pluralism. Pedersen and Wright characterize pluralism about truth as the thesis that “there is more than one way of being true” (Pedersen et al. 2023). More precisely, the view is usually understood as the claim that there is more than one truth-property. On a moderate version of the view, there is more than one truth-property, but at least one of these properties is shared by all true sentences. On a stronger version, there is more than one truth-property and none of them is possessed by all true sentences. Strong pluralism therefore directly denies that all true sentences are true in the same way, whereas moderate pluralism allows that there may be a general truth-property shared by all truths, together with further domain-specific truth-properties.

The main motivation for alethic pluralism is often called the scope problem. Traditional inflationary theories of truth usually attempt to identify certain property such that every true sentence is true in virtue of having that property. Different theories propose different candidates for mentioned properties, such as correspondence, coherence, superassertibility, or some other property. The difficulty is that these candidates appear plausible in some domains of discourse but not in others. Correspondence may seem natural for ordinary empirical claims, while coherence or superassertibility may seem more plausible for mathematical, moral, legal, or fictional discourse. The pluralist response is to reject the demand for a single property that explains truth across all domains, and instead to claim that truth consists in several properties, different ones being appropriate to different regions of discourse.²

The present paper is directed against the need for multiplication of truth-senses. It does not deny that different domains involve different subject matters, methods of justification, or explanatory relations. Nor does it deny that, for example, the truth of a mathematical proposition may be explained differently from the truth of a proposition about ordinary physical objects. The target is rather the further claim that such differences require us to posit several primitive senses or ways of being true.³

3. The argument

Let us present the argument outlined in the introduction. Fix the following abbreviations:

²This is why truth pluralism is especially relevant to the philosophy of mathematics. If mathematical propositions are not true in the same way as empirical propositions, one might conclude that mathematical truth is not a species of the same general phenomenon as physical truth. For example, a realist about mathematics may hold that mathematical propositions are objectively true in something like the way well-confirmed physical propositions are objectively true. By contrast, a fictionalist or nominalist may hold that mathematical discourse is closer to fiction than to science. The pluralist framework appears to make room for this kind of contrast by allowing mathematical truth, physical truth, fictional truth, and other kinds of truth to be different ways of being true.

³For details regarding specific pluralist positions, the reader is especially referred to Wright's classical book (Wright 1992), but also to: (Asay 2018), (Kölbel 2013), (Lynch 2009), (Lynch 2013), (Pedersen & Wright 2013) and (Tappolet 1997).

Def₁ Proposition ψ has *ostensibly the same truth value* in truth sense s' as proposition ϕ in truth sense s iff: if ϕ is true in sense s , then ψ is true in sense s' ; otherwise, ψ is false in sense s' .

Def₂ All senses of truth are *mutually expressible* iff: for any senses of truth s and s' and any proposition ϕ , there is proposition ψ such that ϕ has truth value in sense s iff ψ has ostensibly the same truth value in s' as ϕ in s .

Since the logic behind the argument is trivial, the argument can be expressed in following manner:

1	(1)	If there are multiple senses of truth, then all senses of truth are mutually expressible.	A
2	(2)	If there are multiple senses of truth and all senses of truth are mutually expressible, then there is no need for multiple senses of truth.	A
1, 2	(3)	If there are multiple senses of truth, then there is no need for multiple senses of truth.	From 1, 2

The argument follows deductively. Hence, one should consider the truth of the premises. We will show that premise (1) holds in the next section. The premise (2) is meant to be a consequence of Occam razor, which we consider obvious.

Note that the argument in this form is aimed at strong pluralism. Namely, since moderate pluralism implies that there is one common truth-property, by Occam's razor it directly follows that there is no need for other truth-properties.

4. Proof of the first premise

In order to prove the first premise of the argument presented in the previous section, we will formulate some sort of *ad hoc* logic.

We will work within the confines of the syntax of classical propositional calculus. Assume that

$$Prop \stackrel{def}{=} \{t_i \mid i < \omega\}$$

is our set of propositional letters (variables). We shall use p, q, r , with or without index, as schematic letters for propositional letters. Also, take $\neg$ and $\wedge$ as primitive propositional connectives. The set of formulas *Form* is defined in the usual manner, as are the other usual connectives. We shall use ϕ, ψ, χ , with or without index, as schematic letters for formulas. Note that $Sub(\phi)$, for $\phi \in Form$, designates set of all subformulas of formula ϕ and that

$$\phi[q_0, \dots, q_{n-1}/\psi_0, \dots, \psi_{n-1}]$$

is formula ϕ in which all occurrences of $q_0, \dots, q_{n-1}$ are, respectively, simultaneously substituted by formulas $\psi_0, \dots, \psi_{n-1}$.

The semantics for this *ad hoc* logic is a bit more complicated than the usual semantics for classical propositional calculus, but the main idea stems from the semantics for classical propositional logic. We will first give a rough idea of this semantics with natural language examples and then continue by presenting formal apparatus.

As an example, assume that the truths of mathematics and the truths about the physical world are true in different senses. So, some may argue, the proposition: “There is an empty set” is true mathematically but not physically. So, the proposition: “There is an empty set” has multiple truth values – one for each sense in which something may be true. View the nonempty set S as the set of these senses in which something may be true. The most natural way to define surrogates for truth values in these formal considerations is to view them as ordered pairs of the form $\langle s, 1 \rangle$ – true in the sense s – and $\langle s, 0 \rangle$ – false in the sense s , – for each $s \in S$. However, in classical propositional logic, value of a formula is its truth value at the same time. This kind of solution leads to some problems regarding the assignment of values to formulas in our case, i.e., the relation that assigns values would not be a function. So, we will take that values are sets of truth values, or, more precisely, functions from ${}^S\{0, 1\}$. So, value of some formula φ may, for example, be the function $\{\langle s_0, 1 \rangle, \langle s_1, 0 \rangle, \langle s_2, 0 \rangle\}$ and its truth values are elements of its value, provided that $S = \{s_0, s_1, s_2\}$.

Definition 1. Assume that S is nonempty set. Every function of the form $v: Prop \rightarrow {}^S\{0, 1\}$ is called *basic valuation grounded on S* .

As is plain, set S from the *Definition 1* plays the role of the set of senses in which a proposition may be true. To each propositional variable, basic valuation grounded on S assigns some value which elements are to be viewed as truth values in the appropriate sense.

However, we yet need to settle the question of value of more complex formulas:

Definition 2. Assume that S is nonempty set and that v is basic valuation grounded on S . *Valuation grounded on S* is function $V: Form \rightarrow {}^S\{0, 1\}$ defined by recursion on the rank of $\varphi \in Form$ in following manner:

- if $\varphi \in Prop$, then $V(\varphi) = v(\varphi)$;
- if $\varphi = \neg\psi$, for $\psi \in Form$, then $V(\neg\psi)(s) = 1 - V(\psi)(s)$, for every $s \in S$;
- if $\varphi = \psi \wedge \chi$, for $\psi, \chi \in Form$, then $V(\psi \wedge \chi)(s) = V(\psi)(s) \cdot V(\chi)(s)$, for every $s \in S$.

Lemma 1. Assume that S is nonempty set and that v is basic valuation grounded on S . Fix arbitrary $s_0, s_1 \in S$ and formula $\varphi \in Form$. If $v(p)(s_0) = v(p)(s_1)$, for all $p \in Prop \cap Sub(\varphi)$, then $V(\varphi)(s_0) = V(\varphi)(s_1)$.

Proof.

Easy proof by induction on the rank of φ from the formulation of the lemma. $\diamond$

Theorem 1. Assume that S is nonempty set and that v is basic valuation grounded on S . Fix arbitrary $s_0, s_1 \in S$ and formula $\varphi \in Form$. If $V(\varphi)(s_0) \neq V(\varphi)(s_1)$, then there are $q_0, \dots, q_{n-1} \in Prop \cap Sub(\varphi)$ such that

$$V(\varphi)(s_0) = V(\varphi[q_0, \dots, q_{n-1}/\neg q_0, \dots, \neg q_{n-1}])(s_1)$$

Proof.

Assume that S is nonempty set and that v is basic valuation grounded on S . Fix arbitrary $s_0, s_1 \in S$ and formula $\varphi \in \text{Form}$. Also, assume that $V(\varphi)(s_0) \neq V(\varphi)(s_1)$. Let $\text{Prop} \cap \text{Sub}(\varphi) = \{q_i \mid i < m\}$. If for all $i < m$, $v(q_i)(s_0) = v(q_i)(s_1)$, then $V(\varphi)(s_0) = V(\varphi)(s_1)$ (*Lemma 1*). Hence, set

$$D = \{q \in \text{Prop} \cap \text{Sub}(\varphi) \mid v(q)(s_0) \neq v(q)(s_1)\}$$

is nonempty. Say that q_i , for $i < n \leq m$ are all elements of D . By induction on rank of φ , show that

$$V(\varphi)(s_0) = V(\varphi[q_0, \dots, q_{n-1}/\neg q_0, \dots, \neg q_{n-1}])(s_1)$$

If $\varphi \in \text{Prop}$, then φ is some q and $n = 1$. Without loss of generality, we can assume that $V(q)(s_0) = 1$. Hence, $V(q)(s_1) = 0$ and, therefore, $V(q[q/\neg q])(s_1) = V(\neg q)(s_1) = 1$.

If φ is of the form $\neg\psi$, assume that the claim holds for ψ . Then:

$$\begin{aligned} V(\neg\psi)(s_0) &= 1 - V(\psi)(s_0) \\ &= 1 - V(\psi[q_0, \dots, q_{n-1}/\neg q_0, \dots, \neg q_{n-1}])(s_1) \\ &= V(\neg\psi[q_0, \dots, q_{n-1}/\neg q_0, \dots, \neg q_{n-1}])(s_1) \end{aligned}$$

Proof is similar in the case that φ is of the form $\psi \wedge \chi$. $\diamond$

Let us give an example of the application of this theorem. Consider the following mathematically true proposition:

There is an empty set and there is a singleton containing empty set.

In a sense in which physical propositions are true, this proposition is false. However, it is mathematically true. *Theorem 1* says that there is an awkward way of speaking that bypasses the fact that it is false in the physical sense. Essentially, one only needs to substitute simplest false propositions by their negations:

An empty set does not exist and there is no singleton containing empty set.

Now, observe that the last is physically true.

By proving *Theorem 1*, conclusive argument for the first premise has been given.

5. Conclusion

In this paper, we considered the view according to which there are multiple senses in which something may be true. Against this view, we argued that if there are multiple senses of truth, then they are mutually expressible. If this is so, then the postulation of multiple senses of truth is unnecessary. By Occam's razor, there is no need to multiply senses of truth beyond what is required. Therefore, although there may be many different domains of discourse and many different ways in which truths are explained, there is no need to suppose that there are many senses in which propositions

are true.

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Dorđe Vukašinović

Oštrica brijača: o jedinstvenosti istina

(Rezime)

U ovom radu argumentujemo protiv stanovišta prema kojem postoji više različitih smislova u kojima nešto može biti istinito. Naš argument protiv ovog stanovišta može se sažeti na sledeći način. Možemo proizvoljno odabrati jedan smisao u kojem nešto može biti istinito i isto to izraziti u svim ostalim smislovima. U skladu s Okamovim brijačem, nema potrebe za višestrukim smislovima u kojima nešto može biti istinito.

KLJUČNE REČI: pluralizam u pogledu istine; aletički pluralizam; istina; Okamova oštrica; monizam u pogledu istine.



Marijana Vitorović

Astropsihologija kao savremena pseudonauka: Teorijske osnove i psihološki mehanizmi verovanja

Apstrakt: *Astropsihologija predstavlja savremeni pokušaj spajanja astroloških koncepata sa psihološkim teorijama o ličnosti i ponašanju. Iako se često promovise kao inovativan pristup razumevanju psihičkih procesa, njen status pseudonauke ostaje nesporan zbog odsustva empirijske potvrde, nepostojanja teorijske koherentnosti i nemogućnosti da zadovolji osnovne naučne kriterijume. Ovaj rad ispituje razloge zbog kojih astropsihologija ne može zadobiti naučni legitimitet oslanjajući se na psihologiju, budući da se zasniva na astrološkim postulatima koji su empirijski i logički neodrživi. Posebna pažnja posvećena je kognitivnim pristrasnostima koje doprinose njenoj popularnosti. Rad takođe razmatra sociokulturne faktore i uticaj digitalnih platformi na širenje astroloških praksi, uključujući popularne aplikacije i personalizovane horoskope. Zaključuje se da astropsihologija, i pored svoje privlačnosti i subjektivne ubedljivosti, ostaje u suprotnosti sa naučnim standardima psihologije i predstavlja značajan primer pseudonaučnog diskursa u savremenom društvu.*

Ključne reči: *astropsihologija, astrologija, pseudonauka, kognitivne pristrasnosti, heuristike.*

Uvod

Astropsihologija predstavlja hibridni model koji kombinuje astrologiju sa dubinskom, humanističkom i transpersonalnom psihologijom. Teoretičari i praktičari ove oblasti pretenduju da astropsihologiju predstave kao sveobuhvatnu teoriju ličnosti. Perry (2012) smatra da ona ispunjava sve zahteve za modernu teoriju ličnosti jer je „jedinstvena po svojoj sposobnosti da otkrije kako se struktura i dinamika psihe ogledaju u spoljašnjim uslovima koji pružaju podsticaj psiho-duhovnom rastu“. Prema njegovom mišljenju, astropsihologija ima široku inkluzivnu strukturu, naglasak na razvoju, sistematsku preciznost, dubinu fokusa i duhovni značaj. Takođe, Perry (2012) ističe da se astropsihologija može koristiti kao dijagnostički alat, tako što pomaže praktičaru da dobije uvid u osnovne, karakterne probleme, koji uzrokuju patnju i oštećenja.

Astropsihologija predstavlja pristup koji povezuje astrološke koncepte sa psihološkim teorijama i konstruktima, nastojeći da položaje nebeskih tela poveže sa osobinama ličnosti, psihičkim procesima i obrascima ponašanja. Međutim, postojanje takvih tvrdnji samo po sebi ne podrazumeva njihovu naučnu utemeljenost. U tom smislu, važno je razlikovati sadržaj i popularnost određenog sistema verovanja od

njegovog epistemološkog statusa. Dosadašnja empirijska istraživanja nisu pružila ubedljive dokaze da položaji nebeskih tela predviđaju osobine ličnosti ili druge ishode, na način koji astrološke teorije pretpostavljaju (Eysenck & Nias, 1982; Carlson, 1985; Blackmore & Seebold, 2001). Štaviše, teško je identifikovati konzistentan korpus znanja koji se može definisati kao astrologija ili astropsihologija. Kritičari ove oblasti suočavaju se sa promenljivim i često kontradiktornim tvrdnjama, koje počivaju na istorijskim vezama i sličnostima između različitih astroloških tradicija, ali koje ne pokazuju jasan obrazac teorijskog i empirijskog napretka (Edis 2018).

Međutim, utvrđivanje epistemološkog statusa astrologije i astropsihologije predstavlja samo jedno od pitanja relevantnih za njihovo proučavanje. Drugo, podjednako važno pitanje odnosi se na psihološke i sociokulturne mehanizme koji doprinose tome da ljudi prihvataju i održavaju verovanja koja nisu adekvatno podržana empirijskim dokazima. Jedno istraživanje sprovedeno na teritoriji SAD-a pokazuje da oko 29% populacije veruje u astrologiju (Poll 2013). Nažalost, podaci o procentu ljudi koji veruju u astrologiju u Srbiji su ograničeni, ali rasprostranjenost astrologije u javnom i medijskom prostoru i vekovni, duboko ukorenjen kulturološki odnos prema sujeverju, koji se može uočiti kroz folklorne prakse i narodna verovanja, može doprineti porastu interesovanja ljudi ka ovoj oblasti na teritoriji Srbije. Stoga, postavlja se pitanje šta ljude čini sklonim da veruju u tvrdnje iz domena astrologije i posvećuju vreme njenom izučavanju, uprkos nedostatku pouzdanih empirijskih dokaza koji bi podržali njihove ključne pretpostavke.

Pre svega, ljudi mogu biti skloni da u astrologiji pronalaze neku vrstu utehe. Istraživanje pokazuje da astrologija pomaže ljudima da pronađu smisao kada im se život čini suviše kompleksnim (Hamilton 2015). Kroz istoriju, uočava se porast interesovanja ka astrologiji tokom teških vremena, kao u slučaju Velike depresije tokom 1930-ih u SAD ili u vreme između dva svetska rata u Nemačkoj (Padgett & Jorgenson 1982). Takođe, kada pogledamo skorija vremena, astrologija je doživela svoj procvat za vreme pandemije korona virusa (Business Research Insights 2025). Iz ovoga možemo zaključiti da ljudi, na neki način, astrologiju koriste kao mehanizam odbrane u situacijama kada je prisutan visok stepen neizvesnosti i anksioznosti u pogledu budućnosti. Pribegavanje horoskopu i proročanstvima ljudima može davati nadu, doživljaj stabilnosti i utehu da će sve biti u redu, ukoliko njihova natalna karta to pokazuje.

Još jedan od razloga za pridavanje značaja astrologiji može biti uticaj spoljašnjeg lokusa kontrole. Pojedinci sa spoljašnjim lokusom kontrole skloniji su da doživljavaju određene događaje kao funkciju sreće, slučajnosti, sudbine, moćnih drugih ili moći koje su izvan njihove lične kontrole i razumevanja. Kada nešto pođe po zlu, oni će kriviti određene spoljašnje uticaje, dok pozitivne događaje pripisuju uticaju sreće. Istraživanje pokazuje da ove osobe, koje životne događaje doživljavaju kao funkciju sreće ili slučajnosti, u većoj meri koriste i veruju astrologiji od pojedinaca sa unutrašnjim

lokusom kontrole (Sosis, Strickland & Haley 1980).

Astrologija se može posmatrati i kao jedan od načina za validaciju identiteta pojedinca (Clements 2020). Horoskop, odnosno natalna karta, često predstavlja alat kroz koji ljudi stiču doživljaj jedinstvenosti i dobijaju određenu vrstu potvrde sopstvene vrednosti. Ljudi su skloni tome da bolje pamte pozitivne informacije, koje potvrđuju njihovu samopercepciju, u odnosu na negativne informacije (Jelić 2022). Iz tog razloga, nije neobično da su ljudi skloni da prihvate dvosmislene poruke, koje dobijaju kroz horoskop, i tumače ih na način koji im više odgovara. Takođe, upotreba astrologije za razumevanje ličnosti i savete kako postići uspeh i kontrolu nad sopstvenim životom je generalno jaka motivacija za ljude da u nju veruju (Baumeister & Leary 2017).

Čak i ako ljudi veruju u astrologiju, postavlja se pitanje kakve to veze ima sa ljudskim razumevanjem i angažovanjem u nauci. Podaci pokazuju da ova dva pitanja mogu biti povezana, odnosno da značajan deo američke i evropske populacije ne samo da veruje u efikasnost astrologije, već veruje i da je ona naučna (National Science Board 2016). Na osnovu podataka od strane National Science Foundation (NSF) za 2016. godinu procenjeno je da oko 60% američke populacije smatra da je astrologija nenaučna, 29% populacije ima mišljenje da je astrologija „donekle naučna“, dok 8% populacije je smatra „vrlo naučnom“ (3% se izjasnilo da ne zna da li je naučna). U Evropi je, čini se, mišljenje o naučnosti astrologije još rasprostranjenije. Na osnovu podataka dobijenih ispitivanjem javnog mnjenja Evropske Unije dobijeno je da oko 76% ispitanika veruje da je astrologija barem delimično naučna, od kojih 26% smatra da je veoma naučna (European Commission 2005). Ovi podaci ukazuju da pripisivanje naučnog statusa astrologiji nije marginalna pojava i otvaraju dodatno pitanje kriterijuma na osnovu kojih ljudi određeni sistem znanja prepoznaju kao naučan.

Na osnovu rezultata istraživanja koje se bavilo razlozima zbog kojih ljudi astrologiju smatraju naučnom, Allum (2011) zaključuje da tome mogu doprineti nedostatak naučne pismenosti i nedovoljno poznavanje same astrologije. S obzirom na to da podaci ukazuju da značajan broj ljudi astrologiji pripisuje određeni stepen naučnosti, važno je razmotriti kriterijume na osnovu kojih se naučne tvrdnje mogu razlikovati od onih koje to nisu. Stoga će u nastavku rada biti razmotren problem razlikovanja nauke i pseudonauke, kao i različiti kriterijumi koji se u tu svrhu koriste.

Nauka i pseudonauka

Tokom istorije ljudi su verovali u najrazličitije stvari i pojave, koje se iz današnje perspektive čine teško prihvatljivim. Međutim, verovanje u određenu tvrdnju i njena naučna vrednost predstavljaju dve različite stvari. Određena tvrdnja može biti pseudonaučna, čak i ako deluje vrlo uverljivo i veliki broj ljudi u nju veruje, dok, s druge strane, može biti i naučno vredna čak i kada zvuči neverovatno i gotovo niko u nju ne veruje. Stoga, naučna vrednost neke teorije ne zavisi od njenog stepena prihvaćenosti, niti od toga da li je intuitivno razumljiva ili uverljiva. Teorija može biti od vrhunske

naučne vrednosti, čak i ako je skoro niko ne razume niti u nju veruje (Lakatos 1978).

Nauka podrazumeva sistematizovano znanje, izvedeno iz posmatranja, proučavanja, eksperimentisanja i kritičkog ispitivanja. Naučna posmatranja se sprovode u strogo kontrolisanim uslovima, kako bi se minimizirao uticaj pristrasnosti i očekivanja od strane istraživača, kao i slučajnih uticaja iz spoljašnje sredine. Naučne tvrdnje treba da budu izložene mogućnosti provere na osnovu dostupnih dokaza, pri čemu se nastoji da uticaj pristrasnosti i očekivanja istraživača, kao i uticaj i slučajnih faktora budu svedeni na najmanju moguću meru. Javna dostupnost metoda i nalaza, otvorenost prema kritičkoj proceni i mogućnost nezavisnog ponavljanja istraživanja predstavljaju važne elemente naučne prakse. Pojedinačni eksperimenti u nekoj oblasti skoro nikada ne rešavaju važnu naučnu debatu, već je neophodno da postoji više dokaza istraživača iz date oblasti, koji mogu da repliciraju međusobne nalaze. Prilikom iznošenja dokaza na osnovu sprovedenih istraživanja, naučnici pokušavaju da generalizuju pojedinačna zapažanja i formulišu opšte zakone. Nakon što uspostave te zakonite odnose, oni razviju opštija objašnjenja i integrišu ih u teorijske okvire koji se mogu dalje proveravati (Beyerstein 1995).

Sa druge strane, pseudonauka se odnosi na oblasti i sisteme tvrdnji koji nastoje da zadobiju autoritet i prestiž nauke, često preuzimajući njena spoljašnja obeležja i terminologiju, ali ne zadovoljavajući standarde naučnog istraživanja i potvrđivanja znanja. Za pseudonaučne pristupe karakteristična je nedovoljna otvorenost prema kritičkom preispitivanju, slabo uvažavanje suprotnih empirijskih nalaza i odsustvo jasnog kumulativnog napretka u razvoju znanja. Njihova objašnjenja mogu biti u suprotnosti sa dobro potvrđenim naučnim saznanjima, dok nalazi na kojima se zasnivaju često ne izdrže nezavisnu i kompetentnu proveru (Beyerstein 1995). Međutim, navedene karakteristike ne predstavljaju pojedinačne i dovoljne kriterijume na osnovu kojih se može povući jasna granica između nauke i pseudonauke. Upravo zbog toga, razlikovanje nauke od pseudonauke predstavlja složen zadatak, koji je u filozofiji nauke poznat kao problem demarkacije.

Jedan od pokušaja određivanja granice između nauke i pseudonauke zasniva se na kriterijumu falsifikacije, koji je razvio Karl Popper (1963). Prema ovom pristupu, naučna teorija mora biti formulisana tako da postoji mogućnost njenog empirijskog opovrgavanja, odnosno da postoje zamislivi ishodi koji bi bili u suprotnosti sa njenim predviđanjima. Činjenica da se za neku teoriju mogu pronaći potvrđujući primeri sama po sebi nije dovoljna, budući da se gotovo za svaku teoriju mogu pronaći potvrde ukoliko se one ciljano traže. Za naučnu teoriju je zato značajno da iznosi rizična predviđanja, odnosno predviđanja čiji bi neuspeh mogao predstavljati ozbiljan argument protiv teorije. U tom smislu, što teorija preciznije određuje koji ishodi jesu, a koji nisu mogući, to je više izložena empirijskoj proveru i potencijalnom opovrgavanju (Popper, 1963). Popper takođe ukazuje na problem naknadnog prilagođavanja teorije nepovoljnim nalazima. Uvođenje *ad hoc* naknadnih pretpostavki ili reinterpretacija

teorije sa ciljem da se izbegne njeno opovrgavanje može omogućiti da teorija formalno opstane uprkos nepovoljnim rezultatima, ali istovremeno umanjuje njen naučni status (Popper 1963).

Međutim, ni kriterijum falsifikacije sam po sebi ne može biti dovoljan za razlikovanje nauke od pseudonauke. Tokom vremena, status pojedinih teorijskih pretpostavki može se menjati u svetlu novih informacija i dokaza, odnosno sa razvojem naučnih znanja, tehnologija i metoda ispitivanja. Određene hipoteze u trenutku nastanka mogu biti ograničeno proverljive, ali uz dalji napredak nauke mogu postati predmet ozbiljnog naučnog ispitivanja. Stoga, aktuelno odsustvo mogućnosti empirijske provere određene hipoteze ne može se automatski smatrati pokazateljem njenog pseudonaučnog statusa. Takođe, sama mogućnost da se tvrdnja može opovrgnuti nije dovoljna da bi se određeni sistem znanja smatrao naučnim. Potrebno je razmotriti i način na koji se teorija odnosi prema kritici i empirijskim nalazima koji nisu u skladu sa njenim predviđanjima, kao i da li tokom vremena omogućava razvoj novih predviđanja i unapređivanje korpusa znanja.

Lakatos (1976) ukazuje na to da se naučna istraživanja u praksi ne odvijaju tako što se teorije automatski napuštaju nakon svakog neuspešnog predviđanja, već da se razvijaju u okviru širih istraživačkih programa koji uključuju osnovne teorijske pretpostavke i pomoćne hipoteze. Zbog toga, za procenu naučnog statusa nije dovoljno utvrditi da li je neka teorija bila izložena empirijskoj proveri, već je potrebno posmatrati način na koji se istraživački pristup razvija kroz vreme. Naučni program može biti progresivan ukoliko dovodi do novih predviđanja i teorijskih saznanja koja se potom empirijski potvrđuju, dok postaje degenerativan ukoliko se nove pretpostavke uvode prvenstveno radi zaštite postojećeg teorijskog okvira od nepovoljnih nalaza, bez stvaranja novih, uspešnih predviđanja (Lakatos 1976). Na taj način, naučni status određenog pristupa ne procenjuje se na osnovu jedne izolovane provere, već i na osnovu njegovog istorijskog razvoja, sposobnosti da proizvodi novo znanje i načina na koji reaguje na probleme u domenu empirijske potvrde.

Na osnovu svega navedenog, može se zaključiti da razlikovanje nauke od pseudonauke zahteva razmatranje šireg skupa karakteristika naučnog istraživanja. Pored mogućnosti empirijske provere i potencijalnog opovrgavanja tvrdnji, važno je uzeti u obzir odnos određenog teorijskog pristupa prema empirijskim dokazima i kritičkom preispitivanju, kao i njegovu sposobnost da kroz vreme proizvodi nova predviđanja, objašnjenja i doprinosi kumulativnom razvoju znanja. U tom smislu, pseudonaučni status određenog pristupa ne proizlazi samo iz odsustva jedne karakteristike naučnog istraživanja, već iz šireg obrasca načina na koji se njegove tvrdnje formulišu, proveravaju, prilagođavaju i razvijaju.

Astropsihologija

Astropsihologija je osmišljena i utemeljena ne samo od strane određenih astrologa,

već i psihologa, koji pokušavaju da povežu zodijačke znakove sa psihološkim osobinama, tvrdeći da položaj planeta i horoskopski znakovi utiču na karakter, ponašanje i sudbinu pojedinca. Kako bismo bolje razumeli astropsihologiju, u nastavku će se biti istaknute neke od činjenica o njenom poreklu, osnovnim idejama i društvenom kontekstu, što će nas dovesti do razmatranja njenog pseudonaučnog statusa.

Istorijski razvoj i osnovni koncepti

Koreni astrologije potiču još iz vremena drevnih naroda, koji su planete doživljavali kao bogove i za koje su verovali da vladaju raznim prirodnim procesima i pojavama. Ljudi su sebe doživljavali kao sudbonosne primaoca kosmičkih sila, koje se mogu umilostiti, ali ne i poreći. Planete su zamišljane kao predajnici misterioznih zraka ili elektromagnetnih sila, koje utiču na pojedinca pri rođenju. Verujući u astrologiju, ljudi su konsultovali zvezde, kao sredstvo koje će im pomoći da izbegnu katastrofalnu sudbinu ili da iskoriste mogućnost za manipulisanje okolnostima u ličnu korist. Tradicionalna astrologija promovisala je verovanje da je pojedinac potencijalna žrtva ravnodušnog univerzuma, nad kojim nema nikakvu kontrolu. Shodno tome, astrolozi su u ovakvoj konstelaciji stvari profitirali kroz davanje ljudima onog što im je bilo potrebno, tj. savete, predviđanja, upozorenja ili jednostavna rešenja za kompleksne psihološke probleme. Ljudi su, verovatno, kroz dobijanje proročanstava od strane astrologa stvarali lažni osećaj kontrole nad budućim, neizvesnim događajima. Takođe, astrološke intervencije uključivale su i opis karaktera pojedinca, koji se uglavnom svodio na površne opise osobina, opterećene moralnim sudovima i površnim savetima. U okviru tradicionalne astrologije, karakter je u nekim slučajevima doživljavan kao statičan i nepromenljiv ili u drugim slučajevima kao lako promenljiv uz savete kosmički informisanog astrologa (Perry 2016).

Neka od ovih shvatanja tradicionalne astrologije zadržana su, u donekle izmenjenoj formi, i u okviru moderne astrologije, koja se razvijala tokom XX veka. Jedan od prvih koji je osnovne koncepte astrologije koristio kao alat za istraživanje ljudske psihe bio je psihoanalitičar Carl Jung. On je tvrdio da astrologija može mnogo doprineti psihologiji i isticao je njen značaj u radu sa klijentima. Kako je sam rekao u jednom od svojih pisama: „Moram reći, da sam vrlo često otkrivao da astrološki podaci razjašnjavaju određene tačke (ličnosti klijenata), koje inače ne bih mogao da razumem“ (Jung 1948). U okviru svoje hipoteze o arhetipovima, smatrao je da astrološki znaci i planete predstavljaju simbole arhetipskih procesa, koji nastaju u kolektivnom nesvesnom. Arhetipovi kolektivnog nesvesnog, prema Jungovom mišljenju, su univerzalni organizujući principi, koji leže u osnovi i motivišu sav psihološki život pojedinca, i individualni, i kolektivni. Arhetip je doživljavan kao subjektivan i objektivni – bio je očigledan i u urođenim idejama ljudske svesti, kao i u fundamentalnim procesima prirode, pa je uticao ne samo na ljudsko iskustvo, već i na kretanje planeta. Prema Jungu, upravo ta, dvostruka priroda arhetipa, omogućila je da natalna karta poveže

unutrašnji karakter sa spoljašnjim događajima, koji su reflektovali taj karakter. Jung je verovao da postoje mnogi primeri analogije između astroloških sazvežđa i psiholoških događaja, odnosno između horoskopa i karakternih dispozicija osobe (Jung 1976). Prema njegovom mišljenju, astrologija predstavlja zbir svih psiholoških znanja antičkog doba (Jung 1971).

Veliki broj astrologa, pa i psihologa pokušali su da unaprede ova Jungova zapažanja i nastave sa produbljivanjem ideja o astrološkoj psihologiji. Najistaknutiji zagovornik i osnivač današnje, postmoderne astrologije (kako je sam naziva) je astrolog i psihoterapeut Glenn Perry. On ima viziju nove, postmoderne astrologije, koja integriše najbolje od tradicionalnih i modernih astroloških sistema. Postmoderna, psihološka astrologija se bavi načinom na koji natalna karta prikazuje psihičku strukturu i procese pojedinca, a ne bavi se proricanjem budućnosti i ishoda. Perry koristi astrološke simbole za razumevanje psihičkih procesa, koje smatra pravim uzrokom ponašanja i događaja. Astropsihologija stavlja odgovornost za životno iskustvo na pojedinca. Njen primarni cilj je da osnaži klijente da budu odgovorni za iskustva koja privlače, kako bi mogli da utiču na stvaranje ishoda koji će ih ispunjavati (Perry 2015). Glenn Perry, ne samo da u svom psihoterapijskom radu koristi koncepte astropsihologije, već je i osnivač i vlasnik Akademije AstroPsihologije, u kojoj obučava svoje polaznike, buduće astropsihologe. Prema Perry (2016), astroterapija se prepoznaje kao primena astroloških koncepata u kliničkoj praksi. Astropsihologija, koja za njega predstavlja teoriju ličnosti i dijagnostički alat, prikazuje psihi kao hijerarhijsku strukturu sastavljenu od arhetipskih potreba, kognitivnih struktura, nastajućih misli i ponašanja, koje utiču na kreiranje odgovarajućih događaja.

Društveni i kulturni kontekst

Posmatrajući istorijski pregled razvoja astrologije i astropsihologije, uočava se da je u različitim vremenskim razdobljima postojala drugačija percepcija njenih osnovnih koncepata, a samim tim se i motivacija za njeno prihvatanje od strane društva dosta razlikovala. Drevni narodi nisu mogli da objasne brojne prirodne pojave, pa su, verovatno, iz tog razloga bili skloni da prihvate ideju o planetama kao bogovima, koji upravljaju ovim pojavama. Kasnije, u vreme različitih ratova, gladi, nemaštine i oskudice, ljudima je astrologija bila potrebna kako bi, na neki način, stekli privid da mogu da kontrolišu buduće događaje. Ukoliko astrolog predvidi da ih čeka loša sudbina, verovatno su verovali da će moći da je spreče, ukoliko imaju znanje da će se određeni događaj odigrati. Ukoliko bi im, ipak, astrolog predvideo lepu budućnost, to im je, verovatno, davalo veću snagu i motivaciju da izdrže surovost svakodnevice.

U današnje vreme, ljudi često mogu biti skloni da prihvate ideje astropsihologije zato što im nudi ideju o uzroku njihovog problema, koji ne potiče od njih samih, već leži u spoljašnjem faktoru, kao što je datum rođenja, horoskopski znak ili naklonost zvezda prema osobi. Ono što astropsihologiju razlikuje od tradicionalne astrologije

je baš ono što je nekim ljudima potrebno, a to su saveti astrologa i astroterapija, koje mogu da promene te negativne karakterne crte, koje su im zvezde i planete dodelile na rođenju. Dosta ljudi ima otpor ka prihvatanju sopstvenog načina funkcionisanja kao razlog za nastanak određenog problema, pa su skloni traženju nevidljivih krivaca za sve svoje brige i nemire. U okviru astropsihologije, oni tog krivca dobijaju u vidu natalnih karti, horoskopa i drugih nevidljivih pojava. Ukoliko pojedinac, koji pridaje određeni značaj astrologiji, ima loš dan, nedelju, mesec, pa čak i godinu, može lako da pronađe odgovor o uzroku u vidu retrogradnog merkura. Ukoliko loše spava, razdražljiv je i anksiozan, krivac može biti pun mesec. Takođe, pojedinac može čekati da „Mars bude u 10. kući“ ili da se desi „Tranzit Venere kroz Devicu“ kako bi situacija za njegove partnerske odnose ili profesionalni uspeh bila povoljna. Ovi karikirani primeri, samo pokazuju koliki značaj za ponašanje osobe može imati astrologija. Jedno istraživanje, koje se bavilo donošenjem odluka pre i nakon čitanja negativnog horoskopa, pokazalo je da 72% ispitanika, koji su verovali u astrologiju, se osećaju manje samouvereno za donošenje odluke nakon čitanja nepovoljnog horoskopa, dok kod ispitanika koji nisu verovali u astrologiju, čitanje horoskopa nije imalo značajan uticaj na donošenje odluka (Patel et al., 2021, prema Dabhade, 2025).

Krajem XX veka došlo je do ekspanzije horoskopa u štampi i časopisima. Dnevni horoskop je počeo da se štampa u skoro svim novinama, kao forma zabave i razonode, koja se održala i u današnje vreme. Postoji podatak da je sredinom XX veka oko 70% dnevnih novina u SAD-u sadržalo astrološke kolumne (Truzzi 1972), a danas je taj procenat verovatno veći. Sa razvojem modernih tehnologija i digitalnog doba, uticaj astrologije je počeo da se širi i kroz društvene mreže i različite aplikacije. Postoje brojni Instagram i TikTok profili, koji su posvećeni astrologiji, u formi razonode, ali i ozbiljnog bavljenja horoskopom. Takođe, u poslednjih 7-8 godina razvijene su i aplikacije poput Co-Star i The Pattern, koje na osnovu vremena i datuma rođenja korisniku automatski daju „personalizovani horoskop“ i omogućavaju mu da uporedi kompatibilnost sopstvenog profila sa profilima prijatelja. Aplikacija The Pattern čak ide i korak dalje, pa pokušava da astrološke podatke preformuliše u psihološki narativ, nudeći određenu vrstu samopomoći (kao online forma astroterapije). Pridavanje važnosti, od strane javnih ličnosti, ovim aplikacijama i astrologiji generalno, značajno doprinosi širenju uticaja njenih osnovnih ideja na mlađe generacije. Istraživanja pokazuju da je ova industrija astroloških aplikacija u stalnom porastu, u smislu popularnosti i broja korisnika, a posebno među mladim generacijama, koje ih doživljavaju kao oblik digitalne zabave (Business Research Insights, 2023). Ovi, digitalni oblici astrologije i astropsihologije, pokazuje visok stepen sposobnosti transformacije, koji im omogućava da opstanu i isprate tehnološki napredak društva, popularizujući na ovaj način svoju filozofiju među novijim generacijama.

Kognitivne pristrasnosti

Do sada smo već razmotrili neke od razloga, koji čine ljude sklonim da veruju u astrologiju, a samim tim da pribegavaju astropsihologiji i traže pomoć astropsihologa u kriznim trenucima. Međutim, verovanje u astrologiju ne potiče samo od proste zablude ljudi, već gledano iz ugla psihologije, njen uzrok možemo posmatrati kroz mehanizme kognitivnih pristrasnosti. Kognitivne pristrasnosti nazivaju se još i *heuristike* i predstavljaju kognitivne prečice koje se koriste da pomognu ljudima prilikom donošenja odluka. One se mogu smatrati kognitivnim pravilom ili smernicom, koju ljudi podsvesno primenjuju na složenu situaciju, kako bi bili efikasniji prilikom donošenja odluka (O'Sullivan & Schofield 2018). Heuristike nam pomažu da se snađemo u složenom svetu, ali nas mogu navesti i na pogrešne zaključke, kao što je verovanje u pseudonaučne tvrdnje, poput astrologije. U nastavku će biti razmotrene neke od kognitivnih pristrasnosti, koje se mogu dovesti u vezu sa sklonošću ka prihvatanju astrologije i verovanju u njenu istinitost ili nam mogu objasniti mehanizam koji se nalazi u osnovi tumačenja horoskopa kod pojedinaca.

- „*Barnum efekat*“

Termin *Barnumov efekat* prvi je upotrebio Paul Meehl (1956) kako bi objasnio fenomen, u okviru kog ljudi dobrovoljno prihvataju tumačenja svoje ličnosti koja se sastoje od nejasnih i dvosmislenih izjava, u kojima se može prepoznati većina populacije. Ove tvrdnje, koje su navodno skrojene po meri osobe, a zapravo su toliko uopštene da se mogu primeniti na širok spektar ljudi, često podsećaju na opise horoskopskih znakova, pa nije ni čudo što se ova kognitivna pristrasnost dovodi u vezu sa verovanjem u astrologiju kod pojedinaca. Istraživanja pokazuju da su astrološki opisi vrlo podobni za pojavu snažnog uticaja *Barnum efekta*. Ljudi su često zadivljeni tačnošću psiholoških opisa, koji su „specijalno namenjeni njima“. Zbog toga, pronađeno je da su identični opisi u istraživanjima doživljeni kao tačniji, kada je ispitanicima rečeno da su u pitanju astrološki opisi, nego kada im je rečeno da su to zapravo tvrdnje koje su generalno istinite za većinu ljudi (Snyder 1974; Snyder, Larsen & Bloom 1976).

- *Samoispunjavajuće proročanstvo*

Ova kognitivna pristrasnost odnosi se na proces u kojem početno pogrešno verovanje vodi ka ponašanju, koje na kraju izaziva ishod koji potvrđuje prvobitno, lažno verovanje. Prividna validnost samoispunjavajućeg proročanstva samo produžava efekat greške (Merton 1948). Ovaj princip se lako može preneti na oblast astrologije i dovesti u vezu sa verovanjem o istinitosti astroloških predviđanja. Verovanja ljudi u svoj horoskopski znak i njegove opise može uticati na njihovo ponašanje i emocije (Dabhade 2025). Na primer, horoskop predviđa turbulentan period na poslu, pa pojedinac može biti obeshrabren, manje motivisan u radu i rezervisaniji u odnosu sa kolegama, što posledično može voditi nepovoljnim okolnostima na poslu, a pojedinac zaključuje da je horoskop dobro predvideo buduća dešavanja.

- *Efekat potvrđivanja*

Efekat potvrđivanja (confirmation bias) je jedna od ključnih pristrasnosti koja nam pomaže da razumemo uzrok verovanja u astrologiju. Ovaj mehanizam pristrasnosti čini ljude sklonim da traže, tumače i pamte informacije na način koji onemogućava verovatnoću da će njihova početna hipoteza biti odbačena (Oswald & Grosjean 2004), tj. značaj se pridaje isključivo informacijama koje su u skladu sa njihovom početnom hipotezom, dok se ostale informacije ignorišu ili se njihov značaj umanjuje. Ako ovo prenesemo na astrološka predviđanja, možemo pretpostaviti da ljudi koji veruju u astrologiju pridaju veći značaj astrološkim predviđanjima koja se ostvare, dok se zanemaruje važnost onih predviđanja koja nisu ostvarena. Na primer, dnevni horoskop predviđa novo poznanstvo. Ukoliko se desi da pojedinac zaista upozna nekoga taj dan, zaključak će biti da horoskop „stvarno radi“. Sa druge strane, ako se desi da pojedinac nikoga ne upozna, on će zanemariti tu informaciju i neće razmišljati o neistinitosti horoskopskog predviđanja.

- *Pristrasnost autoritarnosti*

Verovanje u astrologiju se često dovodi u vezu autoritarnošću, koja podrazumeva osobine pojedinca kao što su netolerancija dvosmislenosti, poštovanje autoriteta i učestala upotreba stereotipa, koja olakšava razumevanje socijalnog sveta (Adorno 1950). Adorno vidi astrologiju kao naglašavanje konformizma i poštovanja prema nekoj vrsti višeg autoriteta (Allum 2011). Astrološka ideologija podseća, u svim svojim glavnim karakteristikama, na mentalitet pojedinaca koji postižu visoke rezultate na dimenziji Autoritarna ličnost (Adorno 1994). Rezultati istraživanja idu u prilog tezi da su autoritarne ličnosti sklonije lakovernosti prema astrologiji (Allum 2011). Adorno (1994) sugerise da deo privlačnosti astrologije leži u tome što se kroz nju formalizuje pojam nekog višeg autoriteta, koji kontroliše životne događaje, ali ne nosi sa sobom eksplicitno restriktivnu strukturu formalnog religioznog pripadanja, koje podrazumeva odlazak u crkvu, post i tako dalje.

Pseudonaučnost

Psihologija se dugo borila da postane prepoznata kao zasebna oblast i dobije svoj naučni status. Bilo je potrebno da se razviju standardizovani testovi, statistički metodi i eksperimentalna metodologija, kako bi se distancirala od pseudonaučnih praksi poput frenologije, mesmerizma i astrologije, koje su takođe pretendovale da objasne uzroke ljudskog ponašanja, ali bez odgovarajuće empirijske provere i naučne potvrde. Kritičko distanciranje od ovakvih tvrdnji je psihologiju i dovelo do današnjeg statusa utemeljene i legitimne naučne discipline. Zbog toga se povezivanje psihologije sa astrološkim konceptima, u vidu astropsihologije, može posmatrati kao problematično ukoliko se astrološkim pretpostavkama pripisuje naučni status koji one same ne mogu da opravdaju. Uvođenje pojmova i metoda iz oblasti psihologije u astropsihologiju ne menja činjenicu da se njene osnovne pretpostavke oslanjaju na astrološki sistem. Stoga je za procenu njenog naučnog statusa potrebno ispitati u kojoj meri astrološke

pretpostavke mogu zadovoljiti kriterijume poput empirijske provere, otvorenosti prema kritici i kumulativnog razvoja znanja.

Jedan od problema sa astrološkim tvrdnjama odnosi se na njihovu prediktivnu validnost i mogućnost jasne empirijske provere. Astrologija ne pruža ubedljive empirijske dokaze da položaji nebeskih tela predviđaju osobine ličnosti, ponašanje ili životne ishode na način koji njene osnovne pretpostavke podrazumevaju. Iako se astrološke tvrdnje često formulišu u vidu tendencija, a ne preciznih predviđanja, pojedine od njih moguće je operacionalizovati i statistički testirati. Na primer, astrologija sugerise da osobe rođene pod uticajem određenih znakova ili planeta imaju veću verovatnoću da se bave određenim zanimanjima, poput tvrdnje da Mars pogoduje vojničkim ili sportskim zanimanjima, dok Venera ima umetnički uticaj. Međutim, rezultati istraživanja na uzorku od 25.000 Francuza nisu pokazali značajnu povezanost između karijera ispitanika i njihovog sunčevog, mesečevog ili ascendentnog znaka (Thagard 1978). Stoga, problem nije samo u tome da li se pojedinačna astrološka tvrdnja može testirati, već i u tome da li astrološki sistem proizvodi dovoljno precizna i rizična predviđanja koja bi omogućila njegovo ozbiljno empirijsko opovrgavanje. Kada su predviđanja nejasna ili dovoljno fleksibilna da budu kompatibilna sa različitim ishodima, neuspeh predviđanja teško se može koristiti kao osnova za odbacivanje same teorije.

Međutim, to nije jedini problem sa kojim se astrologija suočava u pokušaju dobijanja naučnog statusa. Fenomen „precesije ravnodnevica“ predstavlja dodatni problem, jer Zemljina osa nije potpuno stabilna, već se polako pomera kroz vreme. Zbog toga, današnji horoskopski znaci više ne odgovaraju istim sazvežđima kao u vreme kada je astrologija nastala. Astrolozi ovaj fenomen uzimaju u obzir kada najavljuju „Doba Vodolije“, ali ga potpuno zanemaruju kada prave svoje karte. Iz perspektive razvoja naučnog znanja, problem astrologije nije samo u tome što pojedine njene tvrdnje mogu biti teško proverljive, već i u tome što dugotrajno postojanje problema nije praćeno razvojem novih i uspešnih predviđanja ili značajnim unapređivanjem njene objašnjavajuće moći.

Thagard (1978) smatra da su dva glavna razloga za pseudonaučnost astrologije: 1) to što je bila manje progresivna od alternativnih teorija tokom dugog vremenskog perioda i suočavanje sa mnogim nerešenim problemima; 2) zajednica praktičara astrologije se ne trudi da radi na rešavanju ovih problema i ne pokazuje zainteresovanost za upoređivanje sa drugim teorijama. Astrologija je vrlo neprogresivna, jer nije ništa promenila niti dodala u svojoj objašnjavalačkoj moći već 2000 godina. Dodatno, kada se suoči sa nepovoljnim empirijskim nalazima, astrološke tvrdnje mogu biti zaštićene uvođenjem *ad hoc* objašnjenja kojima se neuspeh predviđanja pripisuje dodatnim, naknadno uvedenim faktorima (Edis 2018), što se takođe smatra još jednim kriterijumom pseudonaučnosti. Pored toga, postoje alternativne, psihološke teorije ličnosti i ponašanja, koje svoje tvrdnje zasnivaju na empirijskim činjenicama i dokazima. Zajednica astrologa je generalno selektivna u razmatranju potvrda i opovrgavanja

svojih premisa i nezainteresovana za unapređenje astrologije u smislu davanja odgovora na pitanje u vezi dugogodišnjih problema. U tom smislu, selektivno prihvatanje potvrđujućih nalaza, zanemarivanje nepovoljnih rezultata, kao i odsustvo sistematskog rada na rešavanju dugotrajnih problema i razvoju novih predviđanja ukazuju na to da astrologija ne ispunjava ključne kriterijume naučne discipline, već ju je opravdano smatrati pseudonaučnim sistemom znanja.

Zaključak

Psihologija je naučna disciplina koja koristi validirane psihološke testove kako bi merila dimenzije ličnosti pojedinaca, koje se dovode u vezu sa određenim ponašanjima. Psihološki testovi poseduju prediktivnu validnost, a rezultati istraživanja se mogu proveravati i replicirati. Sa druge strane, astropsihologija koristi simboličke reprezentacije zasnovane na položaju zvezda i planeta kako bi opisala pojedinca i predvidela njegovo ponašanje. Ove simboličke reprezentacije nemaju nikakvu naučnu osnovu, pa astropsihologija spada u domen pseudonauke. Iako čitanje horoskopa može predstavljati vid zabave i razonode za veliki broj ljudi, a astrologija nosi sa sobom određeni kulturni i simbolički značaj, vrlo je važno razumeti da njena primena u praksi nema naučno utemeljenje i značaj. Samim tim, upotreba astropsihologije za rešavanje različitih psiholoških problema može delovati jedino na nivou uticaja placebo. Iz ovih razloga, potrebno je raditi na osveščivanju pojedinaca koji pribegavaju astroterapiji ili drugim psihoterapijskim metodama sumnjive validnosti da ne postoje dokazi da ovakvi vidovi terapije funkcionišu niti da doprinose rešavanju problema. U najboljem slučaju, ove terapije ne utiču na promenu stanja osobe, a, u nekim slučajevima, mogu doprineti pogoršanju problema (Lilienfeld et al., 2014). Zato je vrlo važno da se ljudi prethodno informišu se o formalnom obrazovanju osobe od koje traže pomoć, metodama koje primenjuju u radu, kao i empirijskim dokazima koji se tiču efikasnosti ovih metoda.

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Marijana Vitorović

Astropsychology as a Modern Pseudoscience: Theoretical Frameworks and Psychological Bases of Belief

(Summary)

Astropsychology represents a contemporary attempt to merge astrological concepts with psychological theories of personality and behavior. Although it is often promoted as an innovative approach to understanding psychological processes, its

status as a pseudoscience remains undisputed due to the absence of empirical support, the lack of theoretical coherence, and its inability to meet fundamental scientific criteria. This paper examines the reasons why astropsychology cannot attain scientific legitimacy through its association with psychology, given that it is grounded in astrological assumptions that are empirically and logically unsustainable. Special attention is devoted to the cognitive biases that contribute to its popularity. The paper also considers sociocultural factors and the influence of digital platforms on the spread of astrological practices, including popular applications and personalized horoscopes. It is concluded that astropsychology, despite its appeal and subjective persuasiveness, remains inconsistent with the scientific standards of psychology and represents a significant example of pseudoscientific discourse in contemporary society.

KEYWORDS: astropsychology, astrology, pseudoscience, cognitive biases, heuristics

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Miloš Kovačević

Education, Impartiality, and the Right to an Open Future¹

Summary: *The paper examines whether early specialization in education violates children's autonomy and their right to an open future. It argues that overexposure to narrow content creates individuals whose futures have been foreclosed before they could choose for themselves. In this way, it threatens autonomy of their future choices. The analysis engages Joel Feinberg's autonomy-based framework and its critique from Claudia Mills, who challenges both the desirability and feasibility of neutrality of education. Rejecting an educational binary between superficial breadth and deep immersion, the paper proposes a "proper bite-size" view. This pedagogical approach advocates for minimal yet meaningful exposure to diverse fields and minority lifestyles, balancing variety with substance. The conclusion maintains that, before autonomy is attained, education should prioritize broad, formative experiences, allowing individuals to later specialize as autonomous agents.*

Keywords: *autonomy, paternalism, right to an open future, specialization, education, childhood.*

Introduction

The famous tennis player Novak Đoković never finished high school because he was too busy building his career. He is just one among many high-achievers who were pushed at a very young age to develop their talents, often leaving them with only a rudimentary general education and scarce opportunities to pursue other vocations. Was their freedom of choice violated? Should we be concerned about their autonomy?

One might argue that Novak is too rich and successful for ordinary people to worry about his well-being. If we remove money and fame from the equation, however, we can consider the example of so-called "dual education." For some time now, 15-year-olds predominantly from disadvantaged backgrounds have been pushed into high school programs that, despite the name, have little to do with education. Dual education is essentially a three- to four-year internship program, allowing students to spend up to 80% of their school time doing manual, repetitive work that offers little upward mobility (Reljanović 2023). This is time that would otherwise be spent in the classroom learning a variety of subjects conducive to well-rounded development.

Despite surface differences, the grim reality of a state-sponsored teenage workforce

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and Novak's glamorous tennis career share something in common: both result from overexposure to highly specific content that seriously threatens to foreclose alternative vocational pathways. This problem would be worth examining even if it only affected those on the margins of the educational system. However, I argue that it persists in less obvious forms even within mainstream schooling. The tension between depth and breadth of knowledge is built into the educational system itself. The examples I have given are not exceptions, but rather inherent features of the process of building expertise. Threats to future autonomy are not limited to the exceptionally talented or the disadvantaged. Moreover, the conflict between breadth and depth extends beyond vocational choice to include lifestyle decisions as well.

This paper proceeds in six distinct parts. In the first part, it places the problem of future autonomy in historical context showing that concern for unwarranted foreclosure of vocational options is modern phenomena. In the second part, it establishes the conceptual framework of autonomy and paternalism, examining the philosophical distinction between children and adults. In the third part, it analyses Joel Feinberg's *right to an open future* as a response to paternalism, detailing how it shapes permissible interference and educational obligations. In the fourth part, it engages with major criticism of this view from Claudia Mills focusing on problems of neutrality and depth of education. In the fifth part, it proposes a constructive alternative: *the proper bite-size view*, which balances educational breadth and depth while safeguarding future autonomy. Finally, in the conclusion, the paper reaffirms the priority of meaningful educational breadth over premature specialization that should avoid producing individuals whose options have been foreclosed before they could choose for themselves.

The Development of Education and the Emergence of Vocational Choice

Universities emerged in the 12th and 13th centuries as intellectual guilds of students and teachers—that is, as a special type of guild dedicated to learning (Rashdall 1987). Depending on the university, students had greater or lesser autonomy in choosing the curriculum. At the University of Bologna, for example, which was renowned for the study of Roman law, students were able to decide which texts their teachers would lecture on (Pedersen 1997: 54–59, Cobban 1975: 66–71). In the 16th century, under the influence of the Renaissance, enthusiasm for humanistic education emerged. This form of education was considerably broader, encompassing the study of Latin and Greek, as well as engagement with classical philosophical works, history, poetry, and law. However, this type of education was reserved for the social elite, and in the territory of present-day Serbia, it appeared only in the 19th century. Until then, in parts of Serbia under Ottoman rule, schooling was almost entirely absent, and “education was scarce and negligible; indeed, even as a concept, it was reduced exclusively to literacy” (Parlić-Božović 2011: 565), available under the auspices of the church almost exclusively to the clergy, while the educational situation in the territory of present-day

Vojvodina was somewhat better (Parlić-Božović 2011: 557–558, 561–563).

Aside from universities, monasteries and churches, informal education also took place in households and craft workshops. In pre-capitalist societies, deeply rooted customs regulated people's lives, forming precise expectations and duties at each stage of their lives (Katz 1976: 387). In Northern Europe and America, the sharp distinction between work and education that we know today did not exist. Instead, "what we would consider paid work, and what we would consider education, were seen as largely the same thing, and both were a process of learning self-discipline" (Graeber 2019). In the centuries preceding industrialization, it was common practice for parents to send their adolescent children away for several years to artisans or to other families to work as apprentices or servants.

In the absence of compulsory education and an efficient bureaucracy, guilds took on the responsibility of regulating the educational relationship between artisans—who assumed the role of educators and often the prerogatives of parents—on the one hand, and apprentices and journeymen, who learned the trade through work, on the other. Since they could not directly regulate the educational process itself, guilds regulated its duration. The prescribed length of apprenticeship was meant to ensure that the apprentice acquired knowledge of different production methods and technologies, as well as of the market, competitive standards, and negotiation with other relevant economic actors. Moreover, the duration of apprenticeship ensured that the costs and benefits of the educational process were distributed in such a way that each party had an incentive to participate (Epstein 1998: 687–692).

Apprenticeship was not limited to crafts; the children of farmers also served in the households of other families. In this way, young people not only acquired the human and economic capital necessary for an independent future, but also achieved greater and prolonged actual autonomy (Katz 1976: 396). As David Graeber points out:

(...) they were acquiring both the knowledge and experience needed to manage a household, shop, or farm, and also the wealth needed to acquire one—or, in the case of women, to be able to offer a dowry to a suitor able to do the same. As a result, medieval people married late, usually around thirty, which meant that "youth"—adolescence, a time when one was expected to be at least a little wild, lustful, and rebellious—would often last a good fifteen to twenty years (Graeber 2019).

The development of capitalism created a need for a mobile workforce, which led to a decline in the practice of employing male servants, and subsequently of apprenticeship itself. Thus, the practice of working in other households gradually disappeared, and with it the limited autonomy that young people had achieved in this way (Katz 1976: 387). Technological progress reduced the value of craft skills and thereby precaritized the position of artisans (Katz 1976: 396, 398). The weakening of guild power manifested itself in the declining control over the conditions under

which apprenticeship was conducted. By the end of the 18th century, the number of unfinished apprenticeships had increased. Previously, those who did not complete their apprenticeship found themselves in a particularly vulnerable position, having spent years developing skills specific to a particular trade without obtaining the qualification to practice it independently (Epstein 1998: 691–692). However, as capitalist expansion increased the demand for semi-skilled wage labourers, the incentive to complete an apprenticeship diminished (Epstein 1998: 706; Snell 1985: 241–243, 253–254).

Public institutions developed in the 19th century as a result of the increasing industrialization of the economy, the urbanization of the population, the need for a mobile workforce, and the expansion of the state's role in securing social welfare. Among these, the system of public education played an important role. Its task was to respond to a whole range of social changes, primarily the redefinition of the family institution, rising youth unemployment, perceptions of increasing crime, and the growing cultural heterogeneity of the population. The acquisition of knowledge and training for access to the labour market were of secondary importance. Education was primarily aimed at cultural standardization and the shaping of students' character, because at that time, moral neglect was seen as the root cause of most social problems (Katz 1976: 384, 386, 391–394, 399).

One of the more dramatic changes brought about by industrialization was the separation of the workplace from the place of residence. In modern capitalist societies, young people spent an increasing number of years in education, continued to live with their parents even after finding employment, and remained in the parental home until they married. Although they remained under full parental control for longer, young people increasingly did not follow their fathers in their choice of occupation. Due to the declining value of craft-produced goods, maintaining the achieved standard of living across generations was no longer possible through the mere transmission of skills from parent to child. Instead, parents had to equip their children for more promising professions in trade and the growing state bureaucracy (Katz 1976: 387–388, 398).

In this way, the material preconditions emerged for the problem of autonomy in the choice of profession to gain relevance. The right to an open future can be seen as a response to a specifically modern phenomenon: the emergence of a significant number of individuals who possess a real opportunity to choose a career path. The fact that the minimal institutional preconditions for relocating such a significant decision from the domain of necessity and predestination into the domain of some semblance of choice appeared only in the 19th century for the broader population tells us the extent to which the future autonomy of young people had, until then, been routinely violated—despite the limited actual autonomy they had achieved through the earlier practice of leaving the parental home.

Autonomy and the Child-Adult Distinction

Generally, we presuppose that autonomous agents possess the cognitive capacities needed to make their own decisions and that they are sufficiently independent from others to carry them out. Children are not considered autonomous because they do not meet these conditions (Schapiro 1999). Compared to adults, children have reduced cognitive capacities and are dependent on their guardians (Brighouse and Swift 2014: 58). The degree of autonomy justifies the difference in the treatment of adults and children (Hannan 2019). Interference with an agent's present preferences for the sake of their future good is not blameworthy in the case of children, but is considered so in the case of adults. Therefore, it could be said that from the state's perspective, the future self of a child carries much more weight than the future self of an adult. While in liberal societies it is accepted that an adult's current preferences (almost) always take precedence over their future well-being, due to the specific position of children, the relationship between their present and future remains unresolved.

One straightforward way to resolve the question of how we should treat children is to reject their supposed lack of autonomy altogether. Doing so would shift them from the category of beings rightfully subject to paternalistic interference into the category of autonomous agents they share with adults. The willingness to make such a move depends on the conception of autonomy one endorses. There are more and less demanding conceptions of autonomy, which could have radically different impacts on the ascription of autonomy to children of a certain age. The age at which children fulfil the demands for autonomy could vary widely, ranging from as young as three years old in some liberal conceptions to eighteen years old in the most conservative ones (Schapiro 1999: 734; Mullin 2014: 416). If one endorses a less demanding conception of autonomy, the whole issue could potentially be bypassed by simply allowing children to decide what they want to study—or whether they want to study at all. One important aspect to consider is that autonomy is not purely an internal phenomenon; its level is significantly dependent on external circumstances. In her analysis of children's autonomy, Sarah Hannan reminds us that “in asking whether children of various ages are capable of autonomy, we must not merely look at children as they currently are. We also need to ask whether children could be autonomous under more favourable external conditions” (Hannan 2019: 118). In other words, we should try to avoid what Sally Haslanger calls *epistemic objectification*—the process by which the subordinated status of a group is naturalised due to their contingent epistemic weaknesses (Haslanger 2017). Subordinated agents can find themselves in a vicious cycle of epistemic objectification:

denied epistemic resources; this causes epistemic harms, sometimes in the form of epistemic deficits; contingent epistemic deficits within the group are then objectified, i.e., treated as normal or natural, so not requiring or even allowing intervention; this rationalizes the denial of epistemic

resources and epistemic credibility, thus perpetuating epistemic harm (Haslanger 2017).

Both Haslanger and Hannan note that women and people of colour have experienced a similar self-fulfilling prophecy (Haslanger 2017; Hannan 2019: 122). Many historical examples warn us against the unwarranted denial of autonomy to certain groups. However, it is difficult to determine whether the widespread view that children are not autonomous is simply another form of normalized epistemic objectification. There is no conception of autonomy, however liberal, that would ascribe autonomy to toddlers younger than three. But setting such a low threshold would likely invite reconsideration of the autonomy status of certain animals as well (Hannan 2019: 118). This is, after all, an empirical question. Generally, an answer might be found by placing children in autonomy-enhancing settings and observing at what age most of them achieve the level of autonomy typical of an 18-year-old agent.

One aspect of autonomy in which some circular reasoning is hidden is the condition of independence. Take, for example, economic self-sufficiency, which Marina Oshana takes as an integral part of independence (Oshana 2008: 86–87). Children are not economically self-sufficient under current conditions, but it is questionable whether this would remain the case under a different legal framework and a more robust welfare system. For instance, if universal basic services were introduced (Kovačević 2022), they could ensure children's (at least economic) independence from guardians.

On the other hand, the autonomy of adults is established on grounds much less solid than many assume. Hannan notes that “the point is not simply that many children are capable of more than we give them credit for, but also that adults are often capable of less than we assume” (Hannan 2019: 119). Hugh LaFollette also highlights how the difference between teenagers and adults is often exaggerated, questioning the autonomy of adult chain smokers and helmetless cyclists (LaFollette 1999: 144).

It could be said that although the dichotomy between adults and older teenagers is on shaky ground, a line must be drawn somewhere (Hannan 2019: 120). In other words, when we debate whether 17- or 16-year-olds are autonomous, we are engaged in fruitless sophistic arguments. Such a conclusion is unwarranted, as it fails to account for individual differences. We all deserve not to be treated as collateral damage of heavy-handed policies. Any interference must be justified to the agent as an individual and not as a member of a certain age group.

Even if children are not autonomous, it is sometimes argued that we should treat them as if they are, because in doing so we prepare them for an autonomous future (Brighouse 2002: 46). This means that children could be allowed to choose their level of specialisation not only out of respect for their present degree of autonomy but also as an educational method for developing full autonomy later in life. I will pursue this line of inquiry here, setting aside the difficult—though interesting—question of whether, and when exactly, children are autonomous.

Open-Future View

A valuable contribution to addressing unlimited paternalism toward children was made by Feinberg through his formulation of *the right to an open future*, articulated in an essay of the same name. Feinberg distinguishes between three types of rights: A, C, and A-C rights. A-C rights are those shared by children and adults (e.g., the right to bodily integrity). A rights belong exclusively to adults (e.g., the right to vote), while C rights belong exclusively to children.

Within C rights, Feinberg further distinguishes between two categories. On one hand, there are *dependency rights*, which respond to children's immediate needs and are not typically held by adults (e.g., the right to food and shelter). On the other hand, there are *rights in trust*, collectively termed the *right to an open future*—rights that children possess as potential autonomous agents. These rights parallel the A rights accorded to adults as actual autonomous agents. Even though children cannot yet exercise these future-oriented rights, the right to an open future can be violated during childhood by influences that decisively shape the choices they will make as adults. In essence, this right protects the child from autonomy-diminishing influences until they are capable of making free choices.

The distinction between A-C rights and the right to an open future can also be explained through the lens of a child's interests. A-C rights protect the interests the child holds *as a child*, whereas the right to an open future protects the *future interests* the child may develop upon reaching adulthood. In this sense, the right to an open future preserves the autonomy of an adult who does not yet exist, and it requires preventing influences that would foreclose key options before the child can choose autonomously (Feinberg 1994: 76–77).

This right refines the kind of paternalism deemed acceptable toward children. By guaranteeing the right to an open future, the state commits to conceptualizing children's future well-being in a way that incorporates their autonomy. Furthermore, the state assumes an obligation to protect this right from autonomy-diminishing influences by third parties—even those justified on paternalistic grounds.

If, as Timothy Fowler notes, “schools have a major role to play in opening up options for children,” they can also threaten future autonomy by failing to open up options *enough* (Fowler 2023). Autonomy theorists view impartiality in education as a safeguard against violating the right to an open future. Students must acquire the knowledge and skills that maximize their options regarding future vocations and lifestyle choices (Feinberg 1994: 84–88). For Feinberg, maximizing opportunities for lifestyle choice means parents must “take a chance” with the content to which their children are exposed in school—refraining from withdrawing them simply because they disapprove of certain influences (Feinberg 1994: 88). The presentation of diverse lifestyles by those who practice them is thus recognized as one way schools can protect the right to an open future (Brighouse 2005: 25).

Criticisms of Open-Future View

Mills criticizes Feinberg's open-future view in both the professional and lifestyle domains. Mills points out the problem of quantifying opportunities—which I discuss below—and also questions whether third parties are really able to *close* some options for the child in the future, or whether they can only *discourage* them (Mills 2003: 501). I suppose that with some extreme forms of conditioning and manipulation, third parties can effectively close certain options, but that is not the point here. My reading of Feinberg is that he does not subscribe to an all-or-nothing view about autonomy, and that any degree of closing opportunities, from a bare nudge to total closure, would be blameworthy because it is an unwarranted divergence from the ideal of impartiality.

This is in line with a stronger claim that Matthew Clayton developed in favour of a more robust type of children's independence. While in Feinberg's account children should be protected only from interference that would in advance foreclose certain future options, Clayton argues that any unjust upbringing interference, even those with reversible effects, should be ruled out (Clayton 2012: 358). Since the main problem with children's autonomy is that they are not yet able to give consent to interference that will affect them in the future, Clayton asks what would be a reasonable response to interference with similarly positioned adults. He gives the example of a nose job performed on an unconscious patient without her consent. Even if it happens that the patient likes her new nose, that would not change the fact that her autonomy was violated (Clayton 2012: 362). By analogy, children should be treated as unconscious patients. Since they are unable to consent to interference, any unjust intervention in their life is already a violation of their autonomy, not only those interventions not later endorsed by their future selves.

However, impartiality itself is a matter of Mills' contestation: "it is simply impossible, I maintain, for parents to raise their children without steering them, however imperceptibly, toward one option rather than another. There is no default setting of perfect neutrality [...] To open one door is inescapably to close another." (Mills 2003: 501, 503). Feinberg already responded to this type of objection, saying that we should aspire to come as close as possible to the ideal of neutrality, even though he is aware that in non-ideal circumstances (as with many other values) we would not be able to achieve it fully. Therefore, Feinberg's open-future view should be understood as a concrete harm-reduction model for protecting children's future autonomy.

Mills develops two arguments against the open-future view, even in above interpretation. One is that this approach trivializes choice, offering a shallow and superficial experience of options due to a lack of time. Additionally, for the same reason, it pushes children toward choosing options only according to their pre-existing beliefs, which hinders personal development. Her other argument is that the open-option view is frenetic and exhausting. Mills discusses the choice about one's vocation and religion, but her arguments could be extrapolated to other lifestyle choices as well (e.g., political

affiliation, relationship style, hobbies, etc.). She takes the example of a boy whose parents expose him to various religions, allowing him to choose for himself what his religion will be, or whether he will have one at all (Mills 2003: 502). This is how the open-future view would look in practice: a buffet of options from which the child could freely choose.

Mills argues that in order to properly experience some options, children must devote much more time to them. The sense of belonging and community that religion offers, or the enjoyment of playing an instrument or sport, cannot be experienced through a short introduction. In other words, children must invest much more time and energy into some options to get the best out of them. Besides that, short exposure of children to various options would not allow them to really challenge their pre-existing notions about what does or does not make sense, or what is worthy of their attention. Therefore, the problem is both that children will not have time to experience the essence of some options, and that due to the “shopping around” attitude encouraged by the open-future view, children will discredit options that would enable personal growth (Mills 2003: 502).

When it comes to her critique of the open-future view as frenetic and exhausting, Mills points out that the idea of maximizing opportunities collides with free time and a carefree childhood. Doubling down on the problem of superficiality, she says:

The drive to amass as many experiences as possible can never be complete: Precisely what will be missing from the list is experiences that require time to assimilate and absorb, experiences that are slow and deep, rather than easily scheduled in an escalating spiral of extracurricular activities. (Mills 2003: 506).

I am very sympathetic to Mills’ concern for the depth of education, and her arguments against the open-future view are generally persuasive. However, I do not believe that the problem of children’s future autonomy will be solved best by taking sides, but by finding a fine balance between the depth and breadth of education. It seems that Mills is also interested in finding a compromise solution. Although she criticizes the open-future view, she does not subscribe to the view that parents should have the right to impose a vocation or lifestyle on their children either. However, for Mills, such imposition would be impermissible not because it violates children’s future autonomy, but because it infringes their present autonomy. Imposing a lifestyle or vocation would indicate a lack of parental unconditional love for the child, that is, an unwillingness to accept the child as they are (Mills 2003: 504–507).

The problem with Mills’ solution is that it begs the question. If children have autonomy, then any discussion of paternalism toward them is already off the table because they rightfully can make choices for themselves. But obviously, Mills does not think that children of any age are autonomous. It seems that she thinks that 12-year-olds and older children can autonomously decide about their religion. But what

about the younger ones? For them, Mills does not indicate any limitations on parental paternalism.

Even if her own solution is not satisfying, Mills' arguments against the open-future view must still be addressed. I believe that both of Mills' points could be acknowledged and the open-future view preserved in some form.

Toward a Proper Bite-Size View

On a more general level, Mills' point is that maximizing the possibilities for choice should not be understood exclusively in quantitative terms—as merely increasing the number of available alternatives. Adequate maximization of opportunities also has a qualitative dimension, related to the *variety* and *significance* of those alternatives. For example, if we have a choice of nine types of beer, adding one more type of beer increases the selection, but not as much as adding wine would (Pettit 2012). Variety qualitatively enhances choice even when the number of options remains the same. As for significance, consider a situation where two sets of alternatives are equal in number but differ dramatically in consequence: choosing between two buttons that do nothing, versus two buttons that can cause massive changes in the world. Clearly, the latter offers greater opportunity for meaningful choice.

The number, variety, and importance of available alternatives can come into conflict. There are situations in which introducing a more different or more significant alternative entails a reduction in the total number of options, or where variety is negatively correlated with importance. This conflict, viewed through the prism of vocation, manifests as the classic divide between *specialists*—agents who develop deep expertise in a narrow field—and *generalists*—agents who cultivate basic knowledge across a broad domain. Specialists sacrifice breadth for depth, while generalists sacrifice depth for breadth.

Generally speaking, the educational system is structured so that with each subsequent level, the generality of content decreases. During elementary school, students are exposed to a variety of sciences and arts. Through secondary school, they narrow and deepen their focus in certain areas, a process that repeats in higher education, leading to even greater specialization. At first glance, this appears to be a system that preserves freedom of choice. However, things are not that simple.

As I pointed out earlier in the paper, the educational process for many talented and underprivileged individuals does not follow this model. Yet even students on a standard educational path are not exposed to all sciences and arts during elementary school. Furthermore, it remains an open question whether the *pace* of specialization is adequate. It is possible that the depth of study in a specific discipline too quickly reduces the entire range of areas to which students are exposed during their schooling.

Miloš Kovačević argued that education comes closest to neutrality when schools present only *minority lifestyles* as a counterweight to students' overexposure to majority

lifestyles outside the school context (Kovačević 2024: 54). For better or worse, parents' right to raise their children as they wish is recognized as a human right. Therefore, the choice of which lifestyles children are exposed to is left to parental discretion, resulting in unbalanced influences outside the educational system. On the other hand, the right to education is also a human right, and schools are one of the few places through which the liberal state can intervene to mitigate the effects of parental discretion. In liberal political theory, it is often assumed that education must be neutral because it is a primary way in which students' ethical and political views are shaped (Fowler 2023). However, this view is mistaken because it ignores the fact that lifestyle acquisition occurs everywhere, not only in school. Schools cannot successfully preserve the right to an open future if they base their curriculum on the false assumption that the context outside school is characterized by impartiality. Therefore, schools must make maximum use of their corrective potential by excluding from the curriculum those lifestyles to which children are overexposed within the family, and instead placing greater emphasis on alternative lifestyles.

At the same time, this limitation addresses both the problem of insufficient time to present all options and the danger that children will not be exposed to content that challenges their pre-existing beliefs. By excluding dominant lifestyle options from the curriculum, time is made for deeper engagement with alternative lifestyles. And because they are alternative, they by definition challenge students' pre-existing beliefs. Indeed, that is precisely why they should be presented in school in the first place.

However, the tension between breadth and depth of content acquisition persists even when limited to minority views of the good life. Is the right to an open future better protected through basic familiarity with as many alternative lifestyles as possible, or through in-depth understanding of a few? And what about science and art? Aside from Hollywood movies and pop music, no art, let alone science, has widespread influence on students outside of school. Therefore, we cannot rely solely on the presentation of alternative lifestyles as the only tool for addressing the problems of the open-options view.

Much of the persuasiveness of Mills' arguments stems from the types of examples she chooses. Certainly, one cannot learn much about a sport or a musical instrument in a day or a week. But consider the case of food tasting: I can fully experience the taste of a dish from just one bite, or of a drink from one sip. Additional sips and bites may not bring many new sensations. Yet even a bite can be too small to be representative of the whole dish. If we do not take a *proper* bite, we experience only part of the dish and miss the rest. Returning from the culinary to the educational context, this means that our understanding of autonomy-enhancing education should move beyond the dichotomy of deep immersion versus frenetic superficiality. We should embrace a third way, which I call *proper bite-size experience*. Every science, art, and lifestyle has its own *minimal yet meaningful unit* that students should experience in order for

their right to an open future to be protected. The task of teachers should be to identify the proper bite-size experience for every subject they teach, ensuring it is minimal (so as not to unfairly take time and energy from other subjects) but also meaningful (so that the subject is reasonably presented and students can experience it as distinct from other content they encounter).

Conclusion

In conclusion, let's return to the original purpose of protecting children's right to an open future. Why are we also not protecting the right of adults to an open future? The choice between degrees of specialization persists even at the highest levels of education. We all know people who focus intensely on a single topic and others who work across multiple areas. Have they not closed certain career doors through those decisions? Undoubtedly, they have. But they have done so as autonomous agents. Consequently, they can be held at least partly responsible for the outcomes of those choices.

Because children lack full autonomy, the state must assume responsibility for the consequences of premature specialization. Yet third parties, be they parents, educators, or policymakers, are not epistemically positioned to predict children's future preferences with any certainty. Therefore, during the period in which individuals are not yet autonomous, the meaningful breadth of education should be prioritized over depth. As children mature into autonomous agents, they will gain the ability, and the right, to narrow their own options through deliberate choices regarding vocation and lifestyle.

The proper bite-size model offers a pedagogical path forward: one that respects the right to an open future by exposing students to meaningful, representative experiences across a range of fields, without demanding premature immersion. In doing so, education can resist the slide toward producing one-trick ponies while still fostering the depth needed for genuine engagement and growth.

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Miloš Kovačević

Obrazovanje, nepristrasnost i pravo na otvorenu budućnost

(Rezime)

Rad ispituje da li rana specijalizacija u obrazovanju krši autonomiju dece i njihovo pravo na otvorenu budućnost. Iznosi se teza da prekomerna izloženost uskom sadržaju stvara pojedince čija je budućnost zatvorena pre nego što su imali priliku da sami donose odluke. Na taj način se ugrožava autonomija njihovih budućih izbora. Analiza se oslanja na okvir zasnovan na autonomiji Džoea Fajnberga (Joel Feinberg) i kritiku Kladije Mils (Claudia Mills) koja dovodi u pitanje kako poželjnost tako i izvodljivost neutralnosti u obrazovanju. Odbacujući obrazovnu binarnost između površne raznovrsnosti i duboke uronjenosti, u radu se predlaže pristup „odgovarajućeg zalogaja”. Ovaj pedagoški pristup zalaže se za minimalnu, ali smislenu izloženost različitim oblastima i manjinskim načinima života, uravnotežujući raznovrsnost sa suštinom. U zaključku se tvrdi da, pre nego što se dostigne autonomija, obrazovanje treba da daje prednost širokim, formativnim iskustvima, omogućavajući pojedincima da se kasnije specijalizuju kao autonomni delatnici.

KLJUČNE REČI: autonomija, paternalizam, pravo na otvorenu budućnost, specijalizacija, obrazovanje, detinjstvo.

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Aleksandra Vučković

DOGWHISTLES: PHILOSOPHICAL BACKGROUND AND BELIEF CLUSTERS

Abstract: *The aim of this paper is to examine philosophical assumptions behind dogwhistles, starting with their definition as speech acts that convey different meanings to specific audiences. As such, dogwhistles can be studied through frameworks like Grice's Cooperative Principle, Stalnaker's common ground, or Lewis's conversational score. Our focus here, however, is Saul's classification of overt and covert, as well as intentional and unintentional forms of dogwhistles. Covert dogwhistles are of particular interest because they bring the audience's preexisting attitudes to the foreground, and do so outside the audience's awareness. We propose that the mechanisms underlying covert dogwhistles can be illustrated through the analogy of belief clusters, shifting the focus from meaning ambiguity to interpretative ambiguity.*

Keywords: *dogwhistles, covert dogwhistles, preexisting attitudes, belief clusters*

1 Introduction

Dogwhistles are often considered speech acts or non-verbal forms of communication that have two distinct meanings: one accessible to everyone and another intended for specific groups. These semantically ambiguous messages are frequently employed in political dialogue to signal loyalty to certain groups without attracting the attention of the general public. For instance, a political speech that refers to the importance of “family values” may appear innocuous to most listeners, but it can simultaneously communicate allegiance to right-wing supporters who interpret this concept through a specific ideological lens.

Two primary factors add to the effectiveness of the dogwhistles. The first is their dependence on “multivocal communication” (Albertson, 2015) in which the same words and concepts hold different meanings for different audiences. In our example, the notion of “family values” remains sufficiently vague to avoid negative associations among the general public, yet is specifically appropriated by the right wing to imply, for instance, anti-abortion or anti-divorce policies. The second factor is that dogwhistles are typically effective when such a gap in meaning is recognized¹ and utilized by the speaker to convey the hidden message.

This ambiguity can sway voters from opposing political spectrums to elect the same parties without even knowing that the other side voted for completely different

¹The exceptions are *unintentional* dogwhistles (Saul 2018), which will be explored further in following sections.

reasons (Goodin & Saward 2005). Although dogwhistles are not known to produce new beliefs among recipients, they can be successful in connecting their preexisting attitudes to specific political options (Saul 2018). A research on Twitter accounts that were active during the 2016 presidential campaign in the US revealed that white supremacists employed dogwhistles to both express hatred towards other groups and signal their own identity to like-minded individuals (Bhat & Klein 2020). Therefore, although dogwhistles may not induce new negative attitudes among the general public, they can still contribute to discrimination by enabling coordination among established hate groups.

Implicit messages are not spread only through spoken word, as visual materials can also function as dogwhistles by associating imagery with specific content (Drainville & Saul 2024). For example, presenting videos of minority groups during discussions of the rise in violent crimes may suggest that these groups are responsible, without explicitly stating so. Additionally, even an intentional pause in conversation can act as a “silent dogwhistle” (Klieber 2024) by, for instance, communicating discomfort with the topic through a refusal to continue the discussion.

Still, even in the absence of explicit language, dogwhistles can be examined in terms of the meaning they seek to convey. Beyond the specific words chosen, the effectiveness of implied messages depends on the situations in which they are presented, the intentions of those presenting them, and the attitudes of recipients. In the following section, we present several philosophical analyses of the dogwhistle phenomenon. The third section examines the subset of covert dogwhistles, which are particularly resistant to traditional interpretations. Finally, in the fourth section, we propose a further exploration of mechanisms underlying covert dogwhistles grounded in the concept of belief clusters.

It should be noted that it is unlikely that a single theory will offer a comprehensive interpretation of every form of dogwhistles. Still, by analyzing them through different frameworks, we can gain a better grasp of *how* they work and *why* they are so impactful in the public sphere.

2 Frameworks for the Analysis of Dogwhistles

Following Grice’s distinction between the *utterer’s* (speaker) meaning and *sentence* (semantic) meaning (Grice 1968, 1991), dogwhistles can be understood as the consequence of discrepancies between these two types of meaning (Rinner & Hieke 2024). Sentence-meaning is derived from grammar and the lexical meanings of the constitutive words (word-meaning), and it is, as such, accessible to all language speakers. On the other hand, the utterer’s meaning is grounded in the speaker’s intentions in a specific context, and to fully comprehend it, we need to be aware of these contingent circumstances.

According to Grice (1975), meaningful conversations are possible when partici-

pants follow the *Cooperative Principle*, which comprises the categories of quantity, quality, relation, and manner. The quantity category requires participants to provide sufficient information, while the quality category demands that the information is (believed to be) true and supported by evidence. The relation category emphasizes relevance, and the manner category requires the avoidance of ambiguity and obscurity. Recognizing that intended messages may be implied rather than explicitly stated, Grice (1975) also introduces the notion of *conversational implicature*. Dogwhistles seem to occur when speakers neglect the manner category by using ambiguous language, and most of the audience fails to draw the implicatures. It is, however, unclear whether dogwhistles can be fully explained within this framework, as it assumes that competent listeners can detect violations of the Cooperative Principle.

Drawing on Grice's concepts of the utterer's meaning and conversational implicature, Stalnaker (2002) introduces the notion of *common ground*, which consists of mutually accepted presuppositions in the background of conversations. Within this framework, dogwhistles can be interpreted as acts of exploitation of two distinct common grounds: one shared by the general public and another specific to an ideological group. Furthermore, common ground can change through the process of accommodation when the speaker introduces new assumptions and other conversation participants accept them (Stalnaker 2002). When recognized, dogwhistles influence common ground either by being accommodated into it or by establishing that the dogwhistle utterer maintains particular ideological views (Henderson & McCready 2024).

Grice's and Stalnaker's insights, along with Lewis's notion of *conversational score*², have also influenced the views of Elisabeth Camp (2018), who introduces *insinuation* as a notion akin to implicature. Through insinuation, speakers can communicate their beliefs, opinions, and requests without stating them explicitly or revealing their intentions. This form of communication allows backtracking and deniability if the insinuated content is not well received. Insinuations, thus, do not influence the conversational score precisely because the insinuated content is not asserted and may not even be included in the common ground. Nonetheless, they still communicate certain speaker attitudes. Although Camp's work does not particularly analyze dogwhistles, she acknowledges that at least some insinuations exhibit characteristics similar to those of dogwhistles (Camp 2018: 55).

The first in-depth classification of dogwhistles was introduced by Jennifer Saul (2018). Dogwhistles are categorized as *overt* or *covert*, depending on whether listeners consciously recognize them. For example, the statement "We need to keep our city safe"

²According to Lewis (1979), we can recognize and track changes in the conversational context similarly to how we track scoreboards in sports. The conversational score is updated following and in accordance with each utterance and its associated background assumptions. Consequently, conversational contexts change dynamically until a participant objects.

may seem innocuous to the general public, since concern for safety is widely regarded as virtuous. However, right-leaning individuals may interpret this statement as support for restrictive measures targeting immigrants. This dogwhistle is considered overt because the intended audience (right-wing supporters) is *aware* that the speaker is signalling loyalty to anti-immigrant policies. In contrast, covert dogwhistles influence intended recipients subconsciously. For instance, a politician who criticizes an opponent's weak anti-crime policies while displaying videos of offenders exclusively from marginalized ethnic groups may be dogwhistling to racist audiences. Unlike overt dogwhistles, in this scenario even the intended audience remains *unaware* that dogwhistling is occurring, although they may begin to subconsciously associate their preexisting attitudes toward minorities with specific electoral choices.

Additionally, according to Saul's classification, dogwhistles may be *intentional* or *unintentional*, depending on whether the speaker is communicating them deliberately. Unintentional dogwhistles warrant attention because they frequently contribute to the dissemination of the original intended message. Speakers who are unaware of the underlying implications may repeat phrases from political speeches, resulting in what Saul (2018) calls *amplifier* dogwhistles.

Both overt and covert dogwhistles can be intentional or unintentional. When a politician calls for stronger city safety, they likely engage in intentional overt dogwhistling. However, when the same phrase is repeated by someone unaware of its hidden meaning, it becomes an unintentional overt dogwhistle. Similarly, evocative utterances and materials presented during premeditated political campaigns are likely to function as intentional covert dogwhistles. Conversely, it is possible to unintentionally say or display something that triggers specific unconscious associations in a subset of the audience. When both the speaker and the recipients are unaware of the hidden message, this constitutes an unintentional covert dogwhistle.

3 The Effects of Covert Dogwhistles

Saul (2018) argues that covert dogwhistles are often more influential than overt forms because they are far more resistant to recognition and eventual repudiation. While they lose effectiveness once the intention is revealed, covert dogwhistles spread easily precisely because the intended audience frequently does not register them. Furthermore, they significantly shape the social climate by amplifying preexisting audience attitudes and associating these attitudes with particular social issues. While exposure to covert dogwhistles will not make one become a racist, it can make preexisting racist sentiments more prominent and relevant even in contexts that do not inherently concern racial issues. For instance, voters who hold negative views about minorities may associate their prejudices with political parties that covertly signal support for such attitudes, often without conscious awareness of this influence on their voting behaviour. Torices (2021) describes these amplified attitudes as either

cognitive, such as prejudices, or non-cognitive, such as emotions, and contends that covert dogwhistles are *attitude-foregrounders*.

In attempts to capture the status of covert dogwhistles, Saul (2018) turns to theories grounded in the notions of common ground and conversational score. One such theory is Jason Stanley's account of political propaganda. Stanley (2015) examines how associating particular words with representations of specific groups can introduce entirely new beliefs about these groups into the common ground. He notes that repeated use of the word "welfare" with the images of Black people carries an implied suggestion that Black people are lazier and, thus, dependent on welfare. Alleged laziness is here "not-at-issue" content that is only implicitly, yet effectively, incorporated into the common ground. While Saul (2018) acknowledges that Stanley's explanation captures some aspects of the covert dogwhistles, she contends that it does not fully account for them. First, unlike the forms of propaganda Stanley discusses here, covert dogwhistles do not produce new opinions, but rather bring preexisting ones to salience. Second, covert dogwhistles can activate both negative and positive preexisting attitudes, depending on the recipient. For instance, individuals with positive attitudes toward minorities may interpret the association of Black people with welfare as evidence of systemic injustice rather than laziness. Third, in contrast to Stanley's (2015) pessimistic conclusion that hidden propaganda resists challenging, Saul (2018) argues that covert dogwhistles can be addressed and diminished once they are recognized as such.

The next potential framework for analyzing covert dogwhistles emerges through a combination of insights from Langton and McGowan. Langton's (2012) thesis focuses on hate speech, and she argues that its expressions contain certain presuppositions. With each utterance, these presuppositions implicitly shift the common ground or change the conversational score (depending on whether we analyze this process through Stalnakerian or Lewisian notions), rendering the hate speech effective. Saul (2018) compares this observation to McGowan's (2004) definition of *covert exercitives*, which are speech acts that influence conversational score and do not require that the speaker hold specific authority. Most importantly, the speaker may utter them intentionally or unintentionally, and the audience may or may not register them. For instance, if a speaker expresses a racist sentiment and the audience does not notice or object, subsequent racist observations become permissible within this updated conversational score. This process alters the common ground to include racist presuppositions. If we apply this framework to covert dogwhistles, we could define them as substantial changes to common ground that occur outside of the audience's knowledge. However, Saul (2018) argues that covert dogwhistles lack the capacity to alter the permissibility facts, as they lose their effectiveness once recognized.

Saul (2018), thus, observes that theories based on common ground and conversational score are insufficient for explaining covert dogwhistles. She subsequently introduces her own account of intentional covert dogwhistles, characterizing them

as *covert perlocutionary acts*. As such, they do not need to be specific propositions that alter the conversational score or be part of the common ground. This conceptualization is grounded in Austin's (1975) speech act theory, which distinguishes three components of an utterance: locution (the content of what is said), illocution (the utterance's intended function or purpose), and perlocution (the effects produced on the audience). Covert perlocutionary acts, unlike "standard" perlocutionary acts, lose their effectiveness once the audience recognizes the intention to induce a specific effect (Saul 2018). When a speaker, for example, tells a joke, the perlocutionary act (the listener's laughter) will succeed even when the listener is aware of the speaker's intention to make them laugh. This is a "standard" perlocutionary act. However, when a speaker communicates a covert dogwhistle, its perlocutionary act (triggering the listener's preexisting attitude) is effective only if the listener remains unaware of the speaker's intention to activate that attitude.

When it comes to unintentional covert dogwhistles, Saul (2018) does *not* view them as covert perlocutionary acts. The speaker, in such cases, does not deliberately attempt to trigger the listener's preexisting attitudes. Consequently, there is no underlying intention that, if recognized by the listener, would undermine the perlocutionary act. Instead, unintentional covert dogwhistles are speech acts with unintentional perlocutionary effects of activating the audience's preexisting attitudes, unbeknownst both to the speaker and the audience. In this way, the speaker may accidentally contribute to the dissemination of the ideology they do not even believe in.

4 Covert Dogwhistles and Belief Clusters

As we have seen in the previous section, Saul managed to define covert dogwhistles, but the following excerpt reveals *why* they were so challenging to capture in the first place. Moreover, careful reading of this paragraph may point us toward the exploration of *how* covert dogwhistles activate listeners' preexisting attitudes.

"Covert intentional dogwhistles are substantially more challenging to capture. There are two key reasons for this. First, what is dogwhistled is not a particular proposition. Instead, certain pre-existing attitudes are brought to salience, without the audience being aware of it. This means that any theory relying on the communication (via semantics or pragmatics) of a particular proposition (or even a range of propositions) will fail. Second, this occurs outside of consciousness. Crucially, when an audience becomes conscious of the dogwhistle, it fails to achieve its intended effect." (Saul 2018: 371)

An analogy with Quine's notion of *web of belief* can help us illustrate the mechanisms underlying covert dogwhistles. According to Quine's (1951) *holism* thesis, all beliefs form an interconnected web such that we do not test the hypotheses independently of the comprehensive system of beliefs in which they are incorporated.

When empirical evidence conflicts with the system, this indicates that some of its components are incorrect, yet it is not possible to identify a specific belief as the source of the misalignment. Instead, we need to reevaluate the whole web.

Covert dogwhistles do not directly communicate specific propositions the intended audience will consciously recognize. This raises the question of whether they rely on the interconnected structure of belief systems to activate multiple, seemingly unrelated beliefs simultaneously and without the listener's awareness. However, for covert dogwhistles to function effectively, the general public and the intended audience would need to possess entirely distinct webs of belief. Furthermore, both webs would need to be organized in such a way that the activation of one belief leads to the activation of others within the same system.

This is where it should be noted that the analogy with Quine's web of beliefs has significant limitations. First, the holism thesis argues only that the hypotheses cannot be tested in isolation. Quine does not provide a psychological theory of how beliefs interact with each other, nor does he argue that triggering one of them will result in the activation of others. Second, the presumption that proponents of opposing political theories possess wholly different webs of belief is highly suspicious. Political opinions, unlike mathematical and logical laws, are at the web's periphery. The assumption that political dissenters disagree on mathematics and logic, adhere to fundamentally different ontologies, and, as a result, possess entirely distinct webs of belief is both bold and improbable.³

Nonetheless, a Quinean illustration suggests a useful analogy. Although covert dogwhistles do not communicate a particular proposition, they may operate upon a set of interconnected beliefs within the audience's broader belief system. A more nuanced line of argumentation would require the abandonment of Quine's original holism thesis in favour of the *moderate* version he proposed in later works (Quine 1981, 1990). This weaker form of holism, at least theoretically, allows us to compare smaller belief clusters without engaging with their underlying ontological assumptions. Only then can we compare the differences in clusters of political beliefs without assuming that their proponents subscribe to fundamentally different ontologies in areas unrelated to politics. Secondly, we need to acknowledge that Quine's holism thesis, even in its moderate form, does not account for the interactions between the beliefs. We would, thus, need to include an additional hypothesis that beliefs are clustered in such a way that the activation of one belief could lead to triggers across the whole cluster.

Then, we can imagine covert dogwhistles as speech acts that succeed at the intersection of two belief clusters. The notion of belief here should be understood in the

³It could be argued, though, that at least some political disputes originate in disagreements over ontological beliefs, as political opinions are sometimes justified on religious grounds. However, disagreements concerning logical and mathematical laws are considerably rarer and largely unrelated to political opinions.

broadest sense, as it also comprises preexisting attitudes, interpretive attitudes, associations, and dispositions. The ambiguity, in this context, does not lie in the meaning of the covertly dogwhistled content, but rather in differences in agents' interpretations, depending on their belief clusters.

In the case of intentional covert dogwhistles, the speaker recognizes and exploits the differences in belief clusters between the general public and the target group. Such a speaker may activate innocuous preexisting beliefs in the general audience, yet produce more specific (and sometimes concerning) associations in individuals who share the "ideological" belief cluster. Intentional covert dogwhistles can be, thus, conceived as conscious exploitations of the gap between the "mainstream" and "ideological" belief clusters. Furthermore, if listeners become aware of the speaker's intention to activate particular beliefs through associations, an intentional covert dogwhistle will fail.

Unintentional covert dogwhistles, similar to their intentional counterparts, are effective due to this gap. However, in these cases, the speaker does not have the knowledge of the aforementioned gap. Instead, the speaker likely shares the general public's beliefs but unintentionally triggers different associations among the individuals who share the cluster of "ideological" beliefs. Therefore, unintentional covert dogwhistles can be understood as non-deliberate activations of "ideological" belief clusters.

The proposed illustration is informative, provided we accept certain assumptions about belief formation and the organization of beliefs within clusters. In this context, belief clusters would have to function, at least to some extent, as cohesive units. One possible explanation is that individuals adopt entire belief clusters at once, including the specific relations among constituent beliefs. This explanation is initially persuasive, as individuals who endorse a particular ideological belief are often more likely to hold other beliefs associated with the same ideology. However, the supporting evidence remains anecdotal, given that the same individual may hold divergent ideological views. Another possible explanation is that each belief is acquired independently and subsequently integrated with similar beliefs to form a cluster. Still, empirical research remains limited on the mechanisms by which belief clusters (as proposed in this paper) and their internal relations operate.⁴

5 Conclusions and Implications for Future Research

As dogwhistles continue to shape political discourse and influence voter behaviour, we have seen that they also present a significant challenge for the philosophy of language. Although multiple theories contribute to their analysis, this paper has primarily examined Saul's account of covert dogwhistles and their perlocutionary effects in different audiences. According to Saul, intentional covert dogwhistles are covert perlocutionary acts that deliberately accentuate preexisting attitudes, yet fail once the intention is recognized. Unintentional covert dogwhistles also bring

⁴I thank the anonymous reviewer for recognizing the need for further empirical research.

the audience's preexisting attitudes to salience, but the speaker does not cause this deliberately. We expanded on this conclusion and proposed an illustration of covert dogwhistles as intentional or unintentional triggers of belief clusters.

Our goal here was not to repudiate Saul's conclusions nor to propose a fundamentally novel account of covert dogwhistles. The proposed "belief cluster" model does not posit that dogwhistles semantically encode two meanings. Instead, it explains how a single utterance can produce audience-specific associations by interacting with differently structured clusters of preexisting beliefs. Relying on Quine's holism thesis offers more of an allegorical than factual representation of how political messages are communicated. As is the case with most analogies, the one provided here is not without imperfections. While the concept of belief clusters is inspired by his moderate holism thesis, Quine himself never endorsed it. Furthermore, the idea that beliefs influence each other is ultimately rather psychological than philosophical.

We can, thus, conceptualize the belief cluster as both an epistemological and a psychological construct. From a purely epistemological perspective, belief clusters resemble sets of mutually related hypotheses that constitute a particular theory, insofar as accepting one of them increases the acceptability of the others. However, belief clusters also possess a psychological component uncharacteristic of hypotheses. First, they involve a degree of subjectivity, even when shared among similar-minded individuals. This allows different associations in different people. Second, beliefs within the same cluster influence and interact with each other. This allows them to be activated simultaneously. Third, an agent may adhere to a belief cluster without consciously registering the connections between each and every constitutive belief. These characteristics make covert dogwhistles effective, as they may activate entire clusters even when agents remain unaware. Arguably, explicating the associative connections within belief clusters may reduce the effectiveness of covert dogwhistles. Still, the question remains whether belief clusters affected by covert dogwhistling remain unchanged afterward or are reassessed after being recognized as a fruitful ground for manipulation.

It is worth noting that there are limitations of the proposed "belief cluster" model. Mainly, we still lack empirical research to confirm that belief clusters operate as described. The hypothesis that beliefs within the same cluster interact in ways that activate one another requires further empirical investigation. Additionally, the proposed model raises questions regarding belief formation. Future research could examine the potential continuity between Stanley's analysis of propaganda and Saul's account of dogwhistles. Stanley's framework may explain the formation of ideological beliefs, while Saul's may clarify their activation in particular contexts.⁵ It also remains to be determined whether individuals adopt entire belief clusters simultaneously or accept each belief individually before integrating them into a cluster. For these reasons, our model

⁵I thank the anonymous reviewer for this insight.

should, at least for now, be regarded as an illustration rather than a comprehensive theory of belief formation and activation.

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Aleksandra Vučković

Dogwhistles: filozofska pozadina i klasteri verovanja

(Rezime)

Cilj ovog rada je preispitivanje filozofskih pretpostavki u osnovi dogwhistles, počevši od njihove definicije kao onih govornih činova kojima se različita značenja prenose različitim delovima publike. Kao takve, dogwhistles možemo proučavati kroz okvire poput Grajsovog principa kooperativnosti, Stalnakerove zajedničke osnove ili Luisovog konverzionog rezultata. Ipak, naš fokus je na Solinoj klasifikaciji očiglednih i prikrivenih, kao i namernih i nenamernih oblika dogwhistles. Prikriveni dogwhistles su posebno interesantni jer prethodno postojeće stavove publike ističu u prvi plan i to čine tako da ni sama publika toga nije svesna. Naš predlog je da se mehanizmi u osnovi delovanja dogwhistles ilustruju kroz analogiju sa klasterima verovanja, čime se fokus pomera sa značenjske dvosmislenosti na dvosmislenost u pogledu interpretacije.

KLJUČNE REČI: dogwhistles, prikriveni dogwhistles, postojeći stavovi, klasteri verovanja.



Veljko Košević

Pesimistička indukcija u svetlu razvoja nauke

APSTRAKT: U ovom radu se razmatra Farbahova kritika pesimističke indukcije, prema kojoj istorija smene uspešnih, ali pogrešnih teorija, dovodi u pitanje približnu istinitost savremenih teorija. Farbah tvrdi da eksponencijalni rast naučne aktivnosti, količine i raznovrsnosti radova omogućava neuporedivo viši stepen empirijskog uspeha od onoga koji su imale napuštene teorije. U radu se, međutim, pokazuje da porast broja istraživača, publikacija i podataka ne podrazumeva pouzdanost teorija. Takođe se osporava Farbahova podela na teorije niskog i visokog stepena uspeha, budući da nije zasnovana na jasno određenom i istorijski neutralnom kriterijumu. Stabilnost savremenih teorija ne mora biti posledica njihove približne istinitosti, već može proistići i iz istraživanja koja se odvijaju unutar prihvaćenih teorijskih okvira, kao i iz opadanja produktivnosti naučnog rada. Zaključuje se da Farbahova argumentacija ne opovrgava pesimističku indukciju, jer eksponencijalni rast naučne aktivnosti sam po sebi nije dovoljan pokazatelj epistemološkog napretka niti opravdava povlašćen odnos prema savremenim teorijama.

KLJUČNE REČI: Naučni realizam, pesimistička indukcija, razvoj nauke

1. Uvod u naučni realizam

Jedan od najvažnijih i najpoznatijih argumenata u korist naučnom realizmu je argument o nepostojanju čuda [NMA]. Najčešće se pripisuje Hilari Patnamu (Hilary Putnam), koji smatra da ovaj argument ne čini uspeh nauke čudom (Putnam 1975: 73). Drugim rečima, možemo se pitati: šta bi moglo objasniti uspeh nauke osim činjenice da su (zrele) naučne teorije tačne? Ili, ako naše najbolje teorije nisu barem delimično tačne, uspeh nauke¹ postaje čudo. Objašnjenje čudom je gotovo potpuno neubedljivo, iako logički nije nemoguće da naše teorije pogrešno opisuju svet i da je njihov uspeh samo srećna slučajnost. Pošto niko ne tvrdi da je uspeh nauke čudo, stoga nas realista upućuje da prihvatimo zaključak da je uspeh nauke najbolje objašnjen hipotezom da su naše trenutno najbolje ili zrele teorije (približno) tačne (Saatsi 2018: 38).

Sada ćemo formalizovati NMA. Očekivano, to se može učiniti na različite načine. Kratka formalizacija može izgledati ovako:

1. Teorija T je izuzetno empirijski uspešna.
2. Uspeh T bio bi čudo ako T nije istinita.
3. Čuda ne postoje, stoga, T je istinita (Ruyant 2021: 160).

¹ Pod „uspehom“ naučne teorije podrazumevamo da teorija predviđa i objašnjava različite entitete, uključujući i one koji nisu bili poznati pre njenog formulisanja, kao i da omogućiti pouzdanu eksperimentalnu primenu. Takođe, pogledati i rekonstrukciju Farbahove argumentacije na 5. stranici rada.

Razrađeniya formalizacija, koju ćemo dati, uzimajući u obzir rad Karla Matesona (Carl Matheson), jeste:

1. Nauka je izuzetno uspešna.
2. Naučni realizam pruža najbolje objašnjenje za ovaj uspeh u odnosu na bilo koje drugo stanovište.
3. Trebalo bi da verujemo u stanovište koja najbolje objašnjava činjenice o nauci.
4. Dakle, trebalo bi da verujemo u naučni realizam (Matheson 1998: 263).

Očigledno, argument predstavlja zaključivanje na osnovu najboljeg objašnjenja (IBE) (Ruyant 2021: 161–162), ili, kako je u filozofiji nauke uobičajeno rečeno – abdukcija. Ovakav način zaključivanja se intenzivno koristi u nauci, te se čini da realista primenjuje metode nauke na samu nauku. Prvo, naučnici primenjuju IBE kada usvajaju teorije. Na primer, najbolje objašnjenje za mogućnost postojanja mikrotalasne pozadinske radijacije jeste model Velikog praska. Na osnovu ovoga, između ostalog, naučnici prihvataju ovaj model. Zatim, na narednom nivou, filozofi nauke će primeniti IBE da brane realizam, tako što će tvrditi da aproksimativna istina naših zrelih teorija predstavlja najbolje objašnjenje za njihov uspeh. To je upravo NMA. Međutim, da li je realizam najbolje objašnjenje za uspeh nauke? Postoji veliki broj prigovora, što će značiti da moramo biti selektivni. Ipak, nekoliko njih se zaista izdvajaju, a jedan od njih je pesimistička indukcija (PI).

2. Pesimistička indukcija

Elementarno poznavanje istorije nauke omogućava nam da zaključimo da su postojale promene naučnih teorija. Antirealisti često uzimaju primere iz istorije nauke kao svoj ključni izvor za argumentaciju protiv realizma. Jedan od prvih je bio Tomas Kun, koji je u svojoj čuvenoj knjizi *The Structure of Scientific Revolution* naglašavao diskontinuitet koji karakteriše revolucionarne prelaze u istoriji nauke (Kuhn 1970). To znači da nije priznavao kumulativnost teorija ili naučnog znanja u kontekstu onoga što je nazivao paradigrama. Ipak, među kritičarima realizma, Lari Laudan (Larry Laudan) je ponudio značajno razmatranje iz istorije nauke za našu temu. Predstavio je važan pregled teorija koje su nekada uživale visok stepen poštovanja. PI, koja se upravo odnosi na Laudanov rad „A Computation of Convergent Realism“, zapažanje da nauka nije postepeno akumuliranje znanja, slično kao kod Kuna (Laudan 1981). To znači da se nauka ne gradi samo od prethodno prihvaćenih teorija, jer veoma često dolazi do radikalnih promena – jedna teorija će biti srušena i zamenjena nekom novom teorijom. Dakle, primerima iz istorije nauke, Laudan će ponuditi argumentaciju protiv naučnog realizma, odnosno, kako ga on naziva, konvergentnim epistemološkim realizmom (Laudan 1981: 20–22). Pogledajmo neke primere za PI:

- humoralna teorija u medicini
- flogistonska teorija sagorevanja u hemiji
- kalorijska teorija toplote
- teorije etra u fizici (optika i elektromagnetizam)

- Ptolomejska astronomija
- Larmarkova teorija
- Teorija mešanog nasleđa u genetici

To su samo neke od odbačenih teorija iz prošlosti. Kao što Laudan navodi, „ova lista bi se mogla proširiti u nedogled“ (Laudan 1981: 33). Samim tim, možemo se nadovezati na Pitera Liptona (Peter Lipton), prema kome je istorija nauke „grobje neuspešnih teorija“ (Lipton 2005: 1265). Većina prošlih teorija su netačne i odbačene, i tako bi, prema indukciji, većina naših trenutnih teorija mogle biti netačne i odbačene u budućnosti. Dakle, ne bi trebalo da verujemo da su naše najbolje teorije (približno) tačne. Teorijski entiteti, poput elektrona ili gena, verovatno će imati istu sudbinu kao etar i flogiston. Samim tim, kada pogledamo istoriju nauke, čini se da nema naročito ubedljivih razloga da mislimo da naše trenutne teorije neće biti odbačene u budućnosti.

Lejdiman (J. Ladyman) smatra da se PI koristi kao argument protiv teze realista da „istina objašnjava uspeh nauke“, pokazujući da su naše najbolje trenutne teorije, pa i one buduće, najverovatnije netačne (Ladyman 1999: 186).

Pogledajmo sada kako izgleda jasnije određene PI.

1. Postojale su mnoge empirijski uspešne teorije u istoriji nauke, koje su se kasnije odbacile i čiji teorijski entiteti ne upućuju ni na šta;
2. Naše najbolje trenutne teorije se metodološki ne razlikuju od odbačenih teorija, tako da nemamo razloga da mislimo da one neće na kraju biti zamenjene;
3. Indukcija nam omogućava nam da očekujemo da će naše najbolje trenutne teorije biti pokazane kao netačne i zamenjene novim teorijama (Ladyman 2002: 236–237).

Možemo primetiti da je ovo izraženo u formi induktivnog argumenta. Možda je to samorazumljivo, zato što u samom nazivu PI imamo reč indukcija. Bez obzira na to, moguće je formulisati PI i u formi deduktivnog argumenta. Lions iznosi prigovor realizmu, formulišući ga u vidu *modus tollens* argumenta:

- Ako bi realistički argument bilo istinit, tada ne bismo imali nijednu uspešnu teoriju koja nije približno istinita. (Ako bismo je imali, svaka bi bila „čudo“, što niko od nas ne prihvata.);
- Međutim, postoje uspešne teorije koje nisu približno istinite (videli smo Laudanovu listu);
- Dakle, realistički argument je lažan (Lyons 2016: 566).

Sada ćemo preći na Farbahovu kritiku ove argumentacije.

3. Farbahova kritika PI

Osnovna Farbahova argumentacija je izneta u radu „How the Growth of Science Ended Theory Change“, gde je istakao da su branioci PI prevideli činjenicu da je rast nauke eksponencijalan. Farbah pretpostavlja da možemo meriti količinu naučnog rada obavljenog u određenom periodu gledajući broj objavljenih članaka u časopisima i broj naučnika koji rade (Fahrbach 2011b: 11). Broj članaka objavljenih godišnje

se udvostručuje svakih 15 do 20 godina, kao i broj zaposlenih istraživača, barem u periodu od 1660. do 1960. godine (Ibid). To daje predstavu o brzom rastu. Tri četvrtine ukupnog naučnog rada obavljeno je u poslednjih 30 do 40 godina. Farbah ukazuje da ove stope udvostručavanja impliciraju rezultate koji su nam naročito važni, jer je najmanje 95% svih naučnih radova obavljeno od 1915. godine i najmanje 80% svih naučnih radova obavljeno od 1950. godine (Fahrbach 2011b: 13). Na kraju krajeva, više naučnog rada znači više eksperimenata, više posmatranja, više argumenata koji testiraju naše teorije. To znači i da imamo mogućnost da pokažemo da je određena teorija pogrešna.

Ispitujući Laudanovu listu, vidimo da su sve te teorije napuštene pre više od 100 godina. To podrazumeva da su sve odgovarajuće promene teorija nastale tokom prvih 5% svih ikada napisanih naučnih radova. Na primer, što se tiče teorija svetlosti, sve promene tih teorija dogodile su se pre 1930-ih. Od sedamnaestog do početka dvadesetog veka, teorije svetlosti su se promenile najmanje četiri puta. Za to vreme objavljeno je oko 3 miliona naučnih radova, što znači da je na svaki milion radova došlo do promene neke teorije, pa čak i više. Danas se objavljuje preko 6 miliona radova godišnje, pa bismo očekivali šest promena teorija godišnje, ali to se ne dešava. Zato bi trebalo analizirati poslednjih 50 do 80 godina, jer samo tada možemo utvrditi da li je pretpostavka PI verovatna i da li se u budućnosti može očekivati promena ili stabilnost. Uvid u ovaj period pokazuje da su najbolje naučne teorije ostale potpuno stabilne.²

Nadalje, jedna od ključnih Farbahovih poenti u tekstu „Theory Change and Degrees of Success“ jeste da uspeh naučnih teorija nije binaran (tačno/netačno), već da postoji u različitim stepenima (Fahrbach 2011a). On uvodi koncept „degrees of success“ kako bi odgovorio na prigovor PI. Podsetimo se da PI ističe kako napuštanje nekadašnjih uspešnih teorija upućena na mogućnost da će isto važiti i za savremene. Farbah, međutim, naglašava da je ključna razlika između prošlih i sadašnjih teorija upravo u stepenu njihovog uspeha (Fahrbach 2011a: 1285–1286). Prošle teorije sa Laudanove liste, poput flogistonske teorije ili teorije etra, imale su određene empirijske uspehe, ali su bili ograničeni u obimu, preciznosti i raznovrsnosti podataka. Njihov stepen uspeha bio je nizak jer su testirane u vreme kada su resursi, instrumenti i računarska moć bili minimalni, a prikupljeni podaci oskudni i često nedovoljno precizno. Samim tim, iako su te teorije imale određenu heurističku i eksplanatornu vrednost, njihov uspeh nije bio uporediv sa današnjim naučnim teorijama. Drugim rečima, PI osvetljava primere koji zapravo potiču iz perioda niskog stepena uspeha, pa njihovo odbacivanje ne potkopava princip da uspeh vodi istini (Ibid).

Suprotno tome, današnje teorije, poput teorije evolucije ili teorije relativnosti, imaju ogroman stepen uspeha. Njih podržavaju milioni eksperimentalnih i opservacionih svedočanstava, prikupljenih različitim metodama. Razvoj tehnologije, automatizacije,

²Više o tome u (Fahrbach 2011b).

kao i eksponencijalni rast naučnog rada, omogućili su da se teorije testiraju na način koji ranije nije bio moguć. Farbah naglašava da su u poslednjih 50 do 80 godina naše najbolje teorije ostale stabilne upravo zato što su prošle kroz ogroman broj rigoroznih testova i time postigle visok stepen uspeha (Fahrbach 2011a: 1290). Zaključak je da PI ne potkopava relaciju između uspeha i istina³ jer prošli primeri prikazuju teorije sa niskim stepenom uspeha, dok današnje teorije poseduju daleko viši stepen uspeha. Upravo ta razlika u „stepenu uspeha“ omogućava realistima da zadrže izmenjeni princip uspeh–istina: ako teorija ima veoma visok stepen uspeha, onda je opravdano smatrati je približno istinitom (Fahrbach 2011a: 1291).

Samim tim, možemo sumirati Farbahovu argumentaciju kroz pet indikatora, pomoću kojih pokazuje zašto su današnje teorije daleko uspešnije od prošlih i zašto PI gubi uverljivost (Fahrbach 2011a: 1286–1291).

Farbahovu argumentaciju možemo izložiti na sledeći način:

1. Stepem empirijskog uspeha naučne teorije zavisi od broja, raznovrsnosti i preciznosti njenih uspešnih primena i testova.
2. Eksponencijalni rast naučnog rada, kao i ogroman rast količine i raznovrsnosti podataka, preciznosti merenja i računarskih kapaciteta, ukazuju na to da su savremene teorije bile izložene znatno većem broju preciznijih testova nego ranije teorije, te su zbog toga naše najbolje savremene teorije opstale poseduju daleko viši stepen empirijskog uspeha od napuštenih teorija sa Laudanove liste.
3. Pesimistička indukcija neopravdano zaključuje od sudbine ranijih teorija sa nižim stepenom uspeha na buduću sudbinu savremenih teorija sa znatno višim stepenom uspeha.
4. Stoga istorijski primeri sa Laudanove liste ne pružaju dovoljnu induktivnu osnovu za zaključak da su mnoge ili većina naših najuspešnijih savremenih teorija lažne i da će u budućnosti biti radikalno odbačene.⁴
4. **Slabosti Farbahove argumentacije**

Videli smo da Farbah iz eksponencijalnog rasta naučne aktivnosti izvodi zaključak da su savremene teorije prošle znatno više empirijskih testova i da zato poseduju daleko viši stepen empirijskog uspeha od napuštenih teorija. Međutim, broj istraživanja, publikacija i prikupljenih podataka nije pouzdan pokazatelj evidencijalne snage neke teorije. Taj kvantitativni rast velikim delom može biti posledica povećanja broja istraživača, širenja naučnih institucija i finansiranja, razvoja instrumenata, računara i baza podataka, kao i nastanka novih disciplina. Pored toga, nisu sva istraživanja nezavisni i strogi testovi osnovnih pretpostavki teorije, jer mnoga istraživanja samo primenjuju već prihvaćeni teorijski okvir, ponavljaju slična merenja ili zavise od

³Radi se o relaciji koja je logički nezavisna makar u jednom smeru. Naime, ako je teorija Ta uspešna, ne implicira da je istinita, dok obrnuto važi, tj. ako je teorija Tb istinita, onda će biti i uspešna (vidi fusnotu 1).

⁴Argumentacija se zasniva na (Fahrbach 2011b) i (Fahrbach 2011a).

istih pomoćnih pretpostavki, instrumenata i metoda obrade podataka. Zbog toga se evidencijalna vrednost testova ne može jednostavno sabirati niti iz ukupnog obima naučne aktivnosti neposredno izvesti stepen uspeha određene teorije. Farbah, dakle, nedovoljno razlikuje rast naučnih resursa i produkcije od rasta nezavisne, raznovrsne i diskriminativne evidencije. Blum i ostali istraživači pokazuju da se, uprkos snažnom povećanju broja istraživača i ulaganja u istraživanje, produktivnost istraživačkog rada u različitim oblastima smanjuje. Za održavanje približno iste stope tehnološkog napretka potrebno je angažovati sve više istraživača i resursa. Prema tome, rast naučnih inputa ne proizvodi nužno proporcionalan rast novih ideja i rezultata. Zbog toga se ni eksponencijalno povećanje naučnog rada, na koje se Farbah poziva, ne može bez dodatnog argumenta poistovetiti sa proporcionalnim povećanjem broja epistemološki relevantnih testova i stepena empirijskog uspeha teorija (Bloom et al. 2020). Ujedno, istraživačka produktivnost se prepolovljuje približno svakih 13 godina, pa se istraživački napor mora udvostručavati samo da bi se održala ista stopa napretka (Bloom et al. 2020: 1134–1138).

Nadalje, Farbah pokušava da ublaži snagu PI tvrdeći da većina istorijskih primera na koje se oslanja Laudan potiče iz teorija koje su imale nizak stepen uspeha: ili nisu bile dovoljno testirane, ili su davale ograničene i nepouzdate rezultate. Po njemu, to znači da PI pogađa samo „slabije“ teorije, dok današnje, „visoko uspešne“ teorije, nisu ugrožene. Međutim, nije tačno da su sve Laudanove teorije bile „niskog uspeha“. Njutnova korpuskularna teorija je dugo davala precizne rezultate u objašnjavanju zakona refleksije i refrakcije. Uprkos tome što je kasnije zamenjena talasnom, a zatim kvantnom optikom, teško je osporiti da je imala visok stepen empirijske uspešnosti u svoje vreme. Flogistonska teorija se održala kroz čitav 18. vek i nudila koherentan okvir za razumevanje sagorevanja i oksidacije. Eksperimentalno je dugo bila korišćena kao objašnjenje hemijskih procesa, sve dok Lavoazje nije uveo teoriju oksidacije.

Ako ovim primerima osporimo status „visoko uspešnih“ teorijama, to deluje kao naknadno odnosno anahrono ispisivanje istorije iz perspektive sadašnjeg znanja, a ne koliko-toliko objektivna ocena njihovog učinka u vremenu u kojem su dominirale. Ako Farbah hoće da se osloni na razliku između niskog i visokog uspeha, mora ponuditi jasan i objektivan kriterijum za određivanje praga. Da li je to broj potvrđenih predviđanja, dužina opstanka teorije, nivo preciznosti merenja ili pak domet na koji se teorija primenjivala? Bez jasnog kriterijuma, ova distinkcija ostaje proizvoljna i otvorena za različita tumačenja. Dakle, distinkcija stepenovanja uspešnosti teorija nije dovoljno utemeljena. Ono se oslanja na naknadnu evaluaciju i lako može delovati ad hoc, a time Farbah ne uspeva da oslabi snagu PI, jer ostaje otvorena mogućnost da čak i teorije koje danas smatramo „visoko uspešnim“ da jednog dana završe na „grobju naučne istorije“ – baš kao i flogiston ili Njutnova optika.

Na kraju, istaći ćemo poslednju kritiku. Kao što smo videli, Farbah ističe da poslednjih 300 godina nauka beleži eksponencijalni rast – u broju istraživača, publikacija,

instrumenata i podataka. Iz toga izvodi zaključak, prisetimo se, da današnje teorije, budući testirane u takvom kontekstu, imaju znatno veću stabilnost. Međutim, ovaj zaključak je sporan, jer se pretpostavlja da sam eksponencijalni rast nauke garantuje da će savremene teorije opstati kao (približno) istinite. To je, međutim, projekcija iz savremene perspektive, bez ikakve izvesnosti da će istorijski razvoj nauke potvrditi ovu pretpostavku. Pre 200 godina, njutnovska fizika se smatramo gotovo neoborivom, ali se danas koristi kao praktičan instrument za rešavanje jednostavnijih problema, dok je ulogu sveobuhvatnog teorijskog okvira preuzela savremena fizika, zasnovana na opštoj teoriji relativnosti i kvantnoj mehanici. Slično tome, hemičari 18. veka smatrali su da flogiston pruža pouzdano objašnjenje procesa sagorevanja, ali je kasniji razvoj hemije pokazao da je ta koncepcija bila neodrživa. Dodajmo i da je kalorijska teorija bila pogrešna u tvrđenju da je toplota materijalna supstanca, ali njeni potvrđeni zakoni nisu zavisili od te ontološke greške. Oni su bili približno istiniti i bili su sačuvani i objašnjeni u termodinamici, te zbog toga istorija kalorijske teorije ne podržava La-udanovu pesimističku metaindukciju.⁵ U prilog ovome, na primer, Vrej smatra da, ako se to dogodilo nekadašnjim teorijama, nema razloga da verujemo da se isto ne može ponoviti i sa našim: savremene teorije mogu se za dva veka našim potomcima činiti podjednako zastarelim. (Wray 2013: 4327). Shodno tome, jedna od ključnih pouka iz istorije nauke jeste da su naučnici polagali poverenje u to da su upravo njihove teorije dovoljno proverene. Tako je u antičkom svetu Ptolomejev geocentrični model vekovima važio za potvrđenu sliku kosmosa, potkrepljen složenim i preciznim proračunima.⁶ U 19. veku, Njutnova mehanika je bila toliko uspešna u predviđanjima i praktičnim primenama da je Laplas mogao tvrditi kako je vasiona u determinisana i da se, bar u principu, može potpuno objasniti. U oba slučaja, istorija je pokazala da je osećaj stabilnosti bio varljiv. Stoga nije dovoljno reći da eksponencijalni rast današnje nauke garantuje istinitost – jer i ranije epohe su imale „dobre razloge“ da veruju u istinitost svojih teorija.

Možemo, dakle, zaključiti da Farbahovo pozivanje na eksponencijalni rast nauke ne otklanja prigovor PI. Taj pristup počiva na pretpostavci da će uslovi u kojima nauka danas deluje važiti i u budućnosti, kao i na ponavljanje iste teze koja je pratila sve ranije epohe: uverenje da je upravo njihovo vreme dostiglo konačnu istinu. Ujedno, izneta kritika Farbahovog stanovišta ne podrazumeva afirmaciju PI, već ukazuje isključivo

⁵ Više o tome videti u (Psillos 1994).

⁶ Zanimljivo je primetiti da uspeh neke stare teorije, u ovom slučaju Ptolomejeve geocentrične teorije, možemo objasniti iz perspektive nove teorije (Keplerove teorije) čak i kada nova teorija nije zadržala dovoljno teorijskih elemenata stare teorije koji su proizveli taj uspeh. Drugim rečima, pokazuju da se, polazeći od Keplerovog sistema, kroz niz geometrijskih transformacija može rekonstruisati Ptolomejev model i objasniti zašto je predviđanje položaja Marsa bilo dobro, da se pojasni gde su nastajale Ptolomejeve greške i ujedno zašto je bio mnogo precizniji od ranijih geocentričnih modela. Tako se ispostavlja da je Keplerova teorija omogućava detaljno objašnjenje Ptolomejevog uspeha, iako iz toga ne sledi da su Ptolomejevi teorijski entiteti i mehanizmi bili približno istiniti. Više o ovome videti u (Díez et al. 2022).

na nedovoljnu ubedljivost njegove argumentacije, koja se u najvećoj meri oslanja na drugu premisu rekonstruisanog argumenta – eksponencijalni rast podataka. Stoga se može zaključiti da eksponencijalni rast nije dovoljan uslov za stvarni napredak naučnog znanja. Svakako, ovo ne implicira da je druga premisa potpuno lažna. Može se prihvatiti da su neke savremene teorije empirijski uspešnije od napuštenih teorija sa Laudanove liste. Međutim, Farbah nije pokazao da je ta razlika u uspehu toliko velika i značajna da opravdava različit odnos prema savremenim i ranijim teorijama. Pre svega, nije pokazao da povećanje broja naučnih radova i količine podataka odgovara povećanju broja raznovrsnih, međusobno nezavisnih i strogih testova, sposobnih da otkriju pogrešnost savremenih teorija. Zato dugotrajno opstajanje ovih teorija ne mora predstavljati svedočanstvo izuzetnog epistemološkog napretka, već ono može biti i posledica toga što se istraživanja uglavnom odvijaju unutar postojećih teorijskih okvira, odnosno pokazatelj teorijske stagnacije.

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Veljko Košević

The Pessimistic Induction in Light of the Development of Science

(Summary)

This paper examines Fahrbach's criticism of the pessimistic induction, according to which the historical replacement of successful but false theories calls into question the approximate truth of contemporary theories. Fahrbach argues that the exponential growth of scientific activity, as well as the quantity and diversity of scientific work, allows contemporary theories to achieve a much higher degree of empirical success than abandoned theories. However, this paper argues that the growth in the number of researchers, publications, and data does not necessarily imply that theories are reliable. Fahrbach's distinction between theories with low and high degrees of success is also challenged, since it is not based on a clearly defined and historically neutral criterion. The stability of contemporary theories need not result from their approximate truth, but may also arise from research conducted within accepted theoretical frameworks and from the declining productivity of scientific research. It is concluded that Fahrbach's argument does not refute the pessimistic induction, because the exponential growth of scientific activity is not, by itself, a sufficient indicator of epistemic progress, nor does it justify treating contemporary theories as epistemically privileged.

KEYWORDS: scientific realism, pessimistic induction, growth of science

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TRUTHMAKER, NEGATIVE EXISTENTIAL, AND PRESENTISM IN PARMENIDES

Summary: *There is a long and extensive history of attempts to interpret Parmenides' thought. Based on the complicated interpretive texts, it sometimes seems as if the interpreters are treating his claims like a Rorschach test. It seems to me that his thoughts are much simpler and much clearer if they are interpreted in the context of the modern truthmaker theory of truth, the issue of negative existentials and presentism. I will finally turn to his theory of truth, using the example of the relationship between motion and empty space, and comparing his theory to the atomistic view. Seen from the perspective of Parmenides' theory, atomistic, like other empirical theories, is subject to the criterion of opinion, not the criterion of truth. For him, no empirical theory can satisfy his criterion of truth.*

Keywords: *Parmenides, truthmakers, negative existential, presentism, opinion.*

Introduction

Countless attempts to interpret and understand Parmenides' poem. Isn't it redundant to add one more? Likewise, a zealous historian might accuse me for making an anachronistic error: “truthmaker,” “negative existential,” and “presentism,” cannot be applied to the interpretation of the poem for these are concepts for which there are no adequate expressions in Parmenides' Greek. However, the interpretation of an old work is rarely successful without the use of a modern philosophical dictionary and its conceptual framework, because the interpretation must be comprehensible to today's reader. Interpreting a work only in its own language is as fruitful as reading a text written in black ink on a black background. I hope to contribute to the understanding of the meaning of Parmenides' poem by interpreting it in the conceptual networks of the theories to which the concepts of truthmaker, negative existential, and presentism belong.

Contrary to our usual intuition that beings come into existence and cease to exist, that there are many beings, and they are diverse, that they existed in the past, exist now, and will exist in the future, in the DK¹ B8 fragment of his poem, Parmenides presents the following unusual theses:

What is (being) is ungenerated and deathless, whole and uniform, and still and perfect; but not ever was it, nor yet will it be, since it is now at

¹DK is the usual abbreviation for: Hermann Diels, H./Walter Kranz, *Die Fragmente der Vorsokratiker*, (Zurich/Hildesheim: Weidmann, 1985).

once all, single, continuous (ἀγένητον ἐὸν καὶ ἀνώλεθρόν ἐστιν, ἐστι γὰρ οὐλομελές τε καὶ ἀτρεμές ἢ δ' ἀτέλεστον· οὐδέ ποτ' ἦν οὐδ' ἔσται, ἐπεὶ νῦν ἔστιν ὁμοῦ πᾶν, ἓν, συνεχές). (DK B8.3—6)

Parmenides goes on to give arguments justifying these theses. However, I suggest these arguments cannot be understood if certain assumptions are not taken into account. Those assumptions are presented in the preceding fragments of the poem, and they concern Parmenides' theory of truth. Likewise, his theory of truth cannot be well understood without applying it to the arguments presented in DK B8.² It is clear that no cosmology is possible under the conditions specified in Parmenides' theses. Nor are the cosmologies of the Ionian thinkers who preceded Parmenides possible, as they all deal with changes, the past, the present, the future, the multitude, etc. Thus, they cannot satisfy Parmenides' criteria of truth knowledge.

In this paper, I will try to make the interpretative theses acceptable by first presenting Parmenides' theory of true knowledge and his truthmaker theory. I will then explain his views on the problem of the negative existential and presentism. Finally, I will turn to his theory of truth, using the example of the relationship between movement and empty space and comparing his theory with the atomistic point of view.

Parmenides' Theory of True Knowledge

The idea that our knowledge of the world is not conclusively true was hinted at by some Ionian thinkers. For example, in one preserved fragment, Xenophanes says the following:

Neither ever was nor ever will be a man who would have clear/obvious/true (σαφές)³ knowledge (εἰδώς) of the gods as well as of all that I speak of. For even if he were to happen to say [something that is true], he himself would still not know it (αὐτὸς ὅμως οὐκ οἶδε): for all is swayed by opining (δόκος δ' ἐπὶ πᾶσι τέτυκται). (DK. 21B34).

The most that can be reached in cosmological theories are opinions, according to Xenophanes. Humans, he seems to think, can progress in their attempts to gain some knowledge of the world,⁴ but they cannot reach theoretical knowledge that is

²I will deal with only some of the arguments presented in DK B8. For attempts to explicate the arguments of DK B8 as a whole: Richard McKirahan, "Signs and Arguments in Parmenides B8," in P. Curd and D. W. Graham (ed.), *The Oxford Handbook of Presocratic Philosophy*, (Oxford: Oxford University Press, 2008) 189—229. and Michael V. Wedin, *Parmenides' Grand Deduction: A Logical Reconstruction of the Way of Truth [Grand Deduction]*, (Oxford: Oxford University Press, 2014).

³Sextus, *Adv. Math.* 7.1.50, to whom we owe this fragment, believes Xenophanes under σαφές thinks "what is true and known" known" (Sextus Empiricus. *Adversus Mathematicos*. ed. G. R. Bury. (Cambridge, Mass.: Harvard University Press / London: William Heinemann, 1976).

⁴In a well-known fragment, he says the following: "Certainly, the gods did not reveal everything to mortals from the beginning, but [people themselves], searching over time, saw better (οὗτοι ἀπ' ἀρχῆς πάντα θεοὶ θνητοῖς ὑπέδειξαν, ἀλλὰ χρόνῳ ζητοῦντες ἐφευρίσκουσιν ἄμεινον)" (Arn. Herm. L. Hereen,

conclusively true. And if they guess something that is true, he says, they cannot know that it is true. Although it looks more like an assertion that could stand at the beginning of conditional argumentation, the following fragment could be in line with this theory: “Let’s look at this as if it was originally like that /as if it were true (ταῦτα δεδοξάσθω μὲν εὐκότα τοῖς ἐτύμοισι).”⁵ About the same time, Milesian historian and logographer Haecateus made a similar point: “I write these things as they seem to me to be true. For the tales told by the Greeks are, as it appears to me, many and absurd /ridiculous (τάδε γράφω, ὥς μοι δοκεῖ ἀληθέα εἶναι: οἱ γὰρ Ἑλλήνων λόγοι πολλοί τε καὶ γελοῖοι, ὥς ἐμοὶ φαίνονται, εἰσίν).”⁶

Both Xenophanes and Haecateus seem to believe that what they claim in their theories is worth considering, even though it cannot be true in a strict sense. In other words, both assume some criterion of truth and knowledge. Otherwise, it would be difficult to understand their claims and suggestions. However, no fragments or testimonies have been preserved in which they mention such a criterion. The earliest preserved fragments in which this question is discussed are fragments of Parmenides’ poem.

What Xenophanes calls opinions, Parmenides calls “mortal opinions.” He does this to strictly separate them from true speech. True speech is assigned in the poem to an omniscient being, the Goddess. However, unlike the mythological, say, theogonic pattern, in which the Muses, as divine beings, inspire the poet to simply recite the events around the origin of the gods and the world,⁷ Parmenides’ Goddess offers arguments for her claims. She does not require mere belief. What Parmenides calls truth seems to be what we call justified belief or knowledge today.

Let’s start the argumentation with the promise the Goddess makes to the young man at the beginning of her teaching: “You ought (Χρεὼν) to learn all things, both the unshaken heart of persuasive (εὐπειθέος)⁸ truth and the opinions of mortals, in which there is no persuasive truth (οὐκ ἔνι πίστις)” (DK B1.28b—30).⁹ As mentioned previously, Xenophanes also distinguishes truth or knowledge from opinions, but he

(ed.) *Ioannis Stobaeus Eclogarum Physicarum et Ethicarum*. (Gottingae: Vandenhoeck et Ruprecht, 1742) 1.9.2. DK. 21B18). In DK, it says 1.8.2 incorrectly, and this error is almost regularly found in the literature). In support of the thesis that Xenophanes developed the idea of cognitive progress, see, for example: Leshner, J. H. 1991. “Xenophanes on Inquiry and Discovery: an Alternative to the ‘Hymn to Progress’ Reading of F. 18.” *Ancient Philosophy* vol. 11. 229—48. i Tulin, A. 1993. “Xenophanes Fr. 18 D.—K. and the Origins of the Idea of Progress.” *Hermes* vol. 121. 129—38.

⁵Plutarch. *Moralia* 4. Gregorius N. Bernardakis (ed.). (Leipzig: Teubner, 1892) 746b. DK 21B35.

⁶Wrys H. Roberts (ed.), *Demetrius On style /De elocutione/* (Cambridge: Cambridge University Press, 1902) 1.12.

⁷This is the case, for example, in Hesiod’s *Theogony* 22 and 31—32. Hesiod. *Theogony. Works and Days. Testimonia*. ed. and transl. Most W. Glenn (Cambridge, MA: Harvard University Press, 2018).

⁸Sextus’ “εὐπειθέος” is more convincing than Simplicius’ “εὐκυκλῆος,” because it stands in opposition to “οὐκ ... πίστις,” which is not the case with “εὐκυκλῆος.”

⁹Sextus Empiricus, *Adv. Math.*, 7.111—14.

considers that what he is talking about ("of all that I speak of") cannot be true – it can only be opinion. In his poem, Parmenides considers this difference in much more detail. Broadly stated, he first considers the problem of the criteria for talking and thinking the truth (DK B2—7). He presents the conclusions that can be reached about what is or what exists, while respecting those criteria (DK B8), and concludes the opinions of mortals cannot meet those criteria (DK B9—19).

Already at the beginning of the exposition of the part concerning the criteria of truthfulness (DK B2.2), Parmenides says: "I shall tell...just which ways of inquiry alone there are for thinking." This naturally raises the question: what kind of ways of inquiry is he talking about? We would expect Parmenides to offer us thinkable, true ways to answer the question, "What exists?"¹⁰ Instead, he gives us the two most basic things that can be thought about existence: "what is" (ἐστί) and "what is not" (οὐκ ἐστί). He says the following:

- 1) That what is and that what is not does not exist (ὅπως ἐστί τε καὶ ὥς οὐκ ἐστὶ μὴ εἶναι) (DK B2.3).
- 2) That what is not and that what is ought not to exist (ὥς οὐκ ἐστί τε καὶ ὥς χρεὼν ἐστὶ μὴ εἶναι) (DK B2.5).

There is a long history of translation and interpretation of these two claims. It seems they are not completely understandable to the majority of interpreters without some emendations, which I will not deal with here.¹¹ To me, the most convincing interpretation is to say Parmenides/the Goddess is telling us that two possibilities comprise the basis of every consideration of what is or what exists: 1a) we can think about what is, and in this case, we ought to think that it is or it exists; 1b) we can think about what is not, and in this case, we ought to think that it is not or it does not exist. 2a) we can think that what is not – is or exists, and 2b) that what is does not exist. This, of course, does not imply that we literally cannot think or say it. Rather, what would be said and thought, as will be seen later, would not be true.

I hope the whole matter will become clearer as we go through the Goddess's reasoning, which appears immediately after each claim. About claim 1), the Goddess

¹⁰I agree with Leshner's assertion that "Parmenides' primary objective in his famous poem was to provide a correct account of what exists" (James H. Leshner, 'Parmenides' Critique of thinking', *Oxford Studies in Ancient Philosophy*, 2 (1988), 1—30 at 1).

¹¹Perhaps the most radical reinterpretations of DK B2.3 and DK B2.5 follow from the way these verses are understood and translated by Mullachius and Finkelberg. Mullachius (Fr. Guil. Aug. Mullachius, *Fragmenta philosophorum graecorum* (Paris: Ambrosio Firmin Didot, 1860) p. 117) DK B2.3 translates into Latin as "esse ens, non esse non ens," and DK B2.5 as "non esse ens, ac necessario esse non ens," while Finkelberg (Aryeh Finkelberg, "Parmenides' Foundation of the Ways of Truth" ["Ways of Truth"], *Oxford Studies in Ancient Philosophy*, vol. 6 (1988) 39—68, p. 46—47) the first verse translates as "that to be is and that not to be is not," and second a "that to be is not and that of necessity not to be is." It is interesting to note that such an understanding structurally corresponds to Aristotle's determination of the true and the false in the *Gamma* book of *Metaphysics* (Aristotle, *Metaphysics*, ed. David W. Ross, (Oxford: Oxford University Press, 1953) 1011b.27—28).

says, “What is the persuasive way, because it is in service of the truth (πειθοῦς ἐστι κέλευθος, ἀληθείη γὰρ ὁπηδεῖ¹²)” (DK B2.4). It seems this is the first of the possible “ways of inquiry” into what exists. When it comes to claim 2), things get more complicated. For “what is not” (οὐκ ἔστιν) as a candidate for existence, the Goddess first says, “This, I tell you, is a path wholly without report (τὴν δὴ τοι φράζω παναπευθέα ἔμμεν ἀταρπόν). Then the Goddess states the reason for the claim: “οὔτε γὰρ ἂν γνοίης τό γε μὴ ἔδον — οὐ γὰρ ἀνυστόν — οὔτε φράσαις.” In this context, “τό γε” seems to signify the impossibility of specifying the reference,¹³ to suggest further consideration of the truthmaker for the negative existential. Not being (μὴ ἔδον) cannot be known as something “specifically (exactly) this (τό γε),” as in the case of the truthmaker for the positive existential. This reading also suggests the meaning of “οὔτε φράσαις.” According to Mourelatos, the word “φράζω” “may not (and usually does not) carry the meaning *dico* in early Greek poetry. It has a rather concrete sense, the core or root of which is ‘to point it out’, ‘to show’, ‘to indicate with a gesture’, ‘to appoint’, ‘to instruct’.”¹⁴ Moreover, he thinks the application of this “concrete sense of φράζω will improve the translation of 2.6—8.”

Based on the above analysis, I think DK B2.6—8 is best translated as follows: “You can neither know exactly (τό γε) what is not – for this is not feasible – nor can you show it (οὔτε φράσαις).” In what is not we cannot look for an answer to the question of what exists, because, as stated in the definition 1b), “what is not does not exist.”

Parmenides’ Truthmaker Theory

In DK B2.6—8, Parmenides points to something we find problematic in the modern truthmaker theory of truth. Before discussing this more closely, let us first look at how philosophers currently define the truthmaking principle. Julian Dodd offers the following definition: (TM) Necessarily, if $\langle p \rangle$ is true, then there exists at least one entity α such that α necessitates $\langle p \rangle$ ’s truth.¹⁵

David Armstrong says: “The idea of a truthmaker for a particular truth, then, is just some existent, some portion of reality, in virtue of which that truth is true.... There is something that exists in reality, independent of the proposition in question, which makes the truth true.”¹⁶ Armstrong’s definition of truthmaker accords with Charles Kahn’s veridical meaning of “ἔστιν” in Parmenides: “the primary veridical notion is

¹²For a translation of ὁπηδεῖ see: Alexander Mourelatos, *The Route of Parmenides* [“The Route”]. (Las Vegas: Parmenides Publishing, 2008) p. 158—160.

¹³Mourelatos translates “τό γε” as “specifically this” (A. Mourelatos, “The Route” p. 170).

¹⁴Alexander Mourelatos, “φράζω and Its Derivatives in Parmenides,” *Classical Philology* vol. 60 (1968) p. 261—62 at 261. See also: A. Mourelatos, “The Route,” p. 20 (n. 28).

¹⁵Julian Dodd, “Negative truths and truthmaker principles.” *Synthese* vol. 156 (2007) pp. 383—401 at p. 385.

¹⁶David M. Armstrong, *Truth and Truthmakers*. (Cambridge: Cambridge University Press, 2004) p. 5.

that of fact or state of affairs, what obtains ‘out there,’ independently of what we say or think, and what makes what we say or think true or false.”¹⁷

But what do we do with truthmakers for the negative existential? Is there “some portion of reality, in virtue of which that truth is true”? Contemporary philosophy has not come up with a widely accepted solution to the problem. It is difficult to dispel the impression that this is the meaning of Parmenides’ observation on what is not as a candidate for the answer to the question of what exists. (DK B2.6—8). This problem leads some contemporary philosophers to reject *truthmaker maximalism*, preferring instead to restrict the truthmaking principle to positive propositions.¹⁸

I will later argue that Parmenides is a kind of truthmaker maximalist, even though he points to the truthmaker problem for negative existentials, and his eventual solution to that problem either has not been preserved or he never offered it. To begin the discussion, let us consider Parmenides’ criterion of truth and the question of positive existentials. To better understand what Parmenides has in mind, let’s look at the beginning of DK B6, where he says: Χρή τὸ λέγειν τε νοεῖν τ’ ἐὼν ἔμμεναι· ἔστι γὰρ εἶναι, μὴδὲν δ’ οὐκ ἔστιν. (DK B6.1—2a)

Here, too, the history of translation and interpretation is long and rich. Karsten translates it into Latin as: “Oportet, dicere ac cogitare esse ens.”¹⁹ In his monograph dedicated to Parmenides’ poem, Diels translates it as: “Das Sagen und Denken muss ein Seiendes sein.”²⁰ John Palmer offers the following: “It is necessary to say and to think that What Is is.”²¹ One of the important differences in these translations centers on the understanding of “Χρή.” Various modified, it appears in several places in the preserved parts of the poem. While many of the translations suggest its variants all mean necessity,²² it is quite unusual to translate “Χρή” as “necessary.”

¹⁷Charles Kahn, *Essays on Being* (Oxford: Oxford University Press, 2009) p. 198.

¹⁸See, for example, Hugh D. Mellor, “Metaphysics: Replies”, in H. Lillehammer and G. Rodriguez—Pereyra (ed.), *Real Metaphysics*, (London and New York: Routledge, 2003) p. 213.

¹⁹Simon Karsten, *Parmenidis Eleatae Carminis Reliquiae*. (Amsterdam: J. Müller, 1835). p. 33.

²⁰Hermann Diels, *Parmenides Lergedicht [Lergedicht]* (Berlin: Reimer 1897) p. 35. Kranz’s translation in the standard edition of DK reads: “Nötig ist zu sagen und zu denken das nur das Seinde ist.” In KRS, we find the following translation: “What is there to be said and thought must needs be.” (Geoffrey S. Kirk / John E. Raven / Malcolm Schofield, *The Presocratic Philosophers* [KRS]. (Cambridge: Cambridge University Press, 2009). Owen translates it as: “What can be spoken and thought of must exist.” (Gwilym. E. L. Owen, “Eleatic Questions” [“Eleatic Questions”]. *The Classical Quarterly*. ns vol. 10. no. 1 (1960) 84—102, p. 94).

²¹John Palmer, *Parmenides and Presocratic Philosophy [Parmenides]* (Oxford: Oxford University Press, 2009) p. 367.

²²These places include: “Χρεῶν” (DK B1.28), “χρεῶν” (DK B2.5), “Χρή” (DK B6.1), “χρεῶν” (DK B8.11, 16, 45 and 54). Karsten translates both “Χρεῶν” in DK B1.28 and “Χρή” in B6.1 as the Latin “oportet.” In a similar way, and I think correctly, Diels translates DK B1.28 into German as “sollst [Du].” Karsten translates “χρεῶν” in DK B2.5 as “necesse,” while Diels translates it as “notwendig”; Karsten translates “Χρή” in DK B6.1 as “oportet,” while Diels translates it as “muss.” In KRS, “Χρεῶν” in DK B1.28 is translated as “proper,” “χρεῶν” in DK B2.5, as “needful” and “Χρή” in DK B6.1, as we have

Dictionaries commonly translate it as “ought” or “need.”²³ I argue that in DK B6.1a, this word is best translated as “ought.” Some of the translators and interpreters may have proceeded from the implicit assumption that there is no place for the modal word “ought” in alethic modal logic. However, this word has a place in deontological logic. I rely on it in my interpretation; therefore, I use it in my translation.

From the start of poem, the Goddess tells the young man what he “ought” or “needs” to do, how he “ought” to talk and think. I think the word “Χρή” at the beginning of DK B6 should be translated in this spirit. The Goddess is continuing her teaching and says: “[You] ought to talk and to think that being [what is] exists.” The word “Χρή” is used here deontologically, albeit for pedagogical, not ethical, purposes. It can be considered a kind of a pedagogical hypothetical imperative, in which the Goddess says something like this to the young man: [If you want to get a persuasive truth or knowledge], “[you] ought to talk and to think that being (what is) exists.”

The next sentence reads: “ἔστι γὰρ εἶναι, μὴδὲν δ' οὐκ ἔστιν” (DK B6.1b—2a). The word “γὰρ” obviously has an explanatory function. The Goddess is explaining why the young man ought to talk and think that being (what is) exists. With that in mind, the sentence comprising DK B6.1b—2a can be translated as: “Because what is exists, while nothing does not [exist].” Therefore, the whole of DK B6.1—2a should be translated as follows: [You] ought to talk and to think that being [what is] exists. Because what is exists, while nothing does not [exist].²⁴

Why should a student say and think “what is exists”? Because it is true. But what makes it true? The mere existence of what is thought and said to exist: the statement or thought “what is exists” is *true*, because what is exists. Therefore, the existence of what is is a truthmaker for the assertion of its existence. The relation, which is a characteristic of the truthmaker theory, is asymmetric: what is does not exist because someone says or thinks “what is exists.” But, as we will see, some interpretations and translations regardless suggest symmetry or some kind of identity between the relata of this relation.

To better explain Parmenides' theory of truth, I will refer to a very controversially interpreted fragment of DK B3. We are dealing with a short sentence for which it is not easy to find a context in the poem that would help us understand it. DK B3 reads:

seen, into “must needs.” John Palmer translates “Χρεῶν” in DK B1.28 as “must needs,” “χρεών” in DK B2.5 as “must” and “Χρή” in DK B6.1, as we have also seen, as “It is necessary.” Cordero translates DK B1.28, DK B2.5 and DK B6.1 as “necessary.” (Nestor—Luis Cordero, *By Being, It Is: The Thesis of Parmenides* (Las Vegas: Parmenides Publishing, 2004).

²³For consideration of the meaning of χρή see: Thomas D. Goodell, “XPH and ΔΕΙ,” *The Classical Quarterly* vol. 8. (1914) p. 84—102, and in: Seth Benardete, “XPH and ΔΕΙ in Plato and Others,” *Glotta* vol. 43 (1965) p. 285—98. Benardete offers a number of examples and parallels. See also: A. Mourelatos, *The Route*, p. 277—78.

²⁴In a similar spirit, this fragment is translated by McKirahan as: “It is right both to say and to think that it is what—is: for it is the case that it is, but nothing is not” (Richard D. McKirahan, *Philosophy before Socrates* (Indianapolis/Cambridge: Hackett Publishing Company, 2010)).

“...τὸ γὰρ αὐτὸ νοεῖν ἐστὶν τε καὶ εἶναι.”

According to one reading, Parmenides claims the identity of thinking and being; a proponent of this reading in modern times is Hermann Diels. According to another reading, Parmenides does not claim the identity of thinking and being. Eduard Zeller is usually taken as the main proponent of this type of reading. First, let us look at Diels's translation. He thinks that in DK B3, Parmenides says: “Denn [das Seiende] denken und sein ist dasselbe.”²⁵ Diels gives no explanatory comment for his translation, except to note that the fragment, “höchst wahrscheinlich,”²⁶ comes directly after DK B2. It is hard to see what this could mean. What is usually meant by the identity reading is not covered by this translation. Another version appears later in the standard editions of *Die Fragmente der Vorsokratiker*: “denn dasselbe ist Denken und Sein.”²⁷

Charles Kahn translates the fragment as: “For knowing and being are the same.”²⁸ Why does he think so? “The answer,” he says, “is a simple one: either it is or it is not. Either knowledge and true speech are Being or they are nothing at all.”²⁹ In the same manner, he translates DK B6.1 as “Cognition and statement must be what is (i.e., must be true and real).” Diels seems to have a similar opinion when he translates DK B6.1a as “Das Sagen und Denken muss ein Seiendes sein.”³⁰ Kahn thinks “Parmenides' parallel between saying, knowing, and being suggests a correspondence theory of truth, but, in fact, he identifies these three terms.”³¹ It seems Diels and Kahn have a problem with the real issue of Parmenides' consideration. They think Parmenides is speaking about the existence of talking and thinking, not of the relation between talking and thinking, on the one hand, and being, as an object of talking and thinking, on the other.

Antony Long assumes Parmenides identifies the source and object of thought (being) as thinking.³² The main problem of Long's interpretation is his assumption that Parmenides thinks the *what is not* is not sayable or thinkable at all. He says Parmenides argues in favor of the thesis that “all thinking must be of ‘what is.’”³³

²⁵H. Diels, *Lergedicht* p. 33. Here he denotes it as fr. 5.

²⁶H. Diels, *Lergedicht* p. 67.

²⁷The first edition of *Die Fragmente der Vorsokratiker* has the same translation: the emphasis is graphically expressed as: “Denn *das Seiende* denken und sein ist dasselbe.”

²⁸Ch. Kahn, “The Thesis of Parmenides” [“The Thesis of Parmenides”]. *The Review of Metaphysics* vol. 22. (1969) p. 721.

²⁹Ch. Kahn, “The Thesis of Parmenides” p. 722.

³⁰H. Diels, *Lergedicht* p. 35.

³¹Ch. Kahn, “The Thesis of Parmenides” p. 723.

³²Antony A. Long, “Parmenides on Thinking Being” [“Thinking Being”], *Proceedings of the Boston Area Colloquium in Ancient Philosophy*, vol. 12. (1996) p. 136—37. When interpreters talk about identity, they appear to be thinking of some of the variants of the theory of identity developed in the contemporary philosophy of mind. Although interpreters do not state this, sometimes they seem to have numerical identity in mind, and sometimes not.

³³A. A. Long, “Thinking Being,” p. 138.

However, I argue Parmenides also deals with the problem of talking and thinking about “what is not.” Moreover, he thinks it is sayable and thinkable.

Sedley also claims Parmenides identifies thinking and being. According to him, the “only natural reading” of DK B3 is the following: “For it is the same to think and to be.” The effect, he says, of a non—identity reading “is to undermine his [Parmenides’] monism, by separating the thinking subject from the object of thought.” Furthermore, Sedley says, “Parmenides does not deny that thinking happens, but since being is all that there is, he must deny that thinking is separate from being. So we must take him to hold that what thinks is, and that what is thinks.”³⁴ It seems Sedley’s motivation in this reading is to save Parmenides’ monism. In fact, it looks like almost all supporters of the identity thesis are concerned with the coherence of Parmenides’ monism. Some of them, as we have seen, translate DK B6.1a as if Parmenides is talking about the very existence of talking and thinking. However, I don’t think Parmenides considers this topic at all.

Some interpreters find inspiration for the non—identity thesis in Eduard Zeller’s reading of DK B3. He translates the fragment and comments on it in the following way: “Das heisst aber nicht: ‘Denken und Sein ist dasselbe’, sondern es ist, wie der Zusammenhang zeigt, ἐστίν zu lesen und zu übersetzen: ‘denn dasselbe kann gedacht werden und sein’, nur das, was sein kann, lässt sich denken.”³⁵ Many interpreters seem to consider only the translation and ignore Zeller’s brief interpretation, yet the way Zeller understands Parmenides precludes the hypothesis of the identity of thinking and being. He writes: “Von dem Seiendem ist auch das Denken nicht verschieden, denn es gibt nicht ausser dem Seienden, und alles Denken ist Denken des Seienden.”³⁶ As we have seen, many proponents of the identity thesis offer similar interpretations. It seems he too is trying to save Parmenides’ monism at all costs. However, as I go on to show, Parmenides’ monism, his argumentation in support of the thesis that being is continuous, homogeneous, and one, does not concern the monism of thinking and being, only the monism of being or what is.

It is curious that none of these interpretations of DK B3 considers the importance of the word “γὰρ” in establishing the meaning of the whole verse. It is plausible to assume it has an explanatory function, making it important to discover what it actually explains. It is “höchst wahrscheinlich,” to use Diels’ words in this context, in that it explains Parmenides’ general theory of truth. As we have seen, in DK B6.1, Parmenides says: “[You] ought to talk and to think that being [what is] exists, because what is exists.” DK B6.1 expresses the truthmaking relation: to think or to say <being

³⁴David Sedley, “Parmenides and Melissus” [“Parmenides”], A. A. Long (ed.) *The Cambridge Companion to Early Greek Philosophy* (Cambridge: Cambridge University Press, 1999) 113—33, p.120.

³⁵Eduard Zeller, *Die Philosophie der Griechen in ihrer geschichtliche Entwicklung* [Entwicklung] (Leipzig: Fues, 1869) p. 470.

³⁶E. Zeller, *Entwicklung*, p. 470.

exists> (truthbearer) is true, because being exists (truthmaker).

Why is it true? Although there is no place for DK B3 in DK B6, given the way I understood and translate it, it can explain what is said in DK B6.1a: "... because (γὰρ) the same what is thought of, also exists (τὸ γὰρ αὐτὸ νοεῖν ἐστίν τε καὶ εἶναι)." Recall that one of the meanings of the Greek construction "τε καὶ" is "not only ... but also." Having this in mind, DK B3 can be understood as follows: "... because this same is not only what one thinks, but it also exists." Based on the connection between DK B6.1a and DK B3, the entire reasoning could be expressed as follows: if one thinks "being exists," and if it is indeed the case that it exists, then one's thought is true, because it [the existence of a being] is not only what one thinks or says; it is also the case that it exists. According to DK B3, this relation, although cross—categorical, is symmetrical. However, it is asymmetrical in DK B6.1: the existence of a being is a condition for the truth of a claim or thought about its existence, but the thought of a being is not a condition for its existence, as suggested by some interpretations of the poem.³⁷

Negative Existential

The whole construction of DK B6.1—2a also suggests the Goddess implicitly tells the young man the following: "[You] ought to talk and to think that" what is not does not exist. "Because ... nothing [what is not] does not [exist]." But as we have seen, in DK B2.6—8, Parmenides claims that we cannot have a truthmaker for negative existentials or for what is not. This does not imply that negative existentials cannot be true. Therefore, Parmenides assumes it is true to think and talk of "what is not" as not existing, even though nowhere in the preserved fragments does he discuss in detail the problem of the truthmaker in thinking of its non—existence.³⁸ It can be assumed that Parmenides' claim that you can "neither know exactly (τό γε) what is not – for this is not feasible – nor can you show it (οὔτε φράσαις)" means that precisely because of this, what is not cannot help to give us an answer to the question of what exists.³⁹

Michael Wedin, like many others, may be wrong when he says at the beginning of his book dedicated to Parmenides: "He notoriously rejects thought about what is not

³⁷This idea is explicated by Jennifer Hornsby: "How things are truly said or thought to be depends upon how they are, but how things are does not depend upon how they are said or thought to be" (Jennifer Hornsby, "Truth without Truthmaking Entities," in Helen Beebe and Julian Dodd (ed.), *Truthmakers: The Contemporary Debate* (Oxford: Clarendon Press, 2005) p. 44).

³⁸For a contemporary discussion of the problem, see Joan Pagès, "Truthmakers for Negatives," *Theoria* vol. 24, no. 1 (2009).

³⁹MacBride writes that "it seems more adequate to regard" negative existentials "as true not in virtue of any truth—maker of their own, but simply in virtue of the fact that the corresponding positive sentences have no truth—maker" (Fraser MacBride, "Truthmakers," *The Stanford Encyclopedia of Philosophy* (Fall 2022 Edition), Edward N. Zalta & Uri Nodelman (eds.), URL = <https://plato.stanford.edu/archives/fall2022/entries/truthmakers/>).

and, indeed, impugns the very possibility that such a thing [what is not] might be.”⁴⁰ I agree that Parmenides “impugns the very possibility that such a thing [what is not] might be,” but it is not true that he “notoriously rejects thought about what is not.” Montgomery Furht, for example, thinks “the central Parmenidean doctrine” is that “it cannot be said that anything is not.”⁴¹ Later in the exposition, in the same spirit, he says, “What can be thought or spoken of must be; what is not (= nothing) is both unthinkable and unnameable.”⁴²

I defend the thesis that Parmenides does not “reject thought about what is not.” Rather, he rejects the thought that what is not exists, and accordingly, what is not in this respect is the same as what is. For example, consider what Parmenides says in DK B8.8b—9a: “οὐ γὰρ φατὸν οὐδὲ νοητὸν ἔστιν ὅπως οὐκ ἔστι.” In *Parmenides Lehrgedicht*, Diels translates the passage as: “Denn es ist unsagbar und undenkbar, wie es nicht existieren solle.”⁴³ In the standard DK edition, the sentence appears as: “Denn unaussprechbar und undenkbar ist, daß NICHT IST ist.” Guthrie’s translation goes as follows: “For it is not to be said or thought that ‘it is not’.”⁴⁴ In KRS, the sentence is translated as: “For it is not to be said nor thought that it is not.” Jonathan Barnes gives a similar translation: “For it is not sayable nor thinkable that it is not.”⁴⁵ Finkelberg translates it as “It is not possible to say and to think ‘what—is is not’,”⁴⁶ and Palmer translates it as “For not to be said and not to be thought is it that it is not.”⁴⁷ In any case, the claim is directed against the equating of what is and what is not or against the assumption according to which what is not in some sense exists, and not against the possibility that what is not can’t be thought or said at all.⁴⁸

This reading is compatible with what the Goddess says, for example, in DK B7.1 and DK B8.46a. More specifically, in the former, she says, “For never will this prevail, [that] not beings exist (Οὐ γὰρ μήποτε τοῦτο δαμῆ εἶναι μὴ ἔόντα),” and in the latter, “For neither not being [what is not] exists (Οὔτε γὰρ οὐκ ἐὼν ἔστι).” Recall that in

⁴⁰M. V. Wedin, *Grand Deduction*. p. 1.

⁴¹Montgomery Furht, “Elements of Eleatic Ontology” [“Eleatic Ontology”], *Journal of the History of Philosophy* vol. 6 no. 2 (1968), p. 111.

⁴²M. Furht, “Eleatic Ontology,” pp. 11—132, p. 119.

⁴³H. Diels, *Lergedicht*, p. 37.

⁴⁴William K. C. Guthrie, *A History of Greek Philosophy* vol. 2 (Cambridge: Cambridge University Press, 1969) p. 26.

⁴⁵Jonathan Barnes, *The Presocratic Philosophers* (London and New York: Routledge, 1982) p. 140.

⁴⁶A. Finkelberg, “Ways of truth,” p. 53.

⁴⁷J. Palmer, *Parmenides*, p. 143.

⁴⁸Arnold Hermann notes the following: “(a) Whenever ‘Not—Being’ and ‘Is Not’ are used in the Poem, they are used strictly in an existential sense. (b) Parmenides does not realize he is transgressing his own rule for inexpressibility he employs the terms ‘Not—Being’ and ‘Is Not’ ” (Arnold Hermann, “Parricide of Heir? Plato’s Uncertain Relationship to Parmenides,” Nestor—Luis Cordero (ed.), *Parmenides, Venerable and Awesome* (Las Vegas: Parmenies Publishing, 2011) 147—163 at 153.). Parmenides does not defend such a rule, and therefore cannot transgress it.

DK B6.1—2a, the Goddess implicitly claims it is true to say and to think what is not (nothing, not being) does not exist. In DK B2.3b, she says, “What is not does not exist (οὐκ ἔστι μὴ εἶναι).” This is supported by what Plato writes in the *Sophist*. His Stranger from Elea says Parmenides thinks it is false to say that “beings do not exist, or that not—beings exist (τά τε ὄντα ... μὴ εἶναι καὶ τὰ μὴ ὄντα εἶναι).”⁴⁹ After all, Eleatic says he could be accused of the parricide of Parmenides by claiming that “not—being, in some respect, exists, and that, on the other hand, being, in a way, does not exist (μὴ ὄν ὡς ἔστι κατὰ τι καὶ τὸ ὄν αὖ πάλιν ὡς οὐκ ἔστι πῃ).”⁵⁰ Plato may be right, as this is precisely the subject of Parmenides’ criticism; in fact, this whole consideration begins as an opposition to the Sophist who “assumes that not—being exists (ὑποθέσθαι τὸ μὴ ὄν εἶναι).” Therefore, contrary to common interpretations, Parmenides believes not—being is sayable and thinkable. Sanders, for example, notes that Parmenides, by explicitly forbidding the “speaking or thinking about what—is—not... would already suggest that such a thing could be said or thought. It makes little sense to prohibit what is not possible.”⁵¹

However, I argue Parmenides is guilt of no such self—referential inconsistency. After all, the way what is not (οὐκ ἔστιν) is explicated in DK B2.5a is also one of two possible ways “for thinking” (DK B2.2), or which can be thought of.⁵² This is best illustrated by some of the arguments in DK B8. In DK B8.46—49a, for example, Parmenides argues:

For neither not being [what is not] exists, which would stop it [being] from reaching the same [being], nor is there any way in which being would be more here and less there (Οὔτε γὰρ οὐκ ἐὸν ἔστι, τό κεν παύοι μιν ἰκνεῖσθαι εἰς ὁμόν, οὔτ' ἐὸν ἔστιν ὅπως εἴη κεν ἐόντος τῇ μᾶλλον τῇ δ' ἥσσον).

From this, he concludes that being exists “everywhere equal to itself” and “remains homogenous.” At the same time, this argument is directed against the thesis of the discontinuity of beings. Being will not be continuous if and only if something prevents and interrupts its continuity. What could prevent being from being and, therefore, from being continuous? Parmenides’ answer is that it is the assumption that not being exists (DK B8.46a). More precisely, being will not reach being if and only if a not being (what is not) exists between them. But according to Parmenides’ criteria of truthfulness, a not being (what is not) does not exist, and therefore, it cannot stop being

⁴⁹ Plato, *Sophist* [“*Sophist*”] (Cambridge MA: Harvard University Press, 1967) 241 A.

⁵⁰ Plato, “*Sophist*,” 241 D.

⁵¹ Kirk R. Sanders, “Much Ado About ‘Nothing’ ”[“Nothing”]: μηδὲν and τό μὴ ἐὸν,” *Apeiron* vol. 35 no. 2 (2002), p. 102.

⁵² This thesis is defended by Sanders, when he says: “If what—is—not is literally inexpressible and unthinkable, as Reinhardt, Taran, and many others take lines [DK B8] 8—9 to indicate, then surely Parmenides should not be found making the claim in lines [DK B8] 12—13 that what—is—not cannot generate anything except what (itself)—is—not” (K. R. Sanders, “Nothing,” p. 92).

from reaching being.⁵³

Parmenides' claim "nor is there any way in which being would be more here and less there" rests on the same criterion: what is not does not exist. This is, by all accounts, a criticism of the theory of change through condensation and rarefaction of the fundamenal substance attributed to the Ionian physicists, above all to Anaximenes. If being is more "here" than "there," and the natural assumption is that "here" and "there" are in some sense (spatially) equal, then it is more condensed "here" than "there." This theory of change suggests that where there are more beings, there are fewer non—beings (and it is denser), and where there are fewer beings, there are more non—beings (and that is rarer), and this would imply the unacceptable conclusion that non—being exists.

According to this theory, changes by way of condensation and rarefaction of substance at a fundamental level emerge in the change of the properties of things at the phenomenal level, at the level accessible to our senses. According to this theory, not only do the things of the phenomenal world differ according to the degree of density of the fundamental substance, but they also change its properties due to the change in the degree of its density. It seems Parmenides points us to this change when he writes: "[T]o it [what is] all things have been given as names, all that mortals have established in their conviction that they are true (τῷ πάντ' ὄνομ' ἔσται, ὅσσα βροτοὶ κατέθεντο πεποιθότες εἶναι ἀληθῆ)." As an example, he mentions "exchanging brilliant colour (χρῶα φανὸν ἀμείβειν) (DK B8 38b—41)."⁵⁴ The change in color in this theory is based on the change in the density of the fundamental substance.

The above arguments, based on the impossibility of the existence of what is not, concern the spatial dimension of being or what exists. According to them, being is continuous and homogeneous. If it is continuous and homogeneous, it is necessarily one. All in all, it is clear that without the assumption that Parmenides thinks it is true to say that "what is not does not exist," these arguments would be incomprehensible.

Parmenides' Presentism

Let's look at another argument from DK B8, which I believe cannot be understandable without the previously presented interpretation of Parmenides' theory of truth. At the beginning of DK B8, Parmenides presents, among others, the thesis for being:

⁵³ Parmenides supports the same thesis in DK B8.22—24 when he says: "Nor is it divisible, since it is all alike, and not at all more in any respect, which would keep it from holding together, or at all inferior, but it is all full of what is" (I use the McKirahan translation here). If the being is "all full of what is," then there is no not—being which would prevent it "from holding together." Parmenides seems to have a similar problem in mind when he says the following in DK B4.2—3: "for you will not cut off what is from holding fast to what is, neither dispersing everywhere every way in a world order (οὐ γὰρ ἀπομήξει τὸ ἐὸν τοῦ ἐόντος ἔχεσθαι οὔτε σκιδνάμενον πάντῃ πάντως κατὰ κόσμον)."

⁵⁴ My interpretation is indirectly inspired by Palmer, because he correctly moves the verses DK B8 38b—41 after DK B8 51 (J. Palmer, *Parmenides* p. 368—369).

“not ever was it, nor yet will it be, since it is now at once all (οὐδέ ποτ' ἦν οὐδ' ἔσται, ἐπεὶ νῦν ἔστιν ὁμοῦ πᾶν)” (DK B8.5).

I leave aside the debate about whether Parmenides has in mind a tense or tenseless theory of time.⁵⁵ It seems to me that he is using tense terminology (“was”/past, “will be”/future, and “now”/present), and his argumentation would not be understandable without that terminology. Modern theorists usually distinguish three types of tense theory: eternalism, gradualism, and presentism. Stoneham, for example, explains them as follows:

An eternalist believes in a block universe: past, present and future events are all on an equal footing. A gradualist believes in a growing block: he agrees with the eternalist about the past and the present but not about the future. A presentist believes that what is present has a special status.”⁵⁶

Sometimes this special status is called the priority of presentism.⁵⁷ According to this point of view, only what exists now fundamentally exists, and what was and will be must somehow be based on what exists now. Craig Bourne, for example, thinks presentism “must accommodate the truth—value links between various times.”⁵⁸

I think Parmenides’ “now” [present] has no special status or priority, nor “must [it] accommodate the truth—value links between various times.” I argue that Parmenides is the strictest possible presentist. Stoneham succinctly defines the position of presentism as follows: “What there is only includes what exists now.”⁵⁹ If we read this statement literally, it would imply an ontology from which everything that does not exist now is excluded from existence. Parmenides says what is or being “is now at once all.” This would make the criterion of truthful speech about what is or what exists more stringent. In that case, what Parmenides says in DK B8.1—2a should be supplemented: “[You] ought to talk and to think that being [what is] exists. Because what is exists [now], while nothing does not [exist] [now].” So according to Parmenides’ criteria of truth, it is true for what is *now* to say that it exists, and for what is not *now* to say that it does not exist. If a truthmaker is a condition for true speech and thought about what exists, then claims about the past and future existence cannot be true, since we cannot have truthmakers for them.

⁵⁵For a representative discussion of the problem, see: Gwilym E. L. Owen, “Plato and Parmenides on Timeless Present,” *The Monist* vol. 50 no. 3 (1966) pp. 317—340 and Leonard Taran, “Perpetual Duration and Atemporal Eternity in Parmenides and Plato,” *The Monist* vol. 62 no. 1 (1979) pp. 43—53. For the difference in the meaning of present time in Parmenides and Plato, see: Huaiyuan Zhang, “The Instant between Time and Eternity: Plato’s Revision of the Parmenidean Now in the *Parmenides*,” *The Review of Metaphysics*, vol. 76, no. 3 (2023), pp. 425–46.

⁵⁶Tom Stoneham, “Time and Truth: The PresentismEternalism Debate” [“PresentismEternalism”], *Philosophy* vol. 84, no. 328. (2009) pp. 201—218, p. 201.

⁵⁷See about it in: Sam Baron, “The priority of the now,” *Pacific Philosophical Quarterly*, vol. 96, no. 3 (2015) pp. 325—348.

⁵⁸Craig Bourne, *A Future for Presentism* (Oxford: Oxford University Press, 2006) p. 39.

⁵⁹T. Stoneham, “PresentismEternalism,” p. 202.

The poem does not offer a direct argumentation for presentism, or the argumentation has not been preserved, but it can be derived from Parmenides' argumentation against the existence of generation and perishing. After asking the often analyzed question, "And indeed what need could have aroused it [being] later rather than before, beginning from nothing, to grow?," Parmenides says: "Thus it must either be altogether [now] or not at all" (DK B8.9—11). He seems to think, *inter alia*, that mortals assume generation is a process which begins with what is not and ends with what is. If we ignore the violation of the principle of sufficient reason or the principle *ex nihilo nihil fit*,⁶⁰ and assume generation, the following question remains: what happens [now, during generation] between the beginning and the end of the process? Are we dealing [now] with what is not or with what is? Or might mortals think that what grows (increases, or is in the process of generation now) is, in part, what is and, in part, what is not? During the process of generation, does what is not decrease and what is increase? This is impossible, because, says Parmenides, "It [what is] must either be altogether [now] or not at all." This rule seems to be an implication of the principle of excluded middle that the Goddess sets out in DK B8.15b—16a: "And the decision about these matters lies in this: [something] exists or does not exist (ἡ δὲ κρίσις περὶ τούτων ἐν τῷδ' ἔστιν· ἔστιν ἢ οὐκ ἔστιν)." The rule suggests the conclusion that the disjunction is exclusive and exhaustive. Thus, because of the violation of these rules, the opinion of mortals that there is a generation of being is not true.

The Goddess continues her argument in DK B8.18—20a where she says: "And how could what is be hereafter? How can it come to be? For if it [what is] came to be, it is not [it does not exist] (εἰ γὰρ ἔγενετ', οὐκ ἔστι), nor if ever it is going to be (οὐδ' εἴ ποτε μέλλει ἔσεσθαι)." Based on this, Parmenides concludes: "Thus, generation is extinguished and perishing unheard of" (DK B8.21). Let us assume that generation or coming to be exists. A fuller argument against this assumption would most likely have to run something like this: if what is came to be [now], and if what came to be does not exist [now], then what comes to be exists [now] and does not exist [now]. If according to the criteria of truthfulness, it is true to say that what is exists and that what is not does not exist, then the reasoning of mortals violates the principle of the excluded middle, for according to mortal reasoning (in the process of the present generation) what is exists [now] and does not exist [now]. Because of this, the existence of generation and perishing must be rejected. Mortals make a crucial mistake, which Parmenides describes as follows: "They [mortals] conceive that to exist and not exist is the same and yet not same (οἷς τὸ πέλειν τε καὶ οὐκ εἶναι ταὐτὸν νενόμισται κοῦ ταὐτόν)" (DK B6.8—9a).

The assumption is that mortals, say Ionian physicists and cosmologists, think their

⁶⁰The principle *ex nihilo nihil fit* Parmenides clearly expresses in DK B8.12—13a: "Nor ever from not being will the force of conviction allow something to come to be beyond it (Οὐδὲ ποτ' ἐκ μὴ ἔόντος ἐφήσει πίστιος ἰσχύς γίνεσθαι τι παρ' αὐτό)."

theories are true. Xenophanes claims these theories cannot reach the true/clear (σαφές) knowledge. Parmenides tries to build the criteria of truth and present arguments according to which those theories do not meet those criteria. According to Xenophanes and Parmenides, we can only have opinions about generation and perishing. Parmenides says, “According to opinion, these things have grown and now are and afterwards after growing up will come to an end. And upon them humans have established a name to mark each one” (DK B19). But according to Parmenides' criteria of truth, generation and perishing do not exist, and with those names, mortals mark as existing something that does not exist.

Conclusion

As we have seen, Parmenides' strict criteria preclude that any cosmological, empirical theory could be considered true. However, even the later Greek philosophers did not give up the pretension of considering their cosmological theories to be true. This is best seen in Aristotle's testimony about the atomistic theory:

Leucippus and his companion Democritus say the elements are the full and the empty [or void], calling them being and not being; of these the full and the solid are being [what—is], the empty not being [what is not] (Λεύκιππος δὲ καὶ ὁ ἑταῖρος αὐτοῦ Δημόκριτος στοιχεῖα μὲν τὸ πλήρες καὶ τὸ κενὸν εἶναι φασί, λέγοντες τὸ μὲν ὄν τὸ δὲ μὴ ὄν, τούτων δὲ τὸ μὲν πλήρες καὶ στερεὸν τὸ ὄν, τὸ δὲ κενὸν τὸ μὴ ὄν). (DK A6)⁶¹

We find similar testimonies about atomistic teaching in several places in Aristotle's works, as well as in the works of his commentators, Simplicius and Philoponus, including Simplicius's testimony about Aristotle's work dedicated to Democritus.⁶² According to the testimonies, atomists equate the empty or void (κενὸν) with not being (μὴ ὄν). However, Parmenides claims that “what is not (not being) does not exist,” while atomists claim “what is not (not being) exists.” What is the reason for such mutually contradictory claims? Parmenides' claim represents part of the *a priori* criterion of truth. Although Parmenides does not use the words “empty space” (κενὸν) anywhere, it seems that by not being or what is not, he means something similar. From that assumption, it follows that there is no empty space for him.

⁶¹William D. Ross (ed.), *Aristotle's Metaphysics*, (Oxford: Oxford University Press, 1924). Aristotle says something similar in *Physics* (188a 22—24, DK A45): “And Democritus makes the full and the void principles, of which he claims the one exists as being, the other as non—being (καὶ Δημόκριτος τὸ πλήρες καὶ κενόν, ὃν τὸ μὲν ὡς ὄν τὸ δὲ ὡς οὐκ ὄν εἶναι φησιν)” (Aristotle, *Physics*, transl. P. H. Wicksteed, F. M. Cornford. (Cambridge, MA: Harvard University Press, 1957).

⁶²Plutarch's testimony (*Contra Col.* 1109a, DK B156) should be added to this: “[T]hing is no more than not—thing. Democritus calls the body thing, the void not—thing, meaning the latter has a certain nature and reality of its own.” (Plutarch. *Moralia*, vol. 14: *Reply to Colotes in Defence of the Other Philosophers*, transl. Benedict Einarson, Phillip H. De Lacy. (Cambridge, MA: Harvard University Press, 1967).

Atomists reach their conclusions based on experience, but according to them, the existence of empty space (what is not) is not part of our experience (I leave aside the question of the existence of atoms here). Part of our experience is, for example, the movement of beings. Atomists come to the conclusion about the existence of empty space *a posteriori*: based on experience.⁶³ Their reasoning rests on a kind of rudimentary form of a transcendental argument, which would roughly look like this: 1) movement of beings exists (we know this based on our experience), 2) the condition of possibility (*die Bedingung der Möglichkeit*, to use Kant's words) of its existence is the existence of empty space, and, therefore, 3) empty space exists (although we do not know this based on experience).⁶⁴ Although he does not claim this anywhere in the preserved fragments, we can imagine that Parmenides, like the atomists, believes the possibility of movement depends on the possibility of empty space or of what is not. In that case, Parmenides, starting from the truth criterion ("what is not does not exist"), could conclude that there cannot be empty space, and therefore, motion is not possible, while atomists conclude from the experience of motion that there must be empty space (κενόν).

Parmenides' strict criteria of truth leads him to completely disregard experience. Starting from Parmenides' assumptions, by rejecting the logical possibilities of movement, the very fact of movement would be rejected. That we see movement is only an illusion or a deception, Parmenides might say.⁶⁵ It goes without saying that we cannot arrive at irrefutable knowledge (σαφές ... εἰδώς, to use Xenophanes' words) using a theory based on sensory experience. However, this should not imply the conclusion that theories cannot be refuted by experience. The fact of existence of motion should be a falsifier for Parmenides' claim that motion does not exist, and implicitly, a falsifier for the claim that empty space does not exist.

For the later atomist Epicurus, the rejection of sense experience leads to the loss of the criteria of knowledge. More, he writes: "If you fight against all sensations, you will have no standard by which to judge even those of them which you say are false (εἰ μαχῇ πάσαις ταῖς αἰσθήσεσιν οὐχ ἔξεις οὐδ' ἄς ἂν φῆς αὐτῶν διεψεῦσθαι πρὸς τί

⁶³Aristotle (*Physics* 213b4—22, DK A19) testifies to four atomists' arguments in which the conclusions that empty space exists are reached so that the argumentation starts from experience. The last argument reads: "They also point to the evidence from ashes, which absorb an amount of water equal to the empty space of the container they are in".

⁶⁴Philoponus sees atomists' argumentation in a similar fashion, but he transforms it into a *modus ponens* argument: "Now Democritus, taking principles testified of by sensation, that there is division and plurality in things, as well as motion, on the basis of these he introduces the void, constructing the obscure from the evident. For if there is division, he says, and plurality in things, there is void; but there is division and multiplicity; hence there is void." (Philoponus, *Commentary on Aristotle's 'On Generation and Corruption*. ed. H. Vitelli, (Berlin: Reimer 1897) 155.10—14.

⁶⁵The Goddess in DK B7.3—6 advises the student: "[D]o not let habit born of much experience force you along this way, to employ aimless sight and echoing hearing and tongue. But judge by reason the strife—filled critique I have delivered" (Palmer's translation).

ποιούμενος τὴν ἀναγωγὴν κρίνῃς)” (DL 10.146 /23/). Bearing in mind his criteria of truth, Parmenides might argue, to use a quasi—Hegelian phrase, “If the theory does not agree with the facts, so much the worse for the facts.” A theory based on the facts of experience cannot be true. According to his theory of truth, this theory belongs to the opinions of mortals.

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Drago Đurić

Istinitelj (truthmaker), negativni egzistencijal i prezentizam kod Parmenida*(Rezime)*

Postoji duga i opsežna istorija pokušaja tumačenja Parmenidove misli. Na osnovu komplikovanih interpretativnih tekstova, ponekad se čini kao da tumači tretiraju njegove tvrdnje kao Roršahov test. Čini mi se da su njegove misli mnogo jednostavnije i mnogo jasnije ako se tumače u kontekstu moderne teorije istinitelja (truthmaker), pitanja negativnih egzistencijala i prezentizma. Na kraju ću se osvrnuti na njegovu teoriju istine, koristeći primer odnosa između kretanja i praznog prostora, i upoređujući njegovu teoriju sa atomističkim pogledom. Gledano iz perspektive Parmenidove teorije, atomistička, kao i druge empirijske teorije, podleže kriterijumu mišljenja, a ne kriterijumu istine. Za njega, nijedna empirijska teorija ne može zadovoljiti njegov kriterijum istine.

KLJUČNE REČI: Parmenid, istinitelj (truthmaker), negativni egzistencijal, prezentizam, mnjenje.

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Priroda, čovek i država: Margaret Kavendiš o kontinuitetu prirodnog i društvenog poretka

Apstrakt. U ovom radu razmatraju se osnovanost i dometi tumačenja Margaret Kavendiš koja u njenoj filozofiji pronalaze slobodno prenošenje teza iz domena prirodne filozofije u domen morala i politike, i obratno. Na dva primera ovakvog postupka kod Kavendiš pokazuje se da je prevazilaženje granica domena prirode i društva kod nje prisutno i opravdano njenim osnovnim filozofskim stavovima. Istovremeno, postoje ograničenja analoškog zaključivanja između domena, uslovljena specifičnostima ljudske prirode. Ova ograničenja ne ugrožavaju bazičnu sličnost zasnovanu na relaciji simpatije koja postoji u celokupnoj materijalnoj prirodi, pa i u ljudskom društvu. Pokazuje se da tako shvaćena relacija simpatije, po uzoru na međuljudske odnose, može da obezbedi osnovu za potencijalno rešenje nekih problema u okviru Kavendišine teorije uzročnosti.

KLJUČNE REČI: Margaret Kavendiš, priroda, simpatija, okazionalna uzročnost.

1. Uvod

U tumačenjima filozofije Margaret Kavendiš (Margaret Cavendish) neretko se može naići na ocenu kako ona svoje stavove o ustrojstvu prirode proširuje izvan domena prirodne filozofije i primenjuje ih na društvene relacije i strukture. Primera radi, Kavendiš prema autorima sklonim ovom uverenju: generalizuje svoje materijalističke zaključke „daleko izvan prirodnog sveta“ što joj omogućuje da zamišlja različite uslove društvenog organizovanja; koncepciju slobode koju implicira njena prirodna filozofija upotrebljava za kritiku statusa žena u društvu svog vremena; ili uspostavlja analogiju između prirode čoveka i čitave materijalne prirode.¹ Na prvi pogled, Kavendišini metafizički i prirodnofilozofski stavovi daju povoda za tumačenja koja kod nje pronalaze ovakve postupke. Ali nije jednostavno evaluirati ovakve interpretativne tvrdnje jer se Kavendiš nije nikada temeljno posvetila političkoj filozofiji, već se njeni filozofski stavovi o pitanjima koja se tiču najboljeg načina da se urede ljudski odnosi uglavnom mogu naći rasuti u dva sporedna spisa, kao i u njenim dramama čija je filozofska relevantnost kontroverzna.²

¹ Primeri potiču, redom, iz (Sarasohn 2010), (Rogers 1998), i (Ankers 2003).

² Pomenuti spisi su WO i SL. Skraćenice za dela Kavendiš koja ću koristiti u radu su standardne: OEP: *Observations Upon Experimental Philosophy*, (Cavendish 2001); PL: *Philosophical Letters*, (Cavendish 1664b); *Grounds of Natural Philosophy* (Cavendish 1668); PPO 1: *Philosophical and Physical Opinions*, (Cavendish 1655a); PPO2: *Philosophical and Physical Opinions, 2nd ed.*, (Cavendish 1663); WO: *The Worlds Olio*, (Cavendish 1655b); SL: *CCXI Sociable Letters*, (Cavendish 1664a).

U ovom tekstu zanimae me sledeće pitanje: da li je i u kojoj meri opravdano interpretirati Kavendiš kao da njene tvrdnje o prirodi mogu važiti i za društvo, i obratno? Najpre ću ukratko prikazati osnovne Kavendišine prirodno-filozofske stavove i objasniti šta u vezi sa njima zasniva unifikatorska čitanja koja nalaze da je, u njenoj filozofiji, granica između prirode i društva dovoljno porozna da može trpeti prenošenje teza i argumenata iz jednog domena u drugi. U trećem delu ću na dva konkretna primera—radi se o jednom argumentu protiv atomizma i o opštim stavovima o relaciji simpatije—demonstrirati kako izgleda kada Kavendiš zakoračuje preko granice između prirodnog i društvenog poretka. Zatim ću u četvrtom delu skrenuti pažnju na to da postoje ograničenja koja Kavendišina pozicija postavlja slobodi transponovanja tvrdnji između dva domena. Čovekova priroda, iako je Kavendiš ne vidi kao izuzetnu na način na koji je to slučaj sa mnogim drugim njenim savremenikima, ipak uslovljava određene specifičnosti u vezi sa međuljudskim relacijama koje ne važe za sva druga stvorenja. Istovremeno, pokazaće se da relacija simpatije opstaje kao osnov za analogiju između prirode i društva. I prirodu, i ljudsko društvo, i svako pojedinačno telo ponaosob istrajno na okupu održava ova bazična relacija. Zato ću se u petom delu osvrnuti na neke implikacije za Kavendišinu prirodnu filozofiju koje potiču od pridavanja ovakve uloge simpatiji i upravo se služeći analogijom između prirodnih i međuljudskih odnosa predložiću rešenja za neke od teških problema sa kojima se suočava njena teorija uzročnosti.

2. Elementi prirodne filozofije Margaret Kavendiš

Kavendišina prirodna filozofija zasnovana je prevashodno na materijalizmu. U stvorenom svetu, prema njenom mišljenju, sve ono što postoji je telesno i nema, osim takvih entiteta, nikakvih drugih koji bi se razlikovali po svojoj prirodi.³ Materija je pre svega protežna i deljiva, ali to ne iscrpljuje njena bitna svojstva, kao što će uskoro postati jasno.

Za Kavendišino shvatanje materijalnog sveta je od izuzetne važnosti pojam *prirode*. Priroda je večni i beskonačni totalitet svih materijalnih tela, u kojem ne postoje ni prazan prostor, ni najmanji delovi (odnosno atomi). Ovaj enormni materijalni plenum Kavendiš na više mesta poredi sa telom, ili organizmom, a nisu retke prilike u kojima ume da govori i o personifikovanoj prirodi, kao živoj i mudroj vladarki (PL, 302). Fundamentalna činjenica o prirodi, u očima Kavendiš, jeste ta da je priroda *uređena*, odnosno da među njenim delovima—pojedinačnim materijalnim entitetima—vladaju red i sklad. Mnoge tvrdnje koje sačinjavaju Kavendišino filozofsko stanovište zasnovane su na argumentima koji se pozivaju upravo na uređenost prirode.⁴

³Videti, među brojnim drugim primerima, (OEP, 137), (PL, 186-87, 321), (GNP, 239).

⁴Šira interpretativna tvrdnja koju zastupam jeste da Kavendiš razvija svoju poziciju služeći se transcendentnim argumentima koji polaze od osnovne činjenice da je priroda uređena. S obzirom na ovu uređenost Kavendiš izvodi različite zaključke, i pozitivne i negativne, koji formulišu nužne uslove za

Priroda je, dakle, uređena celina, a njenu strukturu konstituišu dve vrste delova. Na osnovnom nivou objašnjenja, Kavendiš razlikuje tri vrste *stepena* materije koje naziva *konstitutivnim* delovima prirode (OEP, 16). U pitanju su, najpre, neživi stepen materije—konstitutivni delovi kojima sasvim nedostaje sposobnost samostalnog kretanja—i dva stepena žive materije koji su pre svega odlikovani sposobnošću „samokretanja“ (odnosno samostalnog kretanja, „self-motion“ u Kavendišinoj terminologiji). Prvi od živih stepena materije Kavendiš naziva racionalnim. Ovo je segment materije koji karakteriše kao slobodan, u najvećoj meri agilan i živ. Drugi je senzitivni stepen materije, onaj segment koji čine delovi materije sposobni za samostalno kretanje, ali istovremeno i zaduženi za pokretanje i razmeštanje neživih delova materije. U ovom „materijalnom trijumviratu“ uloge su strogo raspoređene (GNP, 4-5; OEP, 206-7). Racionalni stepen materije je poput arhitekta ili projektanta koji određuje na koji način različite konfiguracije materije treba da budu realizovane; senzitivni stepen je poput radnika koji sprovodi zamisli racionalne materije oblikovanjem, premeštanjem i raspoređivanjem nežive materije; najzad, neživa materija je poput materijala u kojem se realizuju zamisli racionalne materije. Tri stepena materije su temeljno izmešana kroz čitavo beskrajno telo prirode na takav način da, prema Kavendiš, ne postoji nijedan segment materijalnog plenuma u kojem se ne mogu naći sva tri konstitutivna dela: koje god parče materije da uzmemo u obzir, ma kako ono bilo malo, u njemu moraju biti prisutne sve tri vrste konstitutivnih delova (OEP, 27). Interakcijom i rasporedom tri stepena materije postiže se raznovrsnost materijalnih bića koju susrećemo u svetu. Različita „konfigurativna kretanja“ i relativno stabilne konfiguracije ovako shvaćene materije su zapravo pojedinačni materijalni entiteti—ljudi, životinje, biljke, ali i oni entiteti koje obično svrstavamo u neživu prirodu, kao što su kamenje, planete i oblaci.

Pojedinačni materijalni entiteti konstituišu drugi strukturni nivo prirode. Kavendiš ih naziva *efektivnim* delovima prirode, i oni su domen u okviru kojeg se uređenost prirode najpre manifestuje (OEP, 27). Stabilni i replikabilni procesi nastajanja, formiranja, razvića i razlaganja efektivnih delova, kao i njihove zakonomerne interakcije kojima svedočimo u prirodnom svetu, najočiglednije su manifestacije opšte uređenosti materijalne prirode. Bića koja dele sličnosti u pogledu strukture, izgleda i načina života, ujednačeno, postepeno i po utvrđenom redosledu razvojnih faza prolaze kroz procese koji čine njihov životni ciklus. Pripadnici nekih vrsta se rađaju iz jaja i sposobni su za letenje, dok pripadnici drugih vrsta puštaju duboko korenje u zemlji i jedino kretanje za koje su sposobni je rast. Ova taksonomska raznovrsnost uslovljena je rasporedom i kretanjem efektivnih delova materije koji se različito, od vrste do vrste, razmeštaju, kreću i raspoređuju. Takođe, susreti i interakcije efektivnih delova prirode su zakono-

uređenost prirode. Ovu tezu detaljno branim u svojoj knjizi *Uređenost prirode i granice materijalizma: prilog tumačenju filozofije Margaret Kavendiš* koja će uskoro biti objavljena. U literaturi je prisutna debata o tome na koji je način najbolje razumeti uređenost prirode i šta ona podrazumeva. Osim stavova koje izlažem u knjizi koju pominje prethodna fusnota, o ovoj temi korisno je konsultovati i (McNulty 2018), (Cunning 2016), (Boyle 2018) i (Detlefsen 2007).

merni i uređeni—list koji se otkine sa grane pada na zemlju, a mirujuća bilijarska kugla se stavlja u kretanje posle susreta sa drugom koja se već kreće. Vidovi raznovrsnosti i uređenosti kojima svedočimo na nivou efektivnih delova su nebrojeni, ali su pouzdani i potiču, u krajnjoj liniji, od kretanja konstitutivnih delova.

Važna posledica Kavendišinog razumevanja strukture materije je *vitalizam*. Zahvaljujući sveprisutnim živim konstitutivnim delovima materija je živa, odnosno ispoljava vitalne funkcije. Za racionalni stepen materije je, osim slobode i upravljačke uloge, karakteristično i to da je odgovoran za one vitalne funkcije koje dovodimo u vezu sa intelektom i mišljenjem uopšte: razum, uobrazilja i memorija su sposobnosti koje postoje, u materijalističkoj slici sveta Margaret Kavendiš, zbog toga što u materiji postoji racionalni konstitutivni stepen. Preostale ključne vitalne funkcije—opažanje i kretanje—zasluga su prisustva senzitivne materije. Budući da se oba živa stepena materije nalaze u *svakom* efektivnom delu prirode, bio to kamen, čovek, ili unutrašnji organ kakve životinje, za Kavendiš su svi efektivni delovi prirode sposobni za samostalno kretanje, opažanje i znanje. Na taj način ona nalazi da je u stanju da objasni fenomene tradicionalno dovodene u vezu sa nematerijalnom dušom izbegavajući u potpunosti pozivanje na entitete po prirodi različite od materije. Sve u prirodi je materijalno, i svi prirodni fenomeni mogu se objasniti strukturom i ponašanjem delova materije.⁵

Pripisivanjem života, kretanja i znanja svim delovima materijalnog plenuma, Kavendiš se našla u poziciji da ispravi jednu od ključnih grešaka za koju je optuživala sedamnaestovekovne mehaniciste. Pristalice mehanicističkog objašnjenja prirodnih fenomena smatrali su materiju inertnom i nesposobnom za samostalno iniciranje kretanja i promene. Kavendiš je bila dobro upoznata sa delima Hobsa i Dekarta, dvojice među najprominentnijim tadašnjim pripadnicima mehanicističkog pristupa u prirodnoj filozofiji, i suprotstavljala se njihovim gledištima služeći se širokim spektrom kritičkih zapažanja i argumenata. Njeno nezadovoljstvo se ticalo inertnosti materije, ali je važniji pravac kritike onaj koji se odnosi na mehanicističko objašnjenje promene. Kavendiš je, između ostalog, smatrala da bi uređenost prirode bila nemoguća ukoliko bi bilo istinito mehanicističko uverenje da je promena u osnovi stvar uzajamnog sudaranja, guranja i pritiskanja materijalnih entiteta.⁶ Kako bi nadomestila nedostatke koje je pronašla u mehanicističkom pristupu prirodnim fenomenima, poslužila se vitalističkom idejom o materiji kao inherentno sposobnoj za samostalno kretanje, opažanje i znanje. Sa druge strane, odbacivanjem promene kao rezultata grubih procesa sudaranja i pritiskanja, Kavendiš se našla u obavezi da ponudi alternativno viđenje promene u materijalnom svetu. Za to joj je poslužila *okazionalna* teorija uzročnosti.

⁵ Zbog Kavendišine tvrdnje da su život i znanje prisutni svuda u materijalnom svetu, njeno stanovište se ponekad karakteriše i kao panpsihističko. Videti (Broad 2011) i (Boyle 2018: 17–18).

⁶ Kavendišini argumenti protiv mehanicizma su brojni i raznovrsni. Uključuju, uz opšte razloge navedene u tekstu i artikulisane na brojne načine, i jednu od najranijih ranomodernih instanci kritike distinkcije između primarnih i sekundarnih kvaliteta. O kritici mehanicizma videti više u (James 1999) i (Chamberlain 2019).

U mehanicističkom pogledu na svet, objašnjenja uzročnih relacija se zasnivaju na nekoj vrsti prenošenja kretanja inertnoj materiji. Dodirom, pritiskom i prenosom kretanja telo koje je u mirovanju biva stavljeno u kretanje.⁷ Kavendiš nije bila jedina sedamnaestovekovna autorka nezadovoljna mehanicističkim modelom promene. Neki autori su posegnuli za božjim uticajem kao faktorom neophodnim za objašnjenjem uzročnosti, tvrdeći da je jedino Bog uzročno aktivan u svetu. Drugi su odbacili mogućnost da jedno telo proizvede promenu u drugom, ali bez oduzimanja uzročne aktivnosti telima; njihov predlog je bio da tela sama proizvode sopstvena kretanja, a da je Bog uredio svet tako da su stanja stvari koja identifikujemo kao uzroke i posledice uzajamno usklađena.⁸ Kavendiš ne može da prihvati nijedno od ovih objašnjenja. Mehanicistička promena putem prenosa kretanja je, po njenom mišljenju, beznadežno nekoherentna. Objašnjenja koja se pozivaju na božji uticaj nisu joj dostupna s obzirom na to da je nastupala sa naturalističke pozicije koja diktira da se fenomeni u prirodi moraju objasniti jedino pomoću objekata i svojstava prisutnih u okviru jednoobraznog materijalnog sveta. Objašnjenje za koje se Kavendiš odlučila bilo je sledeće. U slučajevima uzročnih relacija u materijalnoj prirodi, ono što bismo smatrali pravim eficientnim uzrokom promene zapravo je samo *upriličavajući* ili *okazionalni* uzrok, odnosno faktor čijim javljanjem se nekom telu daje prilika da *samostalno* proizvede posledicu. Kada bilijska kugla u kretanju naiđe na kuglu u mirovanju, nastanak kretanja mirujuće kugle nije ni rezultat prenosa kretanja, niti rezultat božje intervencije, već je rezultat samostalnog kretanja te kugle nastalog kao odgovarajuća reakcija na opaženo kretanje prve kugle.⁹ Kavendišin vitalizam i ovde nalazi korisnu primenu. S obzirom na to da su sva tela obdarena opažanjem i samokretanjem, ona su sposobna i za to da uoče položaj i kretanje upriličavajućeg uzroka, kao i da sprovedu sopstveno kretanje koje je usklađeno sa upriličavajućim uzrokom.

Kratka skica Kavendišinih osnovnih metafizičkih i prirodnofilozofskih stavova koju sam ponudio omogućuje da se sagledaju sledeći stavovi koji oblikuju unifikacionističko gledište o prirodi i ljudskom društvu. Na metafizičkom nivou, fundamentalna struktura i karakter svih entiteta i njihovih relacija je svuda ista. Kako je materija jedini konstituent stvarnosti, i pritom je živa, opažajuća i osposobljena za neku vrstu znanja, svi odnosi u stvorenom svetu su odnosi između živih aktera koji imaju slobodu da postupaju na skladan način. Za Kavendiš čak ni čovek nije izuzetan među drugim stvorenjima. Jednom kada se život, samokretanje i znanje pripisu svakom telu, ljudske

⁷Mechanicistička objašnjenja ove vrste često se svrstavaju u širu grupu „teorija influksa“. Videti (O'Neill 1989).

⁸Prva od dve navedene alternative je okazionalizam kakav je branio, na primer, Malbranš. Druga je prestabilirana harmonija, teorija čiji je najpoznatiji primer ponudio Lajbnic. Okazionalizam je važno razlikovati od *okazionalne uzročnosti*, budući da je u prvoj vrsti teorije Bog jedini uzročno efikasan, a u drugoj Bog ne igra nikakvu uzročnu ulogu već se u slučajevima uzročne relacije tela samostalno pokreću. Za ovu distinkciju videti (Nadler 1994).

⁹Videti, na primer, (PL, 444-45).

intelektualne sposobnosti—koje su filozofima oduvek služile kao osnov za izdvajanje čoveka kao izuzetnog stvorenja—postaju samo jedan od izraza sveprisutne materijalne vitalnosti, a ne superioran i jedinstven fenomen u prirodi. Uporedo sa tim, opšta homogenost u pogledu prirode otvara prostor za to da se tvrdnje o fenomenima iz jednog domena stvarnosti prenose i primenjuju na druge domene stvarnosti. Kavendiš, po svemu sudeći, upravo to čini. Stvorenja su kompleksne strukture pojedinačnih materijalnih delova koji interaguju na načine primerene materijalnim telima, ali to je slučaj i sa kompleksnim *udruženjima*, ili *društvima* materijalnih tela. Kao što je ljudsko telo strukturalno kompleksan entitet konstituisan materijalnim delovima, tako je to i ljudsko društvo, a i celokupni materijalni plenum. Stoga onu vrstu zaključaka koje donosimo o jednom možemo primenjivati i na drugo. U sledećem odeljku ću analizirati dva primera koja mogu da posluže da se uoči upravo kako Kavendiš premošćuje rastojanje između domena prirode i domena ljudskog društva prenošenjem uvida o jednom na ono drugo.

3. Društvo i priroda: dva primera

3.1 Živi atomi i njihova društva

Upečatljiv primer toga kako Kavendiš vrši analoški transfer teza iz jednog domena u drugi možemo naći u važnom argumentu koji razvija protiv atomizma. Kavendiš je tokom svog filozofskog stvaralaštva prošla kroz više faza u svom odnosu prema atomizmu.¹⁰ U najranijim delima je bila pristalica atomizma, ali je ubrzo shvatila da mehanicistički atomizam kakav se mogao naći pre svega u delima revitalizatora epikurejske pozicije nije zadovoljavajući. Neživi atomi, čijim kretanjem kroz prazan prostor upravlja slučaj, ne mogu da proizvedu uređenu prirodu, tvrdila je. Ipak, isprva je ostavila mogućnost da bi *vitalistički* atomizam, odnosno atomizam koji postulira atome žive materije, mogao da bude adekvatno stanovište. U drugom izdanju *Filozofskih i fizičkih gledišta*, međutim, raskrstila je i sa ovom verzijom atomizma. Presudan argument za takvu promenu mišljenja nalazimo u sledećim redovima, u kojima Kavendiš tvrdi kako priroda sačinjena od živih atoma

„nije verovatna; jer ako bi svaki atom bio od žive supstancije, i imao jednaku moć, život i znanje, i posledično slobodnu volju i slobodu, te bi tako svaki od njih bio apsolutan kao svaki drugi, oni bi se teško složili u jedinstvenoj vlasti, jednako malo verovatno kao da se više kraljeva složi u jednom kraljevstvu, ili pre kao ljudi, kad bi svaki imao jednaku moć, da formiraju dobru vlast; i ako bi to zavisilo od pristanka i dogovora, kao što je to slučaj sa ljudskom vlašću, bilo bi mnogo promena i zbrkanosti svetova, proizvedenih neslaganjem kao u ljudskim državama i vladama,

¹⁰O ovome pišem detaljno u (Vuletić 2026).

jer nužno mora biti slobode i moći u svakom atomu da se ne slaže kao i da se slaže.“ (*PPO 2*, str. XIX-XX)

Živi atomi bi, shodno vitalističkom materijalizmu, morali imati neku vrstu razuma, čulnosti i slobode—ovo poslednje svojstvo proističe iz sposobnosti samokretanja i teorije okazionalne uzročnosti—te se ne bi radilo o slučajno razbacanim entitetima kojima upravljaju spoljašnje sile. Takvi atomi bi, prema Kavendiš, bili jednaki u pogledu svojih moći, i svako stupanje u složene strukture iziskivalo bi pristanak pojedinačnih atoma na to da budu deo veće celine. Da bi se videlo zašto je tako, treba se podsetiti dva elementa Kavendišinog materijalizma. Najpre, atome ništa ne može *primorati* da se kreću na određeni način, budući da prisilno dejstvo kakvo je karakteristično za mehanicističke atome nije moguće u materijalnom svetu čijim promenama upravlja okazionalna uzročnost. Jedan atom ne može na silu gurnuti drugi i tako ga izmestiti; može jedino da obezbedi upriličavajući uzrok za to da se ovaj drugi samostalno pokrene i zauzme nov položaj. Istovremeno, nijedan atom nije apsolutno slobodan u izboru svojih kretanja budući da je sa svih strana okružen drugim telima na koja ne može da ostvari mehanički uticaj i razgrne ih u cilju ostvarenja svog nameravanog kretanja. Živi atomi bi imali samokretanje, ali ono bi bilo ograničeno *pristankom* okolnih tela koja se moraju samostalno izmaći kako bi kretanje bilo omogućeno.

Ključno za razumevanje Kavendišinog argumentativnog postupka u ovom tekstu je da se ustanovi zbog čega ona misli da su živi atomi nespojivi sa uređenošću prirode—to je ono što Kavendiš tvrdi kada kaže da bi bilo „mnogo promena i zbrkanosti svetova“. U tom pogledu smatram da je korisno uspostaviti vezu sa Hobsovom političkom filozofijom, s obzirom na to da Kavendiš pravi analogiju između atoma i ljudskih individua koje se organizuju u državu. Kavendiš je bila blisko upoznata sa Hobsovim filozofskim spisima i malo je verovatno da su slučajne sličnosti između njegovih stavova o uspostavljanju države i odnosu suverena i podanika sa njenim stavovima o materijalnim entitetima i strukturi prirode. Kod Hobsa nalazimo, naravno, ideju da će se slobodni i u pogledu svojih moći jednaki pojedinci, u slučaju da nisu uredili svoje odnose nekom vrstom dogovora, nalaziti u stanju stalne nesigurnosti, opasnosti i međusobnog sukoba. Kako bi se prevazišlo takvo stanje i postigli mir i sigurnost, mislio je Hobs, neophodno je da se pojedinci potčine jedinstvenom autoritetu i upravljanju „jednog rasuđivanja“, na taj način formirajući državu. Za Hobsa je ključno to da se prilikom udruživanja u državnu zajednicu ne sme računati na to da će se iz mnoštva pojedinačnih volja stvoriti sloga:

„Jedini način da se uspostavi zajednička vlast koja može biti sposobna da ljude brani od zavojevača spolja i od povreda koje jedni drugima nanose i da ih obezbedi (...) jeste da sve svoje moći i svu svoju snagu povere jednom čoveku ili jednom skupu ljudi koji će moći sve njihove volje putem većine glasova da svede na jednu volju. Drugim rečima da odrede jednog čoveka ili jedan skup ljudi koji će biti nosilac njihove ličnosti (...)

i da u tome svaki pojedinac potčini svoju volju volji nosioca zajedničke ličnosti, a svoje rasuđivanje njegovom rasuđivanju.“ (*Levijatan*, 150)¹¹

Upravo ovde nalazimo sličnost sa Kavendišinim redovima o živim atomima: „apsolutni“ živi atomi bi se jednako teško udružili u skladnu i uređenu prirodnu zajednicu kao i slobodni i jednaki ljudi u dobru društvenu zajednicu. U oba slučaja potrebno je da dođe do toga da se nešto od individualne slobode žrtvuje i podredi spoljnom autoritetu sa ciljem osnivanja i ostvarivanja zajednice koja donosi vrednosti koje se drukčije ne mogu postići. Kod ljudi to podrazumeva potčinjavanje suverenu koje donosi mir, red i stabilnost u okviru države, tog velikog „veštačkog tela“ (*Levijatan*, 29). Kod materijalnih tela uopšte, to podrazumeva podređivanje beskonačnoj kontinuiranoj prirodi i sjedinjavanje u njoj kao prirodnom telu.¹²

3.2 Simpatija i jedinstvo

U primeru sa argumentom protiv vitalističkog atomizma videli smo kako se Kavendiš služi uvidima iz domena politike kako bi po analogiji donela zaključke o prirodi. U njenom delu postoji i uticaj koji ide u suprotnom smeru, odnosno rasuđivanje koje vodi od prirodnofilozofskih teza ka analognim zaključcima o društvenim relacijama.

Okvako nešto možemo pronaći u ulozi koju Kavendiš u dva domena namenjuje relaciji *simpatije*. Kada diskutuje o uslovima jedinstva kompleksnih materijalnih tela, Kavendiš se poziva na međusobnu simpatiju, ili privlačnost, koja vlada među delovima tela. Radi se o relaciji čija upotreba ima dugu istoriju u filozofiji, kao i u drugim naukama, i najčešće podrazumeva sličnost između entiteta koji u nju stupaju.¹³ Kod Kavendiš se simpatija—najčešće u paru sa svojim suprotnim korelatom, antipatijom—javlja u upotrebi u različitim kontekstima, od fundamentalno metafizičkog pa do

¹¹ Služim se prevodom *Levijatana* Milivoja Markovića u izdanju Pravnog fakulteta Univerziteta u Beogradu (2011), a paginacija citata odgovara tom izdanju.

¹² Mogla bi se staviti primedba kako u slučaju prirode nema ničega što bi odgovaralo ulozi i položaju apsolutnog monarha, pa da stoga analogija nije dobra i da se ne može zaključiti da Kavendiš preslikava supstantivne zaključke iz jednog domena u drugi. Mislim da se primedba može otkloniti ako shvatimo da je njena namera da brani tezu kako u slučaju prirode mora postojati *jedan uređivački princip*, a ne nužno jedan entitet koji doslovno upravlja prirodnim procesima. Da je to slučaj, može se videti i po tome kako Kavendiš na još jednom mestu koristi analogiju između prirode i države. Trudeći se da pokaže kako nematerijalna duša ne može da utiče na materiju, Kavendiš iskazuje skepticizam prema postojanju takvog uticaja još jednom se pozivajući na sličnost sa društvenim odnosima. Nematerijalna duša i materija bi se toliko često nalazili u suprotnosti i neslaganju, kao što se to dešava sa kraljem i njegovim podanicima, da bi to vodilo narušavanju prirodne uređenosti (PL, 214). Mislim da način na koji Kavendiš koristi analogiju ukazuje na to da je cilj koji želi da postigne taj da pokaže kako je postojanje više od jednog uređivačkog principa nespojivo sa uređenošću—kada su podanici u sukobu sa kraljem, onda nema jednog autoriteta kojem se građani povinuju, baš kao što ne bi bilo jedinstvenog uređivačkog principa kada bi kretanjima u prirodi upravljali i tela i nematerijalni entiteti.

¹³ Za istoriju upotrebe pojma simpatije videti tekstove sakupljene u (Schliesser 2015), a naročito uvodno poglavlje. O mestu Kavendišine upotrebe simpatije u okviru ove tradicije videti (Borcherding 2021).

moralnog i društvenog. Zajedničko raznovrsnim upotrebama je to da simpatija ima ulogu organizacionog i ujedinjujućeg principa. Kavendiš tako simpatiji nalazi ulogu u formiranju nebeskih tela (PPO 1, 98), ali i u formiranju i održavanju uređenog prirodnog poretka koji prožima čitavu prirodu tako što usklađuje kretanja materije (PPO 1, 107). Sa stanovišta potreba ovog teksta, najvažnija je uloga koju simpatiji Kavendiš namenjuje u objašnjavanju individuacije pojedinačnih efektivnih delova prirode i jedinstva njihovih delova. Uzajamna privlačnost i ljubav, kako to Kavendiš često kaže, upravo je ono što uslovljava sjedinjavanje delova materije u relativno trajne konfiguracije koje prepoznajemo kao materijalna bića:

„Ljubav pripada različitim delovima, tako da različiti delovi jednog društva, kao što je jedno stvorenje, imaju ljubav i samoljublje, budući da su simpatetički ujedinjeni [sympathetically united] u jedno društvo. Takođe, ne samo da delovi jednog istog društva mogu imati uzajamnu ljubav, već ljubav postoji i između različitih društava“ (GNP, 68).¹⁴

Naročito je za pomenuto jedinstvo značajno to što među živim konstitutivnim delovima, zahvaljujući njihovoj uzajamnoj sličnosti, postoji izrazito „snažna simpatija“ (GNP, 21). Povrh toga, Kavendiš nalazi da je simpatija zaslužna i za to što se efektivni delovi prirode ponašaju tako da njihova udruživanja i kretanja odgovaraju karakterističnim odlikama pripadnika pojedinih vrsta (GNP, 31-33). Taksonomska raznovrsnost prirode je na taj način takođe uslovljena privlačnim vezama između materijalnih tela. Najzad, Kavendiš je stava da je u pitanju relacija koja je prisutna u celokupnoj prirodi čije nebrojene delove povezuje u jedno:

„Uticaj je ovo: kada telesna kretanja, u različitim vrstama i sortama stvaranja, ili u jednoj istoj sorti ili vrsti, kreću se simpatetično. I mada postoje antipatetička kretanja kao i simpatetička, svi beskonačni delovi materije su složni u svojoj prirodi, budući da su svi materijalni i samokrećući, i s obzirom na to da nema vakuuma, mora nužno postojati uticaj između svih delova prirode“ (GNP, 15-16).

Simpatija je relacija čije je javljanje evidentno, prema Kavendišinom mišljenju, svuda u materijalnoj prirodi, i koja igra ključnu ulogu kohezivnog faktora i na konstitutivnom i na efektivnom nivou organizacije materije. Dodatno, na efektivnom nivou simpatija omogućuje individuiranje pojedinih tela koja stupaju u složne konfiguracije.

Kavendiš eksplicitno proširuje prisustvo simpatije i na domen složenih zajednica, ili društava, koje prepoznajemo na nivou organizacije ljudskih jedinki:

„U svakom normalnom ljudskom društvu postoji ljubav među povezanim delovima, kao među studentima u koledžu, slugama u jednoj kući, srodnicima u jednoj porodici, ili podanicima u jednoj naciji, ili u pripadnicima

¹⁴ U (Lascano and Schliesser 2022) tvrdi se da relacija simpatije nastaje iz jedinstva delova materijalnih tela. Mesta poput ovog, međutim, govore u prilog tome da se jedinstvo zapravo ostvaruje putem relacije simpatije.

jedne crkve. Na taj način i samokrećući delovi ljudskog bića, budući da su povezani, vole jedni druge, i stoga se trude da očuvaju svoje društvo od razlaganja.“ (GNP, 75)

Vidimo da Kavendiš smatra da se mogu povlačiti analogije između prirodnog i ljudskog institucionalnog poretka upravo na osnovu činjenice da je u oba domena prisutna relacija privlačnosti među konstitutivnim članovima.

Ima li, međutim, u ovim primerima ičeg više od ilustracija čije sličnosti su zasnovane na labavoj analogiji? Istorijat filozofskih analogija između složenih prirodnih i društvenih entiteta je bogat i dugačak. Analogije, međutim, nekad mogu biti zasnovane na istovetnosti (ili velikoj sličnosti) poređenih struktura koja potiče iz činjenice da su te strukture zapravo manifestacije *istog* fenomena, dok ponekad ne predstavljaju mnogo više od retoričkog sredstva koje je pogodno za promovisanje određenog pogleda na jedan od domena poređenja. Ima li razloga da se misli da je Kavendišino korišćenje simpatije zasnovano na metafizičkoj strukturi materije koja diktira i činjenice na polju moralnih i društvenih fenomena?

Osim same činjenice da njen temeljni materijalizam načelno *omogućuje* da se društvene strukture posmatraju kao specifične instance udruživanja materijalnih delova, mislim da postoji i konkretan razlog u Kavendišinom tekstu za tumačenje da je upravo reč o supstantivnoj sličnosti između prirode i društva potekloj od metafizičkih uvida. Radi se o sledećem. Dobro je dokumentovano da Kavendiš završava na poziciji sličnoj Hobsovoj kad je u pitanju oblik uređenja države koji preporučuje.¹⁵ Monarhija je oblik vladavine u državi koji omogućuje najbolje šanse za stabilan, uređen i miran poredak među ljudima. Postoji ipak tačka razilaženja između Kavendiš i Hobsa koja je izrazito indikativna sa stanovišta teme ovo teksta. Kavendiš nije bila mišljenja da stvaranju države prethodi prirodno stanje kakvo nalazimo u filozofiji njenog starijeg materijalističkog kolege. Nikakvo pred-društveno stanje koje ljudi napuštaju stupanjem u društveni ugovor ne može se naći u Kavendišinim osvrtima na društvo i državu. Razlog za taj stav možemo pronaći, kao što primećuje Debora Bojl (Deborah Boyle), upravo u Kavendišinom mišljenju da su ljudi materijalna bića između kojih postoji bazična simpatetička relacija. Među ljudima postoji prirodna benevolentnost koja potiče od ove relacije čije je poreklo, kao što smo videli, pre svega metafizičko, i koja postoji nezavisno od toga u kojoj meri i na koji način su prirodna stvorenja organizovana u složene celine.¹⁶

4. Razlike između prirodnog i društvenog poretka

U Kavendišinom opusu postoje, dakle, filozofske osnove za to da se uvidi iz prirodne filozofije transponuju u političku filozofiju, i obratno. Ne treba ipak zaključiti

¹⁵ Za diskusiju i relevantna mesta kod Kavendiš videti (Boyle 2018, 151–54).

¹⁶ Bojl ispravno smatra da je ovo mišljenje implicirano i u Kavendišinim diskusijama o zajednicama životinja poput mrava i pčela; videti (Boyle 2018, 149–50).

da između dva domena kojima se pomenute discipline bave postoji potpuni kontinuitet i da, recimo, o društvima možemo zaključivati *samo* na osnovu metafizičkih i prirodnofilozofskih teza. Iako je za Kavendiš važna poenta da čovek nije izuzetno biće kakvim sebe voli da smatra—neke vrste razuma i znanja ima *svuda* u prirodi pa čovek nije po tome poseban, a Kavendiš je vrlo skeptična kad je u pitanju naša sposobnost da sagledamo kakve sposobnosti pripadaju drugim vrstama—te da njegov položaj u prirodi nije položaj nekoga kome pripada vlast nad drugim stvorenjima, treba biti svestan i toga da u pojedinim bitnim tačkama prirodni i društveni poredak manifestuju različitost. Najvažnije je u ovom pogledu ono što Kavendiš nalazi da je izrazito karakteristično za čoveka: samoljublje koje može biti zloupotrebjeno.

Jedna od funkcija simpatetičke relacije između delova složenih bića je, kao što smo videli, održavanje na okupu u okviru jedinstvene celine. Kavendiš nalazi da je uzajamna simpatija, ili ljubav, ono što održava u zajednici „društva“ delova u okviru pojedinačnih stvorenja.¹⁷ Videli smo u prethodnom odeljku da ona vidi kako i za ljudsko biće važi isto—ljubav između delova je razlog za težnju ka samoočuvanju organizma. Kavendiš je, međutim, osetljiva i za specifičnosti ljudske psihologije koje uslovljavaju izvesne diskontinuitete između prirodnih i ljudskih društava. U ovom kontekstu posebno je važno to što ljubav prema sebi, ili samoljublje (self-love), prema njenom mišljenju, kod čoveka dobija manifestacije koje mogu biti destabilizujuće po ljudsku zajednicu.¹⁸ Iz osnovnog, metafizičkog samoljublja koje ima moralno neutralnu ulogu održavanja jedinke u postojanju, kod čoveka se rađa i vrsta ljubavi prema sebi koja je obeležena ljudskom svešću o sopstvenoj smrtnosti. Kavendiš je smatrala da ljude neizvesnost u pogledu mogućnosti produženja sopstvenog postojanja, do kojeg im je stalo samom činjenicom da su konstituisani simpatetičkim relacijama svojih delova, navodi da teže tome da budu upamćeni i da tako sebi obezbede nastavak bivstvovanja:

„Videvši (...) da njihov život posle smrti ne može biti isti kao sadašnji život, oni se trude (jer ne mogu sprečiti sopstveno društvo [delova] od rastavljanja) da njihovo društvo ostane u sećanju među pojedinačnim i opštim društvima iste vrste stvorenja, koje nazivamo čovečanstvom: i ova namera podstiče sve senzitivne i racionalne delove u jednom društvu da budu preduzimljivi i da ostave neki trag za trajno sećanje među sebi

¹⁷Videti, na primer, (GNP, 32). U diskusiji o ljubavi i samoljublju oslanjam se u velikoj meri na uvide Deборе Bojl iznete u četvrtom poglavlju (Boyle 2018).

¹⁸U sedamnaestom veku dugotrajnom gledištu o čovekovoj superiornosti nad ostatkom stvorenog sveta pridodata je i nova komponenta, oličena u Bejkonovom (Francis Bacon) uverenju da je prvobitnim grehom „čovek pao i iz stanja nevinosti i iz svoje prevlasti [dominion] nad stvorenim. Ali čak i u ovom životu oba ova gubitka mogu se nadoknaditi; predašnji religijom i verom, a potonji veštinama i naukom“ (Nov. Org. II.52). Nauka, prema ovom stanovištu, omogućuje ne samo razumevanje i znanje o svemu što je Bog stvorio, nego čoveku pruža nadu i u povratak prevlasti nad drugim stvorenjima. Kavendišini stavovi o prirodi nespojivi su sa oba ova elementa razumevanja čovekovog mesta u prirodi.

sličnim stvorenjima. Ovo opšte sećanje čovek zove slavom.“ (GNP, 75-6)

Vid postojanja posle smrti koji čovek može pokušati samostalno da obezbedi, naročito u okolnostima u kojima je neizvesno da li će i kako nastaviti da postoji, jeste postojanje kroz pamćenje drugih ljudi.¹⁹ Stoga je za čoveka prirodno i karakteristično da teži slavi. Uprkos tome što težnja ka slavi ima koren u prirodnoj tendenciji ka jedinstvu i istrajnosti u slozi između delova, ona može biti izvitoperena ukoliko se pogrešno usmeri. Kavendiš razlikuje samoljublje koje je „čisto“ i ono koje je „korumpirano“ (SL, 163), povezujući potonje sa težnjom ka prepoznatljivosti i slavi po svaku cenu i nezavisno od brige za razloge koji će dovesti do slave. Čisto samoljublje, sa druge strane, čoveka usmerava ka nastojanju da dobije priznanje zbog dobrih dela i vrlina. Samoljublje koje se izmetne u iskvarenu težnju ka slavi po svaku cenu faktor je nestabilnosti u društvu budući da lako vodi narušavanju društvenih normi ponašanja. Postupanje ove vrste je, u Kavendišinoj terminologiji, „nepravilno“ i njegovim upražnjavanjem individue sklone bezuslovnoj samopromociji narušavaju uređenost društva. U najgorem slučaju ishod narušavanja uređenosti ljudske zajednice biće katastrofalni građanski rat kakav su iskusili Kavendiš i Hobs.

Treba, dakle, upražnjavati oprez kada koristimo sličnosti između prirodnog i društvenog poretka da izvedemo zaključke o Kavendišinih stavovima. U tom pogledu korisno je razmotriti jednu instancu interpretativnog postupanja koje koristi poroznost granica između prirodnog i društvenog poretka. Karen Detlefsen (Karen Detlefsen) je ponudila uticajno, originalno i nadahnuto viđenje Kavendišinog postupka u objašnjavanju prirode. Prema perspektivi koju predlaže Detlefsen, Kavendiš je u objašnjavanju prirodnih fenomena učinila nešto inovativno i sasvim različito od onoga čemu su pribegavali njeni mehanicistički savremenici. U rukama mehanicista, fenomeni koji se tiču funkcionisanja čoveka, njegovog organizma, ali u nekim slučajevima i duše (u zavisnosti od osnovnih metafizičkih obaveza), objašnjavani su pomoću bazičnih mehanizama pritiska, guranja i prenošenja kretanja. Kavendiš, u očima Detlefsen, učinila je nešto sasvim obrnuto: predložila je da fenomene koje smo skloni da vidimo kao osnovne vidove interakcije u svetu—sudare tela, recimo—razumemo po modelima funkcionisanja kakve nalazimo u uzajamnom ljudskom saobraćanju. Tela u prirodi ne guraju i ne pritiskaju jedna druga. Umesto toga, ona postupaju—najčešće i kada stvari funkcionišu kako treba—skladno, složno i prateći namere i sugestije drugih tela. Kavendišin plenistički, vitalistički materijalizam tako nešto zahteva. S obzirom na to da između materijalnih entiteta ne postoji prazan prostor i da su uzročne veze okazionalne, nema kretanja jednog tela koje ne podrazumeva kao svoj neophodni uslov kretanje drugih tela koja će ustupiti svoje mesto prvom. Možemo ovu situaciju uporediti sa kretanjem putnika u punom autobusu. Kada se otvore vrata i nekolicina

¹⁹U protestantskoj tradiciji, uključujući anglikansku, značajno je prisustvo mortalističkih stanovišta koja podrazumevaju da duša po smrti čoveka takođe, na neki način, umire. Dobra filozofska diskusija o ovom stanovištu može se naći u (Johnston 2012).

putnika iz sredine poželi da napusti vozilo, njihovo kretanje zahteva da im drugi putnici ustupe svoja mesta. Uz to, ako bismo putnike posmatrali kao tela u Kavendišinom plenumu, do pomeranja susednih putnika mora doći *njihovom* voljom i zahvaljujući *njihovom* kretanju, a ne delovanjem putnika koji želi da napusti vozilo. Razlog se tiče ključne obaveze koju Kavendiš preuzima na sebe: tela u Kavendišinom materijalnom plenumu ne mogu započeti kretanje drugih tela pritiskom i guranjem, već jedino mogu pružiti ovim drugim telima priliku za samostalno kretanje. Neuljudni putnici, ili putnici zabrinuti zbog izostanka kooperativnosti svojih nepokretnih saputnika, mogu da se posluže guranjem i pritiskanjem susednih tela. Kavendišina materijalna stvorenja to ne mogu. Uzevši ove interpretativne činjenice zajedno, Detlefsen zaključuje da delovi prirode moraju delovati na načine koji su analogni vidovima interakcije karakterističnim za funkcionalne ljudske odnose. Ljudski odnosi teku skladno i uređeno onda kada, između ostalog, postupamo služeći se razmenom racionalnih razloga i argumenata, kada se pozivamo na racionalnu prirodu svog bližnjeg i demonstriramo osetljivost na uverljivost ponuđenih razloga za postupanje. Slično važi, u okviru Kavendišinog sistema, i za sva druga tela, ma koliko nama ona mogla delovati obično ili jednostavno, naročito u poređenju sa čovekom. Za Kavendiš u svetu postoji univerzalnost važenja neophodnosti uređenog i harmonizovanog delovanja mnogih delova prirode, od odnosa između stenja i kiše, do ljudskih društava. Ona vidi da je u materijalnom plenumu žive materije koja operiše okazionalnom uzročnošću tako nešto neophodno:

„svaka aktivnost prirode je perceptivna [perceptive]: jer, pošto ne postoje pojedinačni delovi u prirodi, nego sve ono što ima telo sastoji se od delova, ne može biti ni pojedinačnih aktivnosti, odnosno aktivnosti pojedinačnog dela, već u svim prirodnim aktivnostima postoji razmena, opštenje, ili slaganje delova. Ovo opštenje ili slaganje ne može biti bez opažanja ili uzajamnog znanja: iz ovoga nužno sledi da je svaka aktivnost perceptivna, ili da je opažanje među delovima neophodno u svakoj aktivnosti prirode“ (OEP, 172).

Predstava o svetu koja izranja iz ove interpretacije je jedinstvena i intrigantna. Brišući granicu između društvene i prirodne strukture, Kavendiš kao da ostavlja sebi otvorenim put ka razumevanju celokupnog sveta kao mesta sličnijeg svakodnevnim poljima ljudskog interagovanja nego što bismo to na prvi pogled pomislili. Sudar bilijarskih kugli, bacanje kamena u jezero i guranje pokvarenog automobila događaji su u kojima, uprkos našoj navici da vidimo u njima nešto sasvim različito, na delu imamo upućivanje zahteva i usvajanje zahteva kao razloga za sopstveno kretanje.

Međutim, najmanje dve ključne stvari nisu odmah jasne. Najpre, postavlja se pitanje *kako* jedno telo upućuje sugestiju drugom. Uz to, nije očigledno zbog čega bi zahtev za kretanjem koje je jedno telo uputilo drugom predstavljalo *razlog* za drugo telo da postupi na određeni način. U ostatku teksta hteo bih da ponudim spekulativne odgovore na ova pitanja, u duhu Kavendišinog teksta, koji će pokazati jedan način

da se produbi vrsta interpretacije kakvu nudi Detlefsen, ali i ukazati na probleme koji preostaju za rezultirajuću poziciju.

5. Zahtevi, razlozi i simpatija

U Kavendišinoj prirodi, sasvim ispunjenoj živom materijom i u kojoj se promene odvijaju po principu okazionalne uzročnosti, uzročne interakcije nisu nalik onome kako smo navikli da o njima razmišljamo. Vratću se primeru sa autobusom koji je dobro popunjen putnicima. Putnik koji želi da napusti vozilo oslanja se na saradnju onih koji stoje između njega i vrata autobusa. Ako saradnja izostane, on će morati da druge putnike pogura, ili da apeluje na njih da oslobode prolaz, ili oba. Tela u Kavendišinom materijalnom plenumu *ne mogu* da pribegnu guranju i pritiskanju okolnih tela jer mogu da deluju jedino kao okazionalni uzroci. Ako izostane saradnja okolnih tela, izostaće i kretanje. Osim ako je nekako moguće apelovati na druga tela da se sklone i ako je takav apel nešto što drugim telima daje razlog da se sklone.

Hjum je u osamnaestom veku slavno konstatovao kako je činjenica da sa nekoliko izgovorenih reči možemo drugome stvoriti razlog za postupanje „jedna od najmisterioznijih i najnesхватljivijih operacija koje se mogu zamisliti“ (*Rasprava o ljudskoj prirodi*, 3.2.5.14). Može se očekivati da će objašnjavanje prirodnih fenomena koje je oslonjeno na racionalne relacije upućivanja zahteva i postupanja prema zahtevima biti izrazito izazovno. Međutim, verujem da izvesne korake možemo napraviti u pravcu boljeg razumevanja kako bi takve relacije mogle da funkcionišu u Kavendišinoj prirodi. U tom smislu, važno je naglasiti ulogu simpatije. Budući da u telima postoji simpatija kao bazična relacija uzajamne privlačnosti i naklonosti, u njoj bismo mogli da potražimo onu sponu koja pomaže da se između materijalnih entiteta stvaraju relacije normativnog obavezivanja. Ljubav prema bližnjima daje nam *prima facie* razlog da postupamo u skladu sa njihovim željama i potrebama. U pitanju je *prima facie* razlog, budući da postoje okolnosti u kojima će drugi razlozi i norme diktirati uzdržavanje od postupanja u skladu sa željama naših bližnjih. Ljubav prema nekome može nam dati razlog da postupimo požrtvovano, ali ne može nam dati razlog da ubijemo slučajnog prolaznika. Utoliko su delovi prirode, kao i ljudi od kojih bližnji imaju očekivanja i kojima predočavaju zahteve, slobodni da pod određenim uslovima uskrate saradnju okolnim telima.

Na ovaj način shvaćena normativna uloga relacije simpatije pomaže da učvrstimo razumevanje njenog doprinosa uređenosti i jedinstvu prirode. Delovi prirode, kada postupaju iz ljubavi prema drugim delovima, onda reaguju skladno i pravilno. Kada, pak, postupaju iz drugih pobuda—recimo, zbog svog iskvarenog samoljublja—onda se radi o postupanju koje može voditi nepravilnom kretanju, a time i zbrci i neredu. U ovome ponovo možemo uočiti sličnosti u Kavendišinom tretiranju prirodnih i društvenih zajednica materijalnih delova.

Diskusija o ovom aspektu Kavendišinog objašnjenja uzročnosti nipošto nije is-

crpljena. Važno bi bilo razmotriti koju vrstu *obaveze* proizvode razlozi proistekli iz simpatetičke težnje ka usklađenom kretanju. U najvećem broju slučajeva koje srećemo u domenu ljudskih interakcija, radi se o nesavršenim dužnostima, odnosno o dužnostima čije ispunjavanje se može izvršiti na različite načine. Putnik koji u punom autobusu prepozna obavezu da obezbedi drugome prolaz ka vratima, tu obavezu može da ispuni na više različitih načina; na primer, može da i sam privremeno napusti vozilo, ili može da ostane u mestu ali da se potruži da zauzme svojim telom što manji deo puta ka vratima. U prirodi ne izgleda da su uzročne relacije uređene na takav način. Postoje zakonomernosti kretanja materijalnih tela koje možemo izraziti opštim matematičkim jednačinama. Kada bacim kamen, izgleda da je veoma ograničena njegova sloboda u ispunjavanju dužnosti saobražavanja svog kretanja sa kretanjem moje ruke—leteće parabolichnom putanjom i doleteti do udaljenosti koju možemo pouzdano izračunati ako znamo nekoliko podataka o njegovoj težini, brzini, itd. I tako će se dogoditi svaki put. Drugim rečima, ako u prirodi postoje nesavršene dužnosti da se postupi u skladu sa upriličavajućim uzrokom, gotovo nikada ne svedočimo različitosti načina za njihovo ispunjavanje.

Preostao je problem razumevanja vida komunikacije kojim tela postaju svesna nameravanog kretanja sebi susednih tela. Kojim putem može biti *saopšteno* da je određena vrsta kretanja zahtevana od drugog tela? Putnik koji želi da napusti pun autobus mogao bi da se vozi mnogo duže nego što je planirao ako ne bi susednim putnicima preneo svoju želju. Srećom, govor je vrlo efikasno sredstvo međuljudske komunikacije. Kako bi trebalo da komunikacija izgleda u slučajevima okazionalne uzročnosti u kojima nema očiglednog analognog načina da se saopšti namera? U pojedinim slučajevima Kavendiš može da se osloni na to da delovi prirode opažaju kretanja drugih delova prirode i shodno tome prilagođavaju svoje ponašanje.²⁰ Kada zamahnem rukom da bacim kamen koji držim u šaci, može se reći da kamen zahvaljujući putanji i brzini kretanja ruke prepoznaje njen „zahtev“ da sopstvenim kretanjem poleti u određenom pravcu i na specifičan način. Međutim, ima brojnih slučajeva u kojima tela, po svemu sudeći, nemaju na raspolaganju indikatore ove vrste. Da bih pokrenuo kašičicu uronjenu u šolju čaja—ili, bolje rečeno, da bi se sama kašičica, shodno kretanju moje šake, upustila u sopstveno kretanje—neophodno je da dođe do pomeranja tečnosti koja okružuje kašičicu. Ali sada nema nikakvog prethodnog kretanja kašičice koje bi tečnost mogla da uoči i da se pomeri, ustupajući svoje dotadašnje mesto. Za Kavendišinu poziciju ovo je nezanemarljiv problem koji seže u same korene metafizičkih osnova njenog materijalizma. Postoji nada da se i ovde može prizvati relacija simpatije kao deo načelnog rešenja. Naime, Kavendiš naglašava kako je u pitanju relacija koja uspostavlja naročito snažne veze između racionalnih i senzitivnih

²⁰ Okazionalna uzročnost zahteva postojanje opažanja u svakom delu prirode upravo zbog toga što harmonično postupanje upriličavajućih uzroka i samokrećućih posledica zahteva sposobnost ovih drugih da opažaju kretanje prvih.

delova materije.²¹ I upravo je ta veza ono što omogućuje, po Kavendišinom mišljenju, da možemo zaključiti ponešto o mentalnim stanjima drugih ljudi na osnovu fizičkih stanja njihovih tela:

„[K]o može reći da se čovekove misli ne saznaju na osnovu nastrojenosti njegovog tela? Jer kad bi ljudi naučili različita kretanja tela, što bi mogli većim zapažanjem, onda bi lako mogli da nauče i pokrete duha, i da tako spoznaju misli, koje jesu različita kretanja unutar duha, a čije konfiguracije se obično kreću simpatetički sa kretanjima tela“ (PPO 1, 171).

Vidimo da Kavendiš tvrdi kako postoje simpatičke veze između unutrašnjih kretanja konstitutivnih delova materije i spolja vidljivih kretanja čitavog tela kojem pripadaju. Na osnovu toga je, u principu, pažljivom posmatraču moguće da putem opažanja spoljašnjih kretanja delova tela takođe prepozna i kretnje i konfiguracije kao što su misli, koje se odvijaju unutar drugih ljudi i koje nisu dostupne neposrednom uvidu putem opažanja.

Možemo zaključiti da pribegavanje pozivanju na simpatiju pruža Kavendiš ek-splanatornu osnovu za zastupanje paralelnih teza u pogledu prirode i društva kakvo Detlefsen kod nje pronalazi. Istovremeno, to ne znači da su rezultati takvog rasuđivanja nekontroverzni i bez svojih problema. Priroda same relacije simpatije je u mnogome nejasna—ostaje pre svega nerazjašnjeno da li se radi o primitivnoj relaciji ili ne. Takođe, očito je da za Kavendišine teze, i pod pretpostavkom da simpatija igra ulogu kakvu moji predlozi predviđaju, ostaju teški problemi kada su u pitanju manifestacije uzročne relacije na kakve nailazimo u svetu.

6. Zaključak

Nedugo po rođenju Margaret Kavendiš, engleski pesnik Džon Don (John Donne) zabeležio je besmrtnu reči: „Nijedan čovek nije ostrvo, sadržan u sebi ceo; svaki je čovek komadić kontinenta, delić talasa morskijeh (...) svaka ljudska smrt okrnjuje i delić mene, jer nisam od čovečanstva izuzet“.²² Vojvotkinja od Njukasla je tridesetak godina kasnije formulisala metafizičku i prirodnofilozofsku predstavu o svetu koja Donov sentiment proširuje na celokupnu prirodu. Ne samo da nijedan čovek nije ostrvo, izolovan od ostatka čovečanstva i ravnodušan na sudbinu drugih—nijedno materijalno telo to nije. U okviru prirode postoji osnovna veza između *svih* materijalnih bića ostvarena kroz relaciju simpatije. Utoliko je svako od nas povezan sa sopstvenim okruženjem, ali i sa celokupnom beskrajnom prirodom i svakim prirodnim stvorenjem. Ova veza, zahvaljujući jednoobraznoj materijalnoj prirodi čitave stvarnosti, prožima sve segmente beskrajnog tela prirode, uključujući i složene ljudske institucionalne tvorevine. Kavendiš zato može da primenjuje uvide iz jednog domena stvarnosti u

²¹ Videti, recimo, (GNP, 63).

²² Prevod Donovih *Molitvi u teškim časovima* Zorana Janjića, u izdanju Alexandria Press (2008), str. 127.

drugim domenima, i da tako iznalazi originalne vidove objašnjavanja i u prirodnoj i u političkoj filozofiji. Dobijamo tako argumente o atomima zasnovane na političkim relacijama, svakodnevne uzročne relacije rasvetljene po modelu racionalnih razmenjivanja razloga, i fundamentalne društvene odnose potekle od metafizičkih činjenica koje zasnivaju uređenost stvarnosti. Pa ipak, rezultat moje diskusije, osim pojedinačnih zaključaka o navedenim eksplanatornim relacijama, jeste i uvid da, za Kavendiš, postoje faktori koji uslovljavaju diskontinuitet između prirodnih i društvenih struktura. Glavni među ovim faktorima je čovek, odnosno ljudska priroda—ne kao izuzetna, *sui generis* pojava u stvorenom svetu, već kao vrsta sposobna za svest o sopstvenoj smrtnosti i posledičnu težnju ka besmrtnosti kroz slavu. Radi se o elementu diskontinuiteta koji nas sprečava da bez zadržke prenosimo teze o prirodi u domen društva i obratno, ali koji je istovremeno sopstvenom naravi neraskidivo vezan za ostatak prirodnog poretka, ujedinjenog simpatijom u skladnu celinu.²³

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²³ Ovaj rad je nastao u sklopu projekta Ministarstva za nauku, tehnološki razvoj i inovacije br. 451-03-33/2026-03/200163, a prezentovan je na Međunarodnoj filozofskoj školi „Felix Romuliana“ u septembru 2026.

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Miloš Vuletić

Nature, Man, and the State: Margaret Cavendish on the Continuity Between Natural and Social Order

(Summary)

This paper examines whether and to what extent it is justified to interpret Margaret Cavendish as freely applying insights from natural philosophy to moral and political philosophy, and vice versa. Two examples of such transfer of theses and arguments are examined, and it is established that Cavendish proceeds in such a way on principled

grounds. At the same time, there are limits to the extent to which an analogy between natural and social orders can be applied, most of all due to her views on human nature. These limits do not jeopardize the grounding of the analogy in the relation of sympathy which is instantiated throughout nature, including human societies. It is shown that the relation of sympathy, modeled on human relations, can provide a basis for potential solutions of some problems that arise for Cavendish's theory of causation.

KEYWORDS: Margaret Cavendish, nature, sympathy, occasional causation.

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Filip D. Jevtić

The Geometry of Grace: Measure and Happiness in St. Augustine's De beata vita

ABSTRACT: *St. Augustine's De beata vita has traditionally been read as a Christian modification of Cicero's Tusculanae Disputationes and Plotinus's Ennead I.4. However, a close reading of the dialogue's mathematical ontology suggests Augustine is doing more than syncretism. By defining God as the Summus Modus (Supreme Measure), Augustine seeks to solve the central problem of ancient ethics: the permanence of happiness amidst physical instability. This paper argues that Augustine addresses the Plotinian challenge not by psychological resilience, but by defining happiness as a geometric 'circumscription' by the Divine, a state of objective reality independent of sensory fluctuation.*

KEYWORDS *Augustine, De beata vita, Plotinus, Measure, Trinity, Divine Geometer*

1 INTRODUCTION

St. Augustine's *De beata vita* is principally informed and modelled on Cicero's *Tusculanae Disputationes* and the ideas of Plotinus, expressed in *Ennead I.4*. Yet, the dialogue represents a radical departure from its predecessors through its introduction of a Trinitarian ontology of 'Measure' (*modus*). While critics often argue that Augustine fails to adequately address the issue to which Plotinus devotes most of his work—how is it that happiness, once obtained, doesn't cease under severe physical challenges—this paper proposes that Augustine offers a *geometrical* rather than psychological solution.

By identifying Happiness with the possession of the Supreme Measure, Augustine argues for an objective *mathematical realism* of the soul¹. In the paper cited, Zwier describes Augustine's mathematical realism as the conviction that mathematical truths are not human inventions but discoveries of divine thoughts. Extending this to ethics, Augustine treats the *happy life* not as a subjective emotional state, which would be vulnerable to physical pain, but as an objective alignment with the Divine Measure. Just as the mathematical truth $7+3=10$ remains valid even when the thinker is asleep or in pain, the soul that possesses the *Summus Modus* retains its happiness as an ontological fact, independent of sensory fluctuation.

2 HAPPINESS LITERATURE

St. Augustine's *De beata vita* belongs to the so-called *happiness literature* of

¹Zwier, Paul. Augustine's Mathematical Realism. ACMS Journal, 2004.

Antiquity, the subject of which is the good, i.e., *eudaimonic* life.

While the earliest examples might be taken from the dialogues of Plato and especially Aristotle's lost dialogues on the happy life, as well as his *Nicomachean Ethics*, it is among the Stoics that this literature flourished most—Seneca's *De vita beata*, which Augustine's work references directly in its title, being the most famous. Augustine was also aware and influenced by the works of Cicero (*Hortensius*, *De Finibus Bonorum et Malorum*, *Tusculanae Disputationes*, *De Officiis*) and Plotinus (*Ennead* I.4).

Augustine's engagement with the tradition of happiness literature is in line with the interests of early Christian authors of his time. Elements of it are present in Lactantius's *Institutiones Divinae*, and Augustine's friend and a person instrumental to his conversion to Christianity, St. Ambrose of Milan, wrote *De Iacob et vita beata*.

3 SETTING OF THE DIALOGUE

The dialogue takes place over a three day period starting on November 13, 386—the day of St. Augustine's 32nd birthday. The banquet Augustine has prepared, both for body and soul, is celebrating not only his birthday, but more importantly, his and his son's conversion.

Present are Augustine's mother, St Monica; his brother Navigius; his son, Adeodatus; his uneducated cousins Lartidianus and Rusticus; his students, Tygestius and Licentius.

Though in a form of a dialogue, the work is more a series of monologues. There is no real opposition to St. Augustine's exposition and the work is more akin to a protrepsis than a Platonic dialogue.

4 OUTLINE OF THE DIALOGUE

4A PROLOGUE: THE MOUNTAIN OF PRIDE

St. Augustine opens with common imagery of *three classes of seafaring men* (taken from Seneca), Philosophy as patron, and *Philosophy as a harbour*. He, however, modifies the common image in the light of Christianity by describing the *Port of Philosophy* as *wide open*—the true safety does not lie there but further beyond. Moreover, before the port stands *a mountain that should be feared most intensely and avoided most carefully*—Mountain of the Academicians (of the Third, i.e., Sceptical, Academy).

Crucially, this mountain is not merely an epistemological error but a moral one. The Skepticism of the Academics is rooted in *superbia* (pride)—a refusal to accept a 'Measure' outside of the human mind². By insisting on probability rather than Truth, the Academics attempt to be the measure of themselves, resulting in the vainglory that

²Curley, Augustine J. Augustine's Critique of Skepticism: A Study of Contra Academicos. Peter Lang, 1996.

Augustine warns against.

Augustine also gives brief information on his philosophical biography:

- his falling in love with philosophy upon reading Cicero's *Hortensius*;
- his experience among Manicheans;
- his experience with the Academics;
- his (limited) exposure to works of Plotinus.

4B PROPOSING A FEAST FOR THE SOUL

If food nourishes the body, '*the soul is nourished by nothing other than understanding and knowledge*'. For greater emphasis, St. Augustine says this through the mouth of his mother. At several other points in the dialogue he uses her to the same effect as, besides himself, obviously, she is praised as having '*utterly mastered the very citadel of philosophy*'. In general, St Monica serves as a kind of assistant to Augustine, representing the faculty of *memoria*—the intuitive hold on God that precedes dialectical articulation³.

This dynamic establishes a critical epistemological contrast between the two: Monica represents a form of *theological intuitionism*, grasping the Truth immediately through faith and memory. Augustine, conversely, embodies *philosophical realism* or dialectic—he must laboriously construct the path to Truth using logical steps. While *Augustine the Geometer* uses the compass to circumscribe the Truth, Monica is already standing at the centre. She possesses the *res* (reality) without needing *signa* (signs) of logical demonstration, foreshadowing what we will claim later on: happy life is an ontological position rather than a mere intellectual achievement.

4C PERMANENCE OF HAPPINESS

St. Augustine follows a standard Stoic argument, almost verbatim as that of Cicero, that happiness must be independent of vicissitudes of Fortune and any external good. Happiness is to be found within; it is obtainable only by the wise. Moreover, the claim is stronger: the wise are necessarily happy, and those who are not happy are not wise.

As happiness cannot depend on anything external, and as it is impossible while there exists unfulfilled needs, the question arises what needs need to be met, i.e., what is it that one must possess, in order to achieve happiness. Whereas the Stoic answer would be wisdom, obtained via self-administering the medicine of Philosophy, Augustine, as a Christian, cannot accept such a view of a self-sufficient man. Therefore, his answer must necessarily be: *to have God*.

4D TO HAVE GOD

Three answers to the question 'What it means to have God' are considered.

- Licentius: He has God who lives well.
- Trygetitus: He has god who does what God wants done.
- Adeodatus: He has God who doesn't have an unclean spirit.

³Hochschild, Paige Evelyn. *Memory in the Theological Anthropology of St. Augustine*. Durham University, 2005.

These options are quickly reduced to 'He has God who lives well', where *living well* is nothing other than doing what God wants. This however proves unsatisfactory for answering the initial question of happiness. The problem, as Augustine reveals, is that these definitions remain circular and lack ontological precision. To define 'having God' merely as 'living well' or 'doing God's will' begs the question of what constitutes the *standard* or *measure* of that well-being. Without an independent criterion, 'living well' is susceptible to subjective interpretation (the error of the Stoics) or infinite regress. Augustine seeks a definition that is not merely performative (what one does) but constitutive (what one *is*). Thus, the dialogue must push beyond ethical behaviour to the metaphysical condition that makes such behaviour possible: the possession of Wisdom itself.

4E NEED AND THE APERION

Previous attempts having failed, the discussion returns to the beginning: happiness is the lack of need. The position that '*there's no middle ground between being miserable and being happy*' is reiterated.

'Then neediness of the mind', I said, 'is nothing else but folly [...]'

Following Crombie's analysis of Platonism, Augustine's concept of 'need' (*textitategestas*) should be understood as the *Apeiron* (ἄπειρον) or the *Unlimited Dyad*—a state of fluctuating 'more and less' without a defined limit⁴. In Platonic metaphysics, the *Apeiron* represents the principle of the 'Great and Small', a continuum of indefinite possibility (hotter/colder, larger/smaller, etc.) that lacks distinct being until it is determined by Limit, i.e., *Peras* (πέρας). For Augustine, the foolish soul is trapped in this bad infinity; it is 'needy' not because it lacks external goods, but because it lacks internal definition. It flows into the 'more and less' of desire without a stopping point. Consequently, the opposite of this chaotic flux is not merely satisfaction, but Fullness, i.e., Plenitudo, which implies the imposition of a limit or boundary. To be 'full' is to be 'measured'—to have reached the integer of one's being where the fluctuation of the Dyad ceases.

4F METAPHYSICS OF MEASURE

The condition of happiness has now changed from *having God*, to *having Wisdom*, and wisdom is inseparable from *Measure*.

'The measure of the mind, then, is wisdom... Wisdom, therefore, is fullness. Moreover, in fullness there is measure, and thus the mind's measure is in wisdom.'

This identification of Wisdom with Measure draws deeply from the late Platonic tradition, particularly the metaphysics of the *Philebus*. As Gosling and Crombie observe, for the later Plato, the Good is not merely an abstract Form but the rational order imposed by the Demiurge upon the chaotic *chora* (χώρα)⁵. To be 'good' is to

⁴Crombie, I. M. *An Examination of Plato's Doctrines: II. Plato on Knowledge and Reality*. Routledge, 1963.

⁵Gosling, J.C.B. *Plato*. Routledge & Kegan Paul, 1973.

be a constituent of the intelligible understructure of the universe; it is to possess the correct ratio or *logos* (λόγος) that grants intelligibility to existence. Augustine adopts this cosmological principle and internalizes it: the soul, like the cosmos, requires the imposition of Limit to escape the non-being of the *Apeiron*. Just as the universe is only comprehensible insofar as it is ordered by the Demiurge, the soul is only ‘happy’—and indeed, only truly exists—insofar as it mirrors this divine order. Happiness, then, is the state of being ‘geometrized’ by God—of becoming a microcosm that perfectly reflects the macrocosmic Limit.

Here Augustine advances a strictly Trinitarian metaphysics, which Schmaus identifies as the core of his theological anthropology⁶.

- **The Father as Summus Modus:** God the Father is the Supreme Measure, the Limit from which all reality proceeds. He corresponds to the Platonic *Peras*.
- **The Son as Veritas:** The Son is the Truth that is ‘born’ of the Measure. He is the Form that reveals the Limit.
- **The Spirit as Admonitio:** The Holy Spirit is the internal admonition that leads the soul from the distraction of the world back to the Truth and the Measure.

Finally, happiness can be said to be the perpetual enjoyment and unity with the Truth—that is Christ⁷—which anchors the soul in the Supreme Measure.

5 DIVINE GEOMETER

A common objection to Augustine's early dialogues, often raised in comparison to Plotinus's *Ennead* I.4, is the problem of physical suffering. If happiness is a connection to Truth, does severe pain or unconsciousness break this connection? Plotinus argues the Sage remains happy even in the Bull of Phalaris because his happiness is internal. Augustine seems, at first glance, to evade the question.

However, if we take Augustine's mathematical realism seriously, the objection dissolves. Augustine views mathematical truths (such as $7+3=10$) as independent of the human mind⁸, a view shared by many Church Fathers, such as St Maximus the Confessor⁹. Similarly, Augustine defines Happiness not as a psychological feeling, but as an ontological *status*—the status of being ‘measured’ by God.

This concept finds its visual correlative in the later Christian iconographic tradition of the *God the Geometer*¹⁰. In these images, the Creator is depicted not merely as a King, but as an Architect wielding a compass.

This iconography, to an extent, visualizes exactly what Augustine attempts to

⁶Schmaus, Michael. *Die Denkform Augustins in seinem Werk De trinitate*. Verlag der Bayerischen Akademie der Wissenschaften, 1962.

⁷John 14:6

⁸Zwier, Paul. *Augustine's Mathematical Realism*. ACMS Journal, 2004.

⁹In line with similar approaches to integration of Plato's philosophy into Christianity, for Maximus the Confessor, mathematical world, is part of ‘*perfect and eternal thoughts of the eternal God*’.

¹⁰The idea is present already in Wisdom (of Solomon) 11:20.

articulate in *De beata vita*.

- **The Fixed Point:** The static leg of the compass represents the Father (*Summus Modus*), the unmoving source of all stability.
- **The Moving Leg:** The rotating leg represents the Son (*Veritas*), who goes out to define the form of reality.
- **The Circle:** The resulting shape is the Holy Spirit's domain—the *boundary* within which the soul finds rest.

It is not that the soul becomes oblivious to suffering; rather, once 'measured', the struggles that the flux of the *Apeiron* brings about are experienced on a different ontological plane than the measure of being imposed by knowing God. Pain exists in the unmeasured void *outside* the circle (the *terra inhabitabilis*), a region of pure becoming and dissolution. The Happy Life, conversely, is not the absence of pain in that void, but the *ontological security* of remaining within the compass's arc. Grace functions as an 'inner turn' from the *Apeiron* of the senses to the geometry of the soul¹¹. Thus, the sleeping or suffering Sage remains happy not because he *feels good*, but because he *is measured*; his being is held by the Supreme Measure, a fact as objective and immutable as a geometric Truth.

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¹¹Curley, Augustine J. *Augustine's Critique of Skepticism: A Study of Contra Academicos*. Peter Lang, 1996.

Filip D. Jevtić

Geometrija Blagovesti: mera i sreća u *De beata vita* Sv. Avgustina

(Rezime)

De beata vita Sv. Avgustina se tradicionalno tumači kao hrišćanska modifikacija Tusculanae Disputationes Cicerona i Ennead I.4 Plotina. No, pažljivo čitanje dijalogue matematičke ontologije sugerise da Avgustin čini nešto više on običnog sinkretizma. Braneći Boga kao Summus Modus (Vrhovna Mera), Avgustin traži rešenje centralnog problema antičke etike: permanentnost sreće u nestabilnosti fizičkog sveta. Ovaj rad argumentuje da Avgustin odgovara na Plotinovski izazov ne pozivanjem na psihološku otpornost, već definisanjem sreće kao geometrijskog 'zaokruživanja' Božanstva, kao stanje objektivne realnosti nezavisne od čulnih fluktuacija.

KLJUČNE REČI: Avgustin, *De beata vita*, Plotin, mera, Trojstvo, Božanski Geometar

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HJUMOVE POSLEDNJE REČI

Povodom dvesta pedeset godina od Hjumove smrti

SAŽETAK: Dejvid Hjum je neposredno pre smrti 1776. godine napisao novi pasus u posljednjem delu Dijaloga o prirodnoj religiji. U tom pasusu skeptik Filon tvrdi da je rasprava između teiste i ateiste samo verbalna jer obojica prihvataju da postoji nekakva analogija između uzroka uređenosti u svetu i ljudske inteligencije. U ovom radu sprovodim detaljnu tekstualnu analizu kako bih pokazala da taj novi pasus, prvo, snažno sugerise da takozvani Filonov preokret ne predstavlja drastičnu promenu njegovog mišljenja i, drugo, služi kao ključ za tumačenje Filonovog poslednjeg govora u Dijalogima. Ova analiza takođe utvrđuje da je Hjumovo stanovište, i pored pomirljivog tona koji se javlja u novom pasusu, prevashodno skeptičko.

KLJUČNE REČI: Dejvid Hjum, teizam, ateizam, skepticizam

1. Uvod: šta je Hjum poslednje napisao?

Početkom 1776. godine Dejvid Hjum (David Hume, 1711–1776) je nesumnjivo bio svestan činjenice da je njegova tadašnja bolest neizlečiva i u januaru je sastavio novi testament.¹ Njegova sačuvana prepiska svedoči o tome da se do smrti 25. avgusta iste godine bavio revidiranjem svojih spisa. Iako tokom nešto manje od poslednjih petnaest godina svog života nije napisao ništa novo, Hjum se u nekoliko navrata vraćao svojim prethodno objavljenim delima i prerađivao ih u sklopu priprema novih izdanja za štampanje. Ove izmene su uglavnom obuhvatale popravljanje pravopisa, preraspoređivanje pojedinih delova tekstova i dodavanje napomena. Jedini novi spis koji je Hjum napisao 1776. godine je kratka autobiografija pod naslovom „Moj život“, za koju je želeo da bude objavljena na početku prvog izdanja njegovih dela koje će biti štampano nakon njegove smrti.² Izuzimajući ovaj tekst, poslednje delo koje je Hjum napisao je drugi tom *Istorije Engleske*, koji je objavljen 1762. godine. Međutim, nešto je teže utvrditi koje je njegovo poslednje filozofsko delo. S jedne strane, izgleda da je zbirka eseja *Četiri disertacije*, koja je objavljena 1757. godine, najbolji kandidat za tu titulu, pri čemu iz Hjumovog pisma Vilijamu Strahanu (William Strahan) iz januara 1772. godine saznajemo da je poslednji bio napisan esej „O merilu ukusa“

¹ Realizaciju ovog istraživanja finansijski je podržalo Ministarstvo nauke, tehnološkog razvoja i inovacija Republike Srbije u sklopu finansiranja naučnoistraživačkog rada na Univerzitetu u Beogradu – Filozofskom fakultetu (broj ugovora 451-03-33/2026-03/ 200163).

² Ova Hjumova želja nije uslišena. „Moj život“ se pojavio već u januarskom izdanju *Škotskog magazina* 1777. godine, a zatim ga je i Hjumov izdavač objavio u zasebnom pamfletu pre novog izdanja njegovih dela (Rasmussen 2017: 218).

(HL 2: 253). Komplet spisa pod naslovom *Eseji i rasprave o različitim temama*, koji je u konačnom obliku prvi put objavljen 1758. godine, nije sadržao ništa napisano nakon *Četiri disertacije*, a nakon njega su štampana samo nova izdanja Hjumovih ranije objavljenih dela. S druge strane, izgleda da imamo dobre razloge da tvrdimo da su zapravo *Dijalozi o prirodnoj religiji* (u daljem tekstu: *Dijalozi*), Hjumovo poslednje objavljeno delo, ujedno i poslednje filozofsko delo koje je on *završio*.

Priča o nastanku i objavljivanju *Dijaloga* je veoma interesantna. Hjum je prvu verziju ovog dela napisao 1751. godine, nakon čega je načinio značajne izmene 1757. godine i manje ispravke i dopune 1761. godine, da bi konačno 1776. godine, neposredno pre smrti, ponovo posvetio pažnju istom rukopisu.³ On je uvažavajući savete nekoliko svojih prijatelja i usled objektivnih okolnosti⁴ odustao od objavljivanja *Dijaloga* za vreme svog života, ali je preduzeo sve neophodne mere kako bi osigurao da će ovo delo biti posthumno objavljeno. Hjum je tokom poslednjih nekoliko meseci svog života najpre neuspešno pokušavao da ubedi Adama Smita (Adam Smith) da objavi *Dijaloge* a zatim je taj zadatak poverio Strahanu, ali je u dodatku testamentu koji je sačinio 7. avgusta 1776. godine zahtevao da, ukoliko *Dijalozi* ne budu objavljeni u roku od dve i po godine nakon njegove smrti, rukopis bude vraćen njegovom bratancu radi objavljivanja (HL 2: 453). Naposletku je tri godine nakon Hjumove smrti, 1779. godine, upravo Hjumov bratanac Dejvid Hjum Mlađi objavio prvo izdanje *Dijaloga*.⁵

³Nemamo tačne podatke o tome kada je Hjum počeo da piše *Dijaloge* i kada je završio prvu verziju. U pismu iz marta 1751. godine u kojem od Gilberta Eliota od Minta (Gilbert Elliot of Minto) traži pomoć u vezi sa jednim argumentom iz *Dijaloga*, on svom prijatelju kaže da nema potrebe da ide dalje od trećeg dela kako bi mu pomogao (HL 1: 157). Na osnovu toga znamo da je rukopis u tom trenutku sadržao više od tri dela i, budući da je ceo spis relativno kratak, gotovo je izvesno da je bio dovršen iste godine. Moguće je da je Hjum zatim 1757. godine prerađivao *Dijaloge* zato što je želeo da ih objavi kao nastavak *Prirodne istorije religije* koja je te godine objavljena u okviru *Četiri disertacije*. On je potom 1762. godine načinio još neke izmene i možda je i tada razmišljao o objavljivanju ovog dela ali, kako do toga nije došlo, on po sopstvenom priznanju narednih petnaest godina nije ponovo prolazio kroz rukopis (HL 2: 334) sve dok mu se nije posvetio neposredno pre smrti. Detaljnije informacije o datiranju različitih delova rukopisa *Dijaloga* videti u Stewart 2000: odeljak 4.

⁴Od kada je objavio *Raspravu o ljudskoj prirodi* Hjum se suočavao sa optužbama za ateizam, koje su često cirkulisane putem anonimnih letaka i koje su svojevremeno doprinele tome da ne dobije posao ni na Univerzitetu u Edinburgu ni na Univerzitetu u Glazgovu. Ove optužbe su 1755. i 1756. godine poprimile još ozbiljniji ton. Pre nego što je objavio *Četiri disertacije*, Hjum je nameravao da objavi zbirku od pet eseja među kojima su bili i eseji „O samoubistvu“ i „O besmrtnosti duše“. On je ipak povukao ta dva eseja iz štampe (objavljeni su tek posthumno 1777. godine), ali je neobjavljeni rukopis *Pet disertacija* dospao do ruku biskupa od Glostera koji se zalagao za to da Hjum bude krivično gonjen zbog bogohuljenja ako ih objavi. Pored toga, u to vreme se konzervativna struja unutar Škotske crkve udružila u pokušaju da pred Generalnom skupštinom Hjuma označi kao nevernika i da ga ekskomunicira iz crkve. Umereno sveštenstvo, među kojim je bilo Hjumovih prijatelja, pobrinulo se da do toga ne dođe, ali je Hjum bio svestan činjenice da mu objavljivanje još jednog spornog dela nikako ne bi išlo u prilog. Više o ovom burnom periodu Hjumovog života i optužbama sa kojima se suočavao videti u Mossner 1980: poglavlja 24 i 25.

⁵Za više detalja o Hjumovim pokušajima da ubedi najpre Smita a potom i Strahana da objavi *Dijaloge*

Budući da je planirao da *Dijalozi* budu objavljeni, Hjum se u poslednjim nedeljama života, onda kada je imao snage za to, bavio pripremanjem rukopisa za štampu te je pre svega radio na usklađivanju pravopisa, povezivanju umetnutih delova teksta, numerisanju stranica i pisanju napomena za izdavača, kao što je to činio i za druga dela pre puštanja u štampu. Međutim, posebno je fascinantno i značajno to što je Hjum 1776. godine takođe napisao i jedan sasvim nov pasus u poslednjem odeljku *Dijaloga*. Budući da do smrti nije stigao da unese sve prepravke koje je želeo i da je taj posao završio njegov bratanac, možemo da zaključimo da je pisanje tog pasusa za Hjuma predstavljalo prioritet i da su u njemu sadržane Hjumove poslednje reči iz filozofije. Stoga se prirodno nameće pitanje značenja i značaja tih reči, odnosno, pitanje šta je Hjum njihovim dodavanjem hteo da postigne i kakav je njihov uticaj na naše razumevanje *Dijaloga* u celini. U nastavku rada ću nastojati da pružim odgovor na ova pitanja tako što ću prvo objasniti kontekst u kojem se novi pasus javlja u *Dijalozima*, zatim ću detaljnom analizom teksta pokazati na koji način je on povezan sa drugim važnim delovima ovog spisa i na kraju ću tvrditi da taj pasus, i pored naizgled pomirljivog tona koji se u njemu javlja, podržava skeptičko čitanje *Dijaloga*.

2. Filonov preokret i interpretacija dela XII *Dijaloga*

Pasus koji je Hjum napisao 1776. godine je najduži pasus u *Dijalozima* i sastoji se od 628 reči na engleskom jeziku. On se nalazi u zaključnom, dvanaestom delu *Dijaloga* koji i dalje izaziva razilaženja među interpretatorima. Demeja, predstavnik religiozne ortodoksije, odlazi na kraju dela XI i u razgovoru u delu XII učestvuju samo dvojica glavnih protagonista *Dijaloga*, eksperimentalni teista Kleant i filozofski skeptik Filon. Njih dvojica su bliski prijatelji – Kleant se jednom prilikom Filonu obraća sa „moj genijalni prijatelju“ (DNR 3.9, KS 155), a Filon kaže da sa Kleantom deli „bezrezervnu prisnost“ i veruje da se njih dvojica dobro razumeju (DNR 12.2, KS 214). Ovaj detalj je važan jer se odmah po Demejinom odlasku ton njihovog razgovora menja. Najveći deo diskusije u prethodnih jedanaest delova *Dijaloga* bio je posvećen argumentu o osmišljenosti⁶, pri čemu je Kleant izlagao i branio različite verzije ovog argumenta, dok ih je Filon podvrgnuo oštroj i razornoj kritici. Međutim, na početku poslednjeg dela izgleda da Filon zastupa sasvim drugačije stanovište kada za sebe kaže da, uprkos tome što uživa u filozofskim raspravama o religiji, „niko nema dublji osećaj za religiju utisnut u svoj um niti iko izražava postojanije divljenje prema božanskom biću dok se ono, kroz neobjašnjive izume i veštine prirode, otkriva ljudskom razumu.“ (*ibid.*) On navodi brojne primere svrhovitosti u prirodi i tvrdi da nas „sve nauke gotovo neosetno navode da priznamo prvog inteligentnog tvorca“ (DNR

nakon njegove smrti, kao i o tome zašto nijedan od njih to nije hteo da učini, videti Rasmussen 2017: poglavlje 10.

⁶„Argument o osmišljenosti“ je moj prevod sa engleskog jezika izraza „the argument from design“, što je drugi naziv za teleološki argument za postojanje Boga.

12.2, KS 215). Drugim rečima, deluje da Filon sada prihvata upravo onaj argument koji je prethodno kritikovao i zbog toga je ovo mesto iz *Dijaloga* u literaturi poznato kao „Filonov preokret“. Doduše, važno je primetiti da izgleda da Filon u delu XII *Dijaloga* prihvata da pomoću argumenta o osmišljenosti možemo da dokažemo božje prirodne attribute, i to pre svega atribut inteligencije, ali da je i dalje skeptičan u pogledu mogućnosti da Bogu pripišemo moralne attribute.⁷ Imajući u vidu da je Filon ličnost koju većina interpretatora uzima za Hjumovog glavnog zastupnika u *Dijalozima* i da je to pretpostavka koju i ja usvajam u ovom radu⁸, već ovo njegovo popuštanje pred Kleantom deluje dovoljno ozbiljno i navodi nas da se zapitamo kakva je njegova krajnja pozicija kada je reč o argumentu o osmišljenosti, to jest, kakvo je njegovo gledište o uspešnosti projekta prirodne teologije shvaćenog kao pokušaj da se religija racionalno zasnuje.

Nakon Filonovog neobičnog iskazivanja pobožnosti, diskusija u poslednjem delu *Dijaloga* se nastavlja i ubrzo postaje očigledno da je on i dalje veoma kritički nastrojen prema religiji. On se ne slaže ni sa Kleantovim uverenjem da je „religija, ma koliko bila iskvarena, još uvek bolja od potpunog izostanka religije“ niti sa njegovim tvrđenjem da je vera u zagrobni život „tako snažan i neophodan oslonac za moral“ (DNR 12.10, KS 219), a u razgovoru koji sledi i koji zaprema gotovo polovinu dela XII *Dijaloga* njih dvojica raspravljaju o praktičnim posledicama religije i Filon bez zadržke ukazuje na vrlo štetan uticaj pre svega organizovane religije na ljudsko društvo. Međutim, on takođe tvrdi da istinska religija nema tako pogubne posledice (DNR 12.22, KS 223), a u svom čuvenom poslednjem govoru u *Dijalozima* iznosi sledeće hipotetičko tvrđenje:

„Ako se čitava prirodna teologija [...] svodi na jedno jednostavno, iako donekle nejasno ili barem nedefinisano tvrđenje, da *uzrok ili uzroci reda u svetu verovatno stoje u nekoj udaljenoj analogiji sa ljudskom inteligencijom*; ako ovaj iskaz ne može da se proširi, izmeni ili detaljnije objasni, ako ne dozvoljava nikakav zaključak koji utiče na ljudski život i ako ne može da bude izvor nikakvog delanja niti uzdržavanja; i ako ta analogija, nesavršena kakva jeste, ne može da se proširi ni na šta dalje od ljudske inteligencije i ako ne može da se prenese, sa bilo kakvim izgledom verovatnoće, na druga svojstva uma; ako je ovo stvarno slučaj, šta onda više može najzapitaniji, najzamišljeniji i najreligiozniji čovek da učini

⁷ On tvrdi da „kako prirodne tvorevine imaju mnogo veću sličnost sa posledicama *naše* umešnosti i domišljatosti, nego sa posledicama *naše* dobronamernosti i pravednosti, imamo razloga da zaključimo da prirodne osobine božanstva imaju veću sličnost sa takvim osobinama čoveka, nego što njegov moralni karakter ima sa ljudskim vrlinama.“ (DNR 12.8; KS 219, Hjumov kurziv) Doduše, Filon na tom mestu kaže da to samo dokazuje da su naši moralni atributi manje savršeni od naših prirodnih atributa, ali ipak ne pokazuje nikakve naznake da odustaje od svoje kritike različitih rešenja teološkog problema zla iz delova X i XI *Dijaloga*.

⁸ Tezu da je Filon Hjumov primarni zastupnik u *Dijalozima* ne mogu da branim u ovom radu, ali sam to prethodno uradila u Davidović 2021.

nego da se čisto, filozofski saglasi sa ovim tvrđenjem [...] i da veruje da argumenti, koji ga podupiru, prevazilaze prigovore, koji su upereni protiv njega?“ (DNR 12.33, KS 227, Hjumov kurziv)

Izgleda da Filon na samom kraju *Dijaloga* prihvata da pomoću argumenta o osmišljenosti možemo da dođemo do zaključka da uzrok ili uzroci reda u svetu verovatno stoje u nekoj udaljenoj analogiji sa ljudskom inteligencijom. Istina je da on navodi veliki broj uslova koji se tiču toga kako treba da shvatimo navedeno tvrđenje da bi ga on prihvatio, ali ga naposljetku ipak prihvata. Ovakav kraj *Dijaloga* je naveo jedan broj interpretatora da tvrdi da Filona, pa stoga ni Hjuma, ne možemo da okarakterišemo kao ateistu već da bi trebalo da mu pripišemo neku vrstu teizma.⁹ S druge strane, brojni interpretatori smatraju da Filonov preokret, kada se ispravno shvati, ne predstavlja stvarni ustupak Kleantu te da Filon i na kraju *Dijaloga* ostaje pri svojim prvobitnim kritikama argumenta o osmišljenosti. Ovi autori Filonovo pa stoga i Hjumovo stanovište obično nazivaju ateističkim.¹⁰ Da bismo odgovorili na pitanje kakva je zapravo Filonova pozicija na kraju *Dijaloga* potrebno je da razumemo sa čime se on tačno saglašava kada prihvata navedeno „donekle nejasno ili barem nedefinisano tvrđenje“, a da bismo to razumeli treba da se vratimo na početak dela XII *Dijaloga* i da vidimo kako sam Filon objašnjava ono što se danas naziva njegovim preokretom. Videćemo da se ključni putokazi za razumevanje kraja *Dijaloga* nalaze upravo u poslednjem pasusu koji je Hjum napisao neposredno pre smrti.

3. Verbalne rasprave i spor između teiste i ateiste u *Dijalozima*

Zanimljivo je da se Filon na početku poslednjeg dela *Dijaloga* ne ponaša kao da je iznenada rekao nešto što od njega ne bismo očekivali. Nastavljajući diskusiju sa Kleantom on primećuje da se i u filozofiji i u teologiji često javljaju verbalne rasprave (DNR 12.7, KS 217). Kao jedan primer takve rasprave on navodi, prvo, da mi uočavamo

⁹Pod teizmom mislim na gledište da postoji bar jedno inteligentno biće koje je uzrok uređenosti u svetu. Hjum svakako nije zastupao jače verzije teizma koje podrazumevaju da postoji personalni Bog i posebno providenje, tako da se vrsta teizma koja mu se pripisuje često naziva i deizmom. Verovatno je najpoznatiji među autorima koji brane ovakvu vrstu interpretacije Gaskin (John C. A. Gaskin), koji u knjizi *Hjumova filozofija religije* Hjumu pripisuje „ublaženi deizam (*attenuated deism*)“, poziciju koja je „između deizma i ateizma“ i koja proističe iz osećanja osmišljenosti a „slabu racionalnu osnovu joj pruža uviđanje da bi red koji nalazimo u prirodi *mogao* (ne i morao) da se objasni kao delo jednog uređivača.“ (Gaskin 1988: 221, Gaskinov kurziv) Pored Gaskina, ublaženi deizam u skorije vreme i na osnovu drugačije argumentacije Hjumu pripisuje i Lorkovski (Lorkowski 2014).

¹⁰Pod ateizmom ne mislim samo na gledište da ne postoji inteligentno biće koje je uzrok reda u svetu, nego i na gledište da pomoću argumenta o osmišljenosti ne možemo da dođemo čak ni do minimalno obavezujućeg teističkog zaključka. Dok je ateizam u prvom smislu dogmatsko gledište, ateizam u drugom smislu obuhvata i agnosticizam i skepticizam. Autori koji Hjuma smatraju ateistom ne poriču da Filon na kraju *Dijaloga* potvrđuje „da uzrok ili uzroci reda u svetu verovatno stoje u nekoj udaljenoj analogiji sa ljudskom inteligencijom“, već svoju karakterizaciju Hjumove pozicije pravdaju time da je Filonova koncepcija tog uzroka ili tih uzroka oslobođena svakog religioznog značaja. Za ovu vrstu tumačenja videti, na primer, Cordry 2011, Lemmens 2012 i Russell 2021.

da postoji analogija između prirodnih i ljudskih tvorevina na osnovu čega zaključujemo da i između njihovih uzroka postoji srazmerna analogija, a pošto uočavamo i velike razlike između tih tvorevina i njihovim uzrocima treba da pripišemo srazmerne razlike. Filon zatim tvrdi da se na ovaj način „postojanje božanstva jasno utvrđuje pomoću razuma, i ako neko postavi pitanje da li, zbog ovih analogija, možemo ispravno da ga nazovemo *umom* ili *inteligencijom*, uprkos ogromnoj razlici za koju se razumno može pretpostaviti da postoji između njega i ljudskih umova; šta je to ako ne puki verbalni spor?“ (DNR 12.6, KS 216–217, Hjumov kurziv) Treba priznati da nije sasvim jasna poenta ovog primera. Filon ponavlja ono na čemu je u prethodnim delovima *Dijaloga* insistirao – između delova prirode i naših tvorevina postoje velike razlike te stoga i između njihovih uzroka mora biti srazmerno velikih razlika. Međutim, on sada tvrdi da polazeći od uočene sličnosti između njih možemo da dokažemo postojanje božanstva a to, čak i ako bismo uzrok prirodnih tvorevina nazvali inteligencijom a ne božanstvom, ipak deluje kao velika promena u odnosu na njegova pređašnja tvrđenja. Pored toga, možda je razilaženje u pogledu toga da li je uzrok uređenosti u svetu ispravnije nazvati božanstvom ili inteligencijom u krajnjoj liniji samo rasprava oko reči, ali ne čini se da je razilaženje oko toga da li su sličnosti između delova sveta i ljudskih tvorevina dovoljno velike da mogu da opravdaju zaključak da su i njihovi uzroci dovoljno i u relevantnim aspektima slični samo verbalni spor. A upravo to je predmet spora između branilaca i kritičara argumenta o osmišljenosti.

Sve do 1776. godine ovo je bila jedina verbalna rasprava o kojoj je u glavnom tekstu dela XII *Dijaloga* bilo reči, ali te godine Hjum dodaje novi, a zvanično sedmi, pasus u kojem opisuje još jedan takav spor. U tom pasusu Filon prvo kaže da se ova vrsta debata obično rešava preciznim definisanjem i jasnom i jednoznačnom upotrebom reči koje su predmet spora, ali potom dodaje da one „rasprave koje se tiču stepena bilo kog kvaliteta ili stanja“ ne mogu na ovaj način da se razreše jer ne postoji način da se ti stepeni precizno izmere (DNR 12.7, KS 218). On u nastavku tvrdi „da je rasprava o teizmu ovakve prirode i da je stoga samo verbalna ili, ako je to uopšte moguće, još nepopravljivije dvosmislena“ (*ibid.*), a onaj deo pasusa u kojem objašnjava zašto je ova rasprava verbalna vreda citirati u celini:

„Pitam teistu, zar ne dopušta da postoji velika i neizmerna, jer je neshvatljiva, razlika između *ljudskog* i *božanskog* uma; što je on pobožniji, to će spremnije odgovoriti potvrdno i biće skloniji da tu razliku uveća, on će čak tvrditi da je priroda te razlike takva da se ona ne može preuveličati. Potom se okrećem ateisti, za kojeg tvrdim da je to samo nominalno i da nikada ne može biti ozbiljan [u pogledu svog ateizma], i pitam ga zar, na osnovu povezanosti i očiglednog slaganja u svim delovima ovog sveta, ne postoji izvestan stepen analogije između svih delovanja u prirodi, u svakoj prilici i u svakom dobu; zar truljenje repe, razvitak jedne životinje i struktura ljudske misli nisu energije između kojih verovatno postoji

neka udaljena analogija; nemoguće je da on to može da porekne, on će to spremno prihvatiti. Nakon što sam obezbedio ovaj ustupak, ja ga još dalje pritiskam u njegovom povlačenju i pitam ga, zar nije takođe verovatno da postoji neka udaljena nezamisliva analogija između onog principa koji je prvi uspostavio i još uvek održava red u ovom svetu i drugih delovanja prirode, a među ostalima i uređenja ljudskog uma i misli. Ma koliko oklevao, on mora to da potvrdi. U čemu se onda, uzvikujem obojici protivnika, sastoji predmet vaše rasprave? Teista dopušta da je izvorna inteligencija veoma različita od ljudskog razuma; ateista dopušta da postoji neka udaljena analogija između izvornog principa i njega [ljudskog razuma]. Hoćete li se, gospodo, sporiti oko stepena i upustiti u raspravu koja nema nikakvo precizno značenje, a stoga i nikakvo razrešenje?“ (DNR 12.7, KS 218–219, Hjumov kurziv)

Stjuart kaže da navedeni pasus „zadobija poseban značaj kao njegovo [Hjumovo] poslednje zaveštanje potomstvu.“ (Stewart 2000: 303) I sama sam prethodno istakla da je „vrlo važno ispravno protumačiti taj deo teksta, jer bi on mogao da sadrži putokaze za razumevanje celine.“ (Davidović 2023: 79) Međutim, to ne znači da bi trebalo da očekujemo da ćemo tu naći gotove odgovore. Hjum je *Dijaloge* svesno napisao u tom obliku, kao književno delo u kojem karakteri i ponašanje likova utiču na dinamiku njihovog razgovora što omogućava da se različite teme smenjuju a da ne budu istražene do kraja i da tvrđenja likova ne budu uvek jasna i jednoznačna.¹¹ On je i pre nego što se razboleo znao da će *Dijalozi* biti objavljeni tek posthumno, ali ga to nije podstaklo da izmeni način na koji je ovo delo napisao niti da svoje poente u njemu učini jasnijim i manje dvosmislenim, pa to od njega ne treba očekivati ni u poslednjem pasusu koji je napisao. Novi pasus se stilski i po sadržaju dobro slaže sa ranije napisanim tekstom i odmah zbunjuje čitaoca budući da spor između teiste i ateiste pre deluje kao fundamentalno neslaganje nego kao samo verbalna rasprava. Pa ipak, pokušaću da pokažem da ovaj pasus, kada se pažljivo analizira, na dva ključna načina doprinosi skeptičkoj interpretaciji dela XII *Dijaloga*. Tvrđiću da Hjumov poslednji dodatak ovom delu, prvo, snažno sugerše da Filonov preokret ne predstavlja drastičnu promenu njegovog mišljenja i, drugo, služi kao ključ za tumačenje Filonovog poslednjeg govora u *Dijalozima*.

Rič Foli (Rich Foley) zastupa suprotnu interpretaciju od one koju ću braniti u ovom radu. On tvrdi da se pasus u kojem se govori o verbalnoj raspravi između teiste i ateiste „može ispravnije razumeti u okviru Filonovog preokreta“ i da ova verbalna rasprava

¹¹ Očigledno je da je Hjum smatrao da je dijaloška forma naročito pogodna za bavljenje temama kojima se bavi u *Dijalozima* budući da je i jedanaesto poglavlje *Istraživanja o ljudskom razumu*, u kojem takođe kritikuje argument o osmišljenosti, napisao u formi dijaloga. On to i potvrđuje posredstvom ličnosti Pamfila koji u uvodnom delu *Dijaloga* kaže da svaka doktrina koja je očigledna i značajna kao i svako filozofsko pitanje koje je nejasno i neizvesno treba da budu obrađeni u vidu dijaloga, te da prirodna religija objedinjuje ove četiri karakteristike (DNR 0.3–0.5, KS 128).

„jeste rasprava između Kleanta i Filona tek nakon što se ovaj drugi predomislio.“ (Foley 2006: 92) Međutim, interesantno je da on kao podršku svojoj tezi mahom citira delove iz pasusa u kojem Filon govori o prvoj verbalnoj raspravi koja se tiče toga da li je uzrok sveta ispravnije nazvati božanstvom ili inteligencijom, a ne iz pasusa koji je Hjum kasnije napisao. Foli smatra da tu prvu verbalnu raspravu ima smisla nazvati samo verbalnom jedino ako je Filonov preokret ozbiljan jer se „verbalni spor ne odnosi na to da li tvorac ima um, već na to da li je božanski um dovoljno sličan ljudskom umu da zaslužuje upotrebu iste reči.“ (*ibid.*) On primećuje da „Filon eksplicitno priznaje ‘značajne razlike’ između prirode i proizvoda veštine, a ispravno analoško zaključivanje zahteva ‘srazmernu razliku’ između uzroka ove dve vrste stvari“ (Foley 2006: 90), ali ipak stavlja akcenat na Filonovo priznavanje postojanja analogije između njih. Foli je u pravu da Filonov govor o prvoj verbalnoj debati stvarno deluje kao da je u skladu sa njegovim odobravanjem argumenta o osmišljenosti s početka dela XII *Dijaloga*, ali se situacija značajno menja kada on počne da govori o raspravi između teiste i ateiste. Iako nije sasvim jasno da li ovo mesto predstavlja nastavak i produbljivanje prve ili uvođenje druge verbalne debate, pokazaću da tu Folijsva interpretacija gubi na uverljivosti.

Filon se obraća teisti i ateisti i od obojice traži da naprave izvesne ustupke kako bi pronašli zajedničko tle. Teista prihvata da postoji velika razlika između ljudskog i božanskog uma, a ateista prvo prihvata da postoji analogija između svih delovanja u prirodi, a zatim prihvata da onda postoji i analogija između principa uređenosti u svetu i ljudskog uma i misli. Međutim, neophodno je da obratimo pažnju na Hjumov izbor reči u navedenom pasusu. Teista prihvata da je analogija između ljudskog i božanskog uma „velika i neizmerna“, „neshvatljiva“, pa čak i „takva da se ona ne može preuveličati“. Nasuprot tome, ateista prihvata da između svih delovanja u prirodi postoji „izvestan stepen analogije“, te da i između takvih pojava kao što su truljenje repe, razvitak jedne životinje i struktura ljudske misli „verovatno postoji neka udaljena analogija“, a potom u sledećem koraku prihvata da postoji i „neka udaljena nezamisliva“ analogija između principa uređenosti u svetu i ljudske misli. Kada se na ovaj način prikažu obe strane u debati izgleda da je ustupak koji teista pravi ozbiljniji. Pošto on u okviru prirodne teologije polazi od sličnosti između prirodnih i ljudskih tvorevina i dokazuje da je uzrok prvih sličan uzroku drugih, odnosno, da je neko inteligentno biće osmislilo ovaj svet, tvrđenje da između ljudskog i božanskog uma postoji neizmerna i neshvatljiva razlika predstavlja očigledan problem za uspešno sprovođenje tog projekta. Filon na jednom mestu u *Dijalozima* kaže da „što su sličnije posledice koje opažamo, i što su sličniji uzroci koje izvodimo, to je jači argument“, pri čemu „svako odstupanje na bilo kojoj strani umanjuje verovatnoću i čini eksperiment manje konkluzivnim.“ (DNR 5.1, KS 165) Prihvatajući da između našeg i božanskog uma postoji neizmerna razlika, teista prihvata značajno odstupanje na strani uzroka koji izvodimo pomoću argumenta o osmišljenosti i samim tim umanjuje verovatnoću zaključka do kojeg nastoji da dođe.

S druge strane, za ateistu nije sporno da prihvati da između svih delovanja u prirodi postoji nekakva analogija jer on time samo umanjuje značaj onih analogija na koje se teista poziva u svom argumentu, dok opisivanje analogije koju prihvata kao udaljene i nezamislive dodatno umanjuje težinu njegovog ustupka.

Foli ne komentariše ustupak teiste, ali tvrdi da nam ustupak koji pravi ateista daje „suptilan nagoveštaj dijalektičkih izvijanja“ koja dovode do Filonovog preokreta (Foley 2006: 93). Prema njegovom tumačenju njihove debate, ustupci koje ateista pravi navode Filona da „trijumfalno tvrdi da između skeptika i teiste zapravo nema neslaganja, jer obojica priznaju da je tvorac sveta sličan ljudskoj inteligenciji.“ (*ibid.*)¹² Foli naravno shvata da ateistu nema potrebe primoravati da napravi ovakve ustupke već da će on „spremno prihvatiti da je svet sličan bilo kojem objektu kao način da umanji naše uverenje u bilo koju specifičnu analogiju.“ (*ibid.*) Ako između svih delovanja u prirodi postoji neki stepen analogije, onda je analogija na kojoj se zasniva argument o osmišljenosti samo jedna od mnogo analogija na koje možemo da se pozovemo kada razmišljamo o uzroku uređenosti u svetu. Filon u delu VI *Dijaloga* kao primer jedne moguće alternative navodi da, budući da vidljivi deo sveta ima mnogo sličnosti sa organizmom koji pokreće neki princip života i kretanja, možemo da zaključimo da je svet životinja a božanstvo njegova duša (DNR 6.3, KS 170–171). Međutim, Foli jeva poenta je da sama činjenica da Filon debatu između ateiste i teiste prikazuje kao verbalnu ukazuje na „izuzetne razmere njegovog preokreta“ (*ibid.*), jer on time pokazuje da sada zanemaruje sopstvene argumente iz *Dijaloga* o postojanju drugih uverljivih objašnjenja uređenosti sveta pored hipoteze o inteligentnom tvorcu. Ako Filon u delu XII *Dijaloga* menja svoja gledišta, ta promena zaista poprma izuzetne razmere, ali smatram da Foli nije pružio uverljivu potporu za ovo tvrđenje. Njegova interpretacija, osim što deluje prilično nategnuto, izostavlja detalje koji čine uverljivim sasvim drugačije čitanje Hjumovog poslednjeg pasusa.

Već sam navela da Foli ignoriše činjenicu da je, kako bi debata između teiste i ateiste mogla da se svede samo na raspravu oko reči, potrebno da teista takođe napravi izvesne ustupke, pri čemu Filonov (odnosno Hjumov) izbor reči u datom pasusu sugerise da je upravo teista onaj koji zarad njihovog usaglašavanja mora više da popusti. Ako teista prihvata da je razlika između božanskog i ljudskog uma *neizmerna*, a ateista prihvata da između principa uređenosti u svetu i uređenja ljudskog uma i misli postoji *udaljena* i *nezamisliva* analogija, onda nije tačno da, kao što Foli kaže, „obojica priznaju da je tvorac sveta sličan ljudskoj inteligenciji“ niti to Filon trijumfalno potvrđuje. Prvo, treba primetiti da se u navedenom pasusu tvorac sveta uopšte ne spominje, te da Filon govori o božanstvu samo onda kada se obraća teisti a da

¹²Foli na ovom i na još nekoliko mesta u svom radu govori o skeptiku i teisti umesto o ateisti i teisti. Možda on debatu između ateiste i teiste povezuje sa debatom između skeptika i dogmatika, trećom verbalnom debatom koja se spominje u delu XII *Dijaloga*, ali to nigde nije eksplicirano. U nastavku ću pokazati da to povezivanje nije pogrešno.

kada se obraća ateisti govori o principu uređenosti u svetu. Drugo, u načinu na koji se Filon obraća ateisti nema naznaka da on sada zanemaruje svoje predašnje argumente. On prvo traži da ateista prihvati da između svih delovanja u prirodi postoji neki stepen analogije „na osnovu povezanosti i očiglednog slaganja u svim delovima ovog sveta“. Stenli Tvejman (Stanley Tweyman) ispravno kaže da se „ovaj ustupak ateiste zasniva na Filonovim argumentima u delovima VI, VII i VIII, u kojima je on pokazao da je sva osmišljenost, bez obzira na uzročno poreklo (vegetacija, razmnožavanje, slepa neusmerena sila, itd.) svodiva na odnose svrhovitosti i usklađenost delova.“ (Tweyman 2020: 13) Međutim, ova vrsta sličnosti između svih delovanja u prirodi je toliko generalna da „ne možemo da postignemo nikakvu specifičnost kada pokušavamo da utvrdimo posebnu sličnost među prirodnim procesima.“ (*ibid.*) Zbog toga ateista prihvata da i između toliko različitih pojava kao što su truljenje repe, razvitak jedne životinje i struktura ljudske misli „verovatno postoji neka udaljena analogija“. Filon potom od ateiste traži da prihvati da i između principa uređenosti u svetu i uređenja ljudskog uma postoji „neka udaljena nezamisliva analogija“ što on, s obzirom na prvi načinjeni ustupak, bez problema prihvata. Možda je i glavna poruka novog pasusa u *Dijalozima* to „da su analogije na kojima se zasniva argument o osmišljenosti suviše nejasne da bi vredelo poricati ih, i stoga suviše beznačajne da bi konstituisale značajna tvrđenja.“ (Penelhum 2012: 208) Ako nekakva analogija postoji između truljenja repe i strukture ljudske misli, onda takva vrsta sličnosti verovatno postoji i između principa uređenosti u svetu i drugih delovanja prirode, *među kojima* je uređenje ljudskog uma i misli. Važno je uočiti da za ateistu „misao nema primat“ (Tweyman 2020: 14), odnosno, ne postoji posebna vrsta analogije između principa uređenosti u svetu i uređenja ljudskog uma, već je ljudski um samo jedna od stvari u prirodi koja ima sličnosti sa tim principom. Pored toga, ta analogija je opisana kao *udaljena* i *nezamisliva* a ateista prihvata da ona tek *verovatno* postoji. Sve ovo pokazuje da je njegov ustupak gotovo beznačajan budući da je minimalno obavezujući – on ne prihvata da je princip uređenosti inteligentno biće, već samo da taj princip ima neke nespecifične sličnosti sa drugim delovanjima u prirodi, a između ostalih i sa ljudskim umom.

Dejvid O’Konor (David O’Connor) tvrdi da Filonovo navođenje *truljenja repe* kao primera prirodnog procesa koji je na neki način sličan ljudskoj misli predstavlja „zapanjujuće unižavanje analogije“ o kojoj je u argumentu o osmišljenosti reč (O’Connor 2000: 200). S druge strane, Foli smatra da je korišćenje tog primera „prilično smešno“, pa čak i da ono „možda uključuje Hjumovo signaliziranje čitaocu da je on sam, uprkos Filonovom preokretu, i dalje prezriv prema argumentu o osmišljenosti“, ali on insistira na tome da Filon ne koristi ovaj primer kako bi parodirao argument o osmišljenosti (Foley 2006: 93). Autori koji Filonov preokret ne smatraju iskrenim neretko tvrde da je on na mnogim mestima u delu XII ironičan¹³, a Foli kaže da je to jedan „interpretativni

¹³Na primer, Ton Vink (Ton Vink) kaže da se ironija „može naći širom *Dijaloga*, iako najviše u delu

trik“ koji onemogućava postizanje slaganja kada je u pitanju tumačenje *Dijaloga* (Foley 2006: 94). Iako verujem da je Filon u *Dijalozima* često ironičan, slažem se sa Folijem da je korišćenje ironije teško dokazati i da će interpretacije koje se isuviše oslanjaju na uočavanje ironičnih delova teksta uvek biti podložnije kritici. Međutim, čak i ukoliko ne tvrdimo da je Filon u navedenom pasusu ironičan, nesumnjivo je da korišćenje primera truljenja repe umanjuje značaj ateistinog prihvatanja analogije između principa uređenosti u svetu i ljudske misli time što slikovito ilustruje koliko malo sličnosti je potrebno da bismo tvrdili da postoji analogija između neke dve pojave i za koliko različite stvari se može reći da su na neki način slične. Budući da Foli smatra da je upravo to mesto na kojem se Hjumovi i Filonovi stavovi u *Dijalozima* razlikuju i na kojem nam sam Hjum sugerise da je i dalje kritičan prema argumentu o osmišljenosti, izgleda da i on na isti način shvata svrhu navođenja tog primera.

Ako je ovde izložena analiza novog pasusa ispravna i ako je ustupak teiste stvarno mnogo veći od ustupka ateiste, onda smo imali manje razloga da sumnjamo u Filonov preokret *pre* čitanja navedenog pasusa. To što Filon debatu između teiste i ateiste svodi na spor oko reči tako što teistu navodi na značajan a ateistu na gotovo prazan ustupak pokazuje da on zapravo nije promenio svoje mišljenje. Teista prihvata tvrđenje koje, iako to u *Dijalozima* nije eksplicirano, značajno umanjuje verovatnoću zaključka argumenta o osmišljenosti. Nasuprot tome, ateista prihvata tvrđenje koje, iako na prvi pogled može da izgleda drugačije, ne uvećava verovatnoću tog zaključka. Prihvatajući da postoji neka analogija između principa uređenosti u svetu i uređenja ljudskog uma, ateista samo prihvata jedno moguće objašnjenje uređenosti koju opažamo u svetu, ali ne i da je to objašnjenje verovatnije od drugih alternativnih mogućnosti. Ovo je veoma daleko od zaključka do kojeg jedan teista želi da dođe, ali je sasvim u skladu sa stavovima koje je Filon prethodno izneo u *Dijalozima*. U delu VII on kaže da je naše iskustvo toliko ograničeno da „mi nemamo podatke za uspostavljanje bilo kakvog sistema kosmogonije.“ (DNR 7.8, KS 177) Ako i pored toga želimo da ponudimo neko objašnjenje celine, moramo da se oslonimo na uočavanje sličnosti između delova sveta i pojava ili stvari čije funkcionisanje nam je poznato. Međutim, Filon tvrdi da su razum, instinkt, vegetacija i razmnožavanje principi koji bi, s obzirom na opseg našeg znanja, jednako dobro mogli da objasne uređenost sveta, da sigurno ima još mnogo takvih principa i da „nije manje razumljivo, niti manje u skladu sa iskustvom, reći da je svet nastao vegetacijom iz semena koje je prosuo neki drugi svet, nego reći da je proistekao iz božanskog razuma ili izuma“ (DNR 7.11, KS 178). Ono što ateista u Filonovom prikazu njegove debate sa teistom prihvata je u skladu sa ovim Filonovim tvrdnjama. Na osnovu toga možemo da zaključimo da Filon, kada tvrdi da je ta debata samo verbalna, zapravo ne izražava slaganje sa Kleantom u pogledu argumenta o osmišljenosti, nego podržava svoje ranije tvrđenje da pomoću ovog argumenta ne

XII“ (Vink 1986: 390), a Vilem Lemens (Willem Lemmens) tvrdi da je „Filonovo navodno prihvatanje argumenta o osmišljenosti prožeto ironijom“ (Lemmens 2012: 207).

možemo da dokažemo da je uzrok uređenosti sveta neko inteligentno biće.

Konačno, navedene dve debate nisu jedine koje su u *Dijalogima* opisane kao čisto verbalne. U fusnoti koja se nalazi na kraju pasusa koji sledi za pasusom u kojem Filon tvrdi da je rasprava između teiste i ateiste verbalna nailazimo na tvrđenje da je debata između dogmatičara i skeptika takođe samo verbalna budući da se tiče „stepena sumnje i uverenja sa kojima pristupamo svakom rasuđivanju“ (DNR 12.8 nap. 11, KS 219).¹⁴ Dok filozofski dogmatičar prihvata da postoje logički nerešive teškoće kako u vezi sa čulima tako i u vezi sa svim naukama, skeptik prihvata da su ljudi, uprkos ovim teškoćama, nužno opredeljeni da veruju i da rasuđuju o različitim temama, te je stoga jedina razlika između njih dvojice to što „skeptik, iz navike, hira ili sklonosti, najviše insistira na teškoćama; dogmatičar, iz sličnih razloga, na nužnosti.“ (*ibid.*) Sličnost između načina na koji je Hjum predstavio ovu verbalnu debatu i onu prethodnu je očigledna.¹⁵ Ako pažljivo analiziramo ono što on ovde kaže, videćemo da je za proglašenje rasprave verbalnom potrebno da dogmatičar napravi veći ustupak nego skeptik. On prihvata da ima *nerešivih* teškoća koje pogađaju naša čula i razum, što bi trebalo da dovede u pitanje njegova dogmatska uverenja, dok skeptik uopšte ne mora da odstupi od svojih skeptičkih argumenata kako bi prihvatio da ipak mora da veruje i da rasuđuje kao i svi drugi ljudi. Skeptikova pozicija je predstavljena isto kao i gledište koje Filon brani još u delu I *Dijaloga* kada prvo ukazuje na „slabost, slepilo i uske granice ljudskog razuma“, „greške i obmane naših čula“ i „nepremostive teškoće koje prate prve principe u svim sistemima“ (DNR 1.3, KS 131), da bi potom priznao da čovek mora „da dela, i živi, i razgovara, kao i drugi ljudi; i za ovakvo držanje nije dužan da navede nijedan drugi razlog osim apsolutne nužnosti da tako postupi.“ (DNR 1.9, KS 134) Ovo znači da je za nas neizbežno da verujemo iako skepticizam nije moguće racionalno opovrgnuti. Kada razmatramo debatu između dogmatičara i skeptika možemo da uočimo jasne ličnosti između skeptikove i Filonove pozicije, kao što smo u prethodnoj debati mogli da uočimo sličnosti između gledišta ateiste i Filona. Nasuprot tome, Kleantova pozicija je bliža gledištima teiste i dogmatičara. S obzirom na ovde izloženu analizu i interpretaciju verbalnih debata iz dela XII *Dijaloga*, možemo da zaključimo da nije neophodno da Filon stvarno promeni svoje gledište da bi se

¹⁴U literaturi se ova fusnota često navodi kao jedino mesto gde Hjum govori u svoje ime a ne posredstvom jednog od likova u *Dijalogima* (videti na primer Tweyman 2020: 12), ali ima i autora koji to osporavaju (na primer Gaskin 1988: 216). S jedne strane, ima smisla pripisati reči iz fusnote Filonu pošto je on ranije već govorio verbalnim debatama ali, s druge strane, ima smisla pripisati ih i Hjumu budući da fusnote obično sadrže napomene autora. Međutim, za potrebe sadašnjeg rada nije neophodno da razrešimo tu dilemu već će biti dovoljno da utvrdimo kako ovaj deo teksta doprinosi razumevanju ostatka.

¹⁵Interesantno je da je Hjum novi pasus 1776. godine napisao na istom listu papira na kojem je mnogo ranije napisao ovu napomenu o raspravi između skeptika i dogmatičara (Stewart 2000: 297 i 307–308). To naravno ne mora da znači da ga je baš čitanje prethodno napisane fusnote navelo da napiše novi pasus, ali je moguće da je bio inspirisan načinom na koji je prikazao jednu verbalnu debatu i da je po uzoru na nju predstavio i drugu.

naizgled složio sa Kleantom. On u ključnom pasusu u kojem kaže da je debata između teiste i ateiste samo verbalna pokazuje na koji način je moguće da se *kao ateista* složi sa Kleantovim teističkim tvrđenjem. Kada tome pridodamo njegovu kritiku praktičnih posledica religije iz nastavka poslednjeg dela *Dijaloga*, od Filonovog preokreta osim reči ne ostaje ništa.

4. Filonov poslednji govor u *Dijalozima*

Preostaje da vidimo kako nam prethodno sprovedena detaljna analiza pasusa koji je Hjum napisao 1776. godine pomaže pri tumačenju Filonovog poslednjeg govora u *Dijalozima*. U njemu Filon kaže da je, ako su određeni uslovi ispunjeni, spreman da prihvati tvrđenje da uzrok ili uzroci reda u svetu verovatno stoje u nekoj udaljenoj analogiji sa ljudskom inteligencijom. On to tvrđenje opisuje kao donekle nejasno ili barem nedefinisano i jasno je zbog čega to kaže. Prvo, on ne govori o tvorcu sveta već o uzroku ili uzrocima reda u svetu. Zato Tvejman greši kada kaže da Hjum u delu XII „argumentuje da postoji jedan iskaz o Bogu koji je razumno prihvatiti.“ (Tweyman 2020: 8) Gaskin pravi sličnu grešku kada kaže da „Hjum daje neku vrstu iskrenog pristanka na tvrđenje *da bog postoji*.“ (Gaskin 1988: 221, Gaskinov kurziv) On napominje da u tom pristanku nema nikakvih osnova za verovanje u personalnog Boga nalik hrišćanskom, ali sama činjenica da govori o bogu, kojeg naziva i „uređujućim agentom“ i „prvobitnim izvorom reda“ (*ibid.*), pokazuje da se previše udaljio od Filonovih reči, jer Filon ne pretpostavlja da postoji samo jedan uzrok uređenosti u svetu već ostavlja otvorenom mogućnost da ih ima više. Pored toga, videli smo da Filon u novom pasusu u kojem opisuje navedenu verbalnu debatu govori o božanskom umu kada se obraća teisti, dok kada se obraća ateisti govori o principu koji je prvi uspostavio i održava red u svetu. Drugim rečima, on se na gotovo isti način izražava onda kada od ateiste traži da prihvati da postoji analogija između principa uređenosti u svetu i uređenja ljudskog uma i kada sam na kraju *Dijaloga* prihvata da postoji analogija između (jednog ili više) uzroka reda u svetu i ljudske inteligencije.

Međutim, to nije najupečatljivija sličnost između ova dva dela teksta. Filon najpre navodi ateistu da prihvati da verovatno postoji neka udaljena analogija između svih delovanja u prirodi i da je takođe verovatno da postoji neka udaljena nezamisliva analogija između principa uređenosti u svetu i uređenja ljudskog uma, a zatim na kraju *Dijaloga* sam prihvata da verovatno postoji neka udaljena analogija između uzroka reda u svetu i ljudske inteligencije. Na oba mesta se u tvrđenjima koja treba prihvatiti javljaju reči „verovatno“, „neka“ i „udaljena“, pri čemu su to jedina mesta u Hjumovim delima gde se javlja izraz „udaljena analogija“ i gde se javlja potpuno ista formulacija „verovatno postoji neka udaljena analogija (*probably bear some remote analogy to*)“. Ovo sugerise da Hjumov izbor reči na tim mestima nije slučajan i da bi pasus u kojem se govori o debati između teiste i ateiste i Filonov govor trebalo čitati zajedno.¹⁶ Pre

¹⁶U literaturi se verbalnoj debati između teista i ateista ne pridaje mnogo značaja, a ni autori koji joj

svega, da Hjum neposredno pre smrti nije dodao ovaj pasus, bilo bi teže razumeti na šta se „neka udaljena analogija“ koju Filon spominje u svom zaključnom govoru odnosi. Međutim, čitajući ove pasuse uporedo, možemo da kažemo da Filon na kraju *Dijaloga* prihvata da između uzroka reda u svetu i ljudske inteligencije postoji neka udaljena analogija kakva postoji čak i između truljenja repe i strukture ljudske misli. Pritom Vink ukazuje na činjenicu da se ovde radi o *strukturnalnoj sličnosti* i da reči „ljudska inteligencija“ ne bi trebalo da nas navedu da pomislimo na neko inteligentno biće (Vink 1989: 494). Kada se obraća ateisti Filon govori o *strukтури* ljudske misli i *uređenju* ljudskog uma i misli i kaže da „na osnovu povezanosti i očiglednog slaganja u svim delovima ovog sveta“ nekakva analogija postoji i između njih i drugih delovanja prirode. Iz toga sledi da „nema filozofskog značaja u upoređivanju uzroka ili uzroka reda u svetu sa ljudskom inteligencijom“ jer „neka udaljena, nezamisliva analogija postoji između svih principa reda, budući da su njihovi proizvodi povezani identičnim osobinama osmišljenosti.“ (Tweyman 2020: 15) Drugim rečima, Filon na kraju *Dijaloga* ostavlja otvorenom mogućnost da su uzrok ili uzroci reda u svetu u nekom smislu nalik ljudskoj inteligenciji, ali time o njima ne kaže ništa specifično jer istovremeno dozvoljava da su oni u istom pogledu nalik drugim delovanjima prirode. Sve to pokazuje da se Filon na kraju *Dijaloga* slaže sa ateistom i da je njegov zaključak krajnje skeptički. O svetu, polazeći od sopstvenog iskustva, ne možemo da kažemo ništa više osim toga da je uređen i da uzrok te uređenosti ima izvesne strukturalne sličnosti sa ljudskom inteligencijom kakve deli i sa drugim delovanjima prirode.

Ovakav zaključak argumenta o osmišljenosti ne predstavlja adekvatnu osnovu za izgradnju religije i Hjum to zna. Osim toga, Filon u svom poslednjem govoru kaže da će prihvatiti navedeno tvrđenje samo pod uslovom da ono nema nikakav uticaj na ljudski život i da ne može da predstavlja osnov za naše postupanje i ophođenje. Budući da je prethodno pokazao da religija gotovo uvek ima veoma negativne praktične posledice, on se ograničava na prihvatanje tvrđenja koje ne može da nam posluži kao vodič u svakodnevnom životu. Filon takođe postavlja uslov da taj zaključak ne može da se proširi ni na šta dalje od ljudske inteligencije, a naročito da ne može da se prenese na druga svojstva uma, čime nas podseća da pomoću argumenta o osmišljenosti ne možemo da dokažemo moralne attribute božanstva. Ako su svi navedeni uslovi ispunjeni i ako se prirodna teologija svodi jedino na tvrđenje da uzrok ili uzroci reda u svetu stoje u nekoj udaljenoj analogiji sa ljudskom inteligencijom i još ako ovo tvrđenje ne može nikako da se proširi ili pojasni, onda ga on prihvata. Iako isprva izgleda da Filon na kraju *Dijaloga* prihvata da pomoću argumenta o osmišljenosti možemo da dodemo do nekakvog supstantivnog zaključka o uzroku reda u svetu, analizom Hjumovog

posvećuju pažnju je obično ne povezuju sa Filonovim govorom na kraju *Dijaloga*. Izuzeci su Vink, koji kaže da „bi zaista bilo suludo smatrati da je velika sličnost u formulaciji pozicije ateiste u verbalnoj debati i Filonovog poslednjeg zaključka bilo šta nalik slučajnosti“ i da „verbalna debata sadrži jasno i značajno parafraziranje Filonovog poslednjeg zaključka“ (Vink 1989: 495), kao i Tvejman, koji kaže da se Filon u svojoj poslednjoj izjavi „oslanja na poziciju koja je pripisana ateisti“ (Tweyman 2020: 14).

poslednjeg pasusa i Filonovog zaključnog govora sam pokazala da je tvrđenje koje on prihvata toliko neodređeno i lišeno svakog istinski religioznog sadržaja da deluje najprikladnije okarakterisati ga kao skeptičko. On prihvata da uređenost sveta može da se objasni delovanjem nekog inteligentnog principa, ali kaže i da ona može da se objasni na više drugih načina. Mi jednostavno nemamo dovoljno podataka da bismo u okvirima prirodne teologije tvrdili išta određenije od toga.

5. Zaključak: postoji li mogućnost pomirenja?

Na kraju želim samo ukratko da razmotrim da li Hjum smatra da je pomirenje teiste i ateiste ili dogmatičara i skeptika zaista moguće. Filon se na kraju izlaganja rasprave između teiste i ateiste obojici obraća sledećim rečima: „Razmotrite onda, gde leži prava tačka spora, i ako ne možete da ostavite po strani svoje rasprave, potrudite se barem da se izlečite od svog neprijateljstva.“ (DNR 12.7, KS 219) Lemens kaže da je on sam, iako Filon možda zvuči pomirljivo kada traži da teista i ateista prestanu da se svađaju, ipak sklon tome da čuje „neku ubojitu ironiju u njegovom glasu.“ (Lemmens 2012: 296) S druge strane, Ričard Diz (Richard Dees) smatra da, iako se Filon slaže sa ateistom, on „naglašava svoje tačke slaganja sa Kleantom“ zato što je za njega „održavanje prijateljstva sa Kleantom važnije od insistiranja na ispravnosti svojih metafizičkih gledišta.“ (Dees 2002: 139) Oba ova gledišta se čine ekstremnim iz različitih razloga. Prvo, Filon verovatno jeste ironičan kada tvrdi da je spor između teiste i ateiste samo verbalan, jer ubrzo postaje jasno da je to tačno samo ako teista ozbiljno odstupi od svoje početne pozicije. Međutim, nije jasno zašto bi trebalo da smatramo da je on takođe ironičan onda kada traži da se oni, bez obzira na međusobna razilaženja, odreknu svog neprijateljstva. Drugo, Diz previše ističe Filonov i Kleantov lični odnos. Filon u toku svoje rasprave sa Kleantom počinje da govori o debati između teista i ateista i između skeptika i dogmatičara upravo zato što njih dvojica nisu samo različite ličnosti, već i predstavnici različitih filozofskih pozicija (Tweyman 2020: 12–13). Ako Hjum stvarno poziva na pomirenje u delu XII *Dijaloga*, onda je tu reč o pomirenju različitih filozofskih gledišta a ne likova u njegovom dijalogu. Pored toga, videli smo da je Filonu ipak važno da argumentuje za svoju skeptičku poziciju, doduše na suptilan način.

Filon poziva teistu i ateistu da razmotre šta je prava tačka njihovog spora. Pokazala sam da se ona tiče toga koliko značaja treba pridati analogiji između principa uređenosti sveta i ljudske inteligencije. Njihova rasprava nije stvarno samo verbalna, ali ona jeste nerazrešiva jer rešenje ključne sporne tačke prevazilazi granice našeg iskustva. Mi prosto nemamo dovoljno podataka da se opredelimo između različitih hipoteza o uzroku reda u svetu. Hjum je već na početku *Dijaloga* posredstvom Pamfila nagovestio da su teme kojima se bavimo u okviru prirodne teologije „tako interesantne, da ne možemo da obuzdamo naše nemirno istraživanje u vezi sa njima; iako ništa osim sumnje, neizvesnosti i protivrečnosti, do sada, nije bilo rezultat naših najtačnijih

istraživanja.“ (DNR 0.5, KS 128) On je tada takođe rekao da je „razumnim ljudima dozvoljeno da imaju različita mišljenja tamo gde niko ne može da bude razumno siguran“ jer nam suprotna mišljenja, čak i ako ne možemo da se opredelimo između njih, „pružaju prijatnu razonodu“ i ujedinjuju „dva najveća i najčistija zadovoljstva ljudskog života, učenje i društvo.“ (DNR 0.4, KS 128) Međutim, kao što to rasplet Hjumovih *Dijaloga* pokazuje, a on je to nesumnjivo i tokom života iskusio, razilaženja koja se tiču religije obično izazivaju snažna osećanja i zato u pasusu koji je napisao neposredno pre smrti Hjum prepoznaje da je takve rasprave nemoguće zanemariti i traži jedino da se pristalice suprotstavljenih gledišta uzdrže od neprijateljskih istupa.

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Aleksandra Davidović

Hume’s Last Words: On the Occasion of the 250th Anniversary of Hume’s Death

(Summary)

In 1776, shortly before his death, David Hume added a new paragraph to the last part of the *Dialogues Concerning Natural Religion*. In that paragraph the sceptic Philo claims that the dispute between a theist and an atheist is merely verbal because both of them accept that the cause of order in the universe bears some analogy to human intelligence. In this paper, I conduct a detailed textual analysis in order to show that the new paragraph, firstly, strongly suggests that the so-called Philo’s reversal does not represent a drastic change of his opinion, and secondly, serves as a key to interpreting Philo’s final speech in the *Dialogues*. This analysis also establishes that Hume’s position, despite the conciliatory tone present in the new passage, is primarily skeptical.

KEYWORDS: David Hume, theism, atheism, scepticism

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CHAOS AND THINKING

ABSTRACT: *In this paper, the author examines the case of Deleuze's and Guattari's last published book, *What is Philosophy?* (1991), in which the French thinkers embark on a unique journey with a goal to once again provide the audience with an answer to the monumental question. With the clear notion that it's not only just the philosophers who wanted or demanded this answer, Deleuze and Guattari are trying to demystify and debunk the false idea that philosophy only concerns philosophers. The aforementioned book could have just as easily borne the title: *What is Philosophy, What is Art, and What is Science?* With the three seminal questions in mind, the book is a substantial exercise in showing both how the pure immanence operates and how the three corresponding disciplines – arts, sciences, and philosophy – get in and out of chaos with different results. It shouldn't be emphasized that the notion of chaos needs to be reexamined and revalued today. Deleuze and Guattari help us realize that it is of utmost urgency to comprehend that the chaos mustn't be seen as a negative term, and that it fundamentally lies beyond the mere opposites of positive and negative categories. Our aim here is to show that the connection of thinking and chaos always leads and points to stories or narratives, in which arts, sciences, and philosophy form their unique planes. Thus, the inseparable connection and alliance form between stories, thinking, and chaos.*

KEYWORDS: *chaos, thinking, stories, narrative, philosophy, science, art, immanence*

I Introduction

In trying to summarize some of the core ideas of *What is Philosophy?*, one could start off by telling it in a form of an illustrious joke: a philosopher, a scientist and an artist walk into a bar chaos, and each one of them walks out with different results – a philosopher gets or takes his concepts with the plane of immanence and conceptual personae, a scientist constructs functions and prospects on a plane of reference, and an artist creates percepts and affects, along with aesthetic figures, on a plane of composition. All three of those planes, along with the mentioned accompanying elements, could be simultaneously considered as a punchline to the (no) joke. In other words, that is where and how Deleuze and Guattari explain the results of the battleground upon which all thinking emerges, or, to put it better, *within* the chaos. Thinking touches upon chaos in a manner where this production of different planes happens. So, it can be said that arts, sciences, and philosophy are just modes of

expressing thought while encountering chaos. They are the “three daughters” of chaos, “depending on the plane that cuts through it” (Deleuze & Guattari 1994: 208). But it can also be said that this is an ongoing or never-ending activity. Philosopher, scientist, or artist are vehicles through which chaos speaks. And how does chaos speak? In the form of narratives/stories. This is the moment where the elements of Wilhelm Schapp’s *Geschichtenphilosophie* come into focus. One of Schapp’s fundamental claims is that *there is no way out of the story (stories)*. We are entangled in stories up to the point where we realize that everything exists only within a story, and that each story connects to another in rhizomatic fashion: “We could perhaps also say that every story is prepared within the horizon of every other story or that in the horizon of every story there is a place for the others stories” (Schapp 2012: 94).¹ Following this route, we could say that, in a way, chaos *are* stories, because every story or narrative framework in which all these disciplines emerge from is shaped through this double bind – there are no stories of sciences, arts or philosophy without the chaos they touch upon. But we should also keep in mind that chaos doesn’t lie beyond stories. We grasp chaos only through stories of chaos. So, the pure immanence that Deleuze and Guattari were often referring to, can be traced back to stories, or in other words, language as such. And thinking always comes in narrative form.² Even the notion of an individual or subjectivity – like artist, scientist or philosopher – rests upon the multiplicity of stories that intertwine in a manner where it’s inconclusive exactly which one of these stories is the leading one. That shaves off the notion of origin, *Ur-narrativ*, or fundamental story, and that also opens up the reading of the history of philosophy and the history of thinking in general in a much different light. This is also in concordance with Deleuze’s and Guattari’s diagnosis that philosophy is constructivism (Deleuze & Guattari 1994: 35–36). So, while Plato creates concepts in order to show that they were already pre-existing, he is actually pulling out the greatest trick that the Western thought has ever seen – convincing nearly everyone that he wasn’t the one who made these concepts (like the concept of Idea, for example), claiming that the Ideas were allegedly always “there.”

In thinking about the history of philosophy (this is essentially the subject of the philosophy of the history of philosophy, which has a rich tradition in names such as Schelling, Hegel, Dilthey, Nietzsche, Husserl, Hartmann, Gadamer, Heidegger, etc.), there’s a way of displaying or showing this history as a map, or even a jigsaw puzzle of different stories that relate to each other in closer proximity, or, on the contrary, stand apart. “Elements” (planes of immanence) of this jigsaw sometimes fit together

¹ All translations in a text are author’s, unless it is specified otherwise.

² For example, in *What is the Creative Act?*, Deleuze is adamant about the fact that *everything has a story*: “Everything has a story. Philosophy also tells stories. Stories with concepts. Cinema tells stories with blocks of movement I duration. Painting invents an entirely different type of block” (Deleuze 2006: 314).

perfectly, but at times do not match at all. We can analyze adherences and deviations of different concepts in different philosophies/philosophers that institute planes of immanence. But we can also think in terms of how the specific concepts, affects/percepts, or functives overlap. For instance, we can take the concepts/functives of nature, space, time, force, substance and connect them to Plato, Aristotle, Galilei, Descartes, Newton, Leibniz, Spinoza, Schelling, Nietzsche, Husserl, Bergson, Einstein, Bohr, Heisenberg, etc. and analyze the degrees in variety, convergences and divergences. Or, we can also delve deeper into the fabric of various arts and find the (mis)matching ideas that coincide in larger or lesser degree to the ones found in sciences and philosophy. Arts also take the subjects of space, time, force, substance, motion, and representation, to name only a few. Music, film, painting, and literature all have their unique ways of dealing with these phenomena. Thomas Mann or Marcel Proust tackle the challenge of time and offer something extraordinary through literature. Their percepts/affects meet Bergson's or Husserl's concepts, just like Boulez's music, Duchamp's artworks, Dalí's paintings, or Tarkovsky's films. Borges takes on the themes that correspond to quantum mechanics, thus encountering the likes of Bohr, Heisenberg, or Hugh Everett III, etc.

What we will find – and not “in the end,” since there is no hierarchy that supports this “top-bottom” logic, other than in narratives that we often find ourselves in (they are contingent, as we'll see) – so, what we will learn or discover is that chaos, as a form of virtuality, always lends itself in such way that thinking, through stories, creates concepts, percepts/affects, functives/functions and their belonging planes. That is how we have philosophy, arts, and sciences.

II Who speaks, constructs, and creates?

The repercussion of the stance that philosophy is constructivism leaves us with the understanding that philosophy is, just like science and art, a fundamentally creative endeavor. But what remains unclear at first is *who* actually does create these specific, belonging planes, if we are fully aware of the problems of individuality and subjectivity? These problems culminate in 20th-century philosophy. The so-called poststructuralist thinkers inform us with convincing arguments that these are unstable categories, to say the least. The notorious “death of man”, proclaimed by Foucault (Foucault 2002: 373), puts us in a rough situation when we try to address these problems and answer these questions. The ideas of a thinking subject, that Descartes once triumphantly declared, Kant's unity of consciousness (or unity of apperception), Husserl's transcendental subjectivity (to name only a handful of projects), have all amassed the urgency to approach these difficulties differently, and to try and shed a different light on them. One of the ways to do this is by returning to language, that is now understood in a much broader sense: language equals stories. So, the thinking subject is caught up in a distinct web of stories, which rests upon the narrative constel-

lation that allows the production of all these categories: of subjectivity, individuality, objective reality, and various ontologies that follow these footsteps. In other words, *language speaks through stories*. The man comes about only through language/stories. As Schapp would say, “die Geschichte steht für den Mann,” “the story stands for the man” (Schapp 2012: 100). We are inscribed in different narrative constellations, and in turn described by stories that form the given constellation in the first place. But the important thing to note is that there’s a pure multiplicity of contingent stories. These heterogeneous narrative constellations constitute the basis for all thinking. So, in trying to answer the questions of *who speaks, or who creates (concepts, percepts/affects, functions, and planes of immanence, reference, composition)*, the short answer would be: stories. A thinking and creative subject exists only at the intersection of different stories. The language is like a great ventriloquist who speaks from the belly of a constellation that it forms in the first place. That is what true immanence gives and accomplishes. And inside of a specific constellation that we can call “Western thinking,” we find these narrative “nodes,” or “hubs” that we call “subjects.”

It is mentioned as particularly interesting the fact that Deleuze and Guattari insist that the philosopher (in other words, *author*) creates concepts. While on the one hand it is an extremely tempting pathway, because with one Aristotelian gesture the Ideas are removed from their pedestal of presupposedness (in the sense of something pre-set, pre-given), on the other hand, the problems of grasping for a certain subjectivity (as if independent of the linguistic-narrative network that establishes that very subjectivity) arise and creep in through the back door. The example with Descartes and Plato (Deleuze & Guattari 1994: 29–30) is highly illustrative here. It is very persuasive, but also potentially problematic. However, if we know how much Deleuze and Guattari emphasized the importance of immanence that is immanent to itself, at the expense of transcendence, things become clearer. The author is a function. But it is a function of language-story, understood as pure immanence. Language is immanent to itself. This is crucial because when Deleuze and Guattari claim that the philosopher as author lays down the plane of immanence and creates concepts, they risk having the whole burden of tradition fall on their backs. Once again, the “Subject” (with the capital “s”) potentially sneaks in instead of acknowledging the eventfulness of stories, a simulacrum of language that is pure immanence. But Deleuze and Guattari sense it, because in the following they recognize that the messenger of philosophy is language itself:

The conceptual personae is not the philosopher’s representative but, rather, the reverse: the philosopher is only the envelope of his principal persona and of all the other personae who are the intercessors [*intercesseurs*], the real subjects of his philosophy. [...] In the same way, the philosophical shifter is a speech-act in third person where it is always a conceptual persona who says “I”: “I think as idiot,” “I will as Zarathustra,” “I dance

as Dionysus,” “I claim as lover.” Even Bergsonian duration has need of a runner (Deleuze and Guattari 1994: 64).

It is particularly interesting that the initiator of philosophy is taken to be nothing but a speech act, which is the starting point in the philosophy of language of figures like Austin and Searle. But it should not mislead, nor should it produce blasphemous thoughts of an attempt to synthesize Deleuze and Searle. That “ordinary language”, referred to by Wittgenstein from the second stage, Austin, Searle, et al., is that area of communication, to which philosophy is reduced, degrading its true, creative impulse. Searle’s “Speech Acts” would most likely be characterized as superfluous thinking of late capitalism, rather than a genuine attempt to establish concepts and planes that correlate with events. The plane of immanence is an image of thought. The plane of immanence would be a peculiar space in which concepts are inscribed. The plane of immanence is a pure variation (Deleuze & Guattari 1994: 39), it is always a single plane, although we are dealing with the upheavals and shifts of the planes of immanence. The plane of immanence creates a characteristic image of the thinking of the epoch. This is, at least for a moment being, irresistibly reminiscent of Foucault’s *épistémè*. Concepts are not deduced from a plane. Concepts must be created, and planes set up (Deleuze & Guattari 1994: 40). But who and what sets them up? Language-story.

III *Ritornello*: the eternal recurrence of stories

In proclaiming that the goal of philosophy is creating new concepts ceaselessly (Deleuze & Guattari 1994: 5), Deleuze and Guattari are simultaneously setting in motion their concept of *ritornello*. Its operating logic rests upon something that made Deleuze (in)famous in the first place – difference and repetition. While one of the tasks of *Difference and Repetition* (1968) was to debunk the core concepts of thinking and categories of Truth, Identity, Factual, Representation, that thinking predominantly rests upon, it also introduced and opened up the space for thinking beyond these coordinates. Repeating something, a concept for example, always entails the repetition of (and in) difference, and never the Same. Deleuze and Guattari are good students of Nietzsche, who was a herald or courier of this idea. That is the reason why we have to be highly suspicious of concepts that we engage with. We can never just take them as they were; we need to critically (re)examine them, and that’s what “philosophizing with the hammer” (Nietzsche 2005: 153) purports and professes. Every single new use of a concept requires a new iteration. They always return differently. But how exactly do we trace these iterations? The answer is: always through stories. It shouldn’t come as a surprise that the concept of *ritornello* is closely connected to chaos (Deleuze & Guattari 2005: 312).

The idea of philosophy as a creative endeavor thus confirms itself. In this particular situation, it returns enriched with the vision of the narrative form it always takes. And the story is never the Same. The dimension of language upon which everything depends

simply cannot be excluded or overlooked. The fabrication of concepts requires an artisan who is nevertheless always inscribed within the narrative web that makes this possible in the first place. That is also what makes possible the escape from a trap that is typical for thinking that emerges in romanticism, for example: by putting an artist and artistic creation on a pedestal, there's a risk of opening the Pandora's box, by propelling hierarchies once again, and essentially endangering the idea of pure immanence *in toto*. We can agree that a philosopher is the potentiality of a concept (Deleuze and Guattari 1994: 5) only if we recognize where they are both situated in a narrative framework that bears the name of pure immanence. And *through pure immanence, chaos lends itself to different orderings*. These orderings are the results of arts, sciences, and philosophies. A philosopher, scientist, or artist is nothing but a meeting point, a node of a rhizome that language orders and disorders. Chaos and language/stories are sides of the same coin. Thinking emerges on these specific planes.

But there is no way of escaping the question of what unity is there, if the concepts are tirelessly changing all the time, and returning never the same. Deleuze and Guattari are fully aware of that problem, and they openly ask that question (Deleuze & Guattari 1994: 8). This question above is one of the key ones concerning the sciences: whether we are talking about the "functions" of time and space, matter, substance, atoms and subatomic particles, or, on the other hand, about philosophical concepts – the question of changeability or immutability of these instances arises. Let's not forget the artistic understanding, the thinking about time in Mann, or Proust, for example. Is science more aware of these changes? Is Newton's absolute time comparable to Einstein's relative time, or Schelling's, Bergson's, or Husserl's time? We know that Newton's concept of absolute time is fundamentally different from Einstein's (that is, the other way around), but the question here is something else – is the original difference, the non-identity of time, between one and the other accounted for? (Deleuze and Guattari would leave time to science, while they would attribute "intermediate" time to philosophy.) Because, in the case of, say, Newton and Einstein, the difference is the assumed Identity of time, where in one it appears as absolute (Newton), and in the other as relative (Einstein). But what remains in the "dead corner" of thought here is the potential impossibility of a starting point, or the stronghold, where the signifier "time" would have other signifieds (notions, concepts, ideas) in each of these cases. Therefore, Deleuze and Guattari go on to ask whether, if philosophical concepts are of a changing character, this also applies to the sciences and arts.

In any case, the concept is understood as multiplicity. And this calls for an intervention immediately. Every concept or name is a header of a story, or stories: "These names represent worlds, world images, or worldviews. The relationship of these names to the stories is obvious. The names are headings for stories. Only through stories is there access to them and, at the same time, to their worlds" (Schapp 2015: 47). Each concept contains a multiplicity of contingent stories, and this pure

multiplicity represents starting points for concepts. It is the “chiffre” that Deleuze and Guattari speak of, which refers to a multitude of combinations. But there’s also an element of *ritornello* that should be recognized in this as well, since Deleuze and Guattari are feeding on the stories and notions that Nietzsche and Mallarmé accentuated accordingly: the Nietzschean eternal recurrence leads or points to Mallarmé’s throwing of the dice; the activity whose offspring of non-choosing, or chance is *chaosmos*. *Un coup de dés jamais n’abolira le hasard* (*A Throw of the Dice will Never Abolish Chance*).

We could take, for example, the concept of time and put it in different epochs, different contexts, ideas, sciences, and understandings. What is it that survives as the so-called “lead,” or “end” story? We can, or even must, go backwards: from Schapp, and through the philosophy of stories, we could show the conceptual apparatus that is characteristic of every paradigmatic story/narrative, say in the field of the history of physics. And then we move along the following paths/stories: the ancient conception of time in the context of the philosophy of stories, the Middle Ages, the New Age, German idealism, the 20th and 21st centuries. From Plato, Aristotle, Augustine, Descartes, Leibniz, Spinoza, to Kant, Schelling, Hegel, Nietzsche, Husserl, Bergson, Einstein, Ricœur, etc. The story is, just like the concept, “a fragmentary whole” (Deleuze & Guattari 1994: 16). The way Deleuze and Guattari are describing how concepts are being formed corresponds to the way we can elucidate the way stories/narratives are configured. The question of why we create concepts is also the question of why philosophy exists – it is about offering different strategies concerning the structure of reality, or concerning how historically problems have been “ill-perceived” or “ill-posed”. Deleuze and Guattari begin analyzing the way concepts are created in philosophy, starting with the concept of the Other. And it can be a reference to Hegel and the *Phenomenology of Spirit*. The Other is understood here primarily as one possible world. Deleuze and Guattari state that one can start from any other concept and carry out this type of analysis, pedagogy of the concept. It also suggests that there is no original concept or a story. Stories and concepts are constituted simultaneously. And they always return differently.

IV The story of chaos is never the same

Is there a criterion that determines what constitutes a bad thought, concept, or story? We can think and process chaos in different ways, through arts, sciences, and philosophy, and each discipline does it according to the narrative constellation that makes these renderings of chaos possible. But these are not absolute. There are good and bad concepts and planes of immanence. However, even if some planes of immanence and concepts formed or constructed before were good, it does not mean that they will have that status forever. They will be appropriate (or inappropriate) for the era or period in which they appear, but there is no guarantee of their success in

the future. The only “warrantor” would be the operative logic that the said narrative constellation brings about. The weaving of stories creates the specific web which resembles the Foucauldian notion of *épistémè* (Foucault 1972: 191–192). But this also means that concepts can be revitalized. Those who once aimed at something and were hitting a target may now prove inadequate in their attempt to account for some state of affairs. It can even happen that an initially poorly “designed” concept becomes functional when it is put into operation in another period, a different narrative fabric that constitutes forms of knowledge and thinking in an era. Even a bad concept can be resuscitated, brought back to life, depending on a set of conditions that would enable the concept or a story to flourish once again. In this sense, there are no absolute, transhistorically good and bad concepts. The same can be said about functions/functionives in sciences – how many times has it happened that the so-called intuition of some earlier philosophers/scientists proved to be more appropriate in the future time (regardless of whether it was “disproved” again later). The list goes on: Bruno, Leibniz, Descartes, Spinoza, Schelling, Plato, among many others.

Whatever concordance we may find with Deleuze’s and Guattari’s idea that we can strictly delineate between philosophy, science, and art, we still have to remind ourselves that this demarcation was not always plausible (some might argue that it’s not viable or achievable even today). If we compare, for example, Plato’s, Aristotle’s, Newton’s, Descartes’, Leibniz’s *concepts* with their *functions/functionives*, we would find it extremely difficult to find any sort of “cut” between them, and to differentiate the parts that belong strictly to philosophy or sciences. The very idea of what constitutes or counts as progress in philosophy or sciences comes from a different angle:

Cartesian concepts can only be assessed as a function of their problems and their plane. In general, earlier concepts were able to prepare a concept but not constitute it, it is because their problem was still trapped within other problems, and their plane did not yet possess its indispensable curvature or movements. And concepts can only be replaced by others if there are new problems and another plane relative to which (for example) “I” loses all meaning, the beginning loses all necessity, and the presuppositions lose all difference – or take on others. A concept always has the truth that falls to it as a function of the condition of its creation (Deleuze & Guattari 1994: 27).

We can say that this is precisely what happens to Deleuze and Guattari, who do not create a story about stories, although they explain their modes of functioning on a current level. The first sentence of the previous quote shows the difficulty of accounting for true or false narratives. The story of the Cartesian subject is true inasmuch as it aptly captures the structure of language in a corresponding epoch, in an almost photographic sense. It is a pure snapshot of a certain nature of things as it manifests itself at a given moment. But the moment is not eternal, because narrative

horizons and constellations change. Thus, even the story of Descartes' cogito cannot withstand the burden of the processual character of language, which transposes that story through further genesis of the story, retelling it "always" (now) in difference. No philosophy can be claimed to be forever outmoded or obsolete, useless, because it solely depends on how different conceptual networks will suit which age, whether they will be able to be put into operation or not. Following the footsteps of great philosophers should be such that we do not repeat what they said, but repeat what they did.

Touching upon the subject of plane of immanence and its corresponding concepts, it should be noted that the plane of immanence must not be confused with the concepts that occupy it. The plane of immanence is pre-philosophical if philosophy begins with the creation of concepts (Deleuze & Guattari 1994: 40). That observation goes hand in hand with the affinity of the understanding of Foucault's *épistémè* and plane. Concepts inhabit planes that are like deserts that move themselves (Deleuze & Guattari 1994: 41). We can also make a certain analogy between the plane of immanence and Saying (Saga), as a pre-concept in Heidegger (Heidegger 1971: 122). There is an undeniable affinity that runs across the idea of nonphilosophical, on the one hand, and the poetic character of thinking, on the other: "The nonphilosophical is perhaps closer to the heart of philosophy than philosophy itself, and this means that philosophy cannot be content to be understood only philosophically or conceptually, but is addressed essentially to nonphilosophers as well" (Deleuze & Guattari 1994: 41). The plane of immanence is presupposed, it is a pre-philosophical plan.

What is crucial here is to understand how the relationship between chaos and thinking emerges. Deleuze and Guattari show that the plane of immanence is "like a section of chaos and acts like a sieve" (Deleuze & Guattari, 1994: 42). They also point out that we should sort of re-arrange our thinking about chaos, or to put it differently, that we should let other stories about chaos form, while simultaneously keeping in mind that the very forming we are referring to is made possible by the pure immanence of the chaos-language-story-plane axis that produces all thinking and creating, at a specific moment:

In fact, chaos is characterized less by the absence of determinations than by the infinite speed with which they take shape and vanish. This is not a movement from one determination to the other but, on the contrary, the impossibility of a connection between them, since one does not appear without the other having already disappeared, and one appears as disappearance when the other disappears as outline. Chaos is not an inert or stationary state, nor is it a chance mixture. Chaos makes chaotic and undoes every consistency in the infinite. [...] By making a section of chaos, the plane of immanence requires a creation of concepts (Deleuze & Guattari 1994: 42).

This is why we are faced with the multiplicity of contingent stories. There is no ascension or descent into chaos, *stricto sensu*. There are only creative activities of different planes that plunge through chaos, bringing home the products of arts, sciences, and philosophies. If a plane acts like a sieve, then the hand holding a machete that moves through the jungle of chaos is nothing but the fabric of thinking through stories that create such metaphors (taken as metamorphoses). Thinking “raids” chaos, but it never plunders it completely, because there’s always a new story to be told, a new way in which chaos shifts, lending itself to not only different variations, but also to be fundamentally rethought. There’s even a constant danger of ceasing all thought and thinking, stopping the flow of stories. In a way, this has to be done because if stories are not flirting with their disappearance, they are condemned to *doxa*, and the stoppage of creative flows that make them possible in the first place. Stories endanger themselves if they are not constantly and effortlessly put to the test and on the brink of extinction. That is how and why we have new sciences, philosophies, or arts. And that should be the only way of talking about progress, the only criterion of assessing the results. Otherwise, we are putting in danger the whole line of reasoning which undercuts hierarchies and transcendences of all sorts. The despotic regimes of thought are constantly lurking, trying to stop this flow and to institute one way of thinking, one narrative horizon as Absolute and True. But the chaos, or rather stories of chaos, are having none of it, at least for the time being. That is how contingency works:

[T]he Greeks were the first to conceive of a strict immanence of Order to a cosmic milieu that sections chaos in the form of a plane. If we call such a plane-sieve Logos, the logos is far from being like simple “reason” (as when one says the world is rational). Reason is only a concept, and a very impoverished concept for defining the plane and the movements that pass through it. In short, the first philosophers are those who institute a plane of immanence like a sieve stretched over the chaos (Deleuze & Guattari 1994: 43).

The treasure hunt in which stories filter chaos, in the form of heterogeneous planes, shows a multitude of variations. In trying to look at a broader picture at all times, we find all sorts of different narratives that shape and inform all of our thinking, being, knowledge, and constitute apparatuses we once thought were unchangeable. But just as the plane of immanence passes over the field of consciousness (it is the notion of *cogito* that, starting from Descartes, through Kant, to Husserl allows the plane of immanence to be interpreted as a field of consciousness), we need to see transcendence as stopping of the movement of the infinite. It is “metaphysical” in its essence, if we understand metaphysics as an attempt to stop processuality. What distinguishes a philosopher from a sage is the institution of a plane of immanence and the creation of concepts. Perhaps the sky is another metaphor for the plane of immanence, while the concepts are stars, celestial bodies, etc. Interestingly, the reason for Nietzsche’s

introduction of the figure of Zarathustra can be glimpsed here – just as philosophy in its beginnings had to somehow disguise itself in a way that resembled the elements of the sage’s thoughts, so a new type of philosophy, and that means thinking, must go hand in hand with this disguise in the figure of Zarathustra. Zarathustra does not create concepts in the classical philosophical sense, but he constantly flirts with them. Zarathustra is both a conceptual persona and an aesthetic figure. This also calls for recognizing the multitude of planes of immanence, the multitude of contingent stories, their layerings, and the way they correspond to chaos. If there are multiple planes and stories, then we should approach the history of philosophy as an art of creating portraits (Deleuze & Guattari 1994: 55). But strictly speaking, philosophy should be seen more as a coexistence of planes, rather than a history (Deleuze & Guattari 1994: 59). And this coexistence of planes points out again to the narrative horizon which holds them in close proximity, or moves them apart, rendering them important or obsolete, at a given time. So “the supreme act of philosophy” (Deleuze & Guattari 1994: 59) thus becomes not so much to think the narrative horizon from which the stories emerge (nor the plane of immanence), but to recognize and show that they there, “unthought in every plane, and to think it in this way as outside and inside of thought, as the non-external outside and the not-internal inside, which cannot be thought and yet must be thought, which was thought once, as Christ was incarnated once, in order to show, that one time, the possibility of the impossible” (Deleuze & Guattari 1994: 59–60).

V The status and place of narratives/stories, narrative horizon(s), and narrative constellation(s)

What emerges as the next question is how do we further assess the relationship between narrative/story, narrative horizon, narrative constellation, web of stories, on the one hand, and the planes of immanence, reference, and composition, with their belonging concepts and conceptual personae, percepts/affects and aesthetic figures, functives and prospects, on the other hand. For instance, Socrates is the main conceptual persona of Platonism. But what about *Timaeus*? It’s a dialogue that differs from most of Plato’s, and in that it is almost a monologue (Socrates in the *Laws*, Plato’s last dialogue, does not have any protagonism at all). But not all characters, say in Plato’s dialogues, are conceptual personae. Some are built as likeable, amiable, charming, while others are unsympathetic or repulsive; some reflect the views of the author, others the views of his opponents. What is characteristic of *Timaeus* is that, written in the monologue form, where Socrates is not the main protagonist, Plato seems to want to inaugurate another conceptual persona, who would be Socrates’ competitor. The *Timaeus* has long been considered the main work of Plato, which gives it special significance. We could even argue that *Timaeus* is in touch with three different planes – immanence (philosophy), reference (science), and composition (art). And this is

perhaps its substantial importance. The last plane, that of art, is visible through the dramaturgical-rhetorical implications of introducing the figure of the Demiurge. The plane of reference can be sensed through the process begun by the pre-Socratics, who had “already grasped the essential point for a determination of science, valid right up to our own time, when they made physics a theory and their different types” (Deleuze & Guattari 1994: 126–127). In any case, Plato’s plane of immanence is constructed in such a way that the ontology it inaugurates tries to hide the traces that we were dealing with the plane of immanence all along.

One might argue that Deleuze and Guattari would consider narrative, as presented, to correspond to a conceptual persona. Others could be inclined to think about it in terms of a concept, or even a plane of immanence. But this is where things demand that we refrain from myopic views and from jumping into conclusions. The narrative constellation “holds together” narrative horizons, as the most obvious candidates for planes of immanence, reference, and composition. Narrative constellation is a condition for the possibility of speaking about narrative horizons, storytelling personae, planes of immanence, reference, and composition. Everything is summed up in a dance that never steps out of itself, or does not exceed itself, which is nothing but the pure immanence; striding (philosophical, scientific, artistic) of language, which can be seen as a dance. That dance is the movement of atoms from Democritus (atomic swerve) to the Standard Model, but also philosophy and art at the same time. That “striding” and “dancing” are far from being only aesthetic categories, but political, philosophical, and scientific just as equally. A persona is nothing but the intersection of narratives: “I am no longer myself but thought’s aptitude for finding itself and spreading across a plane that passes through me at several places” (Deleuze & Guattari 1994: 64). Where does this “thought’s aptitude” come from, other than language/stories, narrative horizon, and narrative constellation that sets in motion all the forces accountable and unaccountable? Thoughts find themselves only in stories. They touch upon chaos and bring us the arts, sciences, and philosophy. Of course, not all narratives have the same power or serve the same function. Some narratives are capable of fundamentally changing the image of thought. Nietzsche creates a plane of immanence in the form of an infinite will to power and eternal return. This serves as a critique of the will to truth. The repercussions of the said narrative horizon are multiple. The very thinking of Deleuze and Guattari harks back to this narrative trail.

Conceptual personae are analogous to aesthetic figures in the arts and partial observers in the sciences. But Deleuze and Guattari insist on their difference, which consists in the fact that conceptual personae are the power of the concept, while aesthetic figures are the power of affects and percepts. However, the basic question is whether they can be so clearly divided? For Deleuze and Guattari, interference is necessary, but the best results are achieved if everyone sticks to their own activity,

more or less,³ since there are rare examples of successful creation of concepts or affects, and perceptions by scientists, or philosophers who create functions, etc. That is why the debate between Einstein and Bergson resembled a game of Chinese whispers. Bergson wanted Einstein to think in terms of duration, while Einstein had the relativity function of time in mind all the time. None of them can be claimed to have been right or wrong from the point of view of some absolute plane, because there is no such plane. Struggle, or connection with chaos, is their basic impulse, just like in the arts. But chaos is not a plane, it is the absence of one. Both of them spoke from different planes and with different plans – Bergson from the plane of immanence and with the plan of pointing out to the vulgar conception of physical time, and Einstein from the plane of reference and with the plan of pointing out to the facts. But we have also seen with the example of Nietzsche how difficult a task it is to separate planes. Nietzsche the poet, Nietzsche the mythmaker, Nietzsche the philosopher, Nietzsche the scientist. On the other hand, we can see something similar in Plato himself. His conceptual figure of the Demiurge has some of the characteristics of both the power of the concept and the power of affects and percepts. However, it is at this point that it seems crucial to refer to the linguistic-narrative dimension of all these experiences. In this way, the necessary interweaving between the narratives of philosophy, science, and art can be explained.

The peculiar character of art and philosophy, and the answer to the question of what they do, is thus found in recognizing the power of affects and concepts to overcome the basic feelings, perceptions, and everyday opinions. So, the art and philosophy always get entangled with chaos, but in a manner that the starting position is always story-bound, as well as the end result. Thinking with affects and percepts, or concepts and functives, gives different results, with the only constant being ever-changing contingent stories, processual “in nature.” And even this “nature” does not lie beyond the narrative horizon or constellation that puts it in motion. It is the operative logic of a narrative constellation that determines the boundaries of all thinking, at a specific moment, in a certain epoch.

“Art and philosophy crosscut the chaos and confront it, but it is not the same sectional plane; it is not populated in the same way. [...] Art thinks no less than philosophy, but it thinks through affects and percepts” (Deleuze & Guattari 1994: 66). One of the key things for Deleuze and Guattari is that authors like Hölderlin, Rimbaud, Kleist, Mallarmé, Kafka, Michaux, Pessoa, Artaud do not synthesize philosophy and art, but are some kind of tightrope walkers, “athletes,” “hybrid geniuses,” who are placed in the very difference between philosophy and art, which are constantly diverging. For us, the question should also be about the relationship between philosophy and science, and art and science, in this regard. Could Plato be considered such a

³ Should it be understood as a Platonic moment from *The Republic*, where the roles are distributed in advance?

“hybrid genius?” Didn't philosophy, careless of pure immanence, create him as its figure or conceptual persona, for which Plato himself certainly already gave all the reasons? Can we now see it stripped bare, beyond the ballast of the accompanying history of philosophy and out of balance that produces the baseless foundation of ontological pyramidity? Conceptual personae and aesthetic figures (Plato) create conceptual personae and aesthetic figures (Socrates, Demiurge, Immovable Mover), which create... Even Deleuze's and Guattari's *nomad* is both a conceptual persona and an aesthetic figure, but also a psychosocial type. It is clear that concepts, perceptions, and affects can intertwine, creating new images of thinking and planes of immanence. If so, an alliance can be sensed between philosophy and science, i.e., concepts and functions, as well as between science and art. The gesture of rolling the dice that pervades philosophy and thought in general – Pascal, Mallarmé, Deleuze, Guattari – points to a way of creating stories, but is a story in itself. At the same time, we also sense a certain crisis, even a burden of an imperative to create and be creative. Today's hyperproduction is a reflection of that crisis. We lack the creation of new stories and new narrative horizons (plane of immanence) and insight into their contingency: “We do not lack communication. On the contrary, we have too much of it. We lack creation. *We lack resistance to the present.* The creation of concepts in itself calls for a future form, for a new earth and people that do not yet exist” (Deleuze & Guattari 1994: 108). This is also the story that Nietzsche tried to tell. But in Deleuze and Guattari, this story is retold and repeated with difference in mind. That is a breakthrough that becomes fully aware of *becoming* – the perennial becoming of philosophy and the accompanying stories in which new ways of living and possibilities of being are sketched.

VI Chaos story, or the story of chaos

One of the main and first differences between philosophy and science is their individual attitude towards chaos (Deleuze & Guattari 1994: 117–118). On the following pages, Deleuze and Guattari give, on the one hand, a good insight into how science tries to slow down the endless speed of chaos, analyzing the frozen frame. We can compare and contrast how time is approached in the New Age (Newton, Descartes, etc.) and Schelling, Bergson, Husserl, as well as Einstein's leaning on the tradition, however much the theory of relativity has at first glance a radical turn concerning the New Age. Is there a way to *define* chaos at all? Or are we always caught up in the middle of a story that tries to give any sort of account regarding the subject matter? For Deleuze and Guattari:

Chaos is defined not so much by its disorder as by the infinite speed with which every form taking shape in it vanishes. It is a void that is not nothingness but a *virtual*, containing all possible particles and drawing out all possible forms, which spring up only to disappear immediately, without

consistency or reference, without consequence. Chaos is an infinite speed of birth and disappearance (Deleuze & Guattari 1994: 118).

We should add that chaos appears and disappears only in stories. The plane of immanence cuts through the chaos, and so do the planes of reference and composition, creating philosophy, science, and art. They are all different, have at times contrasting or diverging results, but the key thing to keep in mind is that the *virtuality* that chaos is comes not only hand in hand with stories/language, but they are essentially indistinguishable. We can agree that science, art, and philosophy all do different things with chaos, but we could also argue that the metaphysical thinking of philosophy works actively to slow down and “stop the image” as well. What else would be the point of making a distinction between being and becoming? Also, on that track, could it be said that Descartes, Leibniz, Gassendi, Newton, and others just simply switch between concepts and functions, between planes of immanence and reference, when they try to establish an understanding of the world and cosmos that is in agreement with these respective fields? For Deleuze and Guattari, the sciences “slow down” the chaos, establishing a plane of reference, along with the functions of the coordinate system, numbers. The plane of reference is a sort of freeze frame of chaos, with its velocities, forces, energies, particle positions, masses, spin values, etc. But physicists also tend to discover concepts while operating with functions. It must be taken into account that philosophy also often thinks within the framework of these same functions, statements, searching for concepts, and stories, not realizing that the boundaries set by these different reference systems are already written into the plane of immanence of stories. In other words, stories are “rejected” for the sake of celebrating the victory of the accuracy of statements, not realizing that they were made possible by stories in the first place. Philosophy tries to imitate science by placing itself under the patronage of functions, subordinating thought to what is not its vocation in that particular realm. Nevertheless, there’s at least one way of distinguishing between the two. If science is paradigmatic (Kuhn), philosophy is syntagmatic (Deleuze and Guattari). And that speaks volumes about the difference between planes of immanence and planes of reference. It can be said that there are many of both, but planes of immanence can have the actuality and legitimacy of their validity at the same time, while the plane of reference requires a singular paradigm based on which the main research moves. On the other hand, the similarity of philosophy, science, and art lies in creation/creativity, which is the creation of stories while touching upon chaos.

We should also keep in mind that science is not deductive,⁴ just as philosophy is not, and one should not be fooled into believing that there are pre-existing functions, just as there are no pre-existing concepts (the Platonism of philosophy and science that is often presumed and almost tacitly taken over). If we exposed philosophy in its Platonic tendencies to establish transcendence, we must do the same with science.

⁴Cf. Aristotle and the birth of scientific method.

How exactly someone does not know something and what creative actions she or he might use in trying to bridge an assumed and implicitly recognized ignorance becomes the line of demarcation between science, philosophy, and art.

A conceptual persona (philosophy) corresponds to a partial observer (science). For example, this would also explain the kind of New Age characteristic in which conceptual personae and partial observers are mixed. But even if we went much earlier, doesn't antiquity already mix them? Pre-Socratics, just like Plato and Aristotle? Aristotle's *Physics* shows itself as exactly that – the unmoved mover is shown as a conceptual figure that superimposes a partial observer. Galileo, Descartes, Leibniz, and Gassendi constantly move along transversals that cut across the planes of immanence and reference; that is, they do not respect these boundaries between philosophy and science. Newton already shows a determination to use as little as possible of the aforementioned transversal between the plane of reference (science) and the plane of immanence (philosophy). But the question is how much he succeeds in doing so. He is interested in showing the *Mathematical principles of natural philosophy*, but in such a way that he expresses his desire to think of concepts as functions, i.e., to inscribe functions into concepts so that the concept becomes marginalized. And there, in the Renaissance and the New Age, an almost imperceptible violence takes place; but we must not fool ourselves that the concept is innocent in its history; it is the source (or one of the sources) of that violence in the first place. The concept is sometimes a fundamental expression of unfreedom, because it requires the taming of free signifiers; it clings to chaos, forms itself as a story, and often insists on its absolute truth. Not only that, it also demeans chaos, blessing the cosmos. And the very etymology of the word κόσμος [*kósmos*] testifies to this. The verb κοσμεῖν refers to order and arrangement, but also to the establishment of a regime. *Kósmos* is originally ornamental, decorative. No wonder mathematics and physics are persistently and repeatedly searching for an elegant, beautiful formula, from the Pythagorean number 10, to $E=mc^2$. However, something negative is seen in chaos, without the insight that every plane of reference, immanence, and composition is nothing but always a partial attempt to tame chaos. And here is the answer to the very problem of quantum mechanics, which unconsciously or perhaps belatedly mourns the death of god (but also reference), that is, transcendence, which it supposedly calmly accepted and set as its condition for the possibility of development, lest at the very end of the road it discovers that it has not even begun to process that death: “The fact that there is no total observer that, like Laplace’s “demon,” is able to calculate the future and the past starting from a given state of affairs means only that God is no more a scientific observer that he is a philosophical persona” (Deleuze & Guattari 1994: 129). Borges, or Hugh Everett III, coquette with the total observer, but also with philosophical concepts, as much as with philosophical functions or artistic perceptions and affects. Following James Gleick, it should also be clear that science doesn’t just bring order

into chaos, but also has a need to *produce chaos* in its own unique way (Deleuze & Guattari 1994: 229).

Bearing everything in mind, it could also be said that there is no *anti-Platonism* in Deleuze and Guattari. This does not mean that they are Platonist thinkers, but that they are not his opponents (nor could they be), strictly speaking. What they show here is how Plato invented his concepts, such as the concept of the Idea. That gesture surely is *against* Platonism, but it is by no means pismatic against Plato himself. Moreover, Plato is valued as a figure who makes an immeasurable contribution to the development of philosophy; through Plato's incredible ability to hide almost all traces of immanence (while constructing his own plane of immanence), and to establish transcendence, hierarchization, pyramidal *ontos on*, based on pure immanence. Plato thus becomes the greatest master of simulacra, the greatest deceiver in the history of philosophy. His biggest trick was convincing the world that he wasn't the one who invented the Ideas. Sobering up lasted for two millennia, until Nietzsche sought in Dionysus' inebriation a cure for the hangover, caused by intoxication with the Socratic-Apollonian (Platonic) *ratio*. And just as Nietzsche's *Ecce Homo* repeats Pontius Pilate, who refers to Christ as *a man*, Deleuze and Guattari chorus repeat (*ritornello*) the refrain of *here is a concept*. That concept of the Idea is Plato's construct, just as God is a construct of man, and man is a linguistic-narrative construct. And there can be no anti-Platonism here – there can only be mixed feelings about deception and a swindle on such a large scale, where the stakes were and are enormous, and where we still see, despite the significant effect that Nietzschean thought produces, the essentially Platonist *modus vivendi* of almost the entire civilization. Science, philosophy, art – all in turn repeat this pattern of thinking, which demands, seeks the Truth of things in some transcendence, in some scientific theory, in some so-called Truth of the work of art. We can applaud the master of deception who brings all these conceptual personae, partial observers, and aesthetic figures to the stage. First and foremost, this also needs to be recognized as a narrative constellation. Plato as such is insignificant, nor does he exist in that form (“as such”), but is also a product of the creation of chaos-language-stories-thinking, and the various planes. We can constantly refer to the deceptive nature of the operative logic by which we organize our lives. We can also recognize that the categories of truth and deception need each other and provide fuel for the culture in which we find ourselves. But to advocate anti-Platonism is absurd, pointless, and fruitless. How then is it possible to plead for a *return to Plato*, in the footsteps of Badiou? In a way, *we have to do it*. It is one of the most urgent tasks. And perhaps in exactly this way that Deleuze and Guattari are proposing – by going back and stripping, uncovering, exposing everything Platonistic, with all its implications and negative consequences.

Through science and logic, philosophical concepts pass through a kind of filter, and what is left of them is what logic and science try to operate with in the capacity of a rarefied, fragmented, devastated, desolated philosophical concept. Is there a

kind of Platonism creeping in? Because it turns out that the philosophical concept is triply distant from its authentic, philosophical conception. On the part of Deleuze and Guattari, the answer would probably move in the direction of pointing out the following: 1) that supposedly authentic philosophical concept is not transcendent to the diluted concept that passes through the sieve of science and logic; 2) the plane of reference borrows a concept from the plane of immanence and makes it strictly referential, that is, makes of it functions/functions or statements; 3) the concept belongs to a simulacrum in this respect, because the plane of immanence in its non-strictly referential sense can be understood as a space in which, by trimming Platonism, the binary oppositions of true and false, truth and lies, the concept is left to lie as a creative dimension of language itself. In its self-transformative role, the concept is recognized as a title, theme, or header of a story, like a keyword that is associated with a certain narrative, but cannot be completely reduced to it. Why? Because the concept understood in this way literally beckons, it calls out to other narratives, and the character of that call is certainly linguistic. Deleuze and Guattari use the term “conceptual personae,” wishing not to summon all the ghosts from the lamp by labeling philosophers “authors,” “subjects,” or “individuals.” Therefore, these “conceptual personae” are themselves placed in the narrative-linguistic weaving and from here understood as “hubs” through which, with which, and from which concepts-narratives manage to proliferate. Those different planes or zones of science, logic, philosophy, and art certainly intertwine, but there is also something negative in that interdisciplinarity, which Deleuze and Guattari point to. The other side of the coin is an attempt to “restrain” the plane of either philosophy, science, logic, or art and establish a hierarchy between them. Just as much as it is a violent act of repackaging a concept into the framework of a function or statement, so is the stake high in restraining perception or affect with a concept, etc. The concept in actuality looks like an intermediary between narratives, as one into which previous narratives merge, but always in such a way that it opens up from itself, evoking other narratives. However, the concept is not a mediator in the true sense of the word, but only a way station of the narrative, like a bottleneck through which the flood of language-narrative flows and does not stop, but only slows down to gather narratives (planes of immanence) around and within itself so that they correspond to a certain state of affairs. This is how correspondence is established, although it is *stricto sensu* unmediated, because it is immanent. From there, it should be understood that there is nothing outside the narrative. Narrative is an event in the sense that Deleuze and Guattari point out that events are what show the consistency of a concept. Only through story/language is there grasping of chaos, which is not passively waiting to be told in different variations. Stories of chaos come into fruition because chaos and language are those virtualities upon which the narrative tissue forms in thinking.

VII Conclusion: the case of a lost umbrella

Language twists and turns all the time, but its different and differing manifestations are probably most obvious in art. Nietzsche, Kafka, Gherasim Luca, Kleist, Artaud, Joyce: their work is a testimony to the unstoppable force that language, thinking, chaos, and stories create. *Finnegan's Wake* is an example of dislocation of language, deterritorialization of established language modes, of letting language grow like a meadow or a rainforest. Language clears its own paths (Heidegger), but it cannot rightly be said that it equally destroys these paths in return, because by this blocking of the paths to their final unrecognizability, the ground is actually being prepared. Language covers its traces, is distributed along unpredictable axes and planes; at times, its transparency is presented as crystal clear (the light of *nous*), while on the other hand, it drags it into a vortex, hints at chaos, and arranges the conditions for new, different types of creation. He deforms, twists, swallows with an unstoppable force the world he creates at first, lest he spit it out; he can no longer stand his own vision of Apollonian orderliness, which he offered for a moment. That's how language works, woven from folds, which are nothing but the folds of stories/narratives. The hums of the language-narrative are diverse, from the unbearable noise of chaos, which hums, screeches, and pinches the auditory nerves, all the way to the consonant and dissonant tones it produces in the musical piece. What thinking does in art, science, and philosophy is "always confronting chaos, laying out a plane, throwing a plane over chaos" (Deleuze & Guattari 1994: 197). The variation of a (no) joke that we started this paper with can now be complemented – philosopher, scientist, and artist enter chaos, return from the land of the dead, and bring three things: philosopher variations, scientist of variables, artist of varieties (Deleuze & Guattari 1994: 202). They all protect against chaos in some way. This is the basis of suffering that Nietzsche also refers to, arguing that only by accepting this suffering, and not repressing it, can we radically transform our own circumstances:

The struggle with chaos that Cézanne and Klee have shown in action in painting, at the heart of painting, is found in another way in philosophy: it is always a matter of defeating chaos by a secant plane that crosses it. Painters go through a catastrophe, or through a conflagration, and leave the trace of this passage on the canvas, as of the leap that leads them from chaos to composition. Mathematical equations do not enjoy tranquil certainty, which would be like the sanction of a dominant scientific opinion, but arise from an abyss that makes the mathematician "readily skip over calculations," in anticipation of not being able to bring about or arrive at the truth without "colliding here and there." And philosophical thought does not bring its concepts together in friendship without again being traversed by a fissure that leads them back to hatred or disperses them in the coexisting chaos where it is necessary to take them up again, to seek

them out, to make a leap” (Deleuze & Guattari 1994: 203).

Those familiar with Nietzsche’s work might recall a puzzling sentence that he wrote, and that was found in his manuscripts after his death: *Ich habe meinen Regenschirm vergessen (I have forgotten my umbrella)* (Nietzsche 1881: 12[62]). Many authors, including Derrida in his *Spurs: Nietzsche’s Styles* (1979), take this sentence into account and try to analyze its meaning, or lack thereof; they ask themselves different questions regarding how this sentence got its way there. Was it an outline of an idea? Was it a simple reminder? Something he just liked? Was he completely insane when he wrote it? Can we ever get to the bottom of the author’s intentions, in the first place? “Gadamereans” and “Hirscheans” all around are sharpening their pencils... But Deleuze and Guattari are referring to Lawrence, who offers a description of how poetry is being produced: “[P]eople are constantly putting up an umbrella that shelters them and on the underside of which they draw a firmament and write their conventions and opinions. But poets, artists, make a slit in the umbrella, they tear open the firmament itself, to let in a bit of free and windy chaos and to frame in a sudden light a vision that appears through the rent” (Deleuze & Guattari 1994: 203). We ask ourselves: can there be a more convenient metaphor than the one proposed by Deleuze and Guattari, where opinions are like an umbrella that protects against chaos, and where great philosophers, scientists, and artists deliberately cut this umbrella to let in parts of chaos? When Nietzsche says *I have forgotten my umbrella*, things could not be more obvious, or less obvious, depending on which angle we look at it: the last phase of Nietzsche’s life can be characterized as the complete lowering of this umbrella, which is the umbrella of opinions, prosaic, but also philosophical, scientific, artistic thoughts. It is a question of deliberately forgetting the umbrella, as a shield of *ratio*, which is only one of the planes, one of the narrative constellations that tries to tame the chaos. Of course, Nietzsche had to not only make a small slit in the firmament, but forget the whole umbrella and let all the chaos come upon him. What else could a philosopher of suffering do? But it is worth noting that the philosopher, scientist, and artist may sometimes fight less against chaos than against clichéd opinions (Deleuze & Guattari 1994: 204). However, it can be said that the reason is also that chaos in that prosaic and platitudinal form is not and cannot be exhausted. Language-chaos, tamed by language-stories. Nietzschean “gazing into the abyss” could be worked out exactly in the direction of *chaos*, which is of substantial significance for Deleuze and Guattari. This “gazing into the abyss” is sort of a precondition for “skimming” the chaos and ordering it in multiple ways. That is essentially how we relate to chaos (as pure immanence), by taking or peeling off its layers. As a result, we have arts, philosophy, and sciences.

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Haos i mišljenje

(Rezime)

U ovom radu autor razmatra poslednju objavljenu knjigu Deleza i Gatarija, „Šta je filozofija?“ (1991), u kojoj se ovi francuski mislioci upuštaju u jedinstveno putovanje sa ciljem da publici ponovo pruže odgovor na to monumentalno pitanje. Polazeći od jasne svesti da nisu samo filozofi želeli ili zahtevali taj odgovor, Delez i Gatari nastoje da demistifikuju i opovrgnu pogrešnu predstavu da se filozofija tiče isključivo filozofa. Pomenuta knjiga je podjednako mogla nositi naslov: „Šta je filozofija, šta je umetnost i šta je nauka?“. Imajući na umu ova tri ključna pitanja, knjiga predstavlja značajan poduhvat u prikazivanju kako načina na koji deluje čista imanencija, tako i načina na koji tri odgovarajuće discipline – umetnost, nauka i filozofija – ulaze u haos i izlaze iz njega, postižući različite rezultate. Ne treba posebno naglašavati da je danas neophodno preispitati i revalorizovati pojam haosa. Delez i Gatari nam pomažu da shvatimo da je od izuzetne važnosti razumeti da se haos ne sme posmatrati kao negativan pojam, te da se on suštinski nalazi izvan puke suprotnosti pozitivnih i negativnih kategorija. Naš cilj ovde je da pokažemo da veza između mišljenja i haosa uvek vodi ka pričama ili narativima, u kojima umetnost, nauka i filozofija oblikuju svoje jedinstvene ravni. Na taj način se uspostavlja neraskidiva veza i savez između priča, mišljenja i haosa.

KLJUČNE REČI: haos, mišljenje, priče, narativ, filozofija, nauka, umetnost, imanencija

PRIKAZI



Igor Živanović

Prikaz knjige: Kameron, Eva. "Svirepi suparnik. Filozofija raka." Fedon, Beograd (2024)

***DUBOKO U TEBI: EVOLUCIONA ISTORIJA RAKA*¹**

Uzeću slobodu da u ovom prikazu odstupim od uobičajene ortodoksije kada je reč o pisanju ovakvih tekstova i zaobiđem suvoparno, dosadno, taksativno i činovničko nabranje, klasifikovanje i sažimanje onoga što je u knjizi sadržano. Polazim od razumne pretpostavke da svako ko se dohvati ove knjige i zaviri u njene poslednje stranice gde se nalazi sadržaj, to lako može da sazna, tako da nabranje poglavlja, potpoglavlja i prepričavanje njihovog sadržaja smatram izlišnim, zamornim i nezanimljivim. A ova knjiga je sve samo ne dosadna i nezanimljiva. Zapravo, knjiga Eve Kameron, *Svirepi suparnik* je uzbudljivo filozofsko delo, koje na nedvosmisleno jasan način pokazuje kako život i biološki procesi sagledani kroz prizmu nauke i filozofije mogu da budu jednako uzbudljivi ili pak uzbudljiviji od fikcije. Iz tog razloga, pokušaću da na približno zanimljiv način čitaocu upoznam s ovom knjigom, to jest s njenom centralnom temom, koja koliko god da je uzbudljiva, ne može a da bar malo ne onesposobi čitaoca, jer je reč o filozofiji raka, jedne opake bolesti koja godišnje odnese veliki broj života i čije izlečenje ili iskorenjivanje i dalje nije nadohvat ruke.

Verovatno najpoznatija analogija u istoriji filozofije jeste ona koju Platon pravi između duše i polisa u dijalogu *Država*. On poziva svoje sagovornike da najpre ispituju šta je pravda u kontekstu polisa, da bi lakše i bolje mogli da razumeju šta je pravednost na individualnom planu. Ono što je malo je možda najbolje sagledati pod uveličavajućim staklom pod kojim pojedine nejasnoće postaju očiglednije, a prostor potencijalnog nerazumevanja se sužava. Možda nešto slično možemo da učinimo i kada je reč glavnoj temi ove knjige. Iznenadjujuće, to šta rade ćelije raka, možda možemo lakše da sagledamo ako se okrenemo pčelama. U knjizi Eve Kameron ove pojedinosti izostaju, ali mogu da posluže kao ilustrativan primer za njeno razumevanje, kao što je Platonu proučavanje strukture države poslužilo za bolje razumevanje ljudske duše. Čini se da ovaj pristup duhom ne odstupa od osnovnog tona i metoda knjige u kojoj su predstavljeni i kritički ispitani teorijski modeli nastanka malignih promena.

Kod medonosne pčele uglavnom samo matica polaže jaja, a biološki pol je određen

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time da li su jaja oplodena ili ne. Iz oplodjenih jaja se izležu ženke (radilice i matice), dok mužjaci (trutovi) nastaju devičanskim začećem, izležu se iz neoplodjenih jaja partenogenezom. Ako veoma pojednostavljeno pretpostavimo da se matica pari s jednim mužjakom, a s obzirom na specifičan (haplodiploidan) način razmnožavanja i odnose srodstva u pčelinjoj zajednici, gde viši stepen srodstva nalazimo među radilicama (sestrama), nego između matice i kćerki, u genetičkom interesu radilica je da potisnu svoje reproduktivne aspiracije i da pomažu matici da proizvodi sestre. Ali, ako se matica pari s više mužjaka, što je mnogo češći slučaj, samo srodstvo nije dovoljno za potiskivanje sebičnih težnji radilica.

Budući da radilice imaju očuvane reproduktivne organe, ali ne mogu da se pare, one mogu da polažu jaja iz kojih se razvijaju mužjaci. Sebične težnje radilica i proliferacija mužjaka, koja narušava socijalnu harmoniju, mogu da se obuzdaju redarenjem, to jest odstranjivanjem jaja koje polažu radilice. Redarenje u kolonijama socijalnih insekata možemo da tumačimo kao deo socijalnog imuniteta, kojim se uspostavlja društvena ravnoteža i obuzdavaju sebične težnje nekih jedinki. Radilice na osnovu olfaktornih signala (feromona) mogu da otkriju koja jaja je položila matica i da zatim eliminišu ona koja polažu druge radilice. Za razliku od ljudi (i drugih sisara) koji imaju urođeni i stečeni imunitet, društveni insekti imaju urođeni i socijalni imunitet. Za sada je sve dobro, ali kao što je svima dobro poznato, stvari umeju da krenu i po zlu.

Naime, neke parazitske vrste pčela mogu navedene signale da krivotvore, što može da ima pogubne posledice po pčelinju zajednicu. Kod *Apis mellifera capensis*, podvrste obične medonosne pčele koja nastanjuje južnu Afriku, dešava se nešto neobično. Naime, partenogeneza radilica se javlja u obliku telitokije, to jest iz neoplodjenih jaja se ne izležu mužjaci, već one polažu diploidna jaja iz kojih se izležu isključivo ženke. Ove reproduktivno sposobne radilice zauzimaju košnice druge podvrste medonosne pčele, *Apis mellifera scutellata*, i tamo polažu jaja iz kojih se izležu radilice. Kako ova jaja oponašaju ona koja je položila matica, ovi uljezi uspevaju da izbegnu da budu otkriveni i da postanu žrtve socijalnog imuniteta. Ono što se dalje dešava narušava funkcionisanje pčelinjeg društva: domaćini ulažu resurse u odgajanje parazitskog potomstva, narušava se društvena kohezija, produktivnost opada, i konačno dolazi do kolapsa zajednice. Sebične reproduktivne strategije unutar zajednice kooperativnih jedinki mogu da podriju tu zajednicu i u krajnjoj konsekvenci da je unište. Slična vrsta reproduktivne sebičnosti važna je za razumevanje raka, jer ako organizam tumačimo kao kooperativni poduhvat velikih razmera u kome su se pojedine ćelije odrekle dela svoje reproduktivne autonomije, onda svaki neosujećeni pokušaj individualne reprodukcije može da predstavlja pretnju.

Okrenimo se zato od velikog ka malom. Ćelijski ciklus somatskih ćelija prolazi kroz niz faza, nastanka, diferencijacije, deobe i ćelijske smrti. Jedna od osnovnih bioloških zakonitosti jeste ta da somatske ćelije, mogu da se podele samo ograničeni broj puta, nakon čega ih sustiže starost i ćelijska smrt. Ljudske telesne ćelije mogu da se

podele oko 50 puta pre nego što skončaju. Gornja granica potencijalnih ćelijskih deoba, poznata je kao *Hejflikova granica*. Uzrok postojanja gornje granice ćelijske deobe jeste taj što se na krajevima hromozoma nalaze telomere, ponovljeni nizovi DNK molekula koji se skraćuju prilikom svake ćelijske deobe. Telomere bismo mogli da posmatramo kao vrstu biološkog sata koji otkucava. Kada vreme istekne, ćeliju sustiže starost i *apoptoza*, programirana ćelijska smrt. Specifičnost ćelija kancera, između ostalog, jeste prisustvo enzima telomeraza koji omogućava neprekidno obnavljanje telomera i beskonačnu ćelijsku deobu. Zahvaljujući telomerazi, koja nije aktivna u ćelijama odrasle osobe, ali jeste u ćelijama raka, ove ćelije mogu da prekorače *Hejflikovu granicu* i da se, bar potencijalno, dele u beskonačnost.

Da bi viševićijski organizam mogao da se održi kao jedinstvena celina pojedine ćelije su morale da se odreknu reproduktivne autonomije. Ovo je osnova međućelijske kooperacije koja omogućava celovitost viševićijskog organizma. Apoptozu, programiranu ćelijsku smrt, kao važan deo imunskog odgovora organizma, mogli bismo da tumačimo kao intrinzični funkcionalni analogon redarenja unutar kolonija socijalnih insekata. Ovaj mehanizam onemogućava međućelijsku kompeticiju unutar viševićijskog organizma, odnosno sebičnu proliferaciju somatskih ćelija, koja vodi pojavi kancera. Međutim, kao i parazitske pčele, ćelije kancera mogu da izbegnu ove zamke i nastave svoj pohod. Osim toga, ćelije kancera mogu da se kreću što je, pod normalnim okolnostima, karakteristika rezervisana isključivo za krvne ćelije, limfocite i polne ćelije. Ova karakteristika ćelija raka omogućava metastaze, to jest „nastanjivanje“ i razmnožavanje na mestu koje je udaljeno od lokacije nastanka tumora. Tako rak može da proizvede novu koloniju malignih ćelija na drugom mestu. Ćelije koje se iz bilo kog razloga odvoje od ekstracelularne matrice ubrzo sustigne *anoikija*, tihi šapat programirane ćelijske smrti. Međutim, ćelije kancera ne čuju ovaj šapat, nastavljaju da žive i da se razmnožavaju.

S jedne strane ćelije kancera možemo da posmatramo kao vešte odmetnike kojima regulatorni mehanizmi mogu teško da stanu na put. Kad smo već krenuli od Platona, da se poslužim jednom brodskom alegorijom. Zamislite brod na kome se dogodila pobuna. I dok se u početku samo nekolicina mornara odvaži da ustane protiv dominantnog kapetana, kao odgovor na kapetanovu represiju protiv pobunjenika, sada se sve više mornara priključuje pobuni, dok ne svrgnu kapetana, preuzmu brod i sunovrate ga u propast. Sada, što je imunski odgovor organizma snažniji na prisustvo odmetnutih ćelija, to one postaju snalažljivije i nastoje da zaobiđu uobičajene telesne mehanizme za borbu protiv stranog tela, koje u slučaju raka uopšte nije strano, već veoma poznato, ali tek toliko izmenjeno da može da uzurpira telesne mehanizme i na prevaru preuzme kontrolu. U bespoštednoj borbi za opstanak organizam uglavnom gubi, i paradoksalno, poraz organizma kao celine je istovremeno i poraz pobunjenih ćelija, koje nestaju zajedno sa svojim nosiocem. S obzirom na to da se kao veslači svi nalaze u istom čamcu, oni dele zajedničku sudbinu.

Na primeru raka može da se pokaže da nesputana proliferacija ćelija vodi tome da ta populacija ćelija u jednom trenutku bude zbrisana. Ili, drugačije rečeno, ako pretpostavimo da populacija ćelija raka predstavlja populaciju potencijalno besmrtnih organizama koji se neograničeno razmnožavaju, onda možemo da uvidimo da je prirodni rast takve populacije po osnovnim maltusovskim principima samoporažavajući, s obzirom na to da je njihov nosilac, to jest resurs koji koriste i kog bespoštedno iskorišćavaju, ograničen i smrtan. Ovo bi bio alegorični prikaz modela koji ćelije kancera sagledava kao mutirane varalice, pobunjenike i odmetnike, koji narušavaju odnose međućelijske saradnje inherentne složenoj biološkoj organizaciji organizma, i koji uz to uzurpiraju imunski odgovor organizma i koriste ga za svoje sebične ciljeve. Treba imati na umu da se strateško ponašanje ćelija raka odvija na nivou slepih kaulzalnih procesa i da je teleološki jezik ovde samo pomoćno sredstvo koje služi lakšem razumevanju. Sama autorka upozorava na korišćenje metafora i teleološkog jezika u biologiji, ali za potrebe prikaza ove knjige i njenog približavanja budućim čitaocima, oni su prilično zgodno sredstvo.

Međutim, šta ako ćelije raka uopšte nisu pobunjenici; šta ako su one samo one somatske ćelije koje se greškom reprogramiraju i vraćaju u primordijalno stanje koje se vezuje za nastanak prvih višćelijskih organizama pre oko dve milijarde godina, a da je njihov patološki status za organizme nosioce, samo sporedni proizvod ćelijskih funkcija specifičnih za geološko razdoblje i posebno nepovoljne uslove sredine njihovog nastanka. Pretpostavlja se da su ove funkcije potisnute pre više stotina miliona godina, ali da je njihova genetička senka ostala zauvek prisutna kod većine višćelijskih organizama ili u obliku gena koji su „isključeni“, ili pak u vidu nekodirajućeg DNK smeća. Iako je ono što je nastalo kao posebna adaptacije, prestalo je da bude adaptivno u dalekoj evolucionoj istoriji, ipak nije potpuno nestalo. U tom smislu, istraživači ove opake bolesti su ustanovili da su geni čije su mutacije odgovorne za nastanak tumora veoma stari, i da bi mogli da vode poreklo upravo od ovih prvih višćelijskih organizama. Ovo je okosnica takozvanog atavističkog modela za razumevanje kancera, koji čini jedan od posebno zanimljivih delova knjige, jer se tematski ne odnosi samo na razumevanje raka već i na razumevanje velikih evolucionih tranzicija, kakva je ona od jednoćelijskih ka višćelijskim organizmima.

S obzirom na to da priroda ne čini skokove, pretpostavlja se da je u velikoj evolucionoj tranziciji između protozoa, prvih jednoćelijskih eukariota i prvih metazoa, pravih višćelijskih organizama, postojao evolucionni međukorak koji je činila prelazna skupina organizama poznata kao protometazoe, za koje je karakteristično da nisu ni jednoćelijski organizmi, niti su pravi potpuno integrisani višćelijski organizmi. To znači da su ćelije takvih organizama mogle jedno vreme da budu delimično integrisane i diferencirane, da bi u sledećem trenutku, ako su okolnosti i uslovi sredine takvi, mogle da nastave svoj život kao samostalni jednoćelijski organizmi. Ovakvo ponašanje ovih organizama bi moglo da bude adaptacija za posebno nepovoljne uslove životne sredine.

To znači da bismo rak mogli da sagledamo kao zaokret od međućelijske saradnje u višećelijskom organizmu, ka primordijalnoj međućelijskoj kompeticiji jednoćelijskih organizama. (Ovde možda nije zgoreg skrenuti pažnji i na to da je bilo, doduše neuspješnih, pokušaja da se prva besmrtna linija ljudskih ćelija dobijena u laboratorijskim uslovima od ćelija tumora grlića materice Henrijete Laks, a uzimajući u obzir određene osobine ovih ćelija, označi kao primer specijacije i proglasi novom biološkom vrstom.)

U istom kontekstu, posebno značajan aspekt knjige je analiza malignih promena iz perspektive selekcije na više nivoa, koju su zagovornici genocentričnog pogleda na evoluciju, a genocentrizam je dugo godina bio dominantni naučnoistraživački program evolucione biologije, često sagledavali kao podlu i posebno opasnu sestru bliznakinju starovremenske i *Vilijamsovom revolucijom* opovrgnute grupne selekcije. Mada se pokazalo da je selekcija na više nivoa matematički ekvivalentna selekciji u srodničkom krugu i da daje jednaka predviđanja, ona ipak naglašava različite perspektive. Dok teorija inkluzivne adaptivne vrednosti prednost daje genetičkom srodstvu, selekcija na više nivoa, osim unutargrupne dinamike, ističe i načine asortativnog grupisanja bioloških entiteta i intergrupnu dinamiku.

Selekcija na više nivoa polazi od pretpostavke da potpuno objašnjenje evolucionog procesa, zahteva da se različiti nivoi selekcije uzmu u razmatranje. Kako različiti nivoi biološke hijerarhije – geni, hromozomi, ćelije, organi, organizmi, itd. – nisu nastali ni od kuda, već su i sami proizvod evolucionog procesa, pretpostavlja se da je prirodna selekcija delovala na više nivoa, iako danas možda ne deluje na svim tim nivoima. Ovaj eksplanatorni model promenio je, između ostalog, i perspektivu u razumevanju pojma „jedinka“. Pod jedinkom se obično razume pojedinačni višećelijski ili jednoćelijski organizam. Ali, ono što iz jedne perspektive može da izgleda kao jedinka, iz drugačije perspektive može da se tumači kao grupa jedinki. Na primer, ako se krećemo linijom biološke hijerarhije odozgo nadole, onda je pojedinačni višećelijski organizam možemo da tumačimo kao koherentnu i funkcionalnu skupinu ćelija. Ove grupe ćelije su u ovom slučaju jedinke u odnosu na organizam, a ćelija je grupa ćelijskih organela, citoplazme, jedra i hromozoma, koje su jedinke u odnosu na ćelije, dok su ćelije sada grupe navedenih elemenata, itd. Ono što na jednom nivou biološke hijerarhije nazivamo organizam, na drugom bi mogao da bude organ ili neki niži stupanj biološke organizacije. Kolonije socijalnih insekata su primer društvene organizacije u kojima grupa tradicionalno shvaćenih jedinki konstituiše entitet koji je, s obzirom na svoje karakteristike, „superorganizam“ i u kome jedinke zapravo funkcionišu kao organi, mada sam ih ja za potrebe ovog teksta tumačio kao ćelije, što možda i nije daleko od istine, jer bi s obzirom na sve do sada rečeno, u društvima socijalnih insekata posebne kaste bi trebalo da predstavljaju organe, odnosno njihove analogone.

Ako se sada zajedno sa autorkom ove knjige okrenemo raku, selekcija na više nivoa je relevantna s obzirom na mogući evolucioni konflikt među različitim nivoima biološke organizacije: naime, kancer nastaje kada je selekcija na ćelijskom nivou

suprotstavljena interesima organizma kao celine. Ali ni to nije sve, tumor predstavlja grupu tumorskih ćelija koja je okružena drugim zdravim ćelijama koje imaju funkciju sličnu onoj koju prirodna sredina ima za organizam, to jest ove zdrave ćelije čine prirodnu sredinu obolelim ćelijama. Dalje, poznato je da tumorske ćelije karakteriše genetička heterogenost, odnosno, ako tumor posmatramo kao grupu tumorskih ćelija, onda je ta grupa sačinjena od podgrupa genetički različitih ćelija. Za svaku podgrupu tumorskih ćelija, ostale podgrupe predstavljaju s jedne strane prirodnu sredinu, a sa druge strane, podgrupe genetički različitih ćelija se nalaze u kooperativnim i konfliktnim odnosima, jer je kompeticija inherentna svakoj biološkoj organizaciji, pa se ćelije tumora onda takmiče i međusobno i sa drugim zdravim ćelijama, itd. Čitalac ove filozofske knjige o raku i njegovom evolucionom poreklu, ne može da se otme utisku i da ne upotrebi tu donekle prokazanu reč – kompleksnost – koja objašnjava sve i ne objašnjava ništa, i služi kao zgodna izlazna strategija iz objašnjavaćakog ćorsokaka. Ali, rak je uistinu složen, neobičan, fascinant i nažalost smrtonosan fenomen, ili bolest, bez obzira na to što je smrt inherentna samom životu, i što je život zapravo krajnji uzrok smrti. Ne zaboravimo, organizmi ne umiru zato što su bolesni, pa ni zato što su bolesni od raka, već zato što su živi, iako ima i onih srećnika koji su preokrećući svoj životni ciklus unazad, pronašli način da umaknu ovoj zloj sudbini.

Ova filozofkinja nas upućuje na koji način proučavanje jedne teške i uglavnom smrtonosne bolesti, koja je s druge strane fascinant život unutar života i čiji su neposredni uzroci brojni, a krajnji uzroci inherentni samoj hijerarhijskoj organizaciji biološkog organizma, može da nam pomogne da načelno razumemo životne procese, kao i evoluciju viševićijskih organizama. Ona čitaoca vodi kroz lavirint popločan istorijskim aspektima filozofije nauke i posebnih naučnih disciplina, pre svega evolucione biologije, pažljivo i nenametljivo osvetljavajući sve njene značajne aspekte od rane darvinovske biologije i populacione genetike s početka dvadesetog veka, preko problema sebičnosti i altruizma u biologiji i s njima blisko povezanim problemom saradnje, do istančanog razumevanja i objašnjenja osnovnih evolucionih pojmova kao što su prirodna selekcija, genetički drift, adaptacija, prava funkcija, itd., do savremenih teorijskih osvrta na velike tranzicije u evoluciji i nastojanja da se biološki fenomeni razumeju na nivoima koji su jedno vreme, opravdano ili neopravdano, bili zanemareni.

S obzirom na to da je pisana jednostavnim, jasnim i razumljivim jezikom, pristupačnim stilom i bez pretencioznih jezičkih mistifikacija, ona je namenjena ne samo evolucionistima, filozofima upućenim u filozofiju biologije ili lekarima i naučnicima koji se bave rakom, već i obrazovanim laicima zainteresovanim za filozofiju nauke, biologiju i medicinu. Ovim ne bih voleo da čitaoca dovedem u zabludu. Iz činjenice njene parsimoničnosti ne sledi da je ovo knjiga za svakoga, jer njeno čitanje i razumevanje, zbilja iziskuje izvesna prethodna znanja ili bar obaveštenost o problemima kojima se bavi. To pogotovo važi za središnji deo knjige koji je nešto kompleksniji i zahteva posebnu budnost i predznanje. Osim samog sadržaja, poseban kvalitet ove

knjige predstavljaju podnaslovi, koji su veoma zanimljivi, ilustrativni, a poneki i prilično uznemirujući. Na primer, „Duboko u tebi“, da izdvojim samo jedan kojim se opisuje duboka evolucionarna istorija kancera, ne može, a da ne izazove nelagodu, blagi osećaj nesigurnosti i prikrivene pretnje.

Na kraju, neophodna je jedna važna napomena, s obzirom na često postavljano pitanje zašto se filozofi uopšte bave rakom, ili bilo kojom drugom oblašću medicine ili ćelijske biologije, kao posebnih naučnih disciplina, imajući na umu da ekspertiza lekara i biologa umnogome nadmašuje ekspertizu filozofa, koji se mahom bave ontološkim i epistemičkim statusom određenih fenomena ili problemima pojmovne analize, da pomenem samo neke posebno istaknute oblasti filozofije. Kako, onda, oni koji su tako udaljeni od svakidašnjeg posla jednog onkologa ili drugih naučnika koji se bave biomedicinskim istraživanjima, uopšte mogu da doprinesu razumevanju ili borbi protiv ove opake bolesti i zašto se uopšte bave nečim što prevazilazi domen metafizike i opštih problema saznanja, jezika, logike i značenja. Tvrdi se da filozofija ne može da dobaci do onih teorijskih visina, niti pak do kliničkih dubina do kojih može da dopre medicina i da stoga može da pruži samo nedovoljno dobre rezultate u pogledu potencijalnog razumevanja ili doprinosa razvoju uspešnih terapija. Načelno to je tačno, kada je reč o ovako složenim temama važnim za ljudsko zdravlje, u sudaru s medicinom filozofija uglavnom gubi, kao što je trenutni uspon eksperimentalne filozofije, manje ili više doživeo prinudno prizemljenje u sudaru s psihologijom i drugim posebnim naukama. Iako imam određene simpatije prema eksperimentalnoj filozofiji, ni sam ne mogu da se otmem utisku da su u podeli naučnog rada neki naučnici jednostavno bolje obučeni za dizajniranje eksperimenta i baratanje statističkim podacima.

U tom smislu filozofija kancera ne može da zameni biomedicinska istraživanja. Međutim, u pogledu njihovog domašaja, ambicije filozofa se razlikuju od ambicija lekara ili biologa. Verujem, nadam se opravdano, da filozofi koji se dohvate ovih tema ne ciljaju na to da donesu velike prodore u razvijanju terapija i lečenju bolesti. To, na kraju krajeva, uopšte i nije njihov posao, niti samostalan rad bilo kog naučnika ili filozofa može taj posao da privede kraju, koji se nažalost i dalje ne nazire. Gotovo je sasvim izvesno da nijedan od filozofa koji se bavi filozofijom kancera neće da dobije Nobelovu nagradu za medicinu ili neko slično priznanje, ali njihov rad može da dovede do promene perspektive potencijalnih laureata, a malim koracima bi mogao da vodi i ka promeni naučne paradigme. Konačno, ono čega bismo svi morali da budemo svesni i što je posebno važno jeste sinergija i zajednički napor naučnika i filozofa u potrazi za potencijalnim rešenjima problema koji ljudima i dalje ledi krv u žilama.

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